



# XR806 IC 用户手册

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## 版本历史

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# 1 前言

## 1.1 文档简介

本文档介绍了关于 XR806 的具体信息，包括微控制器、存储器、外设模块的功能和寄存器信息的完整描述。芯片的管脚描述、电气特性和封装信息等可以从 XR806 Datasheet 获取。

## 1.2 目标读者

使用 XR806 的应用开发人员。

## 1.3 适用范围

此文档适用于支持 XR806 系列所有芯片产品。

## 1.4 文档约定

### 1.4.1 标志说明

本文档采用各种醒目的标志来表示在操作过程中应该特别注意的地方，这些标志的含义如下：

| 标识                                                                                     | 说明                                        |
|----------------------------------------------------------------------------------------|-------------------------------------------|
|  警告 | 该标志后的说明应给予格外关注，如果不遵守，可能会导致人员受伤或死亡。        |
|  注意 | 提醒操作中应注意的事项。不当的操作可能会损坏器件，影响可靠性、降低性能等。     |
|  说明 | 为准确理解文中指令、正确实施操作而提供的补充或强调信息。              |
|  窍门 | 一些容易忽视的小功能、技巧。了解这些功能或技巧能帮助解决特定问题或者节省操作时间。 |

### 1.4.2 地址与数据描述方法约定

本文档在描述地址、数据时遵循如下约定：

| 符号 | 例子                  | 说明                                                                   |
|----|---------------------|----------------------------------------------------------------------|
| 0x | 0x0200, 0x79        | 地址或数据以 16 进制表示。                                                      |
| 0b | 0b010, 0b00 000 111 | 数据采用二进制表示(寄存器描述除外)。                                                  |
| X  | 00X, XX1            | 数据描述中，X 代表 0 或 1。<br>例如，00X 代表 000 或 001；XX1 代表 001, 011, 101 或 111。 |

### 1.4.3 数值单位约定

本文档在描述数据容量（如 NAND 容量）时，单位词头代表的是 1024 的倍数；描述频率、数据速率等时则代表的是 1000 的倍数。具体如下：

| 类型              | 符号  | 对应数值          |
|-----------------|-----|---------------|
| 数据容量（如 NAND 容量） | 1 K | 1024          |
|                 | 1 M | 1 048 576     |
|                 | 1 G | 1 073 741 824 |
| 频率，数据速率等        | 1 k | 1000          |
|                 | 1 M | 1 000 000     |
|                 | 1 G | 1 000 000 000 |

## 2 系统和存储

### 2.1 XR806 应用子系统

#### 2.1.1 概述

XR806 应用子系统采用了带有 Trustzone-M 功能的 Cortex M33 Star 微控制器（ARMv8-M 架构），支持时钟频率从 32KHz 到 160MHz，集成了 320KB SRAM 和 160KB ROM，同时支持通过 SQPI 接口扩展外部 Flash 容量至最高 16MB，允许代码直接从 Flash 和 PSRAM 内执行，允许读写数据在 PSRAM 中。它还包括许多外围设备，包括 UART, TWI, SPI, PWM, IrDA (T/R)和 GPADC 等。

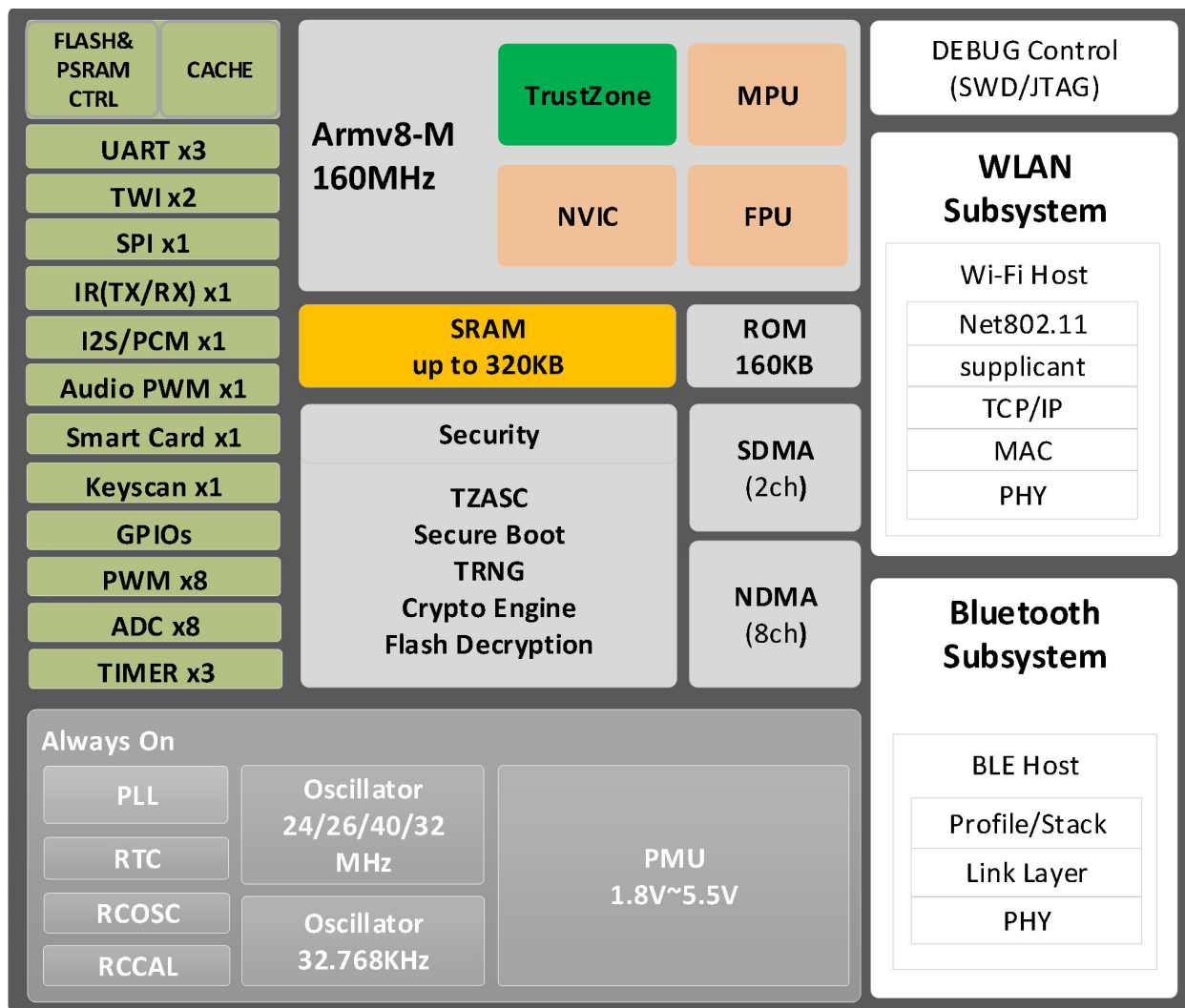
Wi-Fi 子系统包含 802.11b/g/n 基带、MAC、集成了 PA、LNA、Switch 及谐波滤波器的射频部分，可以满足低功耗、高集成度和高性能的网络应用。采用 XRADIOTECH MPD™ 技术设计了一种新型的数字射频发射机，以提供更高的输出功率并保持较高的发射效率，同时保持芯片对天线失配不那么敏感，使得在不同驻波比下始终具有良好的 EVM。

蓝牙子系统包含 Bluetooth 5.0 标准的全部特性，包括射频、调制解调器、基带和规范等。针对长距离和低功耗的应用特别是智能家居网络连接做了专门优化，高性能和更大的 SRAM 允许简单的组网网关设计与集成的 Wi-Fi 系统直接连接到互联网。

为低功耗嵌入式网络应用而设计的 SoC，集成了网络处理器，包含了大量 TCP/IP 协议的 IPv4/IPv6 基础服务集。可以通过连接到外部主机 CPU 的串行 UART/SPI 链路访问这些服务。

## 2.1.2 功能描述

### 2.1.2.1 系统框图



### 2.1.2.2 关键特性

| 特性                   | 描述                                                         | XR806AF2L                              | XR806BF2L          | XR806BM2I          |
|----------------------|------------------------------------------------------------|----------------------------------------|--------------------|--------------------|
| Package              | Trays and tape-in-reel                                     | 4x4mm <sup>2</sup>                     | 5x5mm <sup>2</sup> | 5x5mm <sup>2</sup> |
|                      |                                                            | QFN32                                  | QFN40              | QFN40              |
| Supply voltage       | Power supply from system                                   | 1.8~5.5V                               |                    |                    |
|                      | 5V IO                                                      | Yes                                    |                    |                    |
| PMU                  | LDO for external device (EXT LDO)                          | Yes                                    |                    |                    |
| Clock                | External HOSC                                              | 24/26/32/40MHz                         |                    |                    |
|                      | External LOSC                                              | 32.768KHz                              |                    |                    |
|                      | Internal RCOSC                                             | 30~60KHz                               |                    |                    |
|                      | Internal RCOSC Calibration                                 | YES                                    |                    |                    |
| MCU Core             | Core Type                                                  | Cortex M33 Star (Armv8-M Architecture) |                    |                    |
|                      | Core clock maximum frequency                               | 160MHz                                 | 160MHz             | up to 240MHz       |
| Memory               | Internal ROM                                               | 160KB                                  |                    |                    |
|                      | Internal RAM                                               | up to 320KB                            |                    |                    |
|                      | Internal Flash with XIP                                    | 2MB                                    | 2MB                | -                  |
|                      | External Flash with XIP                                    | Max. 16MB                              |                    |                    |
|                      | Internal PSRAM                                             | -                                      | -                  | 2MB                |
| Backup register      | Backup register for power save                             | 16B                                    |                    |                    |
| Secure boot          | -                                                          | Yes                                    |                    |                    |
| Trustzone-M          | Default size: 28KB                                         | Yes                                    |                    |                    |
| Crypto Engine        | AES, DES, 3DES, SHA-1, MD5, TRNG, CRC32/16, SHA256         | Yes                                    |                    |                    |
| TRNG                 | Provide random number seed                                 | Yes                                    |                    |                    |
| WDG reset protection | Protect specified peripherals been reset by watchdog reset | Yes                                    |                    |                    |
| BOR                  | BOR Detection                                              | Yes                                    |                    |                    |
| Wi-Fi                | 802.11 b/g/n                                               | Yes                                    |                    |                    |
| Bluetooth            | BLE 5.0                                                    | Yes                                    |                    |                    |
|                      | BLE Configure                                              | Yes                                    |                    |                    |
|                      | BLE Mesh                                                   | Yes                                    | Yes                | Yes                |
| Peripheral           | 1.8/3.3/5V GPIO                                            | General Purpose                        | 4                  | 4                  |

| 特性 | 描述           |                  | XR806AF2L | XR806BF2L | XR806BM2I |
|----|--------------|------------------|-----------|-----------|-----------|
|    | 1.8/3.3 GPIO | General Purpose  | 14        | 22        | 22        |
|    | WAKEUP IO    | From RTC wake-up | 8         | 10        | 10        |
|    | UART         | Max.3Mbps        | 3         | 3         | 3         |
|    | SPI          | Master and slave | 1         | 1         | 1         |
|    | I2C          | Max.400Kbps      | 2         | 2         | 2         |
|    | IR           | TX and RX        | 1         | 1         | 1         |
|    | DAudio       | IIS/PCM          | 1         | 1         | 1         |
|    | Audio PWM    | Playback Channel | 1         | 1         | 1         |
|    | PWM          | Output Channel   | 8         | 8         | 8         |
|    | KEYSCAN      | 8*16             | -         | YES       | YES       |
|    | Smart Card   | ISO7816          | 1         | 1         | 1         |
|    | GPADC        | VBAT channel     | 1         | 1         | 1         |
|    |              | Normal channel   | 3         | 7         | 7         |
|    | NDMA         | 8 Channel        | YES       | YES       | YES       |
|    | SDMA         | 2 Channel        | YES       | YES       | YES       |
|    | Timer        | RTC timer        | 1         | 1         | 1         |
|    |              | Watchdog timer   | 1         | 1         | 1         |
|    |              | Normal timer     | 2         | 2         | 2         |
|    |              | Wakeup timer     | 1         | 1         | 1         |



### 2.1.2.3 NVIC 寄存器列表

关于系统控制、系统定时器（System Tick）、可嵌套中断向量（NVIC）等详细信息的描述，可参考《ARM Cortex-M33 Technology Reference Manual》。以下仅对中断向量部分做简要描述。

| Module Name                          | Base Address    |                                     |
|--------------------------------------|-----------------|-------------------------------------|
| Nested Vectored Interrupt Controller | 0xE000E000      |                                     |
| Register Name                        | Offset Address  | Description                         |
| SYSTICK_CTRL                         | 0x0010          | SysTick Control and Status Register |
| SYSTICK_RELOAD                       | 0x0014          | SysTick Reload Value Register       |
| SYSTICK_CURVAL                       | 0x0018          | SysTick Current Value Register      |
| NVIC_ISER0 – NVIC_ISER7              | 0x0100 – 0x011C | Interrupt Set-Enable Registers      |
| NVIC_ICER0 – NVIC_ICER7              | 0x0180 – 0x019C | Interrupt Clear-Enable Registers    |
| NVIC_ISPR0 – NVIC_ISPR7              | 0x0200 – 0x021C | Interrupt Set-Pending Registers     |
| NVIC_ICPR0 – NVIC_ICPR7              | 0x0280 – 0x029C | Interrupt Clear-Pending Registers   |
| NVIC_IABR0 – NVIC_IABR7              | 0x0300 – 0x031C | Interrupt Active Bit Registers      |
| NVIC_IPR0 – NVIC_IPR59               | 0x0400 – 0x04CF | Interrupt Priority Registers        |
| SCB                                  | 0x0D08 – 0x0D3f | System Control Block                |

### 2.1.3 中断寄存器描述

| 中断号 | 向量   | 中断源    | 优先级          | 描述                                                             |
|-----|------|--------|--------------|----------------------------------------------------------------|
| 0   | 0x00 | -      | -            | Stack top is loaded from first entry of vector table on reset. |
| 1   | 0x04 | -      | -3           | Reset                                                          |
| 2   | 0x08 | -      | -2           | NMI                                                            |
| 3   | 0x0C | -      | -1           | Hardware Fault                                                 |
| 4   | 0x10 | -      | configurable | Memory Management                                              |
| 5   | 0x14 | -      | configurable | Bus Fault                                                      |
| 6   | 0x18 | -      | configurable | Usage Fault                                                    |
| 7   | 0x1C | -      | -            | -                                                              |
| 8   | 0x20 | -      | -            | -                                                              |
| 9   | 0x24 | -      | -            | -                                                              |
| 10  | 0x28 | -      | -            | -                                                              |
| 11  | 0x2C | -      | configurable | SVCall                                                         |
| 12  | 0x30 | -      | configurable | Debug Monitor                                                  |
| 13  | 0x34 | -      | -            | -                                                              |
| 14  | 0x38 | -      | configurable | PendSV                                                         |
| 15  | 0x3C | -      | configurable | System Tick                                                    |
| 16  | 0x40 | DMA    | configurable | Direct Memory Access Controller                                |
| 17  | 0x4C | MSGBOX | configurable | Message Box(IRQ0)                                              |
| 18  | 0x50 | UART0  | configurable | UART Controller 0                                              |
| 19  | 0x54 | UART1  | configurable | UART Controller 1                                              |
| 20  | 0x58 | SPI0   | configurable | SPI Controller 0                                               |
| 21  | 0x60 | TWI0   | configurable | TWI Controller 0                                               |
| 22  | 0x64 | TWI1   | configurable | TWI Controller 1                                               |
| 23  | 0x68 | WDT    | configurable | Watchdog                                                       |
| 24  | 0x6C | TIMER0 | configurable | Timer 0                                                        |
| 25  | 0x70 | TIMER1 | configurable | Timer 1                                                        |
| 26  | 0x74 | ALARM0 | configurable | Alarm 0                                                        |
| 27  | 0x78 | ALARM1 | configurable | Alarm 1                                                        |

|    |      |           |              |                                     |
|----|------|-----------|--------------|-------------------------------------|
| 28 | 0x80 | DAUDIO    | configurable | Digital Audio Controller            |
| 29 | 0x84 | PWM/ECT   | configurable | PWM/ECT Controller                  |
| 30 | 0x88 | SS        | configurable | Secure System Controller            |
| 31 | 0x8C | GPADC     | configurable | GPADC Controller                    |
| 32 | 0x90 | GPIOB     | configurable | GPIO Controller Port B              |
| 33 | 0x98 | IRRX      | configurable | IR Receiver                         |
| 34 | 0x9C | IRTX      | configurable | IR Transmitter                      |
| 35 | 0XA4 | A-WUP     | configurable | APP Wake Up(Timer & IO)             |
| 36 | 0xA8 | FLASHC    | configurable | FLASH Controller                    |
| 37 | 0xAC | UART2     | configurable | UART Controller 2                   |
| 38 | 0xB4 | WLAN      | configurable | WLAN                                |
| 39 | 0xB8 | Audio PWM | configurable | AUDIO PWM                           |
| 40 | 0XC0 | AVS       | configurable | AVS psensor                         |
| 41 | 0XCC | BLE_LL    | configurable | BLE_LL                              |
| 42 | 0XD0 | BTCOEX    | configurable | BTCOEX                              |
| 43 | 0XD4 | KEYSCAN   | configurable | Key scan controller                 |
| 44 | 0XD8 | SMCARD    | configurable | ISO7816 SIM CARD interface          |
| 45 | 0XDC | DMA_SEC   | configurable | Secure DMA memory access controller |
| 46 | 0XE4 | LPUART0   | configurable | LPUART0 wakeup controller           |
| 47 | 0XE8 | LPUART1   | configurable | LPUART1 wakeup controller           |
| 48 | 0XEC | SEC_FAULT | configurable | Security access fault case          |
| 49 | 0XF0 | RCCAL     | configurable | RCOSC calibration                   |



## 说明

1. 详细信息请参考《ARM Cortex-M33 Technology Reference Manual》。

## 2.2 系统控制模块（SYSCTRL）

### 2.2.1 概述

系统控制模块主要包含了部分 SIP FLASH/PSRAM PIN Mapping、Psensor、Debug Register 等信息。

### 2.2.2 SYSCTRL 寄存器列表

| 模块名                     | 基地址        |                                          |
|-------------------------|------------|------------------------------------------|
| System Control          | 0x4000A000 |                                          |
| 寄存器名                    | 偏移地址       | 描述                                       |
| SIP_FLASH_TEST_MAP_REG  | 0x0044     | SIP FLASH Test Mapping Register          |
| PS_CTL_REG              | 0x0050     | Psensor control register                 |
| PS_CNT_REG              | 0x0054     | Psensor counter register                 |
| FLASH_PSARM_PIN_MAP_REG | 0x0060     | FLASH/PSRAM PIN Mapping Control Register |
| GENERAL_DBG0_REG        | 0x00B0     | General Debug 0 Register                 |
| GENERAL_DBG1_REG        | 0x00B4     | General Debug 1 Register                 |

### 2.2.3 SYSCTRL 寄存器描述

#### 2.2.3.1 SIP\_FLASH\_TEST\_MAP\_REG

| 偏移地址：0x0044 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 31:16       | RW | 0x0 | <p>0x429d: SIP_FLASH_TEST/ SIP_PSRAM_TEST filed is writable<br/>other: SIP_FLASH_TEST/ SIP_PSRAM_TEST filed is read only</p> <p>Note: The field of SIP_FLASH_TEST/SIP_PSRAM_TEST only can be changed when these bits are set to 0x429D. In one write operation, the GRPCM will check this field whether equals to 0x429D. If not, the value of SIP_FLASH_TEST/SIP_PSRAM_TEST will be ignored.</p> <p>This field is always read with value 0x0000.</p> <p>Examples:<br/>After write 0x429d0001, then read value = 0x00000001<br/>After write 0x11110000, then read value = 0x00000001<br/>After write 0x429d0000, then read value = 0x00000000</p> |
| 15:1        | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0           | RW | 0x0 | <p>SIP_FLASH_TEST</p> <p>0: SIP FLASH IOs mapping is disabled<br/>1: SIP FLASH IOs are mapped to the external flash PINs</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### 2.2.3.2 PS\_CTL\_REG

| 偏移地址: 0x0050 |      |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------|------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置           | 访问   | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 31:17        | /    | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 16           | RW1C | 0x0 | CLK250M_CNT_RDY<br>clk250M counter ready<br>0: not finish<br>1:finish                                                                                                                                                                                                                                                                                                                                                                        |
| 15:11        | /    | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 10:8         | R/W  | 0x0 | PS_DEV_SEL<br>psensor device sel<br>000: select psensor_0<br>001: select psensor_1<br>Others: reserved                                                                                                                                                                                                                                                                                                                                       |
| 7:6          | /    | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5:4          | R/W  | 0x0 | PS_EN_SEL<br>OSC select<br>00:disable(no dynamic power)<br>01 : rvt_casecode ring osc<br>10: lvt_casecode ring osc<br>11:rvt_normal ring osc                                                                                                                                                                                                                                                                                                 |
| 3:1          | R/W  | 0x0 | PS_N_PRD<br>clk num period.<br>0: 4 --- 2/13 us (26M HOSC), 0.1us (40M HOSC)<br>1: 8 --- 4/13 us (26M HOSC) , 0.2us (40M HOSC)<br>2: 16---8/13 us (26M HOSC) , 0.4us (40M HOSC)<br>3: 32---16/13us (26M HOSC) , 0.8us (40M HOSC)<br>4: 64---32/13 us (26M HOSC) , 1.6us (40M HOSC)<br>5: 128---64/13 us (26M HOSC) , 3.2us (40M HOSC)<br>6: 256---128/13us (26M HOSC) , 6.4us (40M HOSC)<br>7: 512---256/13us (26M HOSC) , 12.8us (40M HOSC) |
| 0            | R/W  | 0x0 | PS_EN<br>psensor Enable.<br>0: disable<br>1: psensor enable                                                                                                                                                                                                                                                                                                                                                                                  |

### 2.2.3.3 PS\_CNT\_REG

| 偏移地址 : 0x0054 |    |     | 默认值: 0x0000_0000                |
|---------------|----|-----|---------------------------------|
| 位置            | 访问 | 默认值 | 描述                              |
| 31:16         | /  | /   | /                               |
| 15:0          | R  | 0x0 | CLK250M_CNT<br>clk 250M counter |

### 2.2.3.4 FLASH\_PSARM\_PIN\_MAP\_REG

| 地址 : 0x0060 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                              |
|-------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                            |
| 31:26       | /  | /   | /                                                                                                                                                                                                                                                                             |
| 25:24       | RW | 0x0 | mode_5vbias<br>Mode control of 5V IO for PB0/1/14/15                                                                                                                                                                                                                          |
| 23:18       | /  | /   | /                                                                                                                                                                                                                                                                             |
| 17:16       | RW | 0x2 | FLASH_PSARM_PIN_MAP1<br>When Flash/Psram access on GPIOB08~GPIOB11 PINs, which controller Chip Select (CS) is corresponding<br>0: reserved<br>1: only CS0 of Flash/Psram controller<br>2: only CS1 of Flash/Psram controller<br>3: both CS0 and CS1 of Flash/Psram controller |
| 15:2        | /  | /   | /                                                                                                                                                                                                                                                                             |
| 1:0         | RW | 0x1 | FLASH_PSARM_PIN_MAP0<br>When Flash/Psram access on GPIOB02~GPIOB05 PINs, which controller Chip Select (CS) is corresponding<br>0: reserved<br>1: only CS0 of Flash/Psram controller<br>2: only CS1 of Flash/Psram controller<br>3: both CS0 and CS1 of Flash/Psram controller |

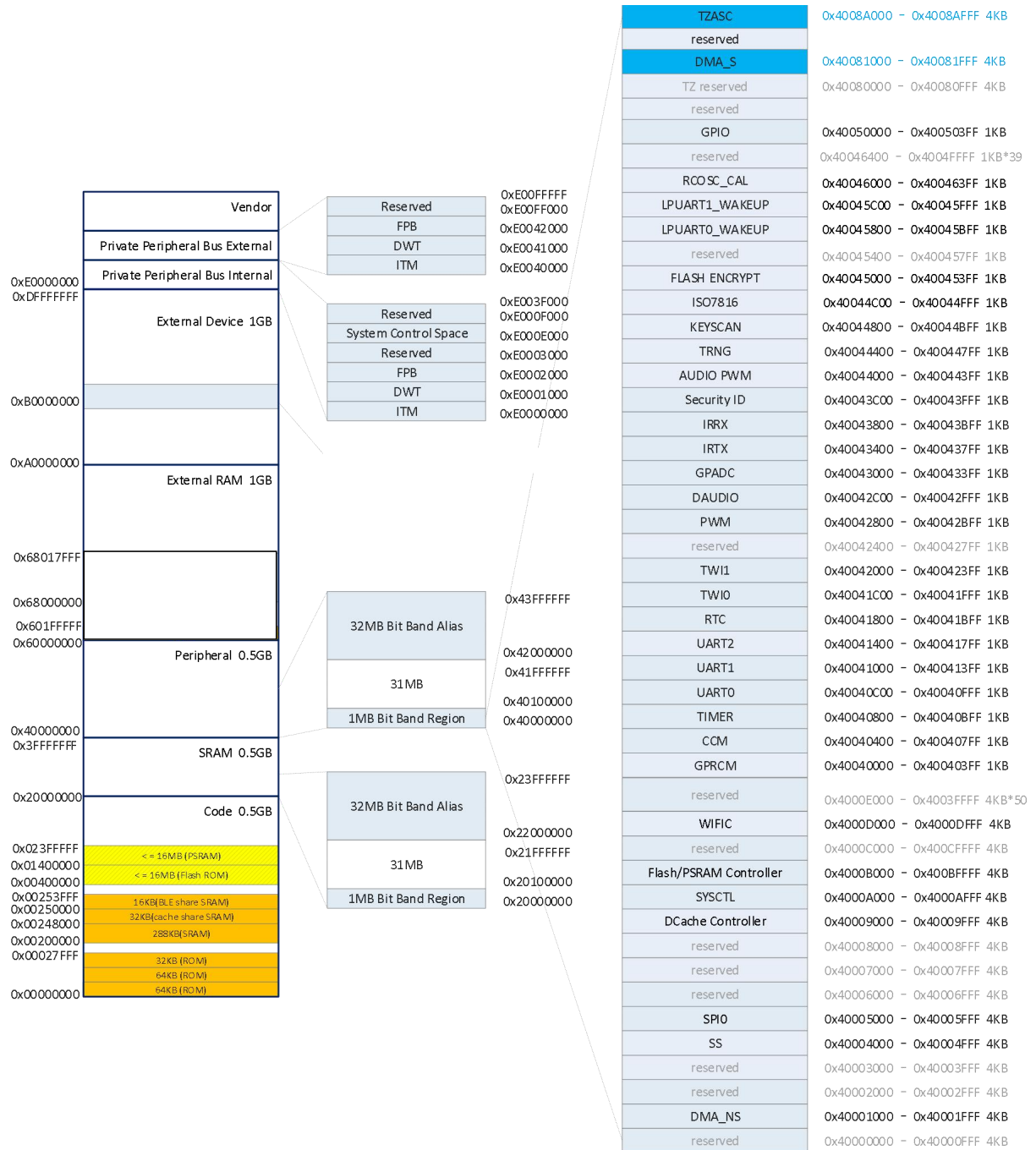
### 2.2.3.5 GENERAL\_DBG0\_REG

| 偏移地址 : 0x00B0 |    |     | 默认值: 0x0000_0000                                                               |
|---------------|----|-----|--------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                             |
| 31:00         | RW | 0   | GENERAL_DBG0<br>General debug register 0 is used for storing flag or some data |

### 2.2.3.6 GENERAL\_DBG1\_REG

| 偏移地址 : 0x00B4 |    |     | 默认值: 0x0000_0000                                                               |
|---------------|----|-----|--------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                             |
| 31:00         | RW | 0   | GENERAL_DBG1<br>General debug register 1 is used for storing flag or some data |

## 2.3 存储空间映射 (Memory Mapping)





## 2.4 存储控制器（Memory Controller）

### 2.4.1 概述

Memory Controller 兼容 Standard SPI /Dual SPI/Quad SPI/QPI(Quad Peripheral Interface)/PSRAM(SQPI)。

Memory Controller 通过 System bus 来进行 Flash 的擦除、读写 Flash/PSRAM、配置内部寄存器。通过 Code bus 读 Flash、读写 PSRAM。System bus 及 Code bus 都是 AHB 接口。SBUS 操作与 CBUS 操作不能同时发生。

Memory Controller 包含 SQPI Controller,支持对 SPI NOR FLASH 进行数据读写的操作控制,支持 16Mb SQPI PSRAM。

Memory Controller 的 IO 口包括：数据端口 IO0~IO3, CS\_FLASH, CS\_PSRAM。Flash 和 PSRAM 共用数据端口,通过 CS 片选,IO 电压必须一致,常见有 3.3V、1.8V。

主要特性如下：

- 支持 SPI 1/2/4 线模式收发
- 支持 SPI Mode0/1/2/3
- 支持 Flash 容量最大  $2^{32}$ Bytes, 常见有 64Mb, 128Mb, 256Mb
- 支持 CPU/DMA 通过系统总线对 Flash 的烧写和读取
- 支持 Flash XIP (Continuous Read)模式（进入/退出）
- 支持 Flash/PSRAM Wrap 模式（进入/退出）
- 支持常见 SPI 接口 Flash 的基本操作
- 支持 16Mb SQPI PSRAM
- 支持异常保护和异常中断
- 支持自动时钟切换

### 2.4.2 编程指引

以下为 Memory Controller 操作注意事项。

1. 合并 PSRAM SQPI(16M)和 FLASH, 共用一个控制器, 两路 CS, 复用一路 IO0~IO3, 支持 FLASH CBUS 读, PSRAM/FLASH CBUS 读/写, SBUS 读/写/发送命令；
2. SBUS 操作时, 需要设置 C\_RW\_EN 为低电平, 再拉高 START\_SEND, 否则触发 NOP 中断；
3. PSRAM SQPI 和 Flash 写共用 wr\_buf 和 wr\_fifo, SBUS 读共用 rd\_buf 和 rd\_fifo。意味着软件 SBUS 操作时, 需要保证切换器件时候 FIFO 为空, 否则可能造成错误。CBUS 的由硬件保证；
4. PSRAM CBUS 访问地址为：0x01400000-23FFFFFF；Flash: 0x00400000 - 0x13FFFFFF。通过 Flash or PSRAM Select 寄存器选择 SBUS 操作器件；
5. 支持 CBUS 单器件访问。设置寄存器 single\_device\_en 和 single\_device\_sel 进行单器件使能和器件选择, 使能后两片地址区间都对 Flash 或 PSRAM 进行操作；
6. 0x00~0x08 为通用设置, 同时对 Flash 和 PSRAM 生效。其他只对 Flash/PSRAM 生效的寄存器已在寄存器描述中标注；

7. CS 拉低时，进行切时钟。典型值：Flash 为 96MHz，PSRAM 为 133MHz。clk\_sel\_en 寄存器默认为 1，设成 0 则禁止切时钟，运行时钟一直为 flash 运行时钟；
8. XIP（Continuous Read Mode）只能用于 Flash，Read\_Latency/Tcem 只用于 PSRAM；
9. PSRAM wrap 功能设置方法与 Flash 一致；
10. Flash 部分命令需要配置 read\_dummy，write\_dummy 作为扩展功能，暂不需要使用也不需要配置 read\_latency；PSRAM 部分命令需要配置 read\_latency，read\_dummy 和 write\_dummy 作为扩展功能，暂不需要使用；

### 2.4.3 Memory Controller 寄存器列表

| 模块名                      | 基地址        |                                               |
|--------------------------|------------|-----------------------------------------------|
| Memory Controller        | 0x4000B000 |                                               |
| 寄存器名                     | 偏移地址       | 描述                                            |
| MEM_COM_CFG_REG          | 0x000      | Memory Controller Common Configuration        |
| SQPI_COM_CFG_REG         | 0x004      | SQPI Controller Common Configuration          |
| CACHE_CFG_REG            | 0x008      | Cache Line Length Configuration               |
| MEM_AC_CFG_REG           | 0x00C      | Memory AC Character Timing Configuration      |
| FLASH_C_READ_CFG_REG     | 0x010      | Flash Code-bus Read Operation Configuration   |
| FLASH_C_WRITE_CFG_REG    | 0x014      | Flash Code-bus Write Operation Configuration  |
| FLASH_C_RD_DUMMY_H_REG   | 0x018      | Flash Code-bus Read Dummy Data Top Half       |
| FLASH_C_RD_DUMMY_L_REG   | 0x01C      | Flash Code-bus Read Dummy Data Bottom Half    |
| FLASH_C_WR_DUMMY_H_REG   | 0x020      | Flash Code-bus Write Dummy Data Top Half      |
| FLASH_C_WR_DUMMY_L_REG   | 0x024      | Flash Code-bus Write Dummy Data Bottom Half   |
| FLASH_C_IO_SWIT_TIME_REG | 0x028      | FLASH Code-bus IO Switch Wait Time            |
| S_RW_CFG_REG             | 0x02C      | System-bus Read/Write Operation Configuration |
| S_ADDR_CFG_REG           | 0x030      | System-bus Address Configuration              |
| S_DUMMY_H_REG            | 0x034      | System-bus Dummy Data Top Half                |
| S_DUMMY_L_REG            | 0x038      | System-bus Dummy Data Bottom Half             |
| S_IO_SWIT_TIME_REG       | 0x03C      | System-bus IO Switch Wait Time                |
| S_WR_NUM_REG             | 0x040      | System-bus Write Byte Number                  |
| S_RD_NUM_REG             | 0x044      | System-bus Read Byte Number                   |

|                        |       |                                       |
|------------------------|-------|---------------------------------------|
| START_SEND_REG         | 0x048 | System-bus Start Send Register        |
| FIFO_TRIG_LEV_REG      | 0x04C | FIFO Trigger Level                    |
| FIFO_STA_REG           | 0x050 | FIFO Status Register                  |
| INT_EN_REG             | 0x054 | Interrupt Enable Register             |
| INT_STA_REG            | 0x058 | Interrupt Status Register             |
| FLASH_MOD_EXE_IND_REG  | 0x05C | XIP/WRAP Mode Executed Indication     |
| START_ADDR0_REG        | 0x080 | Address Field0 Start Position         |
| END_ADDR0_REG          | 0x084 | Address Field0 End Position           |
| BIAS_ADDR0_REG         | 0x088 | Address Field0 Bias                   |
| START_ADDR1_REG        | 0x090 | Address Field1 Start Position         |
| END_ADDR1_REG          | 0x094 | Address Field1 End Position           |
| BIAS_ADDR1_REG         | 0x098 | Address Field1 Bias                   |
| START_ADDR2_REG        | 0x0A0 | Address Field2 Start Position         |
| END_ADDR2_REG          | 0x0A4 | Address Field2 End Position           |
| BIAS_ADDR2_REG         | 0x0A8 | Address Field2 Bias                   |
| START_ADDR3_REG        | 0x0B0 | Address Field3 Start Position         |
| END_ADDR3_REG          | 0x0B4 | Address Field3 End Position           |
| BIAS_ADDR3_REG         | 0x0B8 | Address Field3 Bias                   |
| START_ADDR4_REG        | 0x0C0 | Address Field4 Start Position         |
| END_ADDR4_REG          | 0x0C4 | Address Field4 End Position           |
| BIAS_ADDR4_REG         | 0x0C8 | Address Field4 Bias                   |
| START_ADDR5_REG        | 0x0D0 | Address Field5 Start Position         |
| END_ADDR5_REG          | 0x0D4 | Address Field5 End Position           |
| BIAS_ADDR5_REG         | 0x0D8 | Address Field5 Bias                   |
| S_WDATA_REG_REG        | 0x100 | System-bus Write Data Register        |
| S_RDATA_REG_REG        | 0x200 | System-bus Read Data Register         |
| PSRAM_COM_CFG_REG      | 0x800 | PSRAM Common Configuration            |
| PSRAM_CTRL_COM_CFG_REG | 0x804 | PSRAM Controller Common Configuration |
| PSRAM_CACHE_CFG_REG    | 0x808 | PSRAM Cache Configuration             |

|                          |       |                                              |
|--------------------------|-------|----------------------------------------------|
| PSRAM_C_READ_CFG_REG     | 0x810 | PSRAM Code-bus Read Operation Configuration  |
| PSRAM_C_WRITE_CFG_REG    | 0x814 | PSRAM Code-bus Write Operation Configuration |
| PSRAM_C_RD_DUMMY_H_REG   | 0x818 | PSRAM Code-bus Read Dummy Data Top Half      |
| PSRAM_C_RD_DUMMY_L_REG   | 0x81C | PSRAM Code-bus Read Dummy Data Bottom Half   |
| PSRAM_C_WR_DUMMY_H_REG   | 0x820 | PSRAM Code-bus Write Dummy Data Top Half     |
| PSRAM_C_WR_DUMMY_L_REG   | 0x824 | PSRAM Code-bus Write Dummy Data Bottom Half  |
| PSRAM_C_IO_SWIT_TIME_REG | 0x828 | PSRAM Code-bus IO Switch Wait Time           |
| PSRAM_FORCE_CFG_REG      | 0x86C | PSRAM Force Configure Register               |
| PSRAM_COM_CFG_REG        | 0x870 | PSRAM Common Configure Register              |

## 2.4.4 Memory Controller 寄存器描述

### 2.4.4.1 MEM\_COM\_CFG\_REG

| 偏移地址：0x0000 |    |     | 默认值：0xB555_0082                                                                                                                                            |
|-------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                                         |
| 31:30       | RW | 10  | Flash Address Size Mode<br>00: Address Size 8bit.<br>01: Address Size 16bit.<br>10: Address Size 24bit.<br>11: Address Size 32bit.                         |
| 29:28       | RW | 11  | IO1 Port Vacancy Default Output Value.<br>00: Output 0.<br>01: Output 1.<br>1x: Output z.<br>Note: IO1 Port Vacancy Output Value When Flash At Send State. |
| 27:26       | RW | 01  | IO2 Port Vacancy Default Output Value.<br>00: Output 0.<br>01: Output 1.<br>1x: Output z.<br>Note: IO2 Port Vacancy Output Value When Flash At Send State. |
| 25:24       | RW | 01  | IO3 Port Vacancy Default Output Value.<br>00: Output 0.                                                                                                    |

|       |    |   |                                                                                                                                                                                                                                                                   |
|-------|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |    |   | 01: Output 1.<br>1x: Output z.<br>Note: IO3 Port Vacancy Output Value When Flash At Send State.                                                                                                                                                                   |
| 23:16 | /  | / |                                                                                                                                                                                                                                                                   |
| 15:8  | RW | 0 | System-bus HREADY Adjustable Wait Time Out<br>Note:Default Value Indicates Wait Time Out Level =0xFFFF                                                                                                                                                            |
| 7     | RW | 1 | System-bus HREADY Time Out Enable<br>0: Disable<br>1: Enable                                                                                                                                                                                                      |
| 6     | RW | 0 | Transfer FIFO Reset<br>0: Auto Clear to 0.<br>1: Reset Transfer FIFO.<br><br>Note: Write ‘1’ to this bit will reset the control portion of the transfer FIFO, and auto clear to ‘0’ when completing reset operation, write ‘0’ to this bit has no effect.         |
| 5     | RW | 0 | Receiver FIFO Reset<br>0: Auto Clear to 0.<br>1: Reset Receiver FIFO.<br><br>Note: Write ‘1’ to this bit will reset the control portion of the receiver FIFO, and auto clear to ‘0’ when completing reset operation, write ‘0’ to this bit has no effect.         |
| 4:3   | /  | / | /                                                                                                                                                                                                                                                                 |
| 2     | RW | 0 | XIP(Continuous Read Mode) Mode Enable.(FLASH only)<br>0: Disable.<br>1: Enable.<br><br>Note: Dummy Register Value Must Be Match When CPU Enable XIP Mode.                                                                                                         |
| 1     | RW | 1 | XIP debug<br>0: When Flash CS Disable at Non-Data Period, XIP Mode Could not Exit By Itself.<br>1: When Flash CS Disable at Non-Data Period, XIP Mode Exit By Itself.                                                                                             |
| 0     | RW | 0 | Enable Code-bus Read/Write Operation.<br>0:Configuring Code-bus Read/Write Operation.<br>1:Configure Code-bus Over.<br><br>Note1: When CPU exits Code-bus Read/Write Operation, Clear 0.<br>Note2: When CPU Configure Code-bus Read/Write Operation Over , Set 1. |

#### 2.4.4.2 SQPI\_COM\_CFG\_REG

| 偏移地址 : 0x0004 |    |     | 默认值: 0x0002_0000                                                                                                                                |
|---------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                              |
| 31:14         | /  | /   | /                                                                                                                                               |
| 13:12         | RW | 0   | Flash Wait Half Cycle Number Of Receive Data.<br>00: No Delay.<br>01: Delay 1 Half Cycle.<br>10: Delay 2 Half Cycle.<br>11: Delay 3 Half Cycle. |
| 11:9          | /  | /   | /                                                                                                                                               |
| 8             | RW | 0   | Flash CS Polarity.<br>0: CS Signal is Low Enable.<br>1: CS Signal is High Enable.                                                               |
| 7:5           | /  | /   | /                                                                                                                                               |
| 4             | RW | 0   | First Receive Bit Select.<br>0: MSB First<br>1: LSB First                                                                                       |
| 3:2           | /  | /   | /                                                                                                                                               |
| 1             | RW | 0   | SPI Clock Polarity Control<br>0: Active High Polarity.<br>1: Active Low Polarity.                                                               |
| 0             | RW | 0   | SPI Clock/Data Phase Control<br>0: Phase 0(Leading Edge For Sample Data)<br>1: Phase 1(Leading Edge For Setup Data)                             |

#### 2.4.4.3 CACHE\_CFG\_REG

| 偏移地址 : 0x0008 |    |     | 默认值: 0x0000_0000                                                                                                                                                             |
|---------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                           |
| 31:14         | /  | /   | /                                                                                                                                                                            |
| 13:12         | RW | 10  | Write Cache Line Length Configuration<br>00: 8Bytes (64Bits)<br>01: 16Bytes (128Bits)<br>10: 32Bytes (256Bits)<br>11: 64Bytes (512Bits)<br>It means write cache line length. |
| 11:4          | RW | 0   | Code-bus HREADY Adjustable Wait Time Out                                                                                                                                     |

|     |    |   |                                                                                                                                                                                                                                                                                                                                  |
|-----|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |    |   | Note:Default Value Indicates Wait Time Out Level =0xFFFF                                                                                                                                                                                                                                                                         |
| 3   | RW | 1 | Code-bus HREADY Time Out Enable<br>0: Disable<br>1: Enable                                                                                                                                                                                                                                                                       |
| 2   | RW | 1 | Code-Bus Write Size Select<br>0: Code-Bus Write Request Size is Cache-line length<br>1: Code-Bus Write Request Size is Bus Size length(HSIZE)<br>Note: SQPI Memory Controller Get Cache-line and Return Data Back to Bus At Same Time,So that SQPI Memory Controller Do not Care Whether Code-Bus Read Request From Cache or CPU |
| 1:0 | RW | 1 | Cache Line Length Configuration<br>00: 8Bytes (64Bits)<br>01: 16Bytes (128Bits)<br>10: 32Bytes (256Bits)<br>Else: 16Bytes (128Bits)<br>It means read cache line length.                                                                                                                                                          |

#### 2.4.4.4 MEM\_AC\_CFG\_REG

| 偏移地址 : 0x000C |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                       |
|---------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                     |
| 31:24         | RW | 1   | SPI Flash CS Enable Wait Cycle.(t_SLCH)<br>Note: These Bits Stand For CS Active To Valid CLK Edge Setup Minimum Time.                                                                                                                                                  |
| 23:16         | RW | 0   | SPI Flash CS Disable Wait Cycle.(t_CHSH)<br>Note: These Bits Stand For Valid CLK Edge To CS Active Hold Minimum Time.                                                                                                                                                  |
| 15:8          | RW | 5   | System-bus CS Deselect Wait Cycle(t_SHSL).<br>Note1:These Bits Stand For The CS Deselect Minimum Time;<br>Note2:Different Value from CS Deselect After Read/Write/Erase<br>Note3:More Details in Data Sheet AC Characters: t_SHSL                                      |
| 7:0           | RW | 5   | Code-bus CS Deselect Wait Cycle.(t_SHSL)<br>Note1: When Select Flash,These Bits Stand For CS Minimum Deselect Time After Read.<br>Note2: When Select PSRAM,Different Value from CS Deselect After Read/Write<br>Note3:More Details in Data Sheet AC Characters: t_SHSL |

#### 2.4.4.5 FLASH\_C\_READ\_CFG\_REG

| 偏移地址 : 0x0010 |    |      | 默认值: 0x0311_0001   |
|---------------|----|------|--------------------|
| 位置            | 访问 | 默认值  | 描述                 |
| 31:24         | RW | 0x03 | Read Command Send. |

|       |    |     |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23    | RW | 0   | Enable DMA read through I/D bus(flash/psram).<br>0: Disable. DMA read should through SBUS.<br>1: Enable.                                                                                                                                                                                                                                                                                                                                      |
| 22:20 | RW | 001 | Bit Number Of Command Send Every Cycle.<br>000: No Send Command.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else:No Send Command.<br>Note1: Code-bus Command Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Command.                                                                   |
| 19    | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 18:16 | RW | 001 | Bit Number Of Address Send Every Cycle.<br>000: No Send Address.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else:No Send Address.<br>Note1: Code-bus Address Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Address.<br>Note3: For Each Code-bus Operation, Send Address Is Necessary. |
| 15    | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 14:12 | RW | 0   | Bit Number Of DUMMY Send Every Cycle.<br>000: No Send DUMMY .<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else: No Send DUMMY.<br>Note1: Code-bus Dummy Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Dummy.                                                                           |
| 11    | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 10:4  | RW | 0   | DUMMY Data Bit Number.<br>0: Send 0 Bytes Dummy.<br>8: Send 1 Bytes Dummy.<br>16:Send 2 Bytes Dummy.<br>.....<br>64:Send 8 Bytes Dummy.<br>Note1: Code-bus Send Dummy Bit Number Initial State is 0.                                                                                                                                                                                                                                          |



|     |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |    |   | <p>Note2: Include XIP Mode Bits (M7-M0), WRAP Bits (W7-W0).</p> <p>Note3: Bit Number Value Must Be the Multiples of 8Bits, Max is 64 Bits.</p> <p>Note4: Same Command May Has Different Dummy Number At Different CLK Frequency, More Details See Data Sheet.</p> <p>Note5: If The Dummy Bit Number More Than 8 Bytes , Force to 8Bytes.</p> <p>Example: Dual Send Mode : Dummy 4 clocks, Dummy Bits Number is 8</p> |
| 3   | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2:0 | RW | 1 | <p>Bit Number Of Data Get Every Cycle.</p> <p>000: No Get Data.</p> <p>001: Get 1 Bit Data Every Cycle.</p> <p>010: Get 2 Bit Data Every Cycle.</p> <p>011: Get 4 Bit Data Every Cycle.</p> <p>100: Get 8 Bit Data Every Cycle.</p> <p>Else:No Get Data.</p> <p>Note1: Code-bus Get Read Data Mode Initial State is Single Mode.</p> <p>Note2: For Each Code-bus Operation, Get Data is Necessary.</p>               |

#### 2.4.4.6 FLASH\_C\_WRITE\_CFG\_REG

| 偏移地址 : 0x0014 |    |      | 默认值: 0x0211_0001                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 31:24         | RW | 0x02 | Write Command Send.<br>Default is Normal Write                                                                                                                                                                                                                                                                                                                                                                     |
| 23            | /  | /    | /                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 22:20         | RW | 1    | <p>Bit Number Of Command Send Every Cycle.</p> <p>000: No Send Command.</p> <p>001: Send 1 Bit Data Every Cycle.</p> <p>010: Send 2 Bit Data Every Cycle.</p> <p>011: Send 4 Bit Data Every Cycle.</p> <p>100: Send 8 Bit Data Every Cycle.</p> <p>Else:No Send Command.</p> <p>Note1: Code-bus Command Send Mode Initial State is Single Mode.</p> <p>Note2: These Bits Indicate Whether Or Not Send Command.</p> |
| 19            | /  | /    | /                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 18:16         | RW | 1    | <p>Bit Number Of Address Send Every Cycle.</p> <p>000: No Send Address.</p> <p>001: Send 1 Bit Data Every Cycle.</p> <p>010: Send 2 Bit Data Every Cycle.</p> <p>011: Send 4 Bit Data Every Cycle.</p>                                                                                                                                                                                                             |

|       |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |    |   | 100: Send 8 Bit Data Every Cycle.<br>Else:No Send Address.<br>Note1: Code-bus Address Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Address.<br>Note3: For Each Code-bus Operation, Send Address Is Necessary.                                                                                                                                                                                                                                                                                 |
| 15    | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 14:12 | RW | 0 | Bit Number Of DUMMY Send Every Cycle.<br>000: No Send DUMMY.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else: No Send DUMMY.<br>Note1: Code-bus Dummy Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Dummy.                                                                                                                                                                         |
| 11    | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 10:4  | RW | 0 | DUMMY Data Bit Number.<br>0: Send 0 Bytes Dummy.<br>8: Send 1 Bytes Dummy.<br>16:Send 2 Bytes Dummy.<br>.....<br>64:Send 8 Bytes Dummy.<br>Note1: Code-bus Send Dummy Bit Number Initial State is 0.<br>Note2: Bit Number Value Must Be the Multiples of 8Bits, Max is 64 Bits.<br>Note3: Same Command May Has Different Dummy Number At Different CLK Frequency, More Details See Data Sheet.<br>Note4: If The Dummy Bit Number More Than 8 Bytes , Force to 8 Bytes.<br>Example: Dual Send Mode : Dummy 4 clocks, Dummy Bits Number is 8 |
| 3     | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2:0   | RW | 1 | Bit Number Of Data Send Every Cycle.<br>000: No Send Data.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else:No Send Data.<br>Note1: Code-bus Send Data Mode Initial State is Single Mode.<br>Note2: For Each Code-bus Operation, Get Data is Necessary.                                                                                                                                                                         |

#### 2.4.4.7 FLASH\_C\_RD\_DUMMY\_H\_REG

| 偏移地址：0x0018 |    |     | 默认值: 0x0000_0000                        |
|-------------|----|-----|-----------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                      |
| 31:0        | RW | 0   | Top Half 32bits Of Code-bus DUMMY Data. |

#### 2.4.4.8 FLASH\_C\_RD\_DUMMY\_L\_REG

| 偏移地址：0x001C |    |     | 默认值: 0x0000_0000                           |
|-------------|----|-----|--------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                         |
| 31:0        | RW | 0   | Bottom Half 32bits Of Code-bus DUMMY Data. |

#### 2.4.4.9 FLASH\_C\_WR\_DUMMY\_H\_REG

| 偏移地址：0x0020 |    |     | 默认值: 0x0000_0000                        |
|-------------|----|-----|-----------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                      |
| 31:0        | RW | 0   | Top Half 32bits Of Code-bus DUMMY Data. |

#### 2.4.4.10 FLASH\_C\_WR\_DUMMY\_L\_REG

| 偏移地址：0x0024 |    |     | 默认值: 0x0000_0000                           |
|-------------|----|-----|--------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                         |
| 31:0        | RW | 0   | Bottom Half 32bits Of Code-bus DUMMY Data. |

#### 2.4.4.11 FLASH\_C\_IO\_SWIT\_TIME\_REG

| 偏移地址：0x0028 |    |     | 默认值: 0x0000_0000                                                       |
|-------------|----|-----|------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                     |
| 31:0        | /  | /   | When the OPI Port Exist I/O Switch, Port Trans/Get Data Needs Suspend. |

#### 2.4.4.12 S\_RW\_CFG\_REG

| 偏移地址：0x002C |    |     | 默认值: 0x0000_0000                                                                                                                                                                |
|-------------|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                                                              |
| 31:23       | /  | /   | /                                                                                                                                                                               |
| 22:20       | RW | 0   | Bit Number Of Command Send Every Cycle.<br>000: No Send Command.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle. |

|       |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |    |   | 100: Send 8 Bit Data Every Cycle.<br>Else:No Send Command.<br>Note: These Bits Indicate Whether Or Not Send Command.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 19    | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 18:16 | RW | 0 | Bit Number Of Address Send Every Cycle.<br>000: No Send Address.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else:No Send Address.<br>Note: These Bits Indicate Whether Or Not Send Address.                                                                                                                                                                                                                                     |
| 15    | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 14:12 | RW | 0 | Bit Number Of DUMMY Send Every Cycle.<br>000: No Send DUMMY.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else: No Send DUMMY.<br>Note: These Bits Indicate Whether Or Not Send Dummy.                                                                                                                                                                                                                                            |
| 11    | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 10:4  | RW | 0 | DUMMY Data Bit Number.<br>0: Send 0 Bytes Dummy.<br>8: Send 1 Bytes Dummy.<br>16:Send 2 Bytes Dummy.<br>.....<br>64:Send 8 Bytes Dummy.<br>Note1:System-bus Send Dummy Bit Number Initial State is 0.<br>Note2: Bit Number Value Must Be the Multiples of 8Bits, Max is 64 Bits.<br>Note3: Same Command May Has Different Dummy Number At Different CLK Frequency, More Details See Data Sheet.<br>Note4: If The Dummy Bit Number More Than 8 Bytes , Force to 8 Bytes.<br>Example：Dual Send Mode ： Dummy 4 clocks , Dummy Bits Number is 8 |
| 3     | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2:0   | RW | 0 | Bit Number Of System-bus Data Get/Write Every Cycle.<br>000: Neither Get Nor Send Data Operation.<br>001: Get/Send 1 Bit Data Every Cycle.                                                                                                                                                                                                                                                                                                                                                                                                  |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                   |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | 010: Get/Send 2 Bit Data Every Cycle.<br>011: Get/Send 4 Bit Data Every Cycle.<br>100: Get/Send 8 Bit Data Every Cycle.<br>Else:Neither Get Nor Send Data Operation.<br>Note1: These Bits Indicate Wether Or Not Send/Get Data.<br>Note2: Work With Write/Read Number To Indicate Current Operation is Send or Get Data State As Long As These Bits Are Non-Zero. |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 2.4.4.13 S\_ADDR\_CFG\_REG

| 偏移地址 : 0x0030 |    |     | 默认值: 0x0000_0000                                                                                                     |
|---------------|----|-----|----------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                   |
| 31:0          | RW | 0   | 32 Bits Flash Page Address Of System-bus Operation.<br>Note:If the Flash Address Size is 24 bits, Write Low 24 Bits. |

#### 2.4.4.14 S\_DUMMY\_H\_REG

| 偏移地址 : 0x0034 |    |     | 默认值: 0x0000_0000               |
|---------------|----|-----|--------------------------------|
| 位置            | 访问 | 默认值 | 描述                             |
| 31:0          | RW | 0   | Top Half 32bits Of Dummy Data. |

#### 2.4.4.15 S\_DUMMY\_L\_REG

| 偏移地址 : 0x0038 |    |     | 默认值: 0x0000_0000                  |
|---------------|----|-----|-----------------------------------|
| 位置            | 访问 | 默认值 | 描述                                |
| 31:0          | RW | 0   | Bottom Half 32bits Of Dummy Data. |

#### 2.4.4.16 S\_IO\_SWIT\_TIME\_REG

| 偏移地址 : 0x003C |    |     | 默认值: 0x0000_0000                   |
|---------------|----|-----|------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                 |
| 31:24         | RW | 0   | System-bus Read Latency Wait Cycle |
| 23:0          | /  | /   | /                                  |

#### 2.4.4.17 S\_WR\_NUM\_REG

| 偏移地址 : 0x0040 |    |     | 默认值: 0x0000_0000                                                                     |
|---------------|----|-----|--------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                   |
| 31:0          | WC | 0   | System-bus Write Number(Byte)<br>0:Non-Write.<br>1:Write 1 Byte.<br>2:Write 2 Bytes. |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | 3:Write 3 Bytes.<br>.....<br>Note1: These Bits Indicate Current Operation is Write Data To Flash.<br>Note2: If CPU Execute Write Data Operation, Bit Number Of System-bus Data Get/Write Every Cycle Must Be Configure Non-Zero Value.<br>Note3: If Select Flash,Then CPU Wants Program More Than 256 Bytes Data , Divide Multiple Pages to Program , Every Time CPU Execute Page Program, Needs Check the Flash BUSY Bit. |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 2.4.4.18 S\_RD\_NUM\_REG

| 偏移地址 : 0x0044 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                          |
|---------------|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                                        |
| 31:0          | WC | 0   | System-bus Read Number(Byte)<br>0:Non-Read.<br>1:Read 1 Byte.<br>2:Read 2 Bytes.<br>3:Read 3 Bytes.<br>.....<br>Note1: These Bits Indicate Current Operation is Read Data From Flash.<br>Note2: If CPU Execute Read Data Operation, Bit Number Of System-bus Data Get/Write Every Cycle Must Be Configure Non-Zero Value. |

#### 2.4.4.19 START\_SEND\_REG\_REG

| 偏移地址 : 0x0048 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 31:1          | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0             | WC | 0   | Enable System-bus Operation<br>0: System-bus Operation is Configuring.<br>1: System-bus Operation Configure Over.<br>Note1: If System-bus is Configuring Write Operation, This Bit should be Configured after all the Register Which Write Operation Needs Complete Configuration and Before Writing Data to Write FIFO.<br>Note2: If System-bus is Configuring Read Operation, This Bit should be Configured after all the Register Which Read Operation Needs Complete Configuration and Before Reading Data From Read FIFO.<br>Note3: If This Bit Current Value is 1, Wait Memory Controller Clear to 0, CPU Could Configure Next System-bus Operation. |

#### 2.4.4.20 FIFO\_TRIG\_LEV\_REG

| 偏移地址 : 0x004C |    |      | 默认值: 0x0000_0000                              |
|---------------|----|------|-----------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                            |
| 31:24         | RW | 0x6F | Write FIFO Full Request Trigger Level(Byte).  |
| 23:16         | RW | 0x0F | Write FIFO Empty Request Trigger Level(Byte). |
| 15:8          | RW | 0x6F | Read FIFO Full Request Trigger Level(Byte).   |
| 7:0           | RW | 0x0F | Read FIFO Empty Request Trigger Level(Byte).  |

#### 2.4.4.21 FIFO\_STA\_REG\_REG

| 偏移地址 : 0x0050 |    |     | 默认值: 0x0000_0000                                                                                                                                                                |
|---------------|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                              |
| 31:26         | /  | /   | /                                                                                                                                                                               |
| 25            | R  | 0   | Read Buffer Full Flag.                                                                                                                                                          |
| 24            | R  | 0   | Write Buffer Full Flag.                                                                                                                                                         |
| 23            | R  | 0   | Write Buffer Could or Not Write.                                                                                                                                                |
| 22:20         | R  | 0   | Write Buffer Counter(Byte)<br>These Bits Indicate Number of Bytes in Write Buffer.                                                                                              |
| 19            | R  | 0   | Read Buffer Could or Not Read                                                                                                                                                   |
| 18:16         | R  | 0   | Read Buffer Counter(Byte)<br>These Bits Indicate Number of Bytes in Read Buffer.                                                                                                |
| 15:8          | R  | 0   | Write FIFO Counter(Byte)<br>These Bits Indicate Number of Bytes in Write FIFO.<br>0: 0 Byte in Write FIFO.<br>1: 1 Bytes in Write FIFO.<br>...<br>128: 128 Bytes in Write FIFO. |
| 7:0           | R  | 0   | Read FIFO Counter(Byte)<br>These Bits Indicate Number of Bytes in Read FIFO.<br>0: 0 Byte in Read FIFO.<br>1: 1 Byte in Read FIFO.<br>...<br>128: 128 Bytes in Read FIFO.       |

#### 2.4.4.22 INT\_EN\_REG\_REG

| 偏移地址：0x0054 |    |     | 默认值: 0x0000_0000                                                                                                                                                        |
|-------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                                                      |
| 31:16       | /  | /   | /                                                                                                                                                                       |
| 15          | RW | 0   | When DMA write address was not in the non-cacheable field which were configured in the DCACHE register mr010~mr034, send interrupt to CPU.<br>0: Disable.<br>1: Enable. |
| 14          | RW | 0   | PSRAM read latency Time Out Interrupt Enable<br>0:Disable.<br>1:Enable.                                                                                                 |
| 13          | RW | 0   | System-bus HREADY Time Out Interrupt Enable<br>0:Disable.<br>1:Enable.                                                                                                  |
| 12          | RW | 0   | Transfer Completed Interrupt Enable<br>0: Disable.<br>1: Enable.                                                                                                        |
| 11          | RW | 0   | Write FIFO Underflow Interrupt Enable<br>0: Disable.<br>1: Enable.                                                                                                      |
| 10          | RW | 0   | Write FIFO Overflow Interrupt Enable<br>0: Disable.<br>1: Enable.                                                                                                       |
| 9           | RW | 0   | Read FIFO Underflow Interrupt Enable<br>0: Disable.<br>1: Enable.                                                                                                       |
| 8           | RW | 0   | Read FIFO Overflow Interrupt Enable<br>0: Disable.<br>1: Enable.                                                                                                        |
| 7           | /  | /   | /                                                                                                                                                                       |
| 6           | RW | 0   | Write FIFO Full Interrupt Enable<br>0: Disable.<br>1: Enable.                                                                                                           |
| 5           | RW | 0   | Write FIFO Empty Interrupt Enable<br>0: Disable.<br>1: Enable.                                                                                                          |
| 4           | RW | 0   | Write FIFO Ready Request Interrupt Enable                                                                                                                               |



|   |    |   |                                                                       |
|---|----|---|-----------------------------------------------------------------------|
|   |    |   | 0: Disable.<br>1: Enable.                                             |
| 3 | /  | / | /                                                                     |
| 2 | RW | 0 | Read FIFO Full Interrupt Enable<br>0: Disable.<br>1: Enable.          |
| 1 | RW | 0 | Read FIFO Empty Interrupt Enable<br>0: Disable.<br>1: Enable.         |
| 0 | RW | 0 | Read FIFO Ready Request Interrupt Enable<br>0: Disable.<br>1: Enable. |

#### 2.4.4.23 INT\_STA\_REG\_REG

| 偏移地址：0x0058 |       |     | 默认值: 0x0000_0000                                                                                                                            |
|-------------|-------|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问    | 默认值 | 描述                                                                                                                                          |
| 31:14       | /     | /   | /                                                                                                                                           |
| 13          | W1C/R | 0   | System-bus HREADY Time Out Interrupt Flag<br>0: System-bus HREADY Time Out Not Happens.<br>1: System-bus HREADY Time Out Happens.           |
| 12          | W1C/R | 0   | Transfer Completed Interrupt Flag<br>0: Busy.<br>1: Transfer Completed.                                                                     |
| 11          | W1C/R | 0   | Write FIFO underflow Interrupt Flag<br>Note: When set, this bit indicates that Write FIFO has underflow. Writing “1” to this bit clears it. |
| 10          | W1C/R | 0   | Write FIFO Overflow Interrupt Flag<br>Note: When set, this bit indicates that Write FIFO has overflowed. Writing “1” to this bit clears it. |
| 9           | W1C/R | 0   | Read FIFO Underflow Interrupt Flag<br>Note: When set, this bit indicates that Read FIFO has underflow. Writing “1” to this bit clears it.   |
| 8           | W1C/R | 0   | Read FIFO Overflow Interrupt Flag<br>Note: When set, this bit indicates that Read FIFO has overflowed. Writing “1” to this bit clears it.   |
| 7           | /     | /   | /                                                                                                                                           |
| 6           | W1C/R | 0   | Write FIFO Full Interrupt Flag<br>0: Write FIFO is Not Full.                                                                                |

|   |       |   |                                                                                                                                                                                                                                                               |
|---|-------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |       |   | 1: Write FIFO is Full.<br>Note: This bit is set when the Write FIFO is full. Writing “1” to this bit clears it.                                                                                                                                               |
| 5 | W1C/R | 0 | Write FIFO Empty Request Interrupt Flag<br>0: Write FIFO always contains one or more bytes.<br>1: Write FIFO has been empty.<br>Note: This bit is set if the Write FIFO is empty. Writing “1” to this bit clears it.                                          |
| 4 | W1C/R | 0 | Write FIFO Empty Request Interrupt Flag<br>0: WF_WL > WF_TRIG_LEVEL.<br>1: WF_WL <= WF_TRIG_LEVEL has happened.<br>Note: This bit is set any time if WF_WL <= WF_TRIG_LEVEL. Writing “1” to this bit clears it. Where WF_WL is the water level of Write FIFO. |
| 3 | /     | / | /                                                                                                                                                                                                                                                             |
| 2 | W1C/R | 0 | Read FIFO Full Interrupt Flag<br>0: Not Full.<br>1: Read FIFO has been Full.<br>Note: This bit is set when Read FIFO is full(>= full_trigger_level). Writing “1” to this bit clears it.                                                                       |
| 1 | W1C/R | 0 | Read FIFO Empty Interrupt Flag<br>0: Not Empty.<br>1: Read FIFO has been Empty.<br>Note: This bit is set if the Read FIFO is empty. Writing “1” to this bit clears it.                                                                                        |
| 0 | W1C/R | 0 | Read FIFO Ready Request Interrupt Flag<br>0: RF_WL < RX_TRIG_LEVEL.<br>1: RF_WL >= RX_TRIG_LEVEL has happened.<br>This bit is set any time if RF_WL >= RF_TRIG_LEVEL. Writing “1” to this bit clears it. Where RF_WL is the water level of Read FIFO.         |

#### 2.4.4.24 FLASH\_MOD\_EXE\_IND\_REG

| 偏移地址 : 0x005C |    |     | 默认值: 0x0000_0000                                                                                                                                                |
|---------------|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                              |
| 31:2          | /  | /   | /                                                                                                                                                               |
| 1             | R  | 0   | flash WRAP<br>Psram wrap is always working, so it does not need flag.<br>0:Wrap Mode Has Not Taken Effect in Memory.<br>1:Wrap Mode Has Taken Effect in Memory. |
| 0             | R  | 0   | FLASH ONLY                                                                                                                                                      |

|  |  |  |                                                                                       |
|--|--|--|---------------------------------------------------------------------------------------|
|  |  |  | 0: XIP Mode Has Not Taken Effect in Memory.<br>1: XIP Mode Has Taken Effect in Memory |
|--|--|--|---------------------------------------------------------------------------------------|

#### 2.4.4.25 START\_ADDR0\_REG

| 偏移地址 : 0x0080 |    |     | 默认值: 0x0000_0000                                        |
|---------------|----|-----|---------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                      |
| 31:26         | /  | /   |                                                         |
| 25:4          | RW | 0   | Address Field0 Start Position<br>Note: With unit of 16B |
| 3:0           | /  | /   | /                                                       |

#### 2.4.4.26 END\_ADDR0\_REG

| 偏移地址 : 0x0084 |    |     | 默认值: 0x0000_0000                                      |
|---------------|----|-----|-------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                    |
| 31:26         | /  | /   |                                                       |
| 25:4          | RW | 0   | Address Field0 End Position<br>Note: With unit of 16B |
| 3:0           | /  | /   | /                                                     |

#### 2.4.4.27 BIAS\_ADDR0\_REG

| 偏移地址 : 0x0088 |    |     | 默认值: 0x0000_0000                                                                                                                                     |
|---------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                   |
| 31            | RW | 0   | Address Field0 Bias Enable<br>1: Bias Enabled, add bias when address is between Address Field0 Start Position and End Position.<br>0: Bias Disabled. |
| 30:0          | RW | 0   | Address Field0 Bias                                                                                                                                  |

#### 2.4.4.28 START\_ADDR1\_REG

| 偏移地址 : 0x0090 |    |     | 默认值: 0x0000_0000                                        |
|---------------|----|-----|---------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                      |
| 31:26         | /  | /   |                                                         |
| 25:4          | RW | 0   | Address Field1 Start Position<br>Note: With unit of 16B |
| 3:0           | /  | /   | /                                                       |

#### 2.4.4.29 END\_ADDR1\_REG

| 偏移地址：0x0094 |    |     | 默认值: 0x0000_0000                                      |
|-------------|----|-----|-------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                    |
| 31:26       | /  | /   |                                                       |
| 25:4        | RW | 0   | Address Field1 End Position<br>Note: With unit of 16B |
| 3:0         | /  | /   | /                                                     |

#### 2.4.4.30 BIAS\_ADDR1\_REG

| 偏移地址：0x0098 |    |     | 默认值: 0x0000_0000                                                                                                                                     |
|-------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                                   |
| 31          | RW | 0   | Address Field1 Bias Enable<br>1: Bias Enabled, add bias when address is between Address Field1 Start Position and End Position.<br>0: Bias Disabled. |
| 30:0        | RW | 0   | Address Field1 Bias                                                                                                                                  |

#### 2.4.4.31 START\_ADDR2\_REG

| 偏移地址：0x00A0 |    |     | 默认值: 0x0000_0000                                        |
|-------------|----|-----|---------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                      |
| 31:26       | /  | /   |                                                         |
| 25:4        | RW | 0   | Address Field2 Start Position<br>Note: With unit of 16B |
| 3:0         | /  | /   | /                                                       |

#### 2.4.4.32 END\_ADDR2\_REG

| 偏移地址：0x00A4 |    |     | 默认值: 0x0000_0000                                      |
|-------------|----|-----|-------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                    |
| 31:26       | /  | /   |                                                       |
| 25:4        | RW | 0   | Address Field2 End Position<br>Note: With unit of 16B |
| 3:0         | /  | /   | /                                                     |

#### 2.4.4.33 BIAS\_ADDR2\_REG

| 偏移地址 : 0x00A8 |    |     | 默认值: 0x0000_0000                                                                                                                                     |
|---------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                   |
| 31            | RW | 0   | Address Field2 Bias Enable<br>1: Bias Enabled, add bias when address is between Address Field2 Start Position and End Position.<br>0: Bias Disabled. |
| 30:0          | RW | 0   | Address Field2 Bias                                                                                                                                  |

#### 2.4.4.34 START\_ADDR3\_REG

| 偏移地址 : 0x00B0 |    |     | 默认值: 0x0000_0000                                        |
|---------------|----|-----|---------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                      |
| 31:26         | /  | /   |                                                         |
| 25:4          | RW | 0   | Address Field3 Start Position<br>Note: With unit of 16B |
| 3:0           | /  | /   | /                                                       |

#### 2.4.4.35 END\_ADDR3\_REG

| 偏移地址 : 0x00B4 |    |     | 默认值: 0x0000_0000                                      |
|---------------|----|-----|-------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                    |
| 31:26         | /  | /   |                                                       |
| 25:4          | RW | 0   | Address Field3 End Position<br>Note: With unit of 16B |
| 3:0           | /  | /   | /                                                     |

#### 2.4.4.36 BIAS\_ADDR3\_REG

| 偏移地址 : 0x00B8 |    |     | 默认值: 0x0000_0000                                                                                                                                     |
|---------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                   |
| 31            | RW | 0   | Address Field3 Bias Enable<br>1: Bias Enabled, add bias when address is between Address Field3 Start Position and End Position.<br>0: Bias Disabled. |
| 30:0          | RW | 0   | Address Field3 Bias                                                                                                                                  |

#### 2.4.4.37 START\_ADDR4\_REG

| 偏移地址：0x00C0 |    |     | 默认值: 0x0000_0000                                        |
|-------------|----|-----|---------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                      |
| 31:26       | /  | /   |                                                         |
| 25:4        | RW | 0   | Address Field4 Start Position<br>Note: With unit of 16B |
| 3:0         | /  | /   | /                                                       |

#### 2.4.4.38 END\_ADDR4\_REG

| 偏移地址：0x00C4 |    |     | 默认值: 0x0000_0000                                      |
|-------------|----|-----|-------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                    |
| 31:26       | /  | /   |                                                       |
| 25:4        | RW | 0   | Address Field4 End Position<br>Note: With unit of 16B |
| 3:0         | /  | /   | /                                                     |

#### 2.4.4.39 BIAS\_ADDR4\_REG

| 偏移地址：0x00C8 |    |     | 默认值: 0x0000_0000                                                                                                                                     |
|-------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                                   |
| 31          | RW | 0   | Address Field4 Bias Enable<br>1: Bias Enabled, add bias when address is between Address Field4 Start Position and End Position.<br>0: Bias Disabled. |
| 30:0        | RW | 0   | Address Field4 Bias                                                                                                                                  |

#### 2.4.4.40 START\_ADDR5\_REG

| 偏移地址：0x00D0 |    |     | 默认值: 0x0000_0000                                        |
|-------------|----|-----|---------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                      |
| 31:26       | /  | /   |                                                         |
| 25:4        | RW | 0   | Address Field5 Start Position<br>Note: With unit of 16B |
| 3:0         | /  | /   | /                                                       |

#### 2.4.4.41 END\_ADDR5\_REG

| 偏移地址：0x00D4 |    |     | 默认值: 0x0000_0000                                      |
|-------------|----|-----|-------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                    |
| 31:26       | /  | /   |                                                       |
| 25:4        | RW | 0   | Address Field5 End Position<br>Note: With unit of 16B |
| 3:0         | /  | /   | /                                                     |

#### 2.4.4.42 BIAS\_ADDR5\_REG

| 偏移地址：0x00D8 |    |     | 默认值: 0x0000_0000                                                                                                                                     |
|-------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                                   |
| 31          | RW | 0   | Address Field5 Bias Enable<br>1: Bias Enabled, add bias when address is between Address Field5 Start Position and End Position.<br>0: Bias Disabled. |
| 30:0        | RW | 0   | Address Field5 Bias                                                                                                                                  |

#### 2.4.4.43 S\_WDATA\_REG

| 偏移地址：0x0100 |    |     | 默认值: 0x0000_0000                                              |
|-------------|----|-----|---------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                            |
| 31:0        | RW | 0   | System-bus Write Data Register.<br>Note: Write FIFO Entrance. |

#### 2.4.4.44 S\_RDATA\_REG

| 偏移地址：0x0200 |    |     | 默认值: 0x0000_0000                                        |
|-------------|----|-----|---------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                      |
| 31:0        | R  | 0   | System-bus Read Data Register.<br>Note: Read FIFO Exit. |

#### 2.4.4.45 PSRAM\_COM\_CFG\_REG

| 偏移地址：0x0800 |    |     | 默认值: 0x0000_0000                                                                                        |
|-------------|----|-----|---------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                      |
| 31:30       | RW | 10  | PsrAm Address Size Mode<br>00: Address Size 8bit.<br>01: Address Size 16bit.<br>10: Address Size 24bit. |

|        |   |   |                         |
|--------|---|---|-------------------------|
|        |   |   | 11: Address Size 32bit. |
| [29:0] | / | / | /                       |

#### 2.4.4.46 PSRAM\_CTRL\_COM\_CFG\_REG

| 偏移地址 : 0x0804 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                     |
|---------------|----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                   |
| 31:17         | /  | /   | /                                                                                                                                                                                                                                                                    |
| 16            | RW | 0   | When DMA write address was not in the non-cacheable field which were configured in the DCACHE register mr010~mr034, the write operation should not take into effect.<br>0: write operation still take into effect.<br>1: write operation would not take into effect. |
| 15:0          | /  | /   | /                                                                                                                                                                                                                                                                    |

#### 2.4.4.47 PSRAM\_CACHE\_CFG\_REG

| 偏移地址 : 0x0808 |    |     | 默认值: 0x0000_8000                                                                                                                                                                                                                                                                                         |
|---------------|----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                       |
| 31            | RW | 0   | Auto send data when not meet cache line length but timeout. Used with bit[30:17].<br>For example, when cache line length is 128bits, after receive address 0x0, 0x4, 0x8 from upstream, controller didn't get address 0xC data for a long time, then send 0x0, 0x4, 0x8 when this register configured 1. |
| 30:16         | RW | 0   | Wait Cache Line data timeout cycle.                                                                                                                                                                                                                                                                      |
| 15            | RW | 1   | Auto cut sequential word send into two write operation when address cross page boundary.<br>PSRAM cbus write only!<br>For SQPI 16Mb, page size is 512Byte. Page size is configured in 0x870.                                                                                                             |
| 14:0          | /  | /   | /                                                                                                                                                                                                                                                                                                        |

#### 2.4.4.48 PSRAM\_C\_READ\_CFG\_REG

| 偏移地址 : 0x0810 |    |      | 默认值: 0x0311_0001                                                                                                                           |
|---------------|----|------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                                                                                                         |
| 31:24         | RW | 0x03 | Read Command Send.                                                                                                                         |
| 23            | /  | /    | /                                                                                                                                          |
| 22:20         | RW | 001  | Bit Number Of Command Send Every Cycle.<br>000: No Send Command.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle. |



|       |    |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |    |     | 011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else:No Send Command.<br>Note1: Code-bus Command Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Command.                                                                                                                                                                                                                                                                                                                                                                 |
| 19    | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 18:16 | RW | 001 | Bit Number Of Address Send Every Cycle.<br>000: No Send Address.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else:No Send Address.<br>Note1: Code-bus Address Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Address.<br>Note3: For Each Code-bus Operation, Send Address Is Necessary.                                                                                                                                                 |
| 15    | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 14:12 | RW | 0   | Bit Number Of DUMMY Send Every Cycle.<br>000: No Send DUMMY .<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else: No Send DUMMY.<br>Note1: Code-bus Dummy Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Dummy.                                                                                                                                                                                                                           |
| 11    | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 10:4  | RW | 0   | DUMMY Data Bit Number.<br>0: Send 0 Bytes Dummy.<br>8: Send 1 Bytes Dummy.<br>16:Send 2 Bytes Dummy.<br>.....<br>64:Send 8 Bytes Dummy.<br>Note1: Code-bus Send Dummy Bit Number Initial State is 0.<br>Note2: Include XIP Mode Bits (M7-M0),WRAP Bits (W7-W0).<br>Note3: Bit Number Value Must Be the Multiples of 8Bits, Max is 64 Bits.<br>Note4: Same Command May Has Different Dummy Number At Different CLK Frequency, More Details See Data Sheet.<br>Note5: If The Dummy Bit Number More Than 8 Bytes , Force to 8Bytes.<br>Example：Dual Send Mode ：Dummy 4 clocks, Dummy Bits Number |

|     |    |   |                                                                                                                                                                                                                                                                                                                                                                 |
|-----|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |    |   | is 8                                                                                                                                                                                                                                                                                                                                                            |
| 3   | /  | / | /                                                                                                                                                                                                                                                                                                                                                               |
| 2:0 | RW | 1 | Bit Number Of Data Get Every Cycle.<br>000: No Get Data.<br>001: Get 1 Bit Data Every Cycle.<br>010: Get 2 Bit Data Every Cycle.<br>011: Get 4 Bit Data Every Cycle.<br>100: Get 8 Bit Data Every Cycle.<br>Else:No Get Data.<br>Note1: Code-bus Get Read Data Mode Initial State is Single Mode.<br>Note2: For Each Code-bus Operation, Get Data is Necessary. |

#### 2.4.4.49 PSRAM\_C\_WRITE\_CFG\_REG

| 偏移地址 : 0x0814 |    |      | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------|----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 31:24         | RW | 0x02 | Write Command Send.<br>Default is Normal Write                                                                                                                                                                                                                                                                                                                                                                                                |
| 23            | /  | /    | /                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 22:20         | RW | 1    | Bit Number Of Command Send Every Cycle.<br>000: No Send Command.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else:No Send Command.<br>Note1: Code-bus Command Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Command.                                                                   |
| 19            | /  | /    | /                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 18:16         | RW | 1    | Bit Number Of Address Send Every Cycle.<br>000: No Send Address.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else:No Send Address.<br>Note1: Code-bus Address Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Address.<br>Note3: For Each Code-bus Operation, Send Address Is Necessary. |
| 15            | /  | /    | /                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 14:12         | RW | 0    | Bit Number Of DUMMY Send Every Cycle.                                                                                                                                                                                                                                                                                                                                                                                                         |

|      |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |    |   | 000: No Send DUMMY.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else: No Send DUMMY.<br>Note1: Code-bus Dummy Send Mode Initial State is Single Mode.<br>Note2: These Bits Indicate Whether Or Not Send Dummy.                                                                                                                                                                                                                  |
| 11   | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 10:4 | RW | 0 | DUMMY Data Bit Number.<br>0: Send 0 Bytes Dummy.<br>8: Send 1 Bytes Dummy.<br>16:Send 2 Bytes Dummy.<br>.....<br>64:Send 8 Bytes Dummy.<br>Note1: Code-bus Send Dummy Bit Number Initial State is 0.<br>Note2: Bit Number Value Must Be the Multiples of 8Bits, Max is 64 Bits.<br>Note3: Same Command May Has Different Dummy Number At Different CLK Frequency, More Details See Data Sheet.<br>Note4: If The Dummy Bit Number More Than 8 Bytes , Force to 8 Bytes.<br>Example: Dual Send Mode : Dummy 4 clocks, Dummy Bits Number is 8 |
| 3    | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2:0  | RW | 1 | Bit Number Of Data Send Every Cycle.<br>000: No Send Data.<br>001: Send 1 Bit Data Every Cycle.<br>010: Send 2 Bit Data Every Cycle.<br>011: Send 4 Bit Data Every Cycle.<br>100: Send 8 Bit Data Every Cycle.<br>Else:No Send Data.<br>Note1: Code-bus Send Data Mode Initial State is Single Mode.<br>Note2: For Each Code-bus Operation, Get Data is Necessary.                                                                                                                                                                         |

#### 2.4.4.50 PSRAM\_C\_RD\_DUMMY\_H\_REG

| 偏移地址 : 0x0818 |    |     | 默认值: 0x0000_0000                        |
|---------------|----|-----|-----------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                      |
| 31:0          | RW | 0   | Top Half 32bits Of Code-bus DUMMY Data. |

#### 2.4.4.51 PSRAM\_C\_RD\_DUMMY\_L\_REG

| 偏移地址：0x081C |    |     | 默认值: 0x0000_0000                           |
|-------------|----|-----|--------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                         |
| 31:0        | RW | 0   | Bottom Half 32bits Of Code-bus DUMMY Data. |

#### 2.4.4.52 PSRAM\_C\_WR\_DUMMY\_H\_REG

| 偏移地址：0x0820 |    |     | 默认值: 0x0000_0000                        |
|-------------|----|-----|-----------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                      |
| 31:0        | RW | 0   | Top Half 32bits Of Code-bus DUMMY Data. |

#### 2.4.4.53 PSRAM\_C\_WR\_DUMMY\_L\_REG

| 偏移地址：0x0824 |    |     | 默认值: 0x0000_0000                           |
|-------------|----|-----|--------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                         |
| 31:0        | RW | 0   | Bottom Half 32bits Of Code-bus DUMMY Data. |

#### 2.4.4.54 PSRAM\_C\_IO\_SWIT\_TIME\_REG

| 偏移地址：0x0828                                                                                                                                                               |    |     | 默认值: 0x0000_0000                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|----------------------------------------|
| 位置                                                                                                                                                                        | 访问 | 默认值 | 描述                                     |
| 31:24                                                                                                                                                                     | RW | 0   | PSRAM Code-bus Read Latency Wait Cycle |
| 23:0                                                                                                                                                                      | RW | /   | /                                      |
|  <b>NOTE</b><br>When the OPI Port Exist I/O Switch, Port Trans/Get Data Needs Suspend. |    |     |                                        |

#### 2.4.4.55 PSRAM\_FORCE\_CFG\_REG

| 偏移地址：0x086C |    |     | 默认值: 0x0000_0000                                                                                                                                |
|-------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                              |
| 31:2        | /  | /   |                                                                                                                                                 |
| 1:0         | RW | 0x0 | Psram Wait Half Cycle Number Of Receive Data.<br>00: No Delay.<br>01: Delay 1 Half Cycle.<br>10: Delay 2 Half Cycle.<br>11: Delay 3 Half Cycle. |

#### 2.4.4.56 PSRAM\_COM\_CFG\_REG

| 偏移地址 : 0x0870 |    |       | 默认值: 0x0000_0000                                                                                                                                                                                                                                   |
|---------------|----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值   | 描述                                                                                                                                                                                                                                                 |
| 31:28         | /  | /     | /                                                                                                                                                                                                                                                  |
| 27:16         | RW | 0x1C0 | Maximum CE# low cycle number.<br>CE# should not be kept low for more than 4 us.<br>This value is different for various frequency, default value is for working in 133MHz.<br>When configured 0, it means not checking CE# low width.               |
| 15            | /  | /     |                                                                                                                                                                                                                                                    |
| 14            | RW | 1' b0 | Write or read could start on random address.<br>0: Write or read should start on even address.<br>1: Write or read could start on odd or even address.<br>When write or read mode registers, configure this to 1. When memory access, configure 0. |
| 13:0          | /  | /     | /                                                                                                                                                                                                                                                  |

## 3 时钟和复位

### 3.1 全局电源、复位、时钟管理（GPRCM）

#### 3.1.1 概述

GPRCM 模块管理整个系统的电源、复位和时钟，工作在 AO（always on）电源域。关于 XR806 电源架构、上下电时序、时钟描述等信息可参考《XR806 Datasheet》。



**注意**

GPRCM 相关寄存器被放置在 RTC 电源域，以保证在关断内部 LDO 或 DCDC 使系统掉电时，这些寄存器的值可以保持不变。当检测到 GPIO 或电源等外部中断事件时，系统可以自动启动，并能正确读取这些寄存器的值。

#### 3.1.2 GPRCM 寄存器列表

| 模块名                             | 基地址                   |                                                  |
|---------------------------------|-----------------------|--------------------------------------------------|
| Power, Reset and Clock Manager  | 0x40040000/0xA0040000 |                                                  |
| 寄存器名                            | 偏移地址                  | 描述                                               |
| SYS_TOP_LDO_SMPS_CTRL_REG       | 0x0000                | System TOP_LDO&SMPS Control Register             |
| SYS_EXT_LDO_DIG_LDO_SW_CTRL_REG | 0x0004                | System EXT_LDO&DIG_LDO & Switch Control Register |
| SYS_LFCLK_CTRL_REG              | 0x0008                | System Low Frequency Clock Control               |
| SYS_HOSC_TYPE_REG               | 0x000C                | System HOSC Type                                 |
| SYS_RCOSC_CAL_CTRL_REG          | 0x0010                | System RC-OSC Calibration Control                |
| SYS_PLL_CTRL_REG                | 0x0020                | System PLL Control                               |
| SYS_CLK1_CTRL_REG               | 0x0024                | System Clock 1 Control                           |
| SYS_CLK3_CTRL_REG               | 0x002C                | System Clock 3 Control                           |
| DEV_CLK_CTRL_REG                | 0x0034                | Device Clock Control                             |
| DCXO_CTRL_REG                   | 0x0058                | DCXO control                                     |
| CONN_BOOT_FLAG_REG              | 0x0070                | Connect mode flag                                |
| POWERCTRL_CFG_REG               | 0x0074                | Power control configuration register             |
| SYS1_CTRL_REG                   | 0x0080                | System 1 Control                                 |
| SYS1_STATUS_REG                 | 0x0084                | System 1 Status                                  |
| SYS2_CTRL_REG                   | 0x0088                | System 2 Control                                 |

|                             |        |                                 |
|-----------------------------|--------|---------------------------------|
| SYS2_STATUS_REG             | 0x08C  | System 2 Status                 |
| SYS3_CTRL_REG               | 0x090  | System 3 Control                |
| SYS3_STATUS_REG             | 0x094  | System 3 Status                 |
| SYS1_WAKEUP_CTRL_REG        | 0x098  | System 1 wakeup control         |
| SYS2_WAKEUP_CTRL_REG        | 0x09C  | System 2 wakeup control         |
| SYS1_SLEEP_CTRL_REG         | 0x0A0  | System 1 Sleep control          |
| SYS2_SLEEP_CTRL_REG         | 0x0A4  | System 2 Sleep control          |
| DCXO_STABLE_REF_TIME_REG    | 0x0A8  | DCXO Stable Reference Time      |
| DPLL_STABLE_REF_TIME_REG    | 0x0AC  | DPLL Stable Reference Time      |
| LDO_STABLE_REF_TIME_REG     | 0x0B0  | LDO Stable Reference Time       |
| DIGITAL_SWITCH_REF_TIME_REG | 0x0B4  | Digital Switch Reference Time   |
| SRAM_VOL_CTRL_REG           | 0x0B8  | SRAM Voltage Control            |
| BANDGAP_STABLE_REF_TIME_REG | 0x0BC  | Band Gap Stable Reference Time  |
| DCDC_STABLE_REF_TIME_REG    | 0x0C0  | DCDC Stable Reference Time      |
| BLE_RTC_RST_CTRL_REG        | 0x0D0  | BLE RTC Reset control Register  |
| RFAS_CTRL_REG               | 0x0D8  | RFAS Reset control Register     |
| AR400A_BOOT_FLAG_REG        | 0x0100 | AR400A boot flag                |
| AR400A_BOOT_ADDR_REG        | 0x0104 | AR400A boot address             |
| AR400A_BOOT_ARG_REG         | 0x0108 | AR400A boot argument            |
| AR400N_PRIV_REG0            | 0x010C | AR400A Private Register         |
| AR400N_PRIV_REG1            | 0x0110 | AR400A Private Register         |
| AR400N_PRIV_REG2            | 0x0114 | AR400A Private Register         |
| AR400A_PRIV_REG3            | 0x0118 | AR400A Private Register         |
| AR400A_PRIV_REG4            | 0x011c | AR400A Private Register         |
| AR400A_WAKE_TIMER_CNT_REG   | 0x0120 | AR400A wakeup timer counter     |
| AR400A_WAKE_TIMER_CTRL_REG  | 0x0124 | AR400A wakeup timer control     |
| AR400A_PRIV_REG5            | 0x0128 | AR400A Private Register         |
| AR400A_PRIV_REG6            | 0x012c | AR400A Private Register         |
| AR400A_IO_WAKE_EN_REG       | 0x0130 | AR400A IO Wakeup Enable         |
| AR400A_IO_WAKE_DEB_CLK_REG  | 0x0134 | AR400A IO Wakeup debounce clock |

|                                |        |                                                   |
|--------------------------------|--------|---------------------------------------------------|
| AR400A_IO_WAKE_ST_REG          | 0x0138 | AR400A IO Wakeup Status                           |
| AR400A_IO_HOLD_CTRL_REG        | 0x013c | AR400A IO Hold Control                            |
| AR400A_IO_WUP_GEN_REG          | 0x0140 | AR400A IO Wakeup Global Enable                    |
| AR400A_IO_WAKE_DEB_CYCLES0_REG | 0x0144 | AR400A IO Wakeup debounce clock cycles 0          |
| AR400A_IO_WAKE_DEB_CYCLES1_REG | 0x0148 | AR400A IO Wakeup debounce clock cycles 1          |
| LDO_MODE_SW_SEL_REG            | 0x0150 | LDO Mode Software Select                          |
| DCDC_PARAM_REG                 | 0x0200 | DCDC Parameter Register                           |
| ANA_BANDGAP_REG                | 0x0204 | ANA Bandgap Control Register                      |
| CLK_LDO_PARAM_REG              | 0x0208 | Clock LDO Parameter Register                      |
| DIG_LDO_PARAM_REG              | 0x020C | Digital LDO Parameter Register                    |
| DPLL_ST_REG                    | 0x0210 | DPLL Status Register                              |
| CFG_IO_STATUS_REG              | 0x214  | Config IO Status                                  |
| CPU_RST_SRC_REG                | 0x218  | CPU reset source register                         |
| WLAN_HIF_OV_CTRL_REG           | 0x21c  | WLAN HIF override control register                |
| SRAM_BIST_CFG_REG              | 0x220  | SRAM configure register                           |
| WLAN_SRAM_SHARE_REG            | 0x224  | WLAN SRAM share control register                  |
| BLE_SRAM_SHARE_CTRL_REG        | 0x228  | BLE 16K SRAM share control register               |
| LPUART0_WAKEUP_CTRL_REG        | 0x230  | LPUART0 wakeup control register                   |
| LPUART1_WAKEUP_CTRL_REG        | 0x234  | LPUART1 wakeup control register                   |
| GPADC_CLK_CTRL_REG             | 0x238  | GPADC core clock control register                 |
| CAPSEN_CLK_CTRL_REG            | 0x23C  | CAPSEN core clock control register                |
| WKUP_SRC_BUS_CLK_CTRL_REG      | 0x240  | Wakeup source bus clock control register          |
| WKUP_SRC_RST_CTRL_REG          | 0x244  | Wakeup source reset control register              |
| FLASH_ENCRYPT_AES_NONCE0_REG   | 0x250  | Flash encrypt module AES nonce low bits register  |
| FLASH_ENCRYPT_AES_NONCE1_REG   | 0x254  | Flash encrypt module AES nonce high bits register |
| BLE_RCOSC_CALIB_REG0           | 0x260  | RCOSC calibration control register0               |
| BLE_RCOSC_CALIB_REG1           | 0x264  | RCOSC calibration control register1               |
| BLE_CLK32K_SWITCH_REG0         | 0x268  | Clock 32k auto switch register0                   |
| BLE_CLK32K_SWITCH_REG1         | 0x26c  | Clock 32k auto switch register1                   |



### 3.1.3 GPRCM 寄存器描述

#### 3.1.3.1 SYS\_TOP\_LDO\_SMPS\_CTRL\_REG

| 偏移地址 : 0x0000 |     |     | 默认值: 0x0F00_8000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 31:29         | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 28:25         | R/W | 0x7 | <p>TOP_LDO_VOLTAGE_SEL</p> <p>Lower power top ldo voltage select</p> <p>N: 0~15</p> <p>0: 0.9V</p> <p>1: 1.0V</p> <p>2: 1.1V</p> <p>3: 1.2V</p> <p>4: 1.3V</p> <p>5: 1.4V</p> <p>6: 1.5V</p> <p>7: 1.6V</p> <p>8: 1.7V</p> <p>9: 1.8V</p> <p>10: 1.9V</p> <p>11: 2.0V</p> <p>12: 2.1V</p> <p>13: 2.2V</p> <p>14: 2.3V</p> <p>15: 2.4V</p> <p>the default is 1600mV</p> <p>Note: This bit will only be reset to the default value in power-on-reset operation. When the system is waked up from the deep sleep mode by some external events, this bit will keep the value which is set by the last write operation.</p> |
| 24            | R/W | 0x1 | <p>TOP_LDO_EN</p> <p>Lower power top ldo enable</p> <p>0: enable</p> <p>1: disable</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 23:16         | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 15:12         | R/W | 0x8 | <p>SMPS_VOLTAGE_SEL 1.0V – 2.5V)</p> <p>SMPS(DC-DC) voltage select</p> <p>N: 0~15</p> <p>0: 1.0V</p> <p>1: 1.1V</p> <p>2: 1.2V</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|       |    |   |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |    |   | 3: 1.3V<br>4: 1.4V<br>5: 1.5V<br>6: 1.6V<br>7: 1.7V<br>8: 1.8V<br>9: 1.9V<br>10: 2.0V<br>11: 2.1V<br>12: 2.2V<br>13: 2.3V<br>14: 2.4V<br>15: 2.5V<br><br>the default is 1800mV<br>Note: This bit will only be reset to the default value in power-on-reset operation. When the system is waked up from the deep sleep mode by some external events, this bit will keep the value which is set by the last write operation. |
| 11:10 | /  | / | /                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 9     | RW | 0 | DIGLDO_OFF_SMPS_ON<br>0: When DIG_LDO was turned off, SMPS could turned off either.<br>1: When DIG_LDO was turned off, SMPS remain turned on                                                                                                                                                                                                                                                                               |
| 8     | RW | 0 | SYS_STANDBY_SMPS_OFF<br>0: When system was in standby mode, SMPS could remain turned on<br>1: When system was in standby mode, SMPS should turned off                                                                                                                                                                                                                                                                      |
| 7     | RW | 0 | SMPS_PWM_SEL_EN<br>Enable SMPS_PWM_SEL function.                                                                                                                                                                                                                                                                                                                                                                           |
| 6: 4  | RW | 0 | SMPS_PWM_SEL<br>SMPS_PWM could from the following source:<br>000: clk_RfipDpllStart<br>001: clk_RfipDcxoStart<br>010: WLAN PHY enable<br>011: APP CPU enable<br>100: WLAN CPU enable<br>others: BLE RF enable                                                                                                                                                                                                              |
| 3     | /  | / |                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2     | RW | 0 | OVR_SMPS_DETCT<br>Overwirte SMPS_DETECT by software.                                                                                                                                                                                                                                                                                                                                                                       |
| 1     | RW | 0 | OVR_SMPS_DETCT_VALUE                                                                                                                                                                                                                                                                                                                                                                                                       |

|   |   |  |                                                 |
|---|---|--|-------------------------------------------------|
|   |   |  | Overwrite value of SMPS_DETECT                  |
| 0 | R |  | SMPS_DETECT<br>SMPS detect value from powerctrl |

### 3.1.3.2 SYS\_EXT\_LDO\_DIG\_LDO\_SW\_CTRL\_REG

| 偏移地址 : 0x0004 |     |     | 默认值: 0x0002_0103                                                                                                                                                                                                                                                                |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                              |
| 31:28         | /   | /   | /                                                                                                                                                                                                                                                                               |
| 27:24         | R/W | 0   | DIG_LDO_RET_VOL_SEL<br>LDO1 retention voltage select<br>0 ~ 12: N<br><br>0: 1125mV<br>1: 1025mV<br>2: 925mV<br>3: 825mV<br>4: 725mV<br>5: 625mV<br><br>8: 1175mV<br>9: 1225mV<br>10: 1275mV<br>11: 1325mV<br>12: 1375mV<br>others: reserved<br><br><i>the default is 1125mV</i> |
| 23            | R/W | 0x0 | EXT_LDO_BYPASS<br>0: EXT_LDO not be bypassed<br>1: EXT_LDO be bypassed                                                                                                                                                                                                          |
| 22:21         | R/W | 0x0 | EXT_LDO_SW_TRIM<br>Bit[1:0], option select of the regulate cycle when EXT_LDO in switch mode                                                                                                                                                                                    |
| 20            | R/W | 0x0 | EXT_LDO_SW_MODE<br>0: EXT_LDO not enter switch mode<br>1: EXT_LDO enter switch mode                                                                                                                                                                                             |
| 19:18         | R/W | 0x0 | EXT_LDO_VOL_SEL<br>EXT_LDO voltage select<br>0: 3.3V<br>1: 3.1V                                                                                                                                                                                                                 |

|       |     |     |                                                                                                                                                                                                                                            |
|-------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |     |     | 2: 2.8V<br>3: 2.5V<br><i>the default is 3.3V</i>                                                                                                                                                                                           |
| 17:16 | R/W | 0x2 | EXT_LDO_MODE<br>EXT_LDO mode<br>00: always off<br>01: on/off same as TOP_LDO<br>1x: always on                                                                                                                                              |
| 15    | /   | /   | /                                                                                                                                                                                                                                          |
| 14    | R   | 0   | SW5_STATUS<br>WLAN System Power Switch SW5 status<br><br>1: SW5 is on<br>0: SW5 is off                                                                                                                                                     |
| 13:9  | /   | /   | /                                                                                                                                                                                                                                          |
| 8     | R   | 1   | SW1_STATUS<br>AR400A and Application System Power Switch status<br><br>1: SW1 is on<br>0: SW1 is off<br><i>Note: this bit will only be set by <b>POWERCTRL</b> module.</i>                                                                 |
| 7:4   | R/W | 0   | DIG_LDO_VOL_SEL<br>0 ~ 12: N<br><br>0: 1125mV<br>1: 1025mV<br>2: 925mV<br>3: 825mV<br>4: 725mV<br>5: 625mV<br><br>8: 1175mV<br>9: 1225mV<br>10: 1275mV<br>11: 1325mV<br>12: 1375mV<br>others: reserved<br><br><i>the default is 1125mV</i> |
| 3:2   | /   | /   | /                                                                                                                                                                                                                                          |
| 1     | R/W | 1   | PLL_LDO_EN                                                                                                                                                                                                                                 |

|   |   |   |                                                                                                                                             |
|---|---|---|---------------------------------------------------------------------------------------------------------------------------------------------|
|   |   |   | 1: Write 1 to enable the PLL_LDO_EN.<br>0: write 0 to disable the PLL_LDO_EN.                                                               |
| 0 | R | 1 | DIG_LDO_EN<br><br>1: Write 1 to enable the LDO1.<br>0: write 0 to disable the LDO1.<br>Note: this bit will only be set by POWERCTRL module. |

### 3.1.3.3 SYS\_LFCLK\_CTRL\_REG

| 偏移地址 : 0x0008 |     |     | 默认值: 0xC100_0000                                                                                                                                                                                                                                              |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                            |
| 31            | R/W | 1   | OSC32K_EN<br><br>1: enable the 32768 Oscillator<br>0: disable the 32768 Oscillator                                                                                                                                                                            |
| 30            | R/W | 1   | RC32K_EN<br><br>1: enable the 32768 RC Oscillator<br>0: disable the 32768 RC Oscillator                                                                                                                                                                       |
| 29:25         | /   | /   | /                                                                                                                                                                                                                                                             |
| 24            | R/W | 1   | LFCLK_SRC_SEL<br><br>Low frequency clock source select<br>0: source the LFCLK from the internal 32768 RC oscillator.<br>1: source the LFCLK from the external 32768 crystal oscillator                                                                        |
| 23:8          | R/W | 0   | PAD_CLOCK_OUT_FACTOR_M<br>factor M,<br>M= Factor + 1<br>Output clock frequency Fout = Fsrc/M                                                                                                                                                                  |
| 2:1           | R/W | 0   | PAD_CLOCK_OUT_SEL<br><br>GPIO PAD 32KOSCO clock source select<br>0: source the LFCLK from the internal 32768 RC oscillator<br>1: source the LFCLK from the BLE 32K<br>2: source the RCO_CALIB 32K<br>3: source the HFCLK from the external crystal oscillator |
| 0             | R/W | 0   | PAD_CLOCK_OUT_EN<br><br>GPIO PAD 32KOSCO clock output enable.                                                                                                                                                                                                 |

|  |  |  |                         |
|--|--|--|-------------------------|
|  |  |  | 0: disable<br>1: enable |
|--|--|--|-------------------------|

#### 3.1.3.4 SYS\_HOSC\_TYPE\_REG

| 偏移地址 : 0x000C |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                           |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                         |
| 31:2          | /   | /   | /                                                                                                                                                                                                                                                                                                          |
| 1:0           | R/W | 0   | HOSC_TYPE<br><br>0: System uses 26MHz HOSC<br>1: System uses 40MHz HOSC<br>2: System uses 24MHz HOSC<br>3: System uses 32MHz HOSC<br><br><i>Note: This bit is set by application software and must be consistent with the real situation. A wrong configuration will cause the RCOSC calibration fail.</i> |

#### 3.1.3.5 SYS\_RCOSC\_CAL\_CTRL\_REG

| 偏移地址 : 0x0010 |     |       | 默认值: 0x000C_8001              |
|---------------|-----|-------|-------------------------------|
| 位置            | 访问  | 默认值   | 描述                            |
| 31:28         | /   | /     | /                             |
| 27:8          | R   | 0xc80 | RC-OSC Frequency<br>V=FREQ/10 |
| 7:1           | /   | /     | /                             |
| 0             | R/W | 1     | RC-OSC Calibration Enable     |

#### 3.1.3.6 SYS\_PLL\_CTRL\_REG

| 偏移地址 : 0x0020 |     |     | 默认值: 0x6000_0281 @24MHz OSC,<br>0x1D89_D241 @26MHz OSC,<br>0x6000_0181 @40MHz OSC,<br>0x6000_01E1 @32MHz OSC |
|---------------|-----|-----|--------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                           |
| 31            | R/W | 0x0 | PLL_ENABLE<br><br>Enable or disable the system PLL.<br>0: Disable the system PLL<br>1: Enable the system PLL |

|       |    |        |                                                                                                                                                                          |
|-------|----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |    |        | <b>Note:</b> The PLL frequency $F_{output} = (F_{hosc} * N.f) / M$ , the output clock must be in range of 960MHz.<br>The default value is set for 960MHz with 40MHz OSC. |
| 30:29 | RW | 0x3    | DPLL_DITHER_DISABLE                                                                                                                                                      |
| 28:13 | RW | 0x0000 | DPLL_FRAC<br><b>Note:</b> $V_{FRAC} = f * 2^{16}$                                                                                                                        |
| 12    | RW | 0x0    | DPLL_FRAC_CTRL                                                                                                                                                           |
| 11:4  | RW | 0x18   | DPLL_NDIV                                                                                                                                                                |
| 3:0   | RW | 1      | FACTOR_M<br><br>PLL factor M (1 ~ 8).<br>M = Factor<br><b>Note:</b> when the factor is set to 0, M will be set to 1                                                      |

### 3.1.3.7 SYS\_CLK1\_CTRL\_REG

| 偏移地址 : 0x0024 |     |     | 默认值: 0x0000_0201                                                                                 |
|---------------|-----|-----|--------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                               |
| 31:25         | /   | /   | /                                                                                                |
| 24            | R/W | 0x0 | ROM_READ_TWO_CYCLES_MODE<br>0: one cycle ROM read.<br>1: two cycles ROM read                     |
| 23:18         | /   | /   | /                                                                                                |
| 17:16         | R/W | 0x0 | CPU_CLK_SRC_SEL<br><br>CPU clock source select<br>00: HFCLK(default)<br>01: LFCLK<br>1x: SYSCLK1 |
| 15:11         | /   | /   | /                                                                                                |
| 10:8          | RW  | 0x2 | FACTOR_M<br>PLL factor M (4~7,9)<br>0~3: M= Factor + 4<br>4: M=9<br>5~7: reserved                |
| 7:4           | /   | /   | /                                                                                                |
| 3:0           | RW  | 0x1 | FACTOR_N<br>PLL factor N (1 ~ 16).<br>Factor = 0~15                                              |

|  |  |  |                                                                                                           |
|--|--|--|-----------------------------------------------------------------------------------------------------------|
|  |  |  | $N = \text{Factor} + 1$<br><b>Note:</b> $F_{sysclk1} = 960\text{MHz} / M/N$ , the default value is 80MHz. |
|--|--|--|-----------------------------------------------------------------------------------------------------------|

### 3.1.3.8 SYS\_CLK3\_CTRL\_REG

| 偏移地址 : 0x002C |    |     | 默认值: 0x0000_0000                                                                                                                                                      |
|---------------|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                    |
| 31            | RW | 0   | SYS_CLK3_EN<br><br>1: enable the clock of system 3<br>0: disable the clock of system 3<br><br><b>Note:</b> $F_{sysclk} = F_{src} / N$ , the clock is fixed to 160MHz. |
| 30:1          | /  | /   | /                                                                                                                                                                     |
| 0             | RW | 0   | WLAN_WUP_BY_HOSC<br>1: WLAN wake up from SLEEP mode using HOSC<br>0: WLAN wake up from SLEEP mode using LOSC                                                          |

### 3.1.3.9 DEV\_CLK\_CTRL\_REG

| 偏移地址 : 0x0034 |    |     | 默认值: 0x0003_0001                                                                                                                                                                                                       |
|---------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                     |
| 31            | RW | 0   | CKADC_AUD_EN<br>1: enable the audio codec ADC clock<br>0: disable the audio codec ADC clock                                                                                                                            |
| 30:24         | /  | /   | /                                                                                                                                                                                                                      |
| 23            | RW | 0   | CKCLD_AUD_EN<br>1: enable the audio class-D clock<br>0: disable the audio class-D clock                                                                                                                                |
| 22:21         | /  | /   | /                                                                                                                                                                                                                      |
| 20:16         | RW | 3   | CKCLD_FACTOR_N<br>PLL factor N (4 ~ 32).<br>Factor = 3~31<br>$N = \text{Factor} + 1$<br><b>Note:</b> $F_{ckld} = 960\text{MHz} / 2/N$ , the default value is 120MHz. this clock is used as a source clock for class-D. |
| 15:11         | /  | /   | /                                                                                                                                                                                                                      |
| 10:8          | RW | 0x0 | DEV1_FACTOR_M<br>PLL factor M (4~7,9)<br>0~3: $M = \text{Factor} + 4$                                                                                                                                                  |



|     |    |     |                                                                                                                                                                                                                                   |
|-----|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |    |     | 4: M=9<br>5~7: reserved                                                                                                                                                                                                           |
| 7:4 | /  | /   | /                                                                                                                                                                                                                                 |
| 3:0 | RW | 0x1 | DEV1_FACTOR_N<br><br>PLL factor N (1 ~ 16).<br>Factor = 0~15<br>N = Factor + 1<br><i>Note: <math>F_{devclk1} = 960\text{MHz} / M/N</math>, the default value is 96MHz. this clock is used as a source clock for some devices.</i> |

### 3.1.3.10 DCXO\_CTRL0\_REG

| 偏移地址 : 0x0058 |    |      | 默认值: 0x9409_0B00                                                                                                                  |
|---------------|----|------|-----------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                                                                                                |
| 31            | RW | 1    | DCXO_EN<br>DCXO Enable<br>0: disable<br>1: enable                                                                                 |
| 30            | /  | /    | /                                                                                                                                 |
| 29:28         | RW | 1    | CLK_MODE_SEL<br>Clock Mode Select (When AUTO_DET_EN is disable)<br>00: Internal Clock<br>01: Crystal<br>10: Analog<br>11: Digital |
| 27:20         | RW | 0x40 | FRE_TRI<br>Frequency Trimming                                                                                                     |
| 19            | RW | 1    | ALC_EN<br>ALC Enable<br>0: disable<br>1: enable                                                                                   |
| 18:17         | /  | /    | /                                                                                                                                 |
| 16:12         | RW | 0x10 | ICTRL<br>BIAS Current Control                                                                                                     |
| 11            | RW | 1    | RF_DIG_REF_CLK_EN<br><br>0: disable<br>1: enable                                                                                  |

|     |    |    |                                                                            |
|-----|----|----|----------------------------------------------------------------------------|
| 10  | /  | /  | /                                                                          |
| 9   | RW | 1  | BUFFER_DPLL_EN<br><br>0: disable<br>1: enable                              |
| 8:6 | /  | /  | /                                                                          |
| 5:4 | R  | xx | DCXO_DET<br>00: Internal clock<br>01: Crystal<br>10: Analog<br>11: Digital |
| 3:1 | /  | /  | /                                                                          |
| 0   | RW | 0  | AUTO_DET_EN<br>Auto Detect Enable<br>0: disable<br>1: enable               |

### 3.1.3.11 CONN\_BOOT\_FLAG\_REG

| 偏移地址：0x0070 |    | 默认值: 0x0000_0000 |                                                                       |
|-------------|----|------------------|-----------------------------------------------------------------------|
| 位置          | 访问 | 默认值              | 描述                                                                    |
| 31:1        | /  | /                | /                                                                     |
| 0           | WR | 0                | CONN_BOOT_FLAG<br>Connect mode flag<br>1: Connect mode<br>0: IOT mode |

### 3.1.3.12 POWERCTRL\_CFG\_REG

| 偏移地址：0x0074 |    | 默认值: 0x0000_05CE |                                                                                                                                                         |
|-------------|----|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值              | 描述                                                                                                                                                      |
| 31:14       | /  | /                | /                                                                                                                                                       |
| 13          | RW | 0                | BT_RSTN_HIGHLEVEL_WAKEUP<br>0: Wakeup when found posedge of BT_RSTN.<br>1: Wakeup when found high level of BT_RSTN.                                     |
| 12          | RW | 0                | WLAN_IRQ_HIGHLEVEL_WAKEUP<br>0: WLAN_IRQ could only wakeup powerctrl through WIC.<br>1: Wakeup when found high level of WLAN_IRQ while BT_RSTN was low. |
| 11          | RW | 0                | IGNORE_WICWAKEUP                                                                                                                                        |

|     |    |    |                                                                                                                                                                                      |
|-----|----|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |    |    | 0: WIC WAKEUP could take into effect at any time.<br>1: WIC WAKEUP couldn't take into effect when BT_RSTN was low.                                                                   |
| 10  | RW | 1  | IGNORE_WICENACK<br>0: Use WICENACK in sleep/wakeup process.<br>1: Don't use WICENACK in sleep/wakeup process.                                                                        |
| 9   | RW | 0  | WAKEUP_MODE<br>0: Wakeup when found posedge of interrupt.<br>1: Wakeup when found high level of interrupt.                                                                           |
| 8   | RW | 1  | RTC_ALARM_WUP_TO_POWERCTRL<br>0: RTC alarm0/1 interrupt could not wake up powerctrl directly..<br>1: RTC alarm0/1 interrupt could wake up powerctrl directly.                        |
| 7:6 | RW | 11 | RCOSC_CALIB_START_SRC<br>RCOSC Calibration Start Source.<br><br>00: APP sleep<br>01: WLAN sleep<br>10: APP and WLAN sleep<br>11: LDO1 shut down                                      |
| 5   | RW | 0  | CPU_SLEEP_NO_RETENTION<br>0: When CPU goto sleep mode, APP RAM could be retention.<br>1: When CPU goto sleep mode, no need to retention APP RAM.                                     |
| 4   | RW | 0  | BT_SLEEP_RETENTION_OVR<br>0: BT reset retention should not overwrite CPU retention.<br>1: BT reset retention could overwrite CPU retention.                                          |
| 3   | RW | 1  | BLE_SLEEP_TIMER_IRQ_TO_POWERCTRL<br>0: BLE Sleep Timer interrupt could not wake up powerctrl directly..<br>1: BLE Sleep Timer interrupt could wake up powerctrl directly.            |
| 2   | RW | 1  | WUPTIMER_IRQ_TO_POWERCTRL<br>0: Wakeup Timer interrupt could not wake up powerctrl directly..<br>1: Wakeup Timer interrupt could wake up powerctrl directly.                         |
| 1   | RW | 1  | WUPIO_IRQ_TO_POWERCTRL<br>0: Wakeup IO interrupt could not wake up powerctrl directly..<br>1: Wakeup IO interrupt could wake up powerctrl directly.                                  |
| 0   | RW | 0  | BT_RSTN_WAKEUPPIN<br>0: BT reset PAD could not be used as wake up pin.<br>1: BT reset PAD could be used as wake up pin, positive edge means wake up, negative edge means goto sleep. |

### 3.1.3.13 SYS1\_CTRL\_REG

| 偏移地址：0x0080 |    |     | 默认值: 0x0000_0003                                                                                                                               |
|-------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                             |
| 31:18       | /  | /   | /                                                                                                                                              |
| 17          | R  |     | WLAN_RSTN_STATU<br>The WRSTN PIN status in connect mode<br>0: WLAN reset signal hold<br>1: WLAN reset signal release                           |
| 16          | R  | /   | BLE_RSTN_STATU<br>The BRSTN PIN status in connect mode<br>0: BLE reset signal hold<br>1: BLE reset signal release                              |
| 15:2        | /  | /   | /                                                                                                                                              |
| 1           | R  | /   | AR800A_RSTN_STATU<br>system 1 Reset<br>1: release the global reset signal of the system 1<br>0: hold the global reset signal of the system 1   |
| 0           | R  | /   | SYS1_RST_RSTN_STATU<br>system 1 Reset<br>1: release the global reset signal of the system 1<br>0: hold the global reset signal of the system 1 |

### 3.1.3.14 SYS1\_STA\_REG

| 偏移地址：0x0084 |    |     | 默认值: 0x0000_0000                                                                                                    |
|-------------|----|-----|---------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                  |
| 31:3        | /  | /   | /                                                                                                                   |
| 2           | R  | 0   | CPU_SLEEP<br><br>1: the CPU of system 1 is in SLEEP mode<br>0: the CPU of system 1 is not in SLEEP mode             |
| 1           | R  | 0   | CPU_SLEEPDEEP<br><br>1: the CPU of system 1 is in SLEEPDEEP mode<br>0: the CPU of system 1 is not in SLEEPDEEP mode |
| 0           | R  | 0   | SUBSYS1_ALIVE<br><br>1: system 1 is alive                                                                           |

|  |  |  |                     |
|--|--|--|---------------------|
|  |  |  | 0: system 1 is down |
|--|--|--|---------------------|

### 3.1.3.15 SYS3\_CTRL\_REG

| 偏移地址 : 0x0090 |    |     | 默认值: 0x0000_0000                                                                                                                                   |      |       |
|---------------|----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                 |      |       |
| 31:17         | /  | /   | /                                                                                                                                                  |      |       |
| 16            | RW | 0   | WLAN_RSTN_CONN<br>Wlan SW reset in connect mode<br>1: release the SW reset of WLAN in connect mode<br>0: hold the SW reset of WLAN in connect mode |      |       |
| 15:2          | /  | /   | /                                                                                                                                                  |      |       |
| 0             | RW | 0   | SYS3_RST                                                                                                                                           |      |       |
|               |    |     | System 3 Reset<br>1: release the global reset of the system 3<br>0: hold the global reset of the system 3                                          |      |       |
|               |    |     |                                                                                                                                                    | Read | Write |
|               |    |     | AR400A                                                                                                                                             | R    | W     |

### 3.1.3.16 SYS3\_STA\_REG

| 偏移地址 : 0x0094 |    |     | 默认值: 0x0000_0000                                                                                                    |
|---------------|----|-----|---------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                  |
| 31:3          | /  | /   | /                                                                                                                   |
| 2             | R  | 0   | CPU_SLEEP<br><br>1: the CPU of system 3 is in SLEEP mode<br>0: the CPU of system 3 is not in SLEEP mode             |
| 1             | R  | 0   | CPU_SLEEPDEEP<br><br>1: the CPU of system 3 is in SLEEPDEEP mode<br>0: the CPU of system 3 is not in SLEEPDEEP mode |
| 0             | R  | 0   | SYS3_ALIVE<br><br>1: system 3 is alive<br>0: system 3 is down                                                       |

**Table 3- 1 System 3 Status Register**

### 3.1.3.17 SYS1\_WAKEUP\_CTRL\_REG

| 偏移地址：0x0098 |    |     | 默认值: 0x0000_000F                                                                                                         |
|-------------|----|-----|--------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                       |
| 31:2        | /  | /   | /                                                                                                                        |
| 1           | RW | 1   | VDDIO_SIP_SW_EN<br><br>1: VDDIO_SIP SW is enabled<br>0: VDDIO_SIP SW is closed                                           |
| 0           | RW | 1   | SYS1_WAKEUP_LDO1_ON<br><br>1: LDO1 will be enabled when waking up the system 1<br>0: keep the LDO with its current state |

### 3.1.3.18 SYS1\_SLEEP\_CTRL\_REG

| 偏移地址：0x00A0 |    |     | 默认值: 0x8000_0002                                                                                                                             |
|-------------|----|-----|----------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                           |
| 31          | RW | 1   | WLAN_SRAM_96K_SWM8_MODE<br><br>WLAN SRAM(96k bytes) retention power switch (SWM8) mode<br>0: always on<br>1: SRAM switch maybe off           |
| 30          | RW | 0   | CACHE_BLE_SRAM_48K_SWM7_MODE<br><br>CACHE/BLE SRAM(48k bytes) retention power switch (SWM7) mode<br>0: always on<br>1: SRAM switch maybe off |
| 29          | RW | 0   | SRAM_16K_SWM6_MODE<br><br>SRAM(16k bytes) retention power switch (SWM6) mode<br>0: always on<br>1: SRAM switch maybe off                     |
| 28          | RW | 0   | SRAM_16K_SWM5_MODE<br><br>SRAM(16k bytes) retention power switch (SWM5) mode<br>0: always on<br>1: SRAM switch maybe off                     |
| 27          | RW | 0   | SRAM_32K_SWM4_MODE                                                                                                                           |

|      |    |   |                                                                                                                                                        |
|------|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |    |   | SRAM(32k bytes) switch (SWM4) mode<br>0: always on<br>1: SRAM switch maybe off                                                                         |
| 26   | RW | 0 | SRAM_64K_SWM3_MODE<br><br>SRAM(64k bytes) switch (SWM3) mode<br>0: always on<br>1: SRAM switch maybe off                                               |
| 25   | RW | 0 | SRAM_96K_SWM2_MODE<br><br>SRAM(96k bytes) switch (SWM2) mode<br>0: always on<br>1: SRAM switch maybe off                                               |
| 24   | RW | 0 | SRAM_32K_SWM1_MODE<br><br>SRAM(32k bytes) switch (SWM1) mode<br>0: always on<br>1: SRAM switch maybe off                                               |
| 23:8 | /  | / | /                                                                                                                                                      |
| 7:2  | /  | / | /                                                                                                                                                      |
| 1    | RW | 1 | SYS1_SLEEP_SW1_OFF<br><br>1: SW1 will be disabled when the system 1 goes into the deep-sleep mode<br>0: keep the power switch with its current state   |
| 0    | RW | 0 | SYS1_SLEEP_LDO1_OFF<br><br>1: LDO1 will be disabled when the system 1 goes into the deep-sleep mode<br>0: keep the power switch with its current state |

### 3.1.3.19 DCXO\_STABLE\_REF\_TIME\_REG

| 偏移地址：0x00A8 |    |      | 默认值: 0x8000_0080                                                                                                      |
|-------------|----|------|-----------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值  | 描述                                                                                                                    |
| 31:10       | /  | /    | /                                                                                                                     |
| 9:0         | RW | 0x80 | DCXO Stable Reference Time<br>$REF\_TIME = T_{LFCLK} * n$<br>Typical Value <= 500us<br>Set Value = 14us * 128 = 1.8ms |

### 3.1.3.20 DPLL\_STABLE\_REF\_TIME\_REG

| 偏移地址 : 0x00AC |    |     | 默认值: 0x0000_000C                                                                                                                  |
|---------------|----|-----|-----------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                |
| 31:8          | /  | /   | /                                                                                                                                 |
| 7:0           | RW | 0xC | DPLL Stable Reference Time<br>$REF\_TIME = T_{LFCLK} * n$<br><i>Typical Value</i> <= 50us<br><i>Set Value</i> = 14us * 12 = 168us |

### 3.1.3.21 LDO\_STABLE\_REF\_TIME\_REG

| 偏移地址 : 0x00B0 |    |     | 默认值: 0x0000_030D                                                                                                                      |
|---------------|----|-----|---------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                    |
| 31:12         | /  | /   | /                                                                                                                                     |
| 11: 8         | RW | 0x3 | RfipLdoDig Stable Reference Time<br>$REF\_TIME = T_{LFCLK} * n$<br><i>Typical Value</i> <= 10us<br><i>Set Value</i> = 14us * 3 = 42us |
| 7:0           | RW | 0xD | LDO Stable Reference Time<br>$REF\_TIME = T_{LFCLK} * n$<br><i>Typical Value</i> <= 60us<br><i>Set Value</i> = 14us * 13 = 182us      |

### 3.1.3.22 DIGITAL\_SWITCH\_REF\_TIME\_REG

| 偏移地址 : 0x00B4 |    |        | 默认值: 0x3000_0002                                                                                                                                       |
|---------------|----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值    | 描述                                                                                                                                                     |
| 31:16         | RW | 0x3000 | Reset Up Reference Time<br>$REF\_TIME = T_{HFCLK} * n$<br><i>Typical Value</i> >= 12288 cycles<br><i>Set Value</i> = 25ns * 12288 = 300us (@40MHz OSC) |
| 15:0          | RW | 0x2    | Digital Switch Reference Time<br>$REF\_TIME = T_{HFCLK} * n$<br><i>Typical Value</i> >= 2 cycles<br><i>Set Value</i> = 25ns * 2 = 50ns (@40MHz OSC)    |



### 3.1.3.23 RTC\_LDO\_VOL\_CTRL\_REG

| 偏移地址 : 0x00B8 |    |     | 默认值: 0x0000_0000                                                                                                                         |
|---------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                       |
| 31:19         | /  | /   | 0                                                                                                                                        |
| 18:16         | RW | 0   | RTC LDO work Voltage Control<br>0: 1.075V<br>1: 0.975V<br>2: 0.875V<br>3: 0.775V<br>4: 0.675V<br>5: 0.60V<br>6: 1.175V<br>7: 1.275V      |
| 15:3          | /  | /   | /                                                                                                                                        |
| 2:0           | RW | 0   | RTC LDO Retention Voltage Control<br>0: 1.075V<br>1: 0.975V<br>2: 0.875V<br>3: 0.775V<br>4: 0.675V<br>5: 0.60V<br>6: 1.175V<br>7: 1.275V |

### 3.1.3.24 BG\_STABLE\_REF\_TIME\_REG

| 偏移地址 : 0x00BC |    |      | 默认值: 0x0000_0017                                                                                                          |
|---------------|----|------|---------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                                                                                        |
| 31:8          | /  | /    | /                                                                                                                         |
| 7:0           | RW | 0x17 | LDO Stable Reference Time<br>$REF\_TIME = T_{LFCLK} * n$<br>Typical Value $\leq 100us$<br>Set Value = $14us * 23 = 322us$ |

### 3.1.3.25 DCDC\_STABLE\_REF\_TIME\_REG

| 偏移地址 : 0x00C0 |    |     | 默认值: 0x0080_000C |
|---------------|----|-----|------------------|
| 位置            | 访问 | 默认值 | 描述               |
| 31:26         | /  | /   | /                |

|       |    |      |                                                                                                                                             |
|-------|----|------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 25:16 | RW | 0x80 | LDO to PWM Mode Stable Reference Time<br>$REF\_TIME = T_{LFCLK} * n$<br>Typical Value $\leq 500us$<br>Set Value = $14us * 128 = 1.8ms$      |
| 15:10 | /  | /    | /                                                                                                                                           |
| 9:0   | RW | 0xC  | Other Work Mode Switch Stable Reference Time<br>$REF\_TIME = T_{LFCLK} * n$<br>Typical Value $\leq 50us$<br>Set Value = $14us * 16 = 168us$ |

### 3.1.3.26 BLE\_RTC\_RST\_CTRL\_REG

| 偏移地址：0x00D0 |    |     | 默认值: 0x0000_0000                                                                                                              |
|-------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                            |
| 31:1        | /  | /   | /                                                                                                                             |
| 0           | RW | 0   | BLE_RTC_RST_RST<br><br>BLE RTC Reset<br>1: release the global reset of the BLE RTC<br>0: hold the global reset of the BLE RTC |

### 3.1.3.27 BLE\_RTC\_RST\_CTRL\_REG

| 偏移地址：0x00D4 |    |     | 默认值: 0x0000_0000                                                  |
|-------------|----|-----|-------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                |
| 31:2        | /  | /   | /                                                                 |
| 1           | RW | 0   | BLE_32M_CLK_EN<br>BLE 32M clock enable<br>0: disable<br>1: enable |
| 0           | RW | 0   | BLE_48M_CLK_EN<br>BLE 48M clock enable<br>0: disable<br>1: enable |

### 3.1.3.28 RFAS\_CTRL\_REG

| 偏移地址：0x00D8 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:1        | /  | /   | /                |

|   |    |   |                                                                                                       |
|---|----|---|-------------------------------------------------------------------------------------------------------|
| 0 | RW | 0 | RFAS_RST<br><br>RFAS Reset<br>1: release the global reset of RFAS<br>0: hold the global reset of RFAS |
|---|----|---|-------------------------------------------------------------------------------------------------------|

### 3.1.3.29 AR800A\_BOOT\_FLAG\_REG

| 偏移地址 : 0x0100 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 31:16         | RW | 0   | BOOT_FLAG_WRITE_LOCK<br><br>0x429B: AR800A_BOOT_FLAG filed is writable<br>other: AR800A_BOOT_FLAG filed is read only<br><br><b>Note:</b> The field of AR800A_BOOT_FLAG only can be changed when this bit is set to <b>0x429B</b> . In one write operation, the GRPCM will check this field whether equals to 0x429B. If not, the value of AR400A_BOOT_FLAG will be ignored.<br><br>This field is always read with value 0x0000.<br><br>Examples:<br>1. After write 0x429B0001, then read value = 0x00000001<br>2. After write 0x11110000, then read value = 0x00000001<br>After write 0x429B0000, then read value = 0x00000000 |
| 15:4          | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3:0           | RW | 0   | AR800A_BOOT_FLAG<br><br>0: boot from cold reset(default)<br>1: boot from deep sleep<br>2: boot for system update<br>3: boot for recovery<br>others: reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### 3.1.3.30 AR800A\_BOOT\_ADDR\_REG

| 偏移地址 : 0x0104 |    |     | 默认值: 0x0000_0000 |
|---------------|----|-----|------------------|
| 位置            | 访问 | 默认值 | 描述               |
| 31:0          | RW | 0   | AR800A_BOOT_ADDR |

### 3.1.3.31 AR800A\_BOOT\_ARG\_REG

| 偏移地址：0x0108 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:0        | RW | 0   | AR800A_BOOT_ARG  |

### 3.1.3.32 AR800A\_PRIV\_REG0

| 偏移地址：0x010C |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:0        | RW | 0   | AR800A_PRIV_REG0 |

### 3.1.3.33 AR800A\_PRIV\_REG1

| 偏移地址：0x0110 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:0        | RW | 0   | AR800A_PRIV_REG1 |

### 3.1.3.34 AR800A\_PRIV\_REG2

| 偏移地址：0x0114 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:0        | RW | 0   | AR800A_PRIV_REG2 |

### 3.1.3.35 AR800A\_PRIV\_REG3

| 偏移地址：0x0118 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:0        | RW | 0   | AR800A_PRIV_REG3 |

### 3.1.3.36 AR800A\_PRIV\_REG4

| 偏移地址：0x011C |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:0        | RW | 0   | AR800A_PRIV_REG4 |

### 3.1.3.37 AR800A\_WAKE\_TIMER\_CNT\_REG

| 偏移地址：0x0120 |    |     | 默认值: 0x0000_0000     |
|-------------|----|-----|----------------------|
| 位置          | 访问 | 默认值 | 描述                   |
| 31          | RW | 0   | AR800A_WAKE_TIMER_EN |

|      |   |   |                                                                                                                                                                                                                                                                                                                |
|------|---|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30:0 | R | 0 | AR400A_WAKE_TIMER_CNT<br>Note: This is a free running counter and when the value of this counter equals to the value of AR400A_WAKE_TIMER_VAL, a wakeup signal will be issued to the Wakeup Interrupt Controller.<br>If AR400A_WAKE_TIMER_EN is set to '0', this counter will be reset to 0 and stop counting. |
|------|---|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.1.3.38 AR800A\_WAKE\_TIMER\_VAL\_REG

| 偏移地址 : 0x0124 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 31            | RW | 0   | AR800A_WAKEUP_TIMER_IRQ_STATUS<br>0: the ar800a wakeup timer irq is not pending<br>1: the ar800a wakeup timer irq is pending<br>Note: this is a write-1-to-clear register. This is an asynchronies operation and this bit will not become to '0' immediately after writing '1' to clear the irq state. We should read the register until the value of this bit becomes '0', which means the irq pending state is actually cleared. |
| 30:0          | RW | 0   | AR800A_WAKE_TIMER_VAL                                                                                                                                                                                                                                                                                                                                                                                                              |

### 3.1.3.39 AR800A\_PRIV\_REG5

| 偏移地址 : 0x0128 |    |     | 默认值: 0x0000_0000 |
|---------------|----|-----|------------------|
| 位置            | 访问 | 默认值 | 描述               |
| 31:0          | RW | 0   | AR800A_PRIV_REG5 |

### 3.1.3.40 AR800A\_PRIV\_REG6

| 偏移地址 : 0x012C |    |     | 默认值: 0x0000_0000 |
|---------------|----|-----|------------------|
| 位置            | 访问 | 默认值 | 描述               |
| 31:0          | RW | 0   | AR800A_PRIV_REG6 |

### 3.1.3.41 AR800A\_WAKE\_IO\_EN\_REG

| 偏移地址 : 0x0130 |    |     | 默认值: 0x0000_0000                                                                                |
|---------------|----|-----|-------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                              |
| 31:30         | /  | /   | /                                                                                               |
| 29:16         | /  | 0   | AR800A_WAKEUP_IOx_MODE<br>1: Positive Edge<br>0: Negative Edge<br>bit0- bit14: WUPIO0 – WUPIO13 |
| 15:14         | /  | /   | /                                                                                               |

|      |    |   |                                                                                                                             |
|------|----|---|-----------------------------------------------------------------------------------------------------------------------------|
| 13:0 | RW | 0 | AR800A_WAKEUP_IOx_ENB<br>1: IOx wakeup detection enable<br>0: IOx wakeup detection disable<br>bit0- bit14: WUPIO0 – WUPIO13 |
|------|----|---|-----------------------------------------------------------------------------------------------------------------------------|

### 3.1.3.42 AR800A\_WAKE\_IO\_DEB\_CLK\_REG

| 偏移地址 : 0x0134 |    |     | 默认值: 0x0000_0000                                                                                                                                                     |
|---------------|----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                   |
| 31:28         | RW | 0x0 | AR800A_WAKEUP_DEB_CLK_SET1<br>IO wakeup debounce clk1 select, the DEB_CLK1 frequency<br>CLK_SEL1<=14, Freq=FCLK/(2 <sup>CLK_SEL1</sup> );<br>CLK_SEL1==15, reserved. |
| 27:24         | RW | 0x0 | AR800A_WAKEUP_DEB_CLK_SET0<br>IO wakeup debounce clk0 select, the DEB_CLK0 frequency<br>CLK_SEL0<=14, Freq=FCLK/(2 <sup>CLK_SEL0</sup> );<br>CLK_SEL0==15, reserved. |
| 23:16         | /  | /   | /                                                                                                                                                                    |
| 15:14         | /  | /   | /                                                                                                                                                                    |
| 13:0          | RW | 0x0 | AR800A_WAKEUP_IOx_CLK_SEL<br>0: IOx select DEB_CLK0<br>1: IOx select DEB_CLK1<br>bit0- bit14: WUPIO0 – WUPIO13                                                       |

### 3.1.3.43 AR800A\_WAKE\_IO\_STA\_REG

| 偏移地址 : 0x0138 |    |     | 默认值: 0x0000_0000                                                                                                                                      |
|---------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                    |
| 31:14         | /  | /   | /                                                                                                                                                     |
| 13:0          | RW | 0   | AR800A_WAKEUP_IOx_ST<br>1: wakeup event is detected on IOx<br>0: Nothing<br>bit0- bit14: WUPIO0 – WUPIO13<br><i>Note: Write 1 to clear the status</i> |

### 3.1.3.44 AR800A\_WAKE\_IO\_STA\_REG

| 偏移地址 : 0x013C |    |     | 默认值: 0x0000_0000 |
|---------------|----|-----|------------------|
| 位置            | 访问 | 默认值 | 描述               |
| 31:14         | /  | /   | /                |

|      |       |   |                                                                                                                                                                                                                                                                                                                                                           |
|------|-------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13:0 | R/W1C | 0 | AR800A_WAKEUP_IOx_ST<br>1: IO Configure Hold Enable<br>0: IO Configure Hold Disable<br><br>bit0- bit14: WUPIO0 – WUPIO13<br><br>Note: when enable the IO hold function, the configuration(IEN, OEN, PULL, ODAT) of the relative IO will be hold to the current state even if the GPIO controller is powered down or the origin configuration is modified. |
|------|-------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.1.3.45 AR800A\_IO\_WUP\_GEN\_REG

| 偏移地址 : 0x0140 |    |     | 默认值: 0x0000_0000                                                                     |
|---------------|----|-----|--------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                   |
| 31:1          | /  | /   | /                                                                                    |
| 0             | RW | 0   | AR800A_IO_WUP_GEN<br>1: IO Wakeup interrupt Enable<br>0: IO Wakeup interrupt Disable |

### 3.1.3.46 AR800A\_WAKEUP\_IO\_DEB\_CYCLES\_REG

| 偏移地址 : 0x0144 |    |          | 默认值: 0x0000_0000                                                                                                                                                                                                                                                    |
|---------------|----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值      | 描述                                                                                                                                                                                                                                                                  |
| 31:0          | RW | 0x000000 | AR800A_WAKEUP_IO_DEB_CYCLES0<br>Bit[4N+3:4*N] : IO[N] debounce cycles (N: 0~7)<br>WUPIO0 – WUPIO7<br><br>First must select clock for debounce (configure by register AR800A_WAKE_IO_DEB_CLK), finally the IO input debounce clock cycles is (DEBC_CYCLE[3:0]+1)+16. |

### 3.1.3.47 AR800A\_WAKEUP\_IO\_DEB\_CYCLES1\_REG

| 偏移地址 : 0x0148 |    |          | 默认值: 0x0000_0000                                                                                                                                                                                                                                                               |
|---------------|----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值      | 描述                                                                                                                                                                                                                                                                             |
| 31:24         | /  | /        | /                                                                                                                                                                                                                                                                              |
| 23:0          | RW | 0x000000 | AR800A_WAKEUP_IO_DEB_CYCLES1<br>Bit[4(N-8)+3 : 4*(N-8)] : IO[N] debounce cycles (N:8~13)<br>WUPIO8 – WUPIO13<br><br>First must select clock for debounce (configure by register AR800A_WAKE_IO_DEB_CLK), finally the IO input debounce clock cycles is (DEBC_CYCLE[3:0]+1)+16. |

### 3.1.3.48 LDO\_MODE\_SW\_SEL\_REG

| 偏移地址：0x0150 |    |     | 默认值：0x0000_0000                                                                                                         |
|-------------|----|-----|-------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                      |
| 31:3        | /  | /   | /                                                                                                                       |
| 2           | RW | 0   | SMPS_PFM_MODE<br>1: SMPS select PFM mode<br>0: SMPS select PWM mode                                                     |
| 1           | RW | 0   | DIG_LDO_LQ_MODE<br>1: SYS_LDO select lq mode<br>0: SYS_LDO select active mode                                           |
| 0           | RW | 1   | SMPS_MODE_SW_SEL_ENABLE<br>1: SMPS PWM/PFM mode select by this register<br>0: SMPS PWM/PFM mode select by power control |

### 3.1.3.49 SMPS\_PARAM\_CTRL\_REG

| 偏移地址：0x0200 |    |     | 默认值：0x0000_0000                                                                                                          |
|-------------|----|-----|--------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                       |
| 31:29       | RW | 0   | SMPS_OCP<br>DCDC over current protection threshold tuning<br>default: 400mA                                              |
| 28:24       | RW | 0   | SMPS_BGTR<br>[4:1]: for DCDC BANDGAP trimming<br>[0]: for OCP disable                                                    |
| 23          | /  | /   | /                                                                                                                        |
| 22:16       | RW | 0   | SMPS_FTR<br>[4:0]: for switching frequency tuning<br>[6:5]: for option to decrease frequency                             |
| 15:8        | RW | 0   | SMPS_TEST_ITEM_SEL                                                                                                       |
| 7:4         | RW | 0   | SMPS_TEST_MODE                                                                                                           |
| 3:2         | RW | 0   | SMPS_BYPASS<br>00, DCDC bypass disable<br>01, DCDC power pmos 1/6 bypass when PFM mode<br>1x, DCDC all power pmos bypass |
| 1           | /  | /   | /                                                                                                                        |
| 0           | R  | 0   | SMPS_WORK_MODE<br>0: PWM Mode                                                                                            |



|  |  |  |                                                                                                                                                                |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | 1: PFM Mode<br><i>Note: the DCDC starts up from PWM mode by default.</i><br><i>This field is read-only and can only be changed by <b>POWERCTRL</b> Module.</i> |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.1.3.50 ANA\_BG\_REG

| 偏移地址 : 0x0204 |    |      | 默认值: 0x0010_000X                                                                                            |
|---------------|----|------|-------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                                                                          |
| 31:21         | RW | 0    | /                                                                                                           |
| 20:16         | RW | 0x10 | BG Output Voltage Trimming                                                                                  |
| 15:1          | /  | /    | /                                                                                                           |
| 0             | R  | x    | ANA_BG_PND_1V8<br>BG Power Up Enable<br><i>Note: this field can be only set by <b>POWERCTRL</b> module.</i> |

### 3.1.3.51 DLDO\_PARAM\_REG

| 偏移地址 : 0x020C |    |     | 默认值: 0x0808_0000                                                                                                                                                                       |
|---------------|----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                     |
| 31:20         | /  | /   | /                                                                                                                                                                                      |
| 19            | RW | 1   | TESTPEAK_LDO1<br>enable the auxiliary circuit to avoid output large ripple during select LDO work<br><br>1: enable<br>0: disable<br><i>Note: this field is set by POWERCTRL module</i> |
| 18            | R  | 0   | STARTICC_LDO1<br>decrease over current protection threshold at start up<br><br>1: enable<br>0: disable<br><i>Note: this field is set by POWERCTRL module</i>                           |
| 17            | RW | 0   | TESTICC_LDO<br>over current protection enable<br><br>1: enable<br>0: disable<br><i>Note: this field is set by POWERCTRL module</i>                                                     |
| 16            | R  | 0   | LDO1_LQ_MODE_EN                                                                                                                                                                        |

|       |    |   |                                                                                                                              |
|-------|----|---|------------------------------------------------------------------------------------------------------------------------------|
|       |    |   | LDO1 LQ mode enable<br>1: enable<br>0: disable<br>Note: this field is set by POWERCTRL module                                |
| 15:12 | /  | / | /                                                                                                                            |
| 11:8  | RW | 0 | BG_OUTPUT_TRIM<br>BG output trimming<br>Note: eFuse override                                                                 |
| 7:1   | /  | / | /                                                                                                                            |
| 0     | R  | 0 | BG_PWR<br>0: disable<br>1: enable<br>Note: This field is set by POWERCTRL module and must be enable before enabling DIG_LDO. |

### 3.1.3.52 CFG\_IO\_ST\_REG

| 偏移地址 : 0x0214 |    |     | 默认值: 0x0000_000X                                          |      |       |
|---------------|----|-----|-----------------------------------------------------------|------|-------|
| 位置            | 访问 | 默认值 | 描述                                                        |      |       |
| 31:2          | RW | 0   | /                                                         |      |       |
| 1             | R  | x   | SIP IO<br>0: no flash sip<br>1: with flash sip in package |      |       |
|               |    |     | /                                                         | Read | Write |
|               |    |     | AR400A                                                    | R    | -     |
| 0             | /  | /   | /                                                         |      |       |

### 3.1.3.53 CPU\_RST\_SRC\_REG

| 偏移地址 : 0x0218 |    |     | 默认值: 0x0000_000X                                                                                                                                                                                                                                       |
|---------------|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                     |
| 31:11         | /  | /   | /                                                                                                                                                                                                                                                      |
| 10:9          | RW | 0   | IS_WATCHDOG_CPU_RST<br>The CPU reset source is watchdog(only reset CPU) output. Write 2'b11 clear these bits.<br>00: no<br>01: one time watchdog(only reset CPU)<br>10: two times watchdog(only reset CPU)<br>11: three times watchdog(only reset CPU) |
| 8             | RW | 0   | IS_WATCHDOG_ALL_RST                                                                                                                                                                                                                                    |

|     |    |   |                                                                                                          |
|-----|----|---|----------------------------------------------------------------------------------------------------------|
|     |    |   | The CPU reset source is watchdog(reset all system) output. Write 1 clear this bit.<br>0: no<br>1: yes    |
| 7:2 | /  | / | /                                                                                                        |
| 1   | RW | 0 | IS_PMU_RST<br>The CPU reset source is digital PMU output. Write 1 clear this bit.<br>0: no<br>1: yes     |
| 0   | RW | 0 | IS_PWRON_RST<br>The CPU reset source is POWER ON or P_RESETh. Write 1 clear this bit.<br>0: no<br>1: yes |

### 3.1.3.54 WLAN\_HIF\_OV\_CTRL\_REG

| 偏移地址：0x021C |     |     | 默认值: 0x0000_0000                                                                                                                                                               |
|-------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                             |
| 31:8        | /   | /   | /                                                                                                                                                                              |
| 7           | R/W | 0   | WLAN_IRQ_OVER_HIF: use wlan_irq generated in WLAN_MISC logic to replace the corresponding output of WLAN_HIF<br>0: not override WLAN_HIF output<br>1: override WLAN_HIF output |
| 6           | R/W | 0   | WLAN_WUP_OVR_HIF: use WLAN_WUP(bit2) to override the corresponding output of WLAN_HIF<br>0: not override WLAN_HIF output<br>1: override WLAN_HIF output                        |
| 5           | R/W | 0   | RESET_CPU_OVR_HIF: use RESET_CPU(bit1) to override the corresponding output of WLAN_HIF<br>0: not override WLAN_HIF output<br>1: override WLAN_HIF output                      |
| 4           | R/W | 0   | DISABLE_CPU_CLK_OVR_HIF: use DISABLE_CPU_CLK(bit0) to override the corresponding output of WLAN_HIF<br>0: not override WLAN_HIF output<br>1: override WLAN_HIF output          |
| 3           | /   | /   | Reserved                                                                                                                                                                       |
| 2           | R/W | 0   | WLAN_WUP: wake-up wlan from sleep mode, host firmware need to clear it after WLAN waked up                                                                                     |
| 1           | R/W | 1   | RESET_CPU: reset WLAN CPU<br>0: release WLAN CPU reset<br>1: WLAN CPU is reset                                                                                                 |

|   |     |   |                                                                                                  |
|---|-----|---|--------------------------------------------------------------------------------------------------|
| 0 | R/W | 1 | DISABLE_CPU_CLK: disable WLAN CPU clock<br>0: wlan cpu clock enable<br>1: wlan cpu clock disable |
|---|-----|---|--------------------------------------------------------------------------------------------------|

### 3.1.3.55 SRAM\_BIST\_CFG\_REG

| 偏移地址 : 0x0220 |    |      | 默认值: 0x2222_2222                                                                                                                             |
|---------------|----|------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                                                                                                           |
| 31:24         | /  | /    | /                                                                                                                                            |
| 23:16         | RW | 0x22 | ARM_SINGLE_PORT_SRAM_CFG<br>ram_cfg[7:0]<br>bit [2:0]: connect to EMA<br>bit [4:3]: connect to EMAW<br>others : reserved                     |
| 15:8          | RW | 0x22 | ARM_DAUL_PORT_SRAM_CFG<br>ram_cfg[7:0]<br>bit [2:0]: connect to EMAA<br>bit [6:4]: connect to EMAB<br>others : reserved                      |
| 7:0           | RW | 0x22 | SYNOPSYS_SRAM_CFG<br>ram_cfg[7:0]<br>bit [0]: connect to REME<br>bit [3]: connect to TEST1<br>bit [7:4]: connect to REM<br>others : reserved |

### 3.1.3.56 WLAN\_SRAM\_SHARE\_REG

| 偏移地址 : 0x0224 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                     |
| 31:7          | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                      |
| 6:0           | RW | 0x0 | WLAN_SRAM_SHARE<br>0000000: No share APP SRAM<br>0000001: share 16K bytes APP SRAM<br>0000011: share 32K bytes APP SRAM<br>0000111: share 64K bytes APP SRAM<br>0001111: share 96K bytes APP SRAM<br>0011111: share 128K bytes APP SRAM<br>0100000: share APP SRAM with connect mode, total 128k bytes.<br>WLAN DTCM: 48k bytes (0x04000000~0x0400BFFF)<br>WLAN AHB: 80k bytes (0x0800C000~0x0801FFFF) |

|  |  |  |                                                                                                                 |
|--|--|--|-----------------------------------------------------------------------------------------------------------------|
|  |  |  | WLAN PAS: 48k bytes (0x0900C000~0x09017FFF)<br>1000000: BLE share 128k bytes for debug mode<br>Others: reserved |
|--|--|--|-----------------------------------------------------------------------------------------------------------------|

### 3.1.3.57 BLE\_SRAM\_SHARE\_CTRL\_REG

| 偏移地址 : 0x0228 |    |     | 默认值: 0x0000_0000                                                                             |
|---------------|----|-----|----------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                           |
| 31:1          | /  | /   | /                                                                                            |
| 0             | RW | 0x0 | BLE_SRAM_SHARE_CTRL<br>0: APP(Cache) use shared 16K SRAM<br>1: BLE use shared 16K bytes SRAM |

### 3.1.3.58 LPUART0\_WAKEUP\_CTRL\_REG

| 偏移地址 : 0x0230 |    |     | 默认值: 0x0000_0000                                                                                                                                                                  |
|---------------|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                |
| 31            | RW | 0   | LPUART0_WAKEUP_ENABLE<br>0: the module clock is gated off<br>1: the module clock is released to work                                                                              |
| 30:18         | /  | /   | /                                                                                                                                                                                 |
| 17:16         | RW | 0x0 | LPUART0_WAKEUP_IN_SEL<br>0: LPUART0 wakeup input use UART0 serial in<br>1: LPUART0 wakeup input use UART1 serial in<br>2: LPUART0 wakeup input use UART2 serial in<br>3: reserved |
| 15:1          | /  | /   | /                                                                                                                                                                                 |
| 0             | RW | 0x0 | LPUART0_CLK_SEL<br>0: LPUART0 wakeup clock use LFCLK<br>1: LPUART0 wakeup clock use HFCLK (used for test)                                                                         |

### 3.1.3.59 LPUART1\_WAKEUP\_CTRL\_REG

| 偏移地址 : 0x0234 |    |     | 默认值: 0x0001_0000                                                                                     |
|---------------|----|-----|------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                   |
| 31            | RW | 0   | LPUART1_WAKEUP_ENABLE<br>0: the module clock is gated off<br>1: the module clock is released to work |
| 30:18         | /  | /   | /                                                                                                    |
| 17:16         | RW | 0x1 | LPUART1_WAKEUP_IN_SEL                                                                                |

|      |    |     |                                                                                                                                                          |
|------|----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |    |     | 0: LPUART1 wakeup input use UART0 serial in<br>1: LPUART1 wakeup input use UART1 serial in<br>2: LPUART1 wakeup input use UART2 serial in<br>3: reserved |
| 15:1 | /  | /   | /                                                                                                                                                        |
| 0    | RW | 0x0 | LPUART1_CLK_SEL<br>0: LPUART1 wakeup clock use LFCLK<br>1: LPUART1 wakeup clock use HFCLK (used for test)                                                |

### 3.1.3.60 GPADC\_CLK\_CTRL\_REG

| 偏移地址 : 0x0238 |     |     | 默认值: 0x0000_0000                                                                                                              |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                            |
| 31            | R/W | 0   | MCLK_ENABLE<br><br>0: the module clock is gated off<br>1: the module clock is released to work<br>$CLK_m = CLK_{src} / N / M$ |
| 30:26         | /   | /   | /                                                                                                                             |
| 25:24         | R/W | 0   | MCLK_SRC_SEL<br><br>Clock source selection<br>00: HFCLK<br>01: LFCLK<br>1x: reserved                                          |
| 23:18         | /   | /   | /                                                                                                                             |
| 17:16         | R/W | 0   | CLK_DIV_RATIO_N<br><br>clock pre-divide ratio N.<br>00: N = 1<br>01: N = 2<br>10: N = 4<br>11: N = 8                          |
| 15:4          | /   | /   | /                                                                                                                             |
| 3:0           | R/W | 0   | CLK_DIV_RATIO_M<br><br>Clock divide ratio M.<br>$M = value + 1 (1 \sim 16)$                                                   |

### 3.1.3.61 32K\_CLK\_CTRL\_REG

| 偏移地址：0x023C |     |     | 默认值: 0x0000_0000                                                                                                                           |
|-------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                         |
| 31:4        | /   | /   | /                                                                                                                                          |
| 3           | R/W | 0   | WLAN_32K_SEL<br>0: select rco_losc_mux_clk as wlan function clock<br>1: select rco_calib_clk as wlan function clock                        |
| 2           | R/W | 0   | WUP_TIMER_32K_SEL<br>0: select rco_losc_mux_clk as wake-up timer function clock<br>1: select rco_calib_clk as wake-up timer function clock |

### 3.1.3.62 WKUP\_SRC\_BUS\_CLK\_CTRL\_REG

| 偏移地址：0x0240 |     |     | 默认值: 0x0000_0000                                                                |
|-------------|-----|-----|---------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                              |
| 31:8        | /   | /   | /                                                                               |
| 7           | RW  | 0x0 | RCCAL_CLK_GATING<br>0: the clock is gated off<br>1: the clock is running        |
| 6           | RW  | 0x0 | GPADC_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running    |
| 5           | /   | /   | /                                                                               |
| 4           |     |     | LPUART1_WKUP_CLK_GATING<br>0: the clock is gated off<br>1: the clock is running |
| 3           | /   | /   | /                                                                               |
| 2           | R/W | 0x0 | LPUART0_WKUP_CLK_GATING<br>0: the clock is gated off<br>1: the clock is running |
| 1:0         | /   | /   | /                                                                               |

### 3.1.3.63 WKUP\_SRC\_RST\_CTRL\_REG

| 偏移地址：0x0244 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:8        | /  | /   | /                |

|     |    |     |                                                                                             |
|-----|----|-----|---------------------------------------------------------------------------------------------|
| 7   | RW | 0x0 | RCCAL_RSTN<br>0: the module is in reset state<br>1: the module is released to work          |
| 6   | RW | 0x0 | GPADC_RSTN<br>0: the module is in reset state<br>1: the module is released to work          |
| 5   | /  | /   | /                                                                                           |
| 4   | RW | 0x0 | LPUART1_WAKEUP_RSTN<br>0: the module is in reset state<br>1: the module is released to work |
| 3   | /  | /   | /                                                                                           |
| 2   | RW | 0x0 | LPUART0_WAKEUP_RSTN<br>0: the module is in reset state<br>1: the module is released to work |
| 1:0 | /  | /   | /                                                                                           |

#### 3.1.3.64 FLASH\_ENCRYPT\_AES\_NONCE0\_REG

| 偏移地址：0x0250 |    | 默认值: 0x0000_0000 |                                                                                                                               |
|-------------|----|------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值              | 描述                                                                                                                            |
| 31:0        | RW | 0x0              | FLASH_ENCRYPT_AES_NONCE0<br>Flash encrypt module AES nonce bit[31:0]. These bits only changed one time after system power up. |

#### 3.1.3.65 FLASH\_ENCRYPT\_AES\_NONCE1

| 偏移地址：0x0254 |    | 默认值: 0x0000_0000 |                                                                                                                                |
|-------------|----|------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值              | 描述                                                                                                                             |
| 15:0        | RW | 0x0              | FLASH_ENCRYPT_AES_NONCE1<br>Flash encrypt module AES nonce bit[47:32]. These bits only changed one time after system power up. |

#### 3.1.3.66 BLE\_RCOSC\_CALIB\_REGr0\_REG

| 偏移地址：0x0260 |    | 默认值: 0x0000_0000 |                                                                |
|-------------|----|------------------|----------------------------------------------------------------|
| 位置          | 访问 | 默认值              | 描述                                                             |
| 31:30       | /  | /                | /                                                              |
| 29          | RW | 0x0              | RCO_CALIB_EN<br>RCOSC calibration function enable<br>1: enable |



|       |    |     |                                                                                                                                                                                                                 |
|-------|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |    |     | 0: disable                                                                                                                                                                                                      |
| 28    | RW | 0x0 | RCO_CALIB_RST_PUL<br>When write 1 to this bit, reset RCOSC calibration then run again, when write 0, no effect.<br>It is always read as 0.                                                                      |
| 27    | /  | /   | /                                                                                                                                                                                                               |
| 26:24 | R  | 0x0 | RCO_CALIB_CS<br>RCOSC calibration current state. It is read only.<br>000: IDLE<br>001: WAIT_DCXO_READY<br>011: WAIT_CAL_FINISH<br>110: CAL_FINISH<br>010: WAIT_TIMER<br>Others: reserved                        |
| 23:21 | /  | /   | /                                                                                                                                                                                                               |
| 20    | RW | 0x0 | RCO_CALIB_SW_REQ_PUL<br>When write 1 to this bit, begin RCOSC calibration process at once, when write 0, no effect.<br>It is always read as 0.                                                                  |
| 19:17 | /  | /   | /                                                                                                                                                                                                               |
| 16    | RW | 0x0 | RCO_WUP_TIME_EN<br>RCOSC calibration periodically wake-up function enable<br>1: enable<br>0: disable                                                                                                            |
| 15:13 | /  | /   | /                                                                                                                                                                                                               |
| 12:0  | RW | 0x0 | RCO_WUP_TIME<br>RCO_SLP calibration periodically wake-up time.<br>Configure by software to wake up system, and begin RCO_SLP calibration process. The time unit of RCO_WUP_TIME is 32Khz(divided from RCO_SLP). |

### 3.1.3.67 BLE\_RCOSC\_CALIB\_REG1

| 偏移地址 : 0x0264 |    |     | 默认值: 0x0000_0000                                |
|---------------|----|-----|-------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                              |
| 31:20         | /  | /   | /                                               |
| 19:16         | RW | 0x0 | RCO_SCALE_PHASE2_NUM<br>wake-up times in phase2 |
| 15:12         | RW | 0x0 | RCO_SCALE_PHASE1_NUM<br>wake-up times in phase1 |

|      |    |     |                                                                                                                                                                                                                                                                                |
|------|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11   | /  | /   | /                                                                                                                                                                                                                                                                              |
| 10:8 | RW | 0x0 | RCO_SCALE_PHASE3_WUP_TIMES<br>multiply factor of RCO_WUP_TIME in wake-up phase3<br>0x0: 10 times<br>0x1: 16 times<br>0x2: 20 times<br>0x3: 24 times<br>0x4: 32 times<br>0x5: 40 times<br>0x6: 64 times<br>0x7: 128 times                                                       |
| 7    | /  | /   | /                                                                                                                                                                                                                                                                              |
| 6:4  | RW | 0x0 | RCO_SCALE_PHASE2_WUP_TIMES<br>multiply factor of RCO_WUP_TIME in wake-up phase2<br>0x0: 2 times<br>0x1: 4 times<br>0x2: 6 times<br>0x3: 8 times<br>0x4: 10 times<br>0x5: 12 times<br>0x6: 14 times<br>0x7: 16 times                                                            |
| 3:2  | /  | /   | /                                                                                                                                                                                                                                                                              |
| 1    | RW | 0x0 | RCO_NORMAL_WUP_TIMES_SEL<br>In normal wake up mode, actual wake-up time is the multiply of RCO_WUP_TIME * factor<br>1: select RCO_SCALE_PHASE3_WUP_TIMES as the multiply factor of RCO_WUP_TIME<br>0: select RCO_SCALE_PHASE2_WUP_TIMES as the multiply factor of RCO_WUP_TIME |
| 0    | RW | 0x0 | RCO_WUP_MODE_SEL<br>Scale wake up and Normal wake up are two wake up methods<br>1: select scale wake up method<br>0: select normal wake up method                                                                                                                              |

### 3.1.3.68 BLE\_CLK32K\_SWITCH\_REG0

| 偏移地址 : 0x0268 |    | 默认值: 0x0271_0000 |                      |
|---------------|----|------------------|----------------------|
| 位置            | 访问 | 默认值              | 描述                   |
| 31:26         | /  | /                | /                    |
| 25:16         | RW | 0x271            | DIV_HALFCYCLE_TARGET |

|      |    |     |                                                                                                                                  |
|------|----|-----|----------------------------------------------------------------------------------------------------------------------------------|
|      |    |     | Bit[9:0], The target half cycle number for 32kHz clock divide                                                                    |
| 15:5 | /  | /   | /                                                                                                                                |
| 4    | RW | 0x0 | DIV_CLK_EN<br>The 32KHz divider enable<br>1: enable<br>0: disable                                                                |
| 3    | RW | 0x0 | SYS_32K_SEL<br><br>The source of 32kHz clock select<br>1: the clock from RCOSC Calibration output<br>0: the RCOSC/LFCLK output   |
| 2    | RW | 0x0 | BLE_SEL_RCCAL_32K<br>The source of 32kHz clock select<br>1: the clock from RCOSC Calibration output<br>0: the RCOSC/LFCLK output |
| 1    | RW | 0x0 | DIV_CLK_SRC_SEL<br>The source clock select for divide to 32KHz clock<br>1: 32MHz Clock<br>0: HFCLK                               |
| 0    | RW | 0x0 | CLK32K_AUTO_SW_EN<br>32K clock auto switch enable.<br>1: enable<br>0: disable                                                    |

### 3.1.3.69 BLE\_CLK32K\_SWITCH\_REG1

| 偏移地址 : 0x026C |    |     | 默认值: 0x0000_0000                                                                           |
|---------------|----|-----|--------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                         |
| 31:27         | /  | /   | /                                                                                          |
| 26:16         | R  | 0x0 | CLK32K_SW_OFFSET_DOWN<br>The offset when switch from divide 32k clock to 32KHz input clock |
| 15:11         | /  | /   | /                                                                                          |
| 10:0          | R  | 0x0 | CLK32K_SW_OFFSET_ON<br>The offset when switch from 32KHz input clock to divide 32k clock   |

## 3.2 时钟控制单元 (CCM)

### 3.2.1 概述

CCMU 模块管理各模块的复位和时钟，并提供各模块的配置以调整各模块的时钟。

### 3.2.2 CCM 寄存器列表

| 模块名                     | 基地址        |                                                                      |
|-------------------------|------------|----------------------------------------------------------------------|
| Clock Control Module    | 0x40040400 |                                                                      |
| 寄存器名                    | 偏移地址       | 描述                                                                   |
| CPU_BUS_CLKCFG_REG      | 0x0000     | CPU BUS Clock Configure Register                                     |
| BUS_CLK_GATING_CTRL_REG | 0x0004     | Bus Clock Gating Control Register                                    |
| BUS_DEV_RST_CTRL_REG    | 0x0008     | Bus Device Reset Control Register                                    |
| SPIO_CLK_CTRL_REG       | 0x0020     | SPIO Clock Control Register                                          |
| SS_CLK_CTRL_REG         | 0x002C     | SS Clock Control Register                                            |
| AUDIO_CLK_CTRL_REG      | 0x0034     | Audio Clock Control Register                                         |
| IRRX_CLK_CTRL_REG       | 0x0038     | IRRX Clock Control Register                                          |
| IRTX_CLK_CTRL_REG       | 0x003c     | IRTX Clock Control Register                                          |
| SYSTICK_REFCLK_CTRL_REG | 0x0040     | System Tick Reference Clock Register                                 |
| SYSTICK_CALIB_CTRL_REG  | 0x0044     | System Tick Clock Calibration Register                               |
| PCLK_SPC_MCLK_CTRL_REG  | 0x005C     | Special APB Clock Control Register<br>(UART/GPRCM/RTC APB bus clock) |
| PSRAM_CLK_CTRL_REG      | 0x0060     | PSRAM clock control register                                         |
| KEYSCAN_CLK_CTRL_REG    | 0x0064     | Keyscan clock control register                                       |

### 3.2.3 CCM 寄存器描述

#### 3.2.3.1 CPU\_BUS\_CLKCFG\_REG

| 偏移地址：0x0000 |     |     | 默认值: 0x0000_0001                                         |
|-------------|-----|-----|----------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                       |
| 31:10       | /   | /   | /                                                        |
| 9:8         | R/W | 0x0 | AHB2_CLK_DIV<br><br>00: /1<br>01: /2<br>10: /3<br>11: /4 |
| 7:6         | /   | /   | /                                                        |
| 5:4         | R/W | 0x0 | APB_CLK_SRC_SEL                                          |

|     |     |   |                                                                         |
|-----|-----|---|-------------------------------------------------------------------------|
|     |     |   | APB bus clock source select<br>00: HFCLK<br>01: LFCLK<br>1X: AHB2 Clock |
| 3:2 | /   | / | /                                                                       |
| 1:0 | R/W | 1 | APB_CLK_DIV<br><br>00: /1<br>01: /2<br>10: /4<br>11: /8                 |

### 3.2.3.2 BUS\_CLK\_GATING\_CTRL\_REG

| 偏移地址 : 0x0004 |           |     | 默认值: 0x0000_0000                                                                   |
|---------------|-----------|-----|------------------------------------------------------------------------------------|
| 位置            | 访问        | 默认值 | 描述                                                                                 |
| 31            | R/W<br>S+ | 0   | SMCARD_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running      |
| 30            | R/W<br>S+ | 0   | TRNG_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running        |
| 29            | R/W<br>S+ | 0   | AUDIO_CODEC_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running |
| 28            | R/W<br>S+ | 0   | UART2_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running       |
| 27            | R/W<br>S+ | 0   | GPIO_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running        |
| 26            | R/W       | 0   | KEYSCAN_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running     |

|    |           |   |                                                                                |
|----|-----------|---|--------------------------------------------------------------------------------|
| 25 | /         | / | /                                                                              |
| 24 | R/W       | 0 | IRRX_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running    |
| 23 | R/W       | 0 | IRTX_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running    |
| 22 | R/W<br>S+ | 0 | DAUDIO_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running  |
| 21 | R/W       | 0 | PWM_ECT_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running |
| 20 | /         | / | /                                                                              |
| 19 | R/W<br>S+ | 0 | TWI1_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running    |
| 18 | R/W<br>S+ | 0 | TWI0_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running    |
| 17 | R/W<br>S+ | 0 | UART1_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running   |
| 16 | R/W<br>S+ | 0 | UART0_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running   |
| 15 | R/W       | 0 | BLE_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running     |
| 14 | R/W       | 0 | DMA_SEC_CLK_GATING                                                             |

|      |           |   |                                                                                   |
|------|-----------|---|-----------------------------------------------------------------------------------|
|      | S+        |   | 0: the clock is gated off<br>1: the clock is running                              |
| 13:9 | /         | / | /                                                                                 |
| 8    | R/W<br>S+ | 0 | FLASH_ENCRYPT_CLK_GATING<br>0: the clock is gated off<br>1: the clock is running  |
| 7    | R/W<br>S+ | 0 | TZASC_CLK_GATING<br>0: the clock is gated off<br>1: the clock is running          |
| 6    | R/W       | 0 | DMA_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running        |
| 5:4  | /         | / | /                                                                                 |
| 3    | R/W       | 0 | FLASH_CTRL_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running |
| 2    | R/W       | 0 | SS_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running         |
| 1    | /         | / | /                                                                                 |
| 0    | R/W<br>S+ | 0 | SPIO_CLK_GATING<br><br>0: the clock is gated off<br>1: the clock is running       |



#### 说明

1. The type S+ means this bit can't change by non-secure access when corresponding module has set security in TZASC register.

#### 3.2.3.3 BUS\_DEV\_RST\_CTRL\_REG

| 偏移地址：0x0008 |           |     | 默认值: 0x0000_0000                                  |
|-------------|-----------|-----|---------------------------------------------------|
| 位置          | 访问        | 默认值 | 描述                                                |
| 31          | R/W<br>S+ | 0   | SMCARD_RST<br><br>0: the module is in reset state |

|    |           |   |                                                                                                                                                  |
|----|-----------|---|--------------------------------------------------------------------------------------------------------------------------------------------------|
|    |           |   | 1: the module is released to work                                                                                                                |
| 30 | R/W<br>S+ | 1 | TRNG_RST<br><br>0: the module is in reset state<br>1: the module is released to work. The default value is 1, used for TRNG retention registers. |
| 29 | R/W<br>S+ | 0 | AUDIO_CODEC_RST<br><br>0: the module is in reset state<br>1: the module is released to work                                                      |
| 28 | R/W<br>S+ | 0 | UART2_RST<br><br>0: the module is in reset state<br>1: the module is released to work                                                            |
| 27 | /         | / | /                                                                                                                                                |
| 26 | R/W       | 0 | KEYSCAN_RST<br><br>0: the module is in reset state<br>1: the module is released to work                                                          |
| 25 | /         | / | /                                                                                                                                                |
| 24 | R/W       | 0 | IRRX_RST<br><br>0: the module is in reset state<br>1: the module is released to work                                                             |
| 23 | R/W       | 0 | IRTX_RST<br><br>0: the module is in reset state<br>1: the module is released to work                                                             |
| 22 | R/W<br>S+ | 0 | DAUDIO_RST<br><br>0: the module is in reset state<br>1: the module is released to work                                                           |
| 21 | R/W       | 0 | PWM_ECT_RST<br><br>0: the module is in reset state<br>1: the module is released to work                                                          |
| 20 | /         | / | /                                                                                                                                                |
| 19 | R/W<br>S+ | 0 | TWI1_RST                                                                                                                                         |



|       |           |   |                                                                                           |
|-------|-----------|---|-------------------------------------------------------------------------------------------|
|       |           |   | 0: the module is in reset state<br>1: the module is released to work                      |
| 18    | R/W<br>S+ | 0 | TWIO_RST<br><br>0: the module is in reset state<br>1: the module is released to work      |
| 17    | R/W<br>S+ | 0 | UART1_RST<br><br>0: the module is in reset state<br>1: the module is released to work     |
| 16    | R/W<br>S+ | 0 | UART0_RST<br><br>0: the module is in reset state<br>1: the module is released to work     |
| 15    | R/W       | 0 | BLE_RST<br><br>0: the module is in reset state<br>1: the module is released to work       |
| 14    | R/W<br>S+ | 0 | DMA_SEC_RST<br><br>0: the module is in reset state<br>1: the module is released to work   |
| 13    | R/W       | 1 | DCACHE_RST<br><br>0: the module is in reset state<br>1: the module is released to work    |
| 12:11 | /         | / | /                                                                                         |
| 10    | R/W       | 0 | WLANC_RST<br><br>0: the module is in reset state<br>1: the module is released to work     |
| 9     | /         | / | /                                                                                         |
| 8     | R/W<br>S+ | 0 | FLASH_ENC_RST<br><br>0: the module is in reset state<br>1: the module is released to work |
| 7     | /         | / | /                                                                                         |
| 6     | R/W       | 0 | DMA_RST                                                                                   |

|     |           |   |                                                                                            |
|-----|-----------|---|--------------------------------------------------------------------------------------------|
|     |           |   | 0: the module is in reset state<br>1: the module is released to work                       |
| 5:4 | /         | / | /                                                                                          |
| 3   | R/W       | 0 | FLASH_CTRL_RST<br><br>0: the module is in reset state<br>1: the module is released to work |
| 2   | R/W       | 0 | SS_RST<br><br>0: the module is in reset state<br>1: the module is released to work         |
| 1   | /         | / | /                                                                                          |
| 0   | R/W<br>S+ | 0 | SPIO_RST<br><br>0: the module is in reset state<br>1: the module is released to work       |


**说明**

1. The type S+ means this bit can't change by non-secure access when corresponding module has set security in TZASC register.

#### 3.2.3.4 SPIO\_CLK\_CTRL\_REG

| 偏移地址 : 0x0020 |     |     | 默认值: 0x0000_0000                                                                                                              |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                            |
| 31            | R/W | 0   | MCLK_ENABLE<br><br>0: the module clock is gated off<br>1: the module clock is released to work<br>$CLK_m = CLK_{src} / N / M$ |
| 30:26         | /   | /   | /                                                                                                                             |
| 25:24         | R/W | 0   | MCLK_SRC_SEL<br><br>Clock source selection<br>00: HFCLK<br>01: DEVCLK<br>1x: reserved                                         |
| 23:18         | /   | /   | /                                                                                                                             |
| 17:16         | R/W | 0   | CLK_DIV_RATIO_N                                                                                                               |

|      |     |   |                                                                               |
|------|-----|---|-------------------------------------------------------------------------------|
|      |     |   | clock pre-divide ratio N.<br>00: N = 1<br>01: N = 2<br>10: N = 4<br>11: N = 8 |
| 15:4 | /   | / | /                                                                             |
| 3:0  | R/W | 0 | CLK_DIV_RATIO_M<br><br>Clock divide ratio M.<br>M = value + 1 (1~16)          |

### 3.2.3.5 SS\_CLK\_CTRL\_REG

| 偏移地址 : 0x002C |     |     | 默认值: 0x0000_0000                                                                                                              |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                            |
| 31            | R/W | 0   | MCLK_ENABLE<br><br>0: the module clock is gated off<br>1: the module clock is released to work<br>$CLK_m = CLK_{src} / N / M$ |
| 30:26         | /   | /   | /                                                                                                                             |
| 25:24         | R/W | 0   | MCLK_SRC_SEL<br><br>Clock source selection<br>00: HFCLK<br>01: DEVCLK<br>1x: reserved                                         |
| 23:18         | /   | /   | /                                                                                                                             |
| 17:16         | R/W | 0   | CLK_DIV_RATIO_N<br><br>Clock pre-divide ratio N.<br>00: N = 1<br>01: N = 2<br>10: N = 4<br>11: N = 8                          |
| 15:4          | /   | /   | /                                                                                                                             |
| 3:0           | R/W | 0   | CLK_DIV_RATIO_M<br><br>Clock divide ratio M.<br>M = value + 1 (1~16)                                                          |

### 3.2.3.6 AUDIO\_CLK\_CTRL\_REG

| 偏移地址：0x0034 |     |     | 默认值: 0x0000_0000                                                                                                      |
|-------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                    |
| 31:24       | /   | /   | /                                                                                                                     |
| 23          | R/W | 0   | DAUDIO_CLK_EN<br>Daudio clock enable<br>0: disable<br>1: enable                                                       |
| 22:17       | /   | /   | /                                                                                                                     |
| 16          | R/W | 0   | DAUDIO_CLK_SEL<br>Daudio clock source select<br>0: select ADC_CLK from DPLL<br>1: select CLASSD_CLK from DPLL         |
| 15:8        | /   | /   | /                                                                                                                     |
| 7           | R/W | 0   | CODEC_CLK_EN<br>Audio codec clock (ADC&ClassD clock) enable<br>0: disable<br>1: enable                                |
| 6:3         | /   | /   | /                                                                                                                     |
| 2           | R/W | 0   | CLASSD_CLK_SEL<br>Codec class-D clock source selection<br>0: select CLASSD_CLK from DPLL<br>1: select outer I2S clock |
| 1           | R/W | 0   | ADC_CLK_SEL<br>Codec ADC clock source selection<br>0: select ADC_CLK from DPLL<br>1: select outer I2S clock           |
| 0           | R/W | 0   | OUTER_CLK_SEL<br>Outer I2S clock source selection<br>0: I2S MCLK input<br>1: I2S BCLK input                           |

### 3.2.3.7 IRRX\_CLK\_CTRL\_REG

| 偏移地址：0x0038 |     |     | 默认值: 0x0000_0000 |
|-------------|-----|-----|------------------|
| 位置          | 访问  | 默认值 | 描述               |
| 31          | R/W | 0   | MCLK_ENABLE      |

|       |     |   |                                                                                                            |
|-------|-----|---|------------------------------------------------------------------------------------------------------------|
|       |     |   | 0: the module clock is gated off<br>1: the module clock is released to work<br>$CLK_m = CLK_{src} / N / M$ |
| 30:26 | /   | / | /                                                                                                          |
| 25:24 | R/W | 0 | MCLK_SRC_SEL<br><br>Clock source selection<br>00: HFCLK<br>01: LFCLK<br>1x: reserved                       |
| 23:18 | /   | / | /                                                                                                          |
| 17:16 | R/W | 0 | CLK_DIV_RATIO_N<br><br>clock pre-divide ratio N.<br>00: N = 1<br>01: N = 2<br>10: N = 4<br>11: N = 8       |
| 15:4  | /   | / | /                                                                                                          |
| 3:0   | R/W | 0 | CLK_DIV_RATIO_M<br><br>Clock divide ratio M.<br>M = value + 1 (1~16)                                       |

### 3.2.3.8 IRTX\_CLK\_CTRL\_REG

| 偏移地址 : 0x003C |     |     | 默认值: 0x0000_0000                                                                                                              |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                            |
| 31            | R/W | 0   | MCLK_ENABLE<br><br>0: the module clock is gated off<br>1: the module clock is released to work<br>$CLK_m = CLK_{src} / N / M$ |
| 30:26         | /   | /   | /                                                                                                                             |
| 25:24         | R/W | 0   | MCLK_SRC_SEL<br><br>Clock source selection<br>00: HFCLK<br>01: LFCLK<br>1x: reserved                                          |

|       |     |   |                                                                                                      |
|-------|-----|---|------------------------------------------------------------------------------------------------------|
| 23:18 | /   | / | /                                                                                                    |
| 17:16 | R/W | 0 | CLK_DIV_RATIO_N<br><br>clock pre-divide ratio N.<br>00: N = 1<br>01: N = 2<br>10: N = 4<br>11: N = 8 |
| 15:4  | /   | / | /                                                                                                    |
| 3:0   | R/W | 0 | CLK_DIV_RATIO_M<br><br>Clock divide ratio M.<br>M = value + 1 (1~16)                                 |

### 3.2.3.9 SYSTICK\_REFCLK\_CTRL\_REG

| 偏移地址 : 0x0040 |     |     | 默认值: 0x0000_0000                                                                                                              |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                            |
| 31            | R/W | 0   | MCLK_ENABLE<br><br>0: the module clock is gated off<br>1: the module clock is released to work<br>$CLK_m = CLK_{src} / N / M$ |
| 30:26         | /   | /   | /                                                                                                                             |
| 25:24         | R/W | 0   | MCLK_SRC_SEL<br><br>Clock source selection<br>00: HFCLK<br>01: LFCLK<br>1x: reserved                                          |
| 23:18         | /   | /   | /                                                                                                                             |
| 17:16         | R/W | 0   | CLK_DIV_RATIO_N<br><br>clock pre-divide ratio N.<br>00: N = 1<br>01: N = 2<br>10: N = 4<br>11: N = 8                          |
| 15:4          | /   | /   | /                                                                                                                             |

|     |     |   |                                                                      |
|-----|-----|---|----------------------------------------------------------------------|
| 3:0 | R/W | 0 | CLK_DIV_RATIO_M<br><br>Clock divide ratio M.<br>M = value + 1 (1~16) |
|-----|-----|---|----------------------------------------------------------------------|

### 3.2.3.10 SYSTICK\_CALIB\_CTRL\_REG

| 偏移地址 : 0x0044 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                   |
|---------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                 |
| 31:26         | /   | /   | /                                                                                                                                                                                                                                                                                                  |
| 25            | R/W | 0   | ST_NOREF<br>1: there is no reference clock<br>0: use the reference clock                                                                                                                                                                                                                           |
| 24            | R/W | 0   | ST_SKEW<br><br>Note: set this bit to 0 if the system timer clock, the external reference clock, or FCLK as indicated by ST_NOREF, can guarantee an exact multiple of 10ms. Otherwise, set this bit to 1.                                                                                           |
| 23:0          | R/W | 0   | ST_10MS_COUNTER<br><br>Note: This field provides an integer value to compute a 10ms (100Hz) delay from either the reference clock, or FCLK if the reference clock is not implemented.<br>For example:<br>No reference clock, FCLK=50MHz, CNT = 0x7A11F(49999)<br>REFCLK = 1MHz, CNT = 0x270F(9999) |

### 3.2.3.11 FLASH\_SPI\_CLK\_CTRL\_REG

| 偏移地址 : 0x0054 |     |     | 默认值: 0x0000_0000                                                                                                              |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                            |
| 31            | R/W | 0   | MCLK_ENABLE<br><br>0: the module clock is gated off<br>1: the module clock is released to work<br>$CLK_m = CLK_{src} / N / M$ |
| 30:26         | /   | /   | /                                                                                                                             |
| 25:24         | R/W | 0   | MCLK_SRC_SEL<br><br>Clock source selection<br>00: HFCLK<br>01: DEVCLK                                                         |

|       |     |   |                                                                                                      |
|-------|-----|---|------------------------------------------------------------------------------------------------------|
|       |     |   | 1x: reserved                                                                                         |
| 23:18 | /   | / | /                                                                                                    |
| 17:16 | R/W | 0 | CLK_DIV_RATIO_N<br><br>clock pre-divide ratio N.<br>00: N = 1<br>01: N = 2<br>10: N = 4<br>11: N = 8 |
| 15:4  | /   | / | /                                                                                                    |
| 3:0   | R/W | 0 | CLK_DIV_RATIO_M<br><br>Clock divide ratio M.<br>M = value + 1 (1~16)                                 |

### 3.2.3.12 PCLK\_SPC\_CLK\_CTRL\_REG

| 偏移地址 : 0x005C |     |     | 默认值: 0x0000_0000                                                                                     |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                   |
| 31:26         | /   | /   | /                                                                                                    |
| 25:24         | R/W | 0   | MCLK_SRC_SEL<br><br>Clock source selection<br>00: HFCLK<br>01: DEVCLK<br>10: FLCLK<br>11: reserved   |
| 23:18         | /   | /   | /                                                                                                    |
| 17:16         | R/W | 0   | CLK_DIV_RATIO_N<br><br>clock pre-divide ratio N.<br>00: N = 1<br>01: N = 2<br>10: N = 4<br>11: N = 8 |
| 15:4          | /   | /   | /                                                                                                    |
| 3:0           | R/W | 0   | CLK_DIV_RATIO_M<br><br>Clock divide ratio M.<br>M = value + 1 (1~16)                                 |



### 3.2.3.13 PSRAM\_OPI\_CLK\_CTRL\_REG

| 偏移地址 : 0x0060 |     |     | 默认值: 0x0000_0000                                                                                                              |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                            |
| 31            | R/W | 0   | MCLK_ENABLE<br><br>0: the module clock is gated off<br>1: the module clock is released to work<br>$CLK_m = CLK_{src} / N / M$ |
| 30:26         | /   | /   | /                                                                                                                             |
| 25:24         | R/W | 0   | MCLK_SRC_SEL<br><br>Clock source selection<br>00: HFCLK<br>01: DEV1_CLK<br>1x: reserved                                       |
| 23:18         | /   | /   | /                                                                                                                             |
| 17:16         | R/W | 0   | CLK_DIV_RATIO_N<br><br>clock pre-divide ratio N.<br>00: N = 1<br>01: N = 2<br>10: N = 4<br>11: N = 8                          |
| 15:4          | /   | /   | /                                                                                                                             |
| 3:0           | R/W | 0   | CLK_DIV_RATIO_M<br><br>Clock divide ratio M.<br>$M = value + 1 (1 \sim 16)$                                                   |

### 3.2.3.14 KEYSKAN\_CLK\_CTRL\_REG

| 偏移地址 : 0x0064 |     |     | 默认值: 0x0000_0000                                                                                                              |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                            |
| 31            | R/W | 0   | MCLK_ENABLE<br><br>0: the module clock is gated off<br>1: the module clock is released to work<br>$CLK_m = CLK_{src} / N / M$ |
| 30:26         | /   | /   | /                                                                                                                             |

|       |     |   |                                                                                                      |
|-------|-----|---|------------------------------------------------------------------------------------------------------|
| 25:24 | R/W | 0 | MCLK_SRC_SEL<br><br>Clock source selection<br>00: HFCLK<br>01: LFCLK<br>1x: reserved                 |
| 23:18 | /   | / | /                                                                                                    |
| 17:16 | R/W | 0 | CLK_DIV_RATIO_N<br><br>clock pre-divide ratio N.<br>00: N = 1<br>01: N = 2<br>10: N = 4<br>11: N = 8 |
| 15:4  | /   | / | /                                                                                                    |
| 3:0   | R/W | 0 | CLK_DIV_RATIO_M<br><br>Clock divide ratio M.<br>M = value + 1 (1~16)                                 |

## 3.3 实时时钟(RTC)

### 3.3.1 概述

实时时钟(RTC)用于日历，是围绕一个48位计数器构建的，用于以“年-月-日”(YY-MM-DD)和“时-分-秒”(HH-MM-SS)的形式计算运行时间。当系统断电时，可通过备用电池进行操作。它有一个内置的闰年发生器。

在下电模式或正常运行模式下，闹钟(Alarm)在指定时间产生告警信号。在正常运行模式下，告警中断和电源管理唤醒都被激活。在下电模式下，电源管理唤醒信号被激活。在本节中，有两种闹钟。闹钟0(Alarm 0)是通用警报器，它的计数器是基于秒的。闹钟1(Alarm 1)是以周为单位的实时闹钟。

### 3.3.2 RTC 寄存器列表

| 模块名             | 基地址        |                                 |
|-----------------|------------|---------------------------------|
| Real Time Clock | 0x40041800 |                                 |
| 寄存器名            | 偏移地址       | 描述                              |
| RTC_CTRL_REG    | 0x0000     | RTC Control Register            |
| RTC_YMMDD_REG   | 0x0010     | RTC Year Month Day Register     |
| RTC_HHMMSS_REG  | 0x0014     | RTC Hour Minute Second Register |

|                    |        |                                          |
|--------------------|--------|------------------------------------------|
| ALMO_CNT_REG       | 0x0020 | Alarm 0 Counter Register                 |
| ALMO_CURVAL_REG    | 0x0024 | Alarm 0 Current Value Register           |
| ALMO_EN_REG        | 0x0028 | Alarm 0 Enable Register                  |
| ALMO_IRQEN_REG     | 0x002C | Alarm 0 IRQ Enable Register              |
| ALMO_IRQST_REG     | 0x0030 | Alarm 0 IRQ Status Register              |
| ALM1_WK_HHMMSS_REG | 0x0040 | Alarm 1 Week Hour-Minute-Second Register |
| ALM1_EN_REG        | 0x0044 | Alarm 1 Enable Register                  |
| ALM1_IRQEN_REG     | 0x0048 | Alarm 1 IRQ Enable Register              |
| ALM1_IRQST_REG     | 0x004C | Alarm 1 IRQ Status Register              |
| ALM_WAKEUP_EN_REG  | 0x0050 | Alarm wakeup Enable Register             |
| FRUN_CNT_L_REG     | 0x0060 | Free Running Counter bit[31:0]           |
| FRUN_CNT_H_REG     | 0x0064 | Free Running Counter bit[47:32]          |

### 3.3.3 RTC 寄存器描述

#### 3.3.3.1 RTC\_CTRL\_REG

| 偏移地址：0x0000 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                  |
|-------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                |
| 31          | R/W | 0x0 | RTC_TEST_MODE_CTRL.<br>RTC TEST Mode Control bit.                                                                                                                                                                                                                                                                 |
| 30          | R/W | 0x0 | RTC_DEBUG.<br>RTC Simulation Control bit<br>0: No effect. 1: simulation mode                                                                                                                                                                                                                                      |
| 29:2        | /   | /   | /                                                                                                                                                                                                                                                                                                                 |
| 1           | R   | 0   | RTC_HHMMSS_ACCE.<br><br>RTC HH-MM-SS access.<br>After writing the RTC HH-MM-SS register, this bit is set and it will be cleared until the real writing operation is finished.<br>After the RTC RTC_HHMMSS_ACCE is cleared, the writing operation of HH-MM-SS register should at most delay one sysrtc_32k cycle.. |
| 0           | R   | 0   | RTC_YYMMDD_ACCE.<br><br>RTC YY-MM-DD access.<br>After writing the RTC YY-MM-DD register, this bit is set and it will be cleared until the real writing operation is finished.<br>After the RTC RTC_YYMMDD_ACCE is cleared, the writing operation                                                                  |

|  |  |  |                                                                 |
|--|--|--|-----------------------------------------------------------------|
|  |  |  | of YY-MM-DD register should at most delay one sysrtc_32k cycle. |
|--|--|--|-----------------------------------------------------------------|

### 3.3.3.2 RTC\_YMMDD\_REG

| 偏移地址 : 0x0010 |     |     | 默认值: 0x0000_0000                                                                                                                                                            |
|---------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                          |
| 31:25         | /   | /   | /.                                                                                                                                                                          |
| 24            | R/W | 0x0 | LEAP.<br><br>Leap Year.<br>0: not<br>1: Leap year.<br><br>This bit cannot be set by hardware. It should be set or cleared by software                                       |
| 23:16         | R/W | x   | YEAR.<br><br>Year.<br>Range from 0~256.                                                                                                                                     |
| 15:12         | /   | /   | /                                                                                                                                                                           |
| 11:8          | R/W | x   | MONTH.<br><br>Month.<br>Range from 1~12.                                                                                                                                    |
| 7:5           | /   | /   | /.                                                                                                                                                                          |
| 4:0           | R/W | x   | DAY.<br><br>Day.<br>Range from 1~31.<br><br>Note: If the written value is not from 1 to 31, it turns into 31 automatically. The month field and the year field are similar. |

### 3.3.3.3 RTC\_HHMMSS\_REG

| 偏移地址 : 0x0014 |     |     | 默认值: 0x0000_0000                                            |
|---------------|-----|-----|-------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                          |
| 31:29         | R/W | 0x0 | WEEK_NO.<br><br>Week number.<br>000: Monday<br>001: Tuesday |

|       |     |   |                                                                                          |
|-------|-----|---|------------------------------------------------------------------------------------------|
|       |     |   | 010: Wednesday<br>011: Thursday<br>100: Friday<br>101: Saturday<br>110: Sunday<br>111: / |
| 28:21 | /   | / | /                                                                                        |
| 20:16 | R/W | x | HOUR.<br><br>Range from 0~23                                                             |
| 15:14 | /   | / | /                                                                                        |
| 13:8  | R/W | x | MINUTE.<br><br>Range from 0~59                                                           |
| 7:6   | /   | / | /                                                                                        |
| 5:0   | R/W | x | SECOND.<br><br>Range from 0~59                                                           |

#### 3.3.3.4 ALMO\_CNT\_REG

| 偏移地址：0x0020 |     | 默认值: 0x0000_0000 |                                                 |
|-------------|-----|------------------|-------------------------------------------------|
| 位置          | 访问  | 默认值              | 描述                                              |
| 31:0        | R/W | 0                | ALMO_CNT<br>Alarm 0 Counter is based on second. |

#### 3.3.3.5 ALMO\_CURVAL\_REG

| 偏移地址：0x0024 |    | 默认值: 0x0000_0000 |             |
|-------------|----|------------------|-------------|
| 位置          | 访问 | 默认值              | 描述          |
| 31:0        | R  | x                | ALMO_CURVAL |

#### 3.3.3.6 ALMO\_EN\_REG

| 偏移地址：0x0028 |     | 默认值: 0x0000_0000 |                                                                                              |
|-------------|-----|------------------|----------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值              | 描述                                                                                           |
| 31:1        | /   | /                | /                                                                                            |
| 0           | R/W | 0                | ALMO_EN<br><br>If this bit is set to 1, the alarm 0 counter register' s valid bits will down |

|  |  |  |                                                                                  |
|--|--|--|----------------------------------------------------------------------------------|
|  |  |  | count to 0 and the alarm pending bit will be set to 1<br>0: disable<br>1: enable |
|--|--|--|----------------------------------------------------------------------------------|

### 3.3.3.7 ALMO\_IRQEN\_REG

| 偏移地址 : 0x002C |     |     | 默认值: 0x0000_0000                          |
|---------------|-----|-----|-------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                        |
| 31:1          | /   | /   | /                                         |
| 0             | R/W | 0   | ALMO_IRQEN<br><br>0: disable<br>1: enable |

### 3.3.3.8 ALMO\_IRQST\_REG

| 偏移地址 : 0x0030 |     |     | 默认值: 0x0000_0000                             |
|---------------|-----|-----|----------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                           |
| 31:1          | /   | /   | /                                            |
| 0             | R/W | 0   | ALMO_IRQST<br><br>0: no effect<br>1: pending |

### 3.3.3.9 Alarm 1\_WK\_HHMMSS\_REG

| 偏移地址 : 0x0040 |     |     | 默认值: 0x0000_0000          |
|---------------|-----|-----|---------------------------|
| 位置            | 访问  | 默认值 | 描述                        |
| 31:21         | /   | /   | /                         |
| 20:16         | R/W | X   | HOUR<br>Range from 0-23   |
| 15:14         | /   | /   | /                         |
| 13:8          | R/W | X   | MINUTE<br>Range from 0-59 |
| 7:6           | /   | /   | /                         |
| 5:0           | R/W | X   | SECOND<br>Range from 0-59 |

### 3.3.3.10 ALM1\_EN\_REG

| 偏移地址：0x0044 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                    |
|-------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                  |
| 31:7        | /   | /   | /                                                                                                                                                                                                                                                                                                                   |
| 6           | R/W | 0   | <p>SUN_ALM1_EN</p> <p>Sunday Alarm 1 Enable</p> <p>0: disable</p> <p>1: enable</p> <p>If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC</p> <p>HH-MM-SS bit [31:29] is 6, the alarm IRQ pending bit will be set to “1”.</p> |
| 5           | R/W | 0   | <p>SAT_ALM1_EN</p> <p>Saturday Alarm 1 Enable 0: disable</p> <p>1: enable</p> <p>If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC</p> <p>HH-MM-SS bit [31:29] is 5, the alarm IRQ pending bit will be set to “1” .</p>     |
| 4           | R/W | 0   | <p>FRI_ALM1_EN</p> <p>Friday Alarm 1 Enable 0: disable</p> <p>1: enable</p> <p>If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC</p> <p>HH-MM-SS bit [31:29] is 4, the alarm IRQ pending bit will be set to “1” .</p>       |
| 3           | R/W | 0   | <p>THU_ALM1_EN</p> <p>Thursday Alarm 1 Enable 0: disable</p> <p>1: enable</p> <p>If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC</p> <p>HH-MM-SS bit [31:29] is 3, the alarm IRQ pending bit will be set to “1” .</p>     |
| 2           | R/W | 0   | <p>WED_ALM1_EN</p>                                                                                                                                                                                                                                                                                                  |

|   |     |   |                                                                                                                                                                                                                                                                                                                |
|---|-----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |   | <p>Wednesday Alarm 1 Enable</p> <p>0: disable</p> <p>1: enable</p> <p>If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC</p> <p>HH-MM-SS bit [31:29] is 2, the alarm IRQ pending bit will be set to “1” .</p>           |
| 1 | R/W | 0 | <p>TUE_ALM1_EN</p> <p>Tuesday Alarm 1 Enable 0: disable</p> <p>1: enable</p> <p>If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC</p> <p>HH-MM-SS bit [31:29] is 1, the alarm IRQ pending bit will be set to “1” .</p> |
| 0 | R/W | 0 | <p>MON_ALM1_EN</p> <p>Monday Alarm 1 Enable 0: disable</p> <p>1: enable</p> <p>If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC</p> <p>HH-MM-SS bit [31:29] is 0, the alarm IRQ pending bit will be set to “1” .</p>  |

### 3.3.3.11 ALM1\_IRQEN\_REG

| 偏移地址 : 0x0048 |     |     | 默认值: 0x0000_0000                                     |
|---------------|-----|-----|------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                   |
| 31:1          | /   | /   | /                                                    |
| 0             | R/W | 0   | <p>ALM1_IRQEN</p> <p>0: disable</p> <p>1: enable</p> |

### 3.3.3.12 ALM1\_IRQST\_REG

| 偏移地址 : 0x004C |    |     | 默认值: 0x0000_0000 |
|---------------|----|-----|------------------|
| 位置            | 访问 | 默认值 | 描述               |
| 31:1          | /  | /   | /                |



|   |     |   |                                                                             |
|---|-----|---|-----------------------------------------------------------------------------|
| 0 | R/W | 0 | ALM1_WEEK_IRQST<br>Alarm 1 week(0-7) IRQ Pending 0: No effect<br>1: pending |
|---|-----|---|-----------------------------------------------------------------------------|

### 3.3.3.13 ALM\_WAKEUP\_EN\_REG

| 偏移地址 : 0x0050 |     |     | 默认值: 0x0000_0000                                                                     |
|---------------|-----|-----|--------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                   |
| 31:2          | /   | /   | /                                                                                    |
| 1             | R/W | 0   | ALM1_WAKEUP<br><br>0: disable alarm1 wakeup output<br>1: enable alarm1 wakeup output |
| 0             | R/W | 0   | ALM0_WAKEUP<br><br>0: disable alarm0 wakeup output<br>1: enable alarm0 wakeup output |

### 3.3.3.14 FREERUN\_CNT\_L\_REG

| 偏移地址 : 0x0060 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                      |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                    |
| 31:0          | R/W | 0   | Free Running Counter bit[31:0]<br><br>Note: Free running counter is a 48-bit counter which is driven by LFCLK and starts to count as soon as the system reset is released and the LFCLK is ready. Write will reset the counter value. |

### 3.3.3.15 FREERUN\_CNT\_H\_REG

| 偏移地址 : 0x0064 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                       |
|---------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                     |
| 31:0          | R/W | 0   | Free Running Counter bit[47:32]<br><br>Note: Free running counter is a 48-bit counter which is driven by LFCLK and starts to count as soon as the system reset is released and the LFCLK is ready. Write will reset the counter value. |

## 3.4 通用定时器(Timer0/1)

### 3.4.1 概述

定时器0和1的输入源可以来自内部RC振荡器、外部32768Hz晶体或OSC。它旨在提供最大的准确性和有效的管理，即使系统处于长或短的响应时间。他们提供32位可编程溢出计数器，工作在自动重载模式或无重载模式。

### 3.4.2 Timer 寄存器列表

| 模块名                 | 基地址        |                                 |
|---------------------|------------|---------------------------------|
| Timer               | 0x40040800 |                                 |
| 寄存器名                | 偏移地址       | 描述                              |
| TMR_IRQ_EN_REG      | 0x0000     | Timer IRQ Enable Register       |
| TMR_IRQ_STA_REG     | 0x0004     | Timer Status Register           |
| TMR0_CTRL_REG       | 0x0010     | Timer 0 Control Register        |
| TMR0_INTV_VALUE_REG | 0x0014     | Timer 0 Interval Value Register |
| TMR0_CUR_VALUE_REG  | 0x0018     | Timer 0 Current Value Register  |
| TMR1_CTRL_REG       | 0x0020     | Timer 1 Control Register        |
| TMR1_INTV_VALUE_REG | 0x0024     | Timer 1 Interval Value Register |
| TMR1_CUR_VALUE_REG  | 0x0028     | Timer 1 Current Value Register  |

### 3.4.3 Timer 寄存器描述

#### 3.4.3.1 TMR\_IRQ\_EN\_REG

| 偏移地址：0x0000 |     |     | 默认值: 0x0000_0000                                                                                                     |
|-------------|-----|-----|----------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                   |
| 31:2        | /   | /   | /                                                                                                                    |
| 1           | R/W | 0x0 | TMR1_IRQ_EN.<br><br>Timer 1 Interrupt Enable.<br>0: No effect<br>1: Timer 1 Interval Value reached interrupt enable. |
| 0           | R/W | 0x0 | TMR0_IRQ_EN.<br><br>Timer 0 Interrupt Enable.<br>0: No effect<br>1: Timer 0 Interval Value reached interrupt enable. |

### 3.4.3.2 TMR\_IRQ\_STA\_REG

| 偏移地址：0x0004 |     |     | 默认值: 0x0000_0000                                                                                                                                          |
|-------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                        |
| 31:2        | /   | /   | /                                                                                                                                                         |
| 1           | R/W | 0x0 | <p>TMR1_IRQ_PEND.</p> <p>Timer 1 IRQ Pending. Set 1 to the bit will clear it.</p> <p>0: No effect,<br/>1: Pending, timer 1 interval value is reached.</p> |
| 0           | R/W | 0x0 | <p>TMRO_IRQ_PEND.</p> <p>Timer 0 IRQ Pending. Set 1 to the bit will clear it.</p> <p>0: No effect,<br/>1: Pending, timer 0 interval value is reached.</p> |

### 3.4.3.3 TMRO\_CTRL\_REG

| 偏移地址：0x0010 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                   |
|-------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                                 |
| 31:8        | /   | /   | /                                                                                                                                                                                                                                  |
| 7           | R/W | 0x0 | <p>TMRO_MODE.</p> <p>Timer 0 mode.</p> <p>0: Continuous mode. When interval value reached, the timer will not disable automatically.</p> <p>1: Single mode. When interval value reached, the timer will disable automatically.</p> |
| 6:4         | R/W | 0x0 | <p>TMRO_CLK_PRES.</p> <p>Select the pre-scale of timer 0 clock source.</p> <p>000: /1<br/>001: /2<br/>010: /4<br/>011: /8<br/>100: /16<br/>101: /32<br/>110: /64<br/>111: /128</p>                                                 |
| 3:2         | R/W | 0x1 | TMRO_CLK_SRC.                                                                                                                                                                                                                      |

|   |     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |     | Timer 0 Clock Source.<br>00: LOSC,<br>01: HOSC.<br>10: /<br>11: /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1 | R/W | 0x0 | TMRO_RELOAD.<br>Timer 0 Reload.<br>0: No effect,<br>1: Reload timer 0 Interval value.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 0 | R/W | 0x0 | TMRO_EN.<br><br>Timer 0 Enable.<br>0: Stop/Pause<br>1: Start.<br><br>If the timer is started, it will reload the interval value to internal register, and the current counter will count from interval value to 0.<br>If the current counter does not reach the zero, the timer enable bit is set to “0” ; the current value counter will pause. At least wait for 2*Tcylce, the start bit can be set to 1.<br><br>In timer pause state, the interval value register can be modified. If the timer is started again, and the Software hope the current value register to down-count from the new interval value, the reload bit and the enable bit should be set to 1 at the same time. |

#### 3.4.3.4 TMRO\_INTV\_VALUE\_REG

| 偏移地址 : 0x0014 |     | 默认值: 0x0000_0000 |                                                                                                                                         |
|---------------|-----|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值              | 描述                                                                                                                                      |
| 31:0          | R/W | x                | TMRO_INTV_VALUE.<br><br>Timer 0 Interval Value.<br>Note: the value setting should consider the system clock and the timer clock source. |

#### 3.4.3.5 TMRO\_CUR\_VALUE\_REG

| 偏移地址 : 0x0018 |     | 默认值: 0x0000_0000 |                                               |
|---------------|-----|------------------|-----------------------------------------------|
| 位置            | 访问  | 默认值              | 描述                                            |
| 31:0          | R/W | 0x0              | TMRO_CUR_VALUE.<br><br>Timer 0 Current Value. |

### 3.4.3.6 TMR1\_CTRL\_REG

| 偏移地址 : 0x0020 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                 |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                               |
| 31:8          | /   | /   | /                                                                                                                                                                                                                                                                                                                |
| 7             | R/W | 0x0 | <p>TMR1_MODE.</p> <p>Timer 1 mode.</p> <p>0: Continuous mode. When interval value reached, the timer will not disable automatically.</p> <p>1: Single mode. When interval value reached, the timer will disable automatically.</p>                                                                               |
| 6:4           | R/W | 0x0 | <p>TMR1_CLK_PRE.</p> <p>Select the pre-scale of timer 1 clock source.</p> <p>000: /1</p> <p>001: /2</p> <p>010: /4</p> <p>011: /8</p> <p>100: /16</p> <p>101: /32</p> <p>110: /64</p> <p>111: /128</p>                                                                                                           |
| 3:2           | R/W | 0x1 | <p>TMR1_CLK_SRC.</p> <p>Timer 1 Clock Source.</p> <p>00: LOSC,</p> <p>01: HOSC.</p> <p>10: /</p> <p>11: /</p>                                                                                                                                                                                                    |
| 1             | R/W | 0x0 | <p>TMR1_RELOAD.</p> <p>Timer 1 Reload.</p> <p>0: No effect</p> <p>1: Reload timer 1 Interval value.</p>                                                                                                                                                                                                          |
| 0             | R/W | 0x0 | <p>TMR1_EN.</p> <p>Timer 1 Enable.</p> <p>0: Stop/Pause</p> <p>1: Start.</p> <p>If the timer is started, it will reload the interval value to internal register, and the current counter will count from interval value to 0.</p> <p>If the current counter does not reach the zero, the timer enable bit is</p> |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                  |
|--|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <p>set to “0” ; the current value counter will pause. At least wait for 2*Tcylce, the start bit can be set to 1.</p> <p>In timer pause state, the interval value register can be modified. If the timer is started again, and the Software hope the current value register to down-count from the new interval value, the reload bit and the enable bit should be set to 1 at the same time.</p> |
|--|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 3.4.3.7 TMR1\_INTV\_VALUE\_REG

| 偏移地址 : 0x0024 |     |     | 默认值: 0x0000_0000                                                                                                                                   |
|---------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                 |
| 31:0          | R/W | x   | <p>TMR1_INTV_VALUE.</p> <p>Timer 1 Interval Value.</p> <p>Note: the value setting should consider the system clock and the timer clock source.</p> |

#### 3.4.3.8 TMR1\_CUR\_VALUE\_REG

| 偏移地址 : 0x0028 |     |     | 默认值: 0x0000_0000                                     |
|---------------|-----|-----|------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                   |
| 31:0          | R/W | 0x0 | <p>TMR1_CUR_VALUE.</p> <p>Timer 1 Current Value.</p> |

## 3.5 看门狗定时器(Watchdog)

### 3.5.1 概述

看门狗用于在控制器受到噪音和系统错误等故障干扰时恢复控制器的运行。本质上它是一个递减计数器，允许看门狗的时间长达16秒。它可以产生一个通用的复位或中断请求。

### 3.5.2 Watchdog 寄存器列表

| 模块名              | 基地址        |                                 |
|------------------|------------|---------------------------------|
| Timer            | 0x40040800 |                                 |
| 寄存器名             | 偏移地址       | 描述                              |
| WDOG_IRQ_EN_REG  | 0x00A0     | Watchdog Irq Enable Register    |
| WDOG_IRQ_STA_REG | 0x00A4     | Watchdog Irq Status Register    |
| WDOG_CTRL_REG    | 0x00B0     | Watchdog Control Register       |
| WDOG_CFG_REG     | 0x00B4     | Watchdog Configuration Register |
| WDOG_MODE_REG    | 0x00B8     | Watchdog Mode Register          |

|                      |        |                                  |
|----------------------|--------|----------------------------------|
| WDOG_OUTPUT_CTRL_REG | 0x00BC | Watchdog Output Control Register |
|----------------------|--------|----------------------------------|

### 3.5.3 Watchdog 寄存器描述

#### 3.5.3.1 WDOG\_IRQ\_EN\_REG

| 偏移地址：0x00A0 |     |     | 默认值: 0x0000_0000                                                |
|-------------|-----|-----|-----------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                              |
| 31:0        | R/W | 0x0 | WDOG_IRQ_EN<br><br>0: No effect<br>1: Watchdog interrupt enable |

#### 3.5.3.2 WDOG\_IRQ\_EN\_REG

| 偏移地址：0x00A4 |     |     | 默认值: 0x0000_0000                                                                                                            |
|-------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                          |
| 31:1        | /   | /   | /                                                                                                                           |
| 0           | R/W | 0x0 | WDOG_IRQ_STA<br><br>Watchdog irq pending. Write 1 to clear it<br>0: No effect<br>1: Pending, the interval value is reached. |

#### 3.5.3.3 WDOG\_CTRL\_REG

| 偏移地址：0x00B0 |     |     | 默认值: 0x0000_0000                                                                                                                                 |
|-------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                               |
| 31:13       | /   | /   | /                                                                                                                                                |
| 12:1        | R/W | 0x0 | WDOG0_KEY_FIELD.<br>Watchdog 0 Key Field.<br>Should be written at value 0xA57. Writing any other value in this field aborts the write operation. |
| 0           | R/W | 0x0 | WDOG_RELOAD<br><br>Watchdog Reload<br>0: no effect<br>1: reload the watchdog                                                                     |

### 3.5.3.4 WDOG\_CFG\_REG

| 偏移地址：0x00B4 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                          |
|-------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                        |
| 31:2        | /   | /   | /                                                                                                                                                                                                                         |
| 8           | R/W | 0x0 | CLK_SEL<br><br>0: 32K<br>1: 32.768K                                                                                                                                                                                       |
| 7:4         | /   | /   | /                                                                                                                                                                                                                         |
| 3:2         | R/W | 0x0 | WDOG0_RST_CPU_MODE<br>00: when only reset CPU, watchdog resetn to CPU PORESETn.<br>01: when only reset CPU, watchdog resetn only to CPU SYSRESETn.<br>1x: when only reset CPU, watchdog resetn to CPU PORESETn/SYSRESETn. |
| 1:0         | R/W | 0x1 | WDOG0_CONFIG.<br>00: only reset CPU<br>01: reset whole system<br>10: only interrupt<br>11: /                                                                                                                              |

### 3.5.3.5 WDOG\_MODE\_REG

| 偏移地址：0x00B8 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 31:8        | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7:4         | R/W | 0   | WDOG_INTV_VALUE<br><br>Watchdog Interval Value<br>The watchdog clock source is HOSC/WDOG_PRESCALER. If the HOSC is turned off, the watchdog will not work.<br><br>0000: 16000 cycles (0.5s)<br>0001: 32000 cycles (1s)<br>0010: 64000 cycles (2s)<br>0011: 96000 cycles (3s)<br>0100: 128000 cycles (4s)<br>0101: 160000 cycles (5s)<br>0110: 192000 cycles (6s)<br>0111: 256000 cycles (8s)<br>1000: 320000 cycles (10s) |



|     |     |     |                                                                                                                                                  |
|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |     | 1001: 384000 cycles (12s)<br>1010: 448000 cycles (14s)<br>1011: 512000 cycles (16s)<br>others: /                                                 |
| 3:1 | W   | 0x0 | WDOG_RST_110<br>When watchdog enable (WDOG_EN=1), written at value 3'b110. The watchdog reset should assert promptly. Other values are no valid. |
| 0   | R/W | 0x0 | WDOG_EN<br><br>Watchdog Enable<br>0: no effect<br>1: Enable watchdog                                                                             |

### 3.5.3.6 WDOG\_OUTPUT\_CTRL\_REG

| 偏移地址 : 0x00BC |     |     | 默认值: 0x0000_0000                                                                                        |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                      |
| 31:5          | /   | /   | /                                                                                                       |
| 4:0           | R/W | 0xA | WDOG_OUTPUT_CONTROL<br><br>0~31<br>the number of cycles with which the watch dog holds the reset signal |

## 4 DMA 控制器

### 4.1 安全 DMA (SDMA)

#### 4.1.1 概述

SDMA控制器有2个AHB DMA通道。并且按照优先级顺序排列，同一时刻只能有一个通道传输，高优先级通道传输完成后，低优先级通道传输才能开始。

SDMA控制器支持8位/16位/32位数据宽度。源端和目的端的数据宽度可以不同，但地址必须对齐。虽然SDMA的增加方式应该是地址对齐的，但它的字节计数器不可以是多个。即使SDMA已经启动，其源地址、目的地址、字节计数器寄存器仍可以被修改。

#### 4.1.2 SDMA 寄存器列表

| 模块名                | 基地址               |                                        |
|--------------------|-------------------|----------------------------------------|
| SDMA Controller    | 0x40081000        |                                        |
| 寄存器名               | 偏移地址              | 描述                                     |
| SDMA_INT_CTRL_REG  | 0x0000            | SDMA Interrupt Control Register        |
| SDMA_INT_STA_REG   | 0x0004            | SDMA Interrupt Status Register         |
| SDMA_PTY_CFG_REG   | 0x0008            | SDMA Priority Configure Register       |
| SDMA_CTRL_REG      | 0x0100+N*0x20     | Secure DMA Configuration (N=0,1)       |
| SDMA_SRC_ADDR_REG  | 0x0100+N*0x20+0x4 | Secure DMA Source Address (N=0,1)      |
| SDMA_DEST_ADDR_REG | 0x0100+N*0x20+0x8 | Secure DMA Destination Address (N=0,1) |
| SDMA_BC_REG        | 0x0100+N*0x20+0xC | Secure DMA Byte Counter (N=0,1)        |

#### 4.1.3 SDMA 寄存器描述

##### 4.1.3.1 SDMA\_IRQ\_EN\_REG

| 偏移地址：0x0000 |     |     | 默认值: 0x0000_0000                                                                                |
|-------------|-----|-----|-------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                              |
| 31:4        | /   | /   | /                                                                                               |
| 3           | R/W | 0x0 | SDMA1_END_IRQ_EN<br><br>SDMA Channel 1 End Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |

|   |     |     |                                                                                                 |
|---|-----|-----|-------------------------------------------------------------------------------------------------|
| 2 | R/W | 0x0 | SDMA1_HF_IRQ_EN<br><br>SDMA Channel 1 Half Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |
| 1 | R/W | 0x0 | SDMA0_END_IRQ_EN<br><br>SDMA Channel 0 End Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |
| 0 | R/W | 0x0 | SDMA0_HF_IRQ_EN<br><br>SDMA Channel 0 Half Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |

#### 4.1.3.2 SDMA\_IRQ\_PEND\_STA\_REG

| 偏移地址 : 0x0004 |       |     | 默认值: 0x0000_0000                                                                                                                       |
|---------------|-------|-----|----------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问    | 默认值 | 描述                                                                                                                                     |
| 31:4          | /     | /   | /                                                                                                                                      |
| 3             | R/W1C | 0x0 | SDMA1_END_IRQ_PEND.<br><br>SDMA Channel 1 End Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 2             | R/W1C | 0x0 | SDMA1_HF_IRQ_PEND.<br><br>SDMA Channel 1 Half Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 1             | R/W1C | 0x0 | SDMA0_END_IRQ_PEND.<br><br>SDMA Channel 0 End Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 0             | R/W1C | 0x0 | SDMA0_HF_IRQ_PEND.<br><br>SDMA Channel 0 Half Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |

#### 4.1.3.3 SDMA\_AUTO\_GAT\_REG

| 偏移地址 : 0x0008 |     |     | 默认值: 0x0000_0000                                                                                                                                                          |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                        |
| 31:17         | /   | /   | /                                                                                                                                                                         |
| 16            | R/W | 0x0 | SDMA Auto Clock Gating bit<br><br>0: SDMA auto clock gating enable<br>1: SDMA auto clock gating disable<br>If SDMA works in continuous mode, this bit should be set to 1. |
| 15:0          | /   | /   | /                                                                                                                                                                         |

#### 4.1.3.4 SDMA\_CTRL\_REG

| 偏移地址 : 0x100+N*0x20<br>(N=0,1) |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                              |
|--------------------------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                             | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                            |
| 31                             | R/W | 0x0 | SDMA_LOADING.<br>DMA Loading.<br>If set to 1, DMA will start and load the DMA registers to the shadow registers. The bit will hold on until the DMA finished. It will be cleared automatically.<br>Set 0 to the bit will reset the corresponding DMA channel. |
| 30                             | RO  | 0x0 | DMA Busy Status.<br>0: DMA idle, 1: DMA busy.                                                                                                                                                                                                                 |
| 29                             | R/W | 0x0 | SDMA_CONTI_MODE_EN.<br>DMA Continuous Mode Enable.<br>0: Disable, 1: Enable.                                                                                                                                                                                  |
| 28:26                          | R/W | 0x0 | SDMA Wait State. (n DMA clock cycles)<br>0: 1<br>1: 2<br>2: 4<br>3: 8<br>4: 16<br>5: 32<br>6: 64<br>7: 128                                                                                                                                                    |
| 25:24                          | R/W | 0x0 | SDMA_DEST_DATA_WIDTH.<br>DMA Destination Data Width.<br>00: 8-bit<br>01: 16-bit                                                                                                                                                                               |

|       |     |     |                                                                                                                                                                                                                                                                                                                               |
|-------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |     |     | 10: 32-bit<br>11: /                                                                                                                                                                                                                                                                                                           |
| 23    | R/W | 0x0 | SDMA_DEST_BST_LEN.<br>DMA Destination Burst Length.<br>00: 1,<br>01: 4                                                                                                                                                                                                                                                        |
| 22    | /   | /   | /                                                                                                                                                                                                                                                                                                                             |
| 21    | R/W | 0x0 | SDMA_DEST_ADDR_TYPE.<br>DMA Destination Address Type.<br>0: Increment<br>1: No Change.                                                                                                                                                                                                                                        |
| 20:16 | R/W | 0x0 | SDMA_DEST_DRQ_TYPE.<br>DMA Destination DRQ Type.<br>00000 : SRAM<br>00001 : SPI0 TX<br>00010 : Reserved<br>00011 : UART0 TX<br>00100 : UART1 TX<br>00101 : SS TX<br>00110 : DAUDIO TX<br>00111 : FLASHC TX<br>01000 : Reserved<br>01001 : UART2 TX<br>01010 : AUDIO PWM<br>01011 : Reserved<br>01100 : Reserved<br>others : / |
| 15    | R/W | 0x0 | BC_MODE_SEL.<br><br>BC mode select.<br>0: normal mode (the value read back is equal to the value that is written)<br>1: remain mode (the value read back is equal to the left bytes to be transferred).                                                                                                                       |
| 14:10 | /   | /   | /                                                                                                                                                                                                                                                                                                                             |
| 9:8   | R/W | 0x0 | SDMA_SRC_DATA_WIDTH.<br>DMA Source Data Width.<br>00: 8-bit<br>01: 16-bit<br>10: 32-bit<br>11: /                                                                                                                                                                                                                              |

|     |     |     |                                                                                                                                                                                                                                                                                                                        |
|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7   | R/W | 0x0 | SDMA_SRC_BST_LEN.<br>DMA Source Burst Length.<br>00: 1<br>01: 4                                                                                                                                                                                                                                                        |
| 6   | /   | /   | /                                                                                                                                                                                                                                                                                                                      |
| 5   | R/W | 0x0 | SDMA_SRC_ADDR_TYPE.<br>DMA Source Address Type.<br>0: Increment<br>1: No Change                                                                                                                                                                                                                                        |
| 4:0 | R/W | 0x0 | SDMA_SRC_DRQ_TYPE.<br>DMA Source DRQ Type.<br>00000 : SRAM<br>00001 : SPI0 RX<br>00010 : Reserved<br>00011 : UART0 RX<br>00100 : UART1 RX<br>00101 : SS RX<br>00110 : DAUDIO RX<br>00111 : FLASHC RX<br>01000 : Reserved<br>01001 : UART2 RX<br>01010 : Reserved<br>01011 : GPADC RX<br>01100 : Reserved<br>others : / |

#### 4.1.3.5 SDMA\_SRC\_ADDR\_REG

| 偏移地址 : $0x100+N*0x20+0x4$<br>(N=0,1) |     | 默认值: 0x0000_0000 |                                                            |
|--------------------------------------|-----|------------------|------------------------------------------------------------|
| 位置                                   | 访问  | 默认值              | 描述                                                         |
| 31:0                                 | R/W | 0x0              | SDMA_SRC_ADDR.<br><br>DMA Channel N Source Address.(N=0,1) |

#### 4.1.3.6 SDMA\_DEST\_ADDR\_REG

| 偏移地址 : $0x100+N*0x20+0x8$<br>(N=0,1) |     | 默认值: 0x0000_0000 |                 |
|--------------------------------------|-----|------------------|-----------------|
| 位置                                   | 访问  | 默认值              | 描述              |
| 31:0                                 | R/W | 0x0              | SDMA_DEST_ADDR. |

|  |  |  |                                           |
|--|--|--|-------------------------------------------|
|  |  |  | DMA Channel N Destination Address.(N=0,1) |
|--|--|--|-------------------------------------------|

#### 4.1.3.7 SDMA\_BC\_REG

| 偏移地址 : $0x100+N*0x20+0xC$<br>(N=0,1) |     |     | 默认值: 0x0000_0000                                                                                                                     |
|--------------------------------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                   | 访问  | 默认值 | 描述                                                                                                                                   |
| 31:18                                | /   | /   | /                                                                                                                                    |
| 17:0                                 | R/W | 0x0 | SDMA_BC.<br><br>DMA Channel N Byte Counter.(N=0,1)<br>Note: If Byte Counter=0, DMA will transfer no byte. The maximum value is 128k. |

## 4.2 非安全 DMA (NDMA)

### 4.2.1 概述

NDMA控制器有8个AHB DMA通道。并且按照优先级顺序排列，同一时刻只能同一时刻只能有一个通道传输，高优先级通道传输完成后，低优先级通道传输才能开始。

同SDMA类似，NDMA控制器也支持8位/16位/32位数据宽度。源端和目的端的数据宽度可以不同，但地址必须对齐。虽然NDMA的增加方式应该是地址对齐的，但它的字节计数器不可以是多个。即使NDMA已经启动，其源地址、目的地址、字节计数器寄存器仍可以被修改。

### 4.2.2 NDMA 寄存器列表

| 模块名                | 基地址                 |                                                           |
|--------------------|---------------------|-----------------------------------------------------------|
| NDMA Controller    | 0x40001000          |                                                           |
| 寄存器名               | 偏移地址                | 描述                                                        |
| NDMA_INT_CTRL_REG  | 0x0000              | NDMA Interrupt Control Register                           |
| NDMA_INT_STA_REG   | 0x0004              | NDMA Interrupt Status Register                            |
| NDMA_PTY_CFG_REG   | 0x0008              | NDMA Priority Configure Register                          |
| NDMA_CTRL_REG      | $0x0100+N*0x20$     | Non-Secure DMA Configuration<br>(N=0,1,2,3,4,5,6,7)       |
| NDMA_SRC_ADDR_REG  | $0x0100+N*0x20+0x4$ | Non-Secure DMA Source Address<br>(N=0,1,2,3,4,5,6,7)      |
| NDMA_DEST_ADDR_REG | $0x0100+N*0x20+0x8$ | Non-Secure DMA Destination Address<br>(N=0,1,2,3,4,5,6,7) |
| NDMA_BC_REG        | $0x0100+N*0x20+0xC$ | Non-Secure DMA Byte Counter<br>(N=0,1,2,3,4,5,6,7)        |

### 4.2.3 NDMA 寄存器描述

#### 4.2.3.1 NDMA\_IRQ\_EN\_REG

| 偏移地址：0x0000 |     |     | 默认值：0x0000_0000                                                                                  |
|-------------|-----|-----|--------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                               |
| 31:16       | /   | /   | /                                                                                                |
| 15          | R/W | 0x0 | NDMA7_END_IRQ_EN.<br><br>NDMA Channel 7 End Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |
| 14          | R/W | 0x0 | NDMA7_HF_IRQ_EN<br><br>NDMA Channel 7 Half Transfer Interrupt Enable.<br>0: Disable, 1: Enable.  |
| 13          | R/W | 0x0 | NDMA6_END_IRQ_EN<br><br>NDMA Channel 6 End Transfer Interrupt Enable.<br>0: Disable, 1: Enable.  |
| 12          | R/W | 0x0 | NDMA6_HF_IRQ_EN<br><br>NDMA Channel 6 Half Transfer Interrupt Enable.<br>0: Disable, 1: Enable.  |
| 11          | R/W | 0x0 | NDMA5_END_IRQ_EN<br><br>NDMA Channel 5 End Transfer Interrupt Enable.<br>0: Disable, 1: Enable.  |
| 10          | R/W | 0x0 | NDMA5_HF_IRQ_EN<br><br>NDMA Channel 5 Half Transfer Interrupt Enable.<br>0: Disable, 1: Enable.  |
| 9           | R/W | 0x0 | NDMA4_END_IRQ_EN<br><br>NDMA Channel 4 End Transfer Interrupt Enable.<br>0: Disable, 1: Enable.  |
| 8           | R/W | 0x0 | NDMA4_HF_IRQ_EN<br><br>NDMA Channel 4 Half Transfer Interrupt Enable.<br>0: Disable, 1: Enable.  |
| 7           | R/W | 0x0 | NDMA3_END_IRQ_EN                                                                                 |



|   |     |     |                                                                                                 |
|---|-----|-----|-------------------------------------------------------------------------------------------------|
|   |     |     | NDMA Channel 3 End Transfer Interrupt Enable.<br>0: Disable, 1: Enable.                         |
| 6 | R/W | 0x0 | NDMA3_HF_IRQ_EN<br><br>NDMA Channel 3 Half Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |
| 5 | R/W | 0x0 | NDMA2_END_IRQ_EN<br><br>NDMA Channel 2 End Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |
| 4 | R/W | 0x0 | NDMA2_HF_IRQ_EN<br><br>NDMA Channel 2 Half Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |
| 3 | R/W | 0x0 | NDMA1_END_IRQ_EN<br><br>NDMA Channel 1 End Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |
| 2 | R/W | 0x0 | NDMA1_HF_IRQ_EN<br><br>NDMA Channel 1 Half Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |
| 1 | R/W | 0x0 | NDMA0_END_IRQ_EN<br><br>NDMA Channel 0 End Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |
| 0 | R/W | 0x0 | NDMA0_HF_IRQ_EN<br><br>NDMA Channel 0 Half Transfer Interrupt Enable.<br>0: Disable, 1: Enable. |

#### 4.2.3.2 NDMA\_IRQ\_PEND\_STA\_REG

| 偏移地址：0x0004 |       |     | 默认值: 0x0000_0000                                                                                          |
|-------------|-------|-----|-----------------------------------------------------------------------------------------------------------|
| 位置          | 访问    | 默认值 | 描述                                                                                                        |
| 31:16       | /     | /   | /                                                                                                         |
| 15          | R/W1C | 0x0 | NDMA7_END_IRQ_PEND.<br><br>NDMA Channel 7 End Transfer Interrupt Pending. Set 1 to the bit will clear it. |

|    |       |     |                                                                                                                                        |
|----|-------|-----|----------------------------------------------------------------------------------------------------------------------------------------|
|    |       |     | 0: No effect, 1: Pending.                                                                                                              |
| 14 | R/W1C | 0x0 | NDMA7_HF_IRQ_PEND.<br><br>NDMA Channel 7 Half Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 13 | R/W1C | 0x0 | NDMA6_END_IRQ_PEND.<br><br>NDMA Channel 6 End Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 12 | R/W1C | 0x0 | NDMA6_HF_IRQ_PEND.<br><br>NDMA Channel 6 Half Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 11 | R/W1C | 0x0 | NDMA5_END_IRQ_PEND.<br><br>NDMA Channel 5 End Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 10 | R/W1C | 0x0 | NDMA5_HF_IRQ_PEND.<br><br>NDMA Channel 5 Half Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 9  | R/W1C | 0x0 | NDMA4_END_IRQ_PEND.<br><br>NDMA Channel 4 End Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 8  | R/W1C | 0x0 | NDMA4_HF_IRQ_PEND.<br><br>NDMA Channel 4 Half Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 7  | R/W1C | 0x0 | NDMA3_END_IRQ_PEND.<br><br>NDMA Channel 3 End Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 6  | R/W1C | 0x0 | NDMA3_HF_IRQ_PEND.                                                                                                                     |

|   |       |     |                                                                                                                                        |
|---|-------|-----|----------------------------------------------------------------------------------------------------------------------------------------|
|   |       |     | NDMA Channel 3 Half Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending.                           |
| 5 | R/W1C | 0x0 | NDMA2_END_IRQ_PEND.<br><br>NDMA Channel 2 End Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 4 | R/W1C | 0x0 | NDMA2_HF_IRQ_PEND.<br><br>NDMA Channel 2 Half Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 3 | R/W1C | 0x0 | NDMA1_END_IRQ_PEND.<br><br>NDMA Channel 1 End Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 2 | R/W1C | 0x0 | NDMA1_HF_IRQ_PEND.<br><br>NDMA Channel 1 Half Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 1 | R/W1C | 0x0 | NDMA0_END_IRQ_PEND.<br><br>NDMA Channel 0 End Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |
| 0 | R/W1C | 0x0 | NDMA0_HF_IRQ_PEND.<br><br>NDMA Channel 0 Half Transfer Interrupt Pending. Set 1 to the bit will clear it.<br>0: No effect, 1: Pending. |

#### 4.2.3.3 NDMA\_AUTO\_GAT\_REG

| 偏移地址 : 0x0008 |     |     | 默认值: 0x0000_0000                                                                                        |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                      |
| 31:17         | /   | /   | /                                                                                                       |
| 16            | R/W | 0x0 | NDMA Auto Clock Gating bit<br><br>0: NDMA auto clock gating enable<br>1: NDMA auto clock gating disable |

|      |   |   |                                                                |
|------|---|---|----------------------------------------------------------------|
|      |   |   | If NDMA works in continuous mode, this bit should be set to 1. |
| 15:0 | / | / | /                                                              |

#### 4.2.3.4 NDMA\_CTRL\_REG

| 偏移地址 : 0x100+N*0x20<br>(N=0,1,2,3,4,5,6,7) |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                              |
|--------------------------------------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                         | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                            |
| 31                                         | R/W | 0x0 | NDMA_LOADING.<br>DMA Loading.<br>If set to 1, DMA will start and load the DMA registers to the shadow registers. The bit will hold on until the DMA finished. It will be cleared automatically.<br>Set 0 to the bit will reset the corresponding DMA channel. |
| 30                                         | RO  | 0x0 | DMA Busy Status.<br>0: DMA idle, 1: DMA busy.                                                                                                                                                                                                                 |
| 29                                         | R/W | 0x0 | NDMA_CONTI_MODE_EN.<br>DMA Continuous Mode Enable.<br>0: Disable, 1: Enable.                                                                                                                                                                                  |
| 28:26                                      | R/W | 0x0 | NDMA Wait State. (n DMA clock cycles)<br>0: 1<br>1: 2<br>2: 4<br>3: 8<br>4: 16<br>5: 32<br>6: 64<br>7: 128                                                                                                                                                    |
| 25:24                                      | R/W | 0x0 | NDMA_DEST_DATA_WIDTH.<br>DMA Destination Data Width.<br>00: 8-bit<br>01: 16-bit<br>10: 32-bit<br>11: /                                                                                                                                                        |
| 23                                         | R/W | 0x0 | NDMA_DEST_BST_LEN.<br>DMA Destination Burst Length.<br>00: 1,<br>01: 4                                                                                                                                                                                        |
| 22                                         | /   | /   | /                                                                                                                                                                                                                                                             |
| 21                                         | R/W | 0x0 | NDMA_DEST_ADDR_TYPE.<br>DMA Destination Address Type.                                                                                                                                                                                                         |

|       |     |     |                                                                                                                                                                                                                                                                                                                               |
|-------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |     |     | 0: Increment<br>1: No Change.                                                                                                                                                                                                                                                                                                 |
| 20:16 | R/W | 0x0 | NDMA_DEST_DRQ_TYPE.<br>DMA Destination DRQ Type.<br>00000 : SRAM<br>00001 : SPI0 TX<br>00010 : Reserved<br>00011 : UART0 TX<br>00100 : UART1 TX<br>00101 : SS TX<br>00110 : DAUDIO TX<br>00111 : FLASHC TX<br>01000 : Reserved<br>01001 : UART2 TX<br>01010 : AUDIO PWM<br>01011 : Reserved<br>01100 : Reserved<br>others : / |
| 15    | R/W | 0x0 | BC_MODE_SEL.<br><br>BC mode select.<br>0: normal mode (the value read back is equal to the value that is written)<br>1: remain mode (the value read back is equal to the left bytes to be transferred).                                                                                                                       |
| 14:10 | /   | /   | /.                                                                                                                                                                                                                                                                                                                            |
| 9:8   | R/W | 0x0 | NDMA_SRC_DATA_WIDTH.<br>DMA Source Data Width.<br>00: 8-bit<br>01: 16-bit<br>10: 32-bit<br>11: /                                                                                                                                                                                                                              |
| 7     | R/W | 0x0 | NDMA_SRC_BST_LEN.<br>DMA Source Burst Length.<br>00: 1<br>01: 4                                                                                                                                                                                                                                                               |
| 6     | /   | /   | /                                                                                                                                                                                                                                                                                                                             |
| 5     | R/W | 0x0 | NDMA_SRC_ADDR_TYPE.<br>DMA Source Address Type.<br>0: Increment<br>1: No Change                                                                                                                                                                                                                                               |

|     |     |     |                                                                                                                                                                                                                                                                                                                        |
|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4:0 | R/W | 0x0 | NDMA_SRC_DRQ_TYPE.<br>DMA Source DRQ Type.<br>00000 : SRAM<br>00001 : SPI0 RX<br>00010 : Reserved<br>00011 : UART0 RX<br>00100 : UART1 RX<br>00101 : SS RX<br>00110 : DAUDIO RX<br>00111 : FLASHC RX<br>01000 : Reserved<br>01001 : UART2 RX<br>01010 : Reserved<br>01011 : GPADC RX<br>01100 : Reserved<br>others : / |
|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 4.2.3.5 NDMA\_SRC\_ADDR\_REG

| 偏移地址 : 0x100+N*0x20+0x4<br>(N=0,1,2,3,4,5,6,7) |     | 默认值: 0x0000_0000 |                                                                        |
|------------------------------------------------|-----|------------------|------------------------------------------------------------------------|
| 位置                                             | 访问  | 默认值              | 描述                                                                     |
| 31:0                                           | R/W | 0x0              | NDMA_SRC_ADDR.<br><br>DMA Channel N Source Address.(N=0,1,2,3,4,5,6,7) |

#### 4.2.3.6 NDMA\_DEST\_ADDR\_REG

| 偏移地址 : 0x100+N*0x20+0x8<br>(N=0,1,2,3,4,5,6,7) |     | 默认值: 0x0000_0000 |                                                                              |
|------------------------------------------------|-----|------------------|------------------------------------------------------------------------------|
| 位置                                             | 访问  | 默认值              | 描述                                                                           |
| 31:0                                           | R/W | 0x0              | NDMA_DEST_ADDR.<br><br>DMA Channel N Destination Address.(N=0,1,2,3,4,5,6,7) |

#### 4.2.3.7 NDMA\_BC\_REG

| 偏移地址 : 0x100+N*0x20+0xC<br>(N=0,1,2,3,4,5,6,7) |     | 默认值: 0x0000_0000 |          |
|------------------------------------------------|-----|------------------|----------|
| 位置                                             | 访问  | 默认值              | 描述       |
| 31:18                                          | /   | /                | /        |
| 17:0                                           | R/W | 0x0              | NDMA_BC. |

|  |  |  |                                                                                                                                             |
|--|--|--|---------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <p>DMA Channel N Byte Counter.(N=0,1,2,3,4,5,6,7)</p> <p>Note: If Byte Counter=0, DMA will transfer no byte. The maximum value is 128k.</p> |
|--|--|--|---------------------------------------------------------------------------------------------------------------------------------------------|

## 5 安全模块

### 5.1 加解密算法加速器(Crypto Engine)

#### 5.1.1 概述

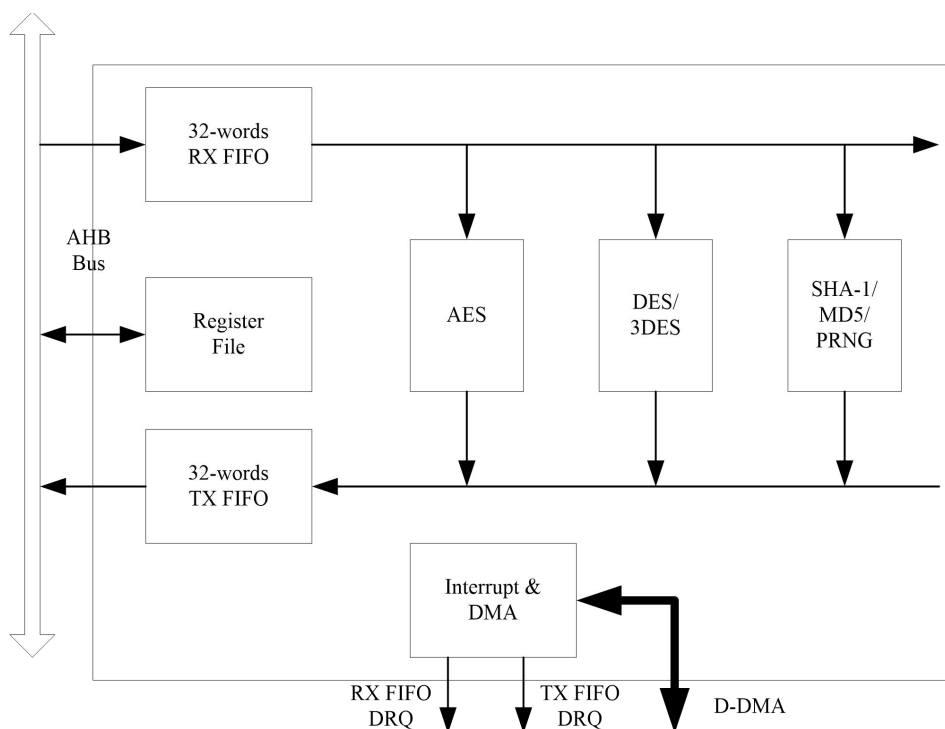
加密引擎(CE)是一种加密/解密算法加速器，它适用于各种应用场合，支持多种模式。不同的应用都支持CPU模式和DMA模式。

主要特性如下：

- 支持 AES、DES、3DES、SHA-1、MD5、PRNG、CRC32/16、SHA256；
- 支持 AES/DES/3DES 的 ECB, CBC, CTR 模式；
- 支持 AES 的 CTS 模式；
- 支持 128 位、192 位和 256 位 AES 密钥大小；
- 支持 160 位硬件 PRNG, 192 位种子；
- 用于高速应用的 32 字 RX FIFO 和 32 字 TX FIFO；
- 支持 CPU 模式和 DMA 模式；

CE模块的内部框架图如下图所示。

图 5-1 CE 模块内部框架图





### 5.1.2 Crypto Engine 寄存器列表

| 模块名                      | 基地址        |                                      |
|--------------------------|------------|--------------------------------------|
| Secure System Controller | 0x40004000 |                                      |
| 寄存器名                     | 偏移地址       | 描述                                   |
| CE_CTL_REG               | 0x00       | CE Control Register                  |
| CE_KEY0_REG              | 0x04       | Input Key 0/ PRNG Seed 0             |
| CE_KEY1_REG              | 0x08       | Input Key 1/ PRNG Seed 1             |
| ...                      | ...        | ...                                  |
| CE_KEY7_REG              | 0x20       | Input Key 7                          |
| CE_IV0_REG               | 0x24       | Initialization Vector 0              |
| CE_IV1_REG               | 0x28       | Initialization Vector 1              |
| CE_IV2_REG               | 0x2C       | Initialization Vector 2              |
| CE_IV3_REG               | 0x30       | Initialization Vector 3              |
| CE_CNT0_REG              | 0x34       | Counter 0/Initialization Vector 4    |
| CE_CNT1_REG              | 0x38       | Counter 1/Initialization Vector 5    |
| CE_CNT2_REG              | 0x3C       | Counter 2/Initialization Vector 6    |
| CE_CNT3_REG              | 0x40       | Counter 3/Initialization Vector 7    |
| CE_FCSR_REG              | 0x44       | FIFO Control/ Status Register        |
| CE_ICSR_REG              | 0x48       | Interrupt Control/ Status Register   |
| CE_MD0_REG               | 0x4C       | SHA1/MD5 Message Digest 0/PRNG Data0 |
| CE_MD1_REG               | 0x50       | SHA1/MD5 Message Digest 1/PRNG Data1 |
| CE_MD2_REG               | 0x54       | SHA1/MD5 Message Digest 2/PRNG Data2 |
| CE_MD3_REG               | 0x58       | SHA1/MD5 Message Digest 3/PRNG Data3 |
| CE_MD4_REG               | 0x5C       | SHA1/MD5 Message Digest 4/PRNG Data4 |
| CE_CTS_LEN_REG           | 0x60       | AES-CTS/CRC text length              |
| CE_CRC_POLY_REG          | 0x70       | CRC Poly Register                    |
| CRC_RESULT_REG           | 0x74       | CRC Result Register                  |
| CE_MD5_REG               | 0xa0       | SHA256 Message Digest Data 5         |

|               |       |                              |
|---------------|-------|------------------------------|
| CE_MD6_REG    | 0xa4  | SHA256 Message Digest Data 6 |
| CE_MD7_REG    | 0xa8  | SHA256 Message Digest Data 7 |
| CE_RXFIFO_REG | 0x200 | RX FIFO input port           |
| CE_TXFIFO_REG | 0x204 | TX FIFO output port          |

### 5.1.3 Crypto Engine 寄存器描述

#### 5.1.3.1 CE\_CTL\_REG

| 偏移地址 : 0x0000 |     |     | 默认值: 0x0000_0000                                                                                                                                                                          |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                        |
| 31            | R/W | 0   | CRC xor out value<br>0: CRC result as CRC out<br>1: CRC result xor 0xffffffff(CRC32) or 0xffff(CRC16) as CRC out                                                                          |
| 30            | R/W | 0   | CRC ref out value<br>0: output word bit sequence not change<br>1: output word bit sequence reverse                                                                                        |
| 29            | R/W | 0   | CRC ref in value<br>0: input data bit sequence not change<br>1: input byte bit sequence reverse                                                                                           |
| 28            | R/W | 0   | CRC init value<br>0: init value is 0<br>1: init value is 0xffffffff for CRC32 or 0xffff for CRC16                                                                                         |
| 27:24         | R/W | 0   | KEY_SELECT<br>AES/DES/3DES key select<br>0: Select input CE_KEYx (Normal Mode)<br>1: Select SID_RKEYx from SID<br>2: /<br>3-10: Select internal Key n (n from 0 to 7)<br>Others: Reserved |
| 18:16         | R   | x   | DIE_ID<br>Die Bonding ID                                                                                                                                                                  |
| 15            | R/W | 0   | PRNG_MODE/CRC_CONT<br>PRNG generator mode or CRC Package Continue<br>0: One-shot mode for PRNG or last package for CRC<br>1: Continue mode for PRNG or non-last package for CRC           |
| 14            | R/W | 0   | IV_MODE<br>IV mode for SHA-1/MD5 constants<br>0: use initial constants defined in FIPS-180                                                                                                |

|       |     |   |                                                                                                                                                                                                                                                                                   |
|-------|-----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |     |   | 1: use input IV                                                                                                                                                                                                                                                                   |
| 13:12 | R/W | 0 | CE_OP_MODE<br>CE Operation Mode<br>00: Electronic Code Book (ECB) mode<br>01: Cipher Block Chaining (CBC) mode<br>10: Counter (CTR) mode<br>11: Cipher Text Stealing (CTS) mode                                                                                                   |
| 11:10 | R/W | 0 | CTR_WIDTH<br>Counter Width for CTR Mode<br>00: 16-bits Counter<br>01: 32-bits Counter<br>10: 64-bits Counter<br>11: 128-bits Counter                                                                                                                                              |
| 9:8   | R/W | 0 | AES_KEY_SIZE<br>Key Size for AES<br>00: 128-bits<br>01: 192-bits<br>10: 256-bits<br>11: Reserved                                                                                                                                                                                  |
| 7     | R/W | 0 | CE_OP_DIR<br>CE Operation Direction<br>0: Encryption<br>1: Decryption                                                                                                                                                                                                             |
| 6:4   | R/W | 0 | CE_METHOD<br>CE Method<br>000: AES<br>001: DES<br>010: Triple DES (3DES)<br>011: SHA-1<br>100: MD5<br>101: PRNG<br>110: SHA256<br>111: CRC                                                                                                                                        |
| 3     | R/W | 0 | CRC_WIDTH<br>0: width 16, use 16 bits generate nominal<br>1: width 32, use 32 bits generate nominal                                                                                                                                                                               |
| 2     | R/W | 0 | SHA1_SHA256_MD5_CRC_END_BIT<br>SHA-1/MD5/SHA256/CRC Data End bit<br><br>Write '1' to tell SHA-1/MD5/SHA256/CRC engine that the text data ends. If RX FIFO is not empty, the engine would process the data in RX FIFO. After finishing message digest, this bit is clear to '0' by |

|   |     |   |                                                                                                                                                                             |
|---|-----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |   | hardware and message digest can be read out from digest registers.<br>Notes: It is only used for SHA-1/MD5/SHA256/CRC engine.                                               |
| 1 | R/W | 0 | PRNG_START<br>PRNG start bit<br><br>In PRNG one-shot mode, write '1' to start PRNG. After generating one group random data (5 words), this bit is clear to '0' by hardware. |
| 0 | R/W | 0 | CE_ENABLE<br>CE Enable bit<br><br>0: Disable: write '0' to tell engine ends<br><br>1: Enable: after configuration, set this bit to '1' to start engine                      |

#### 5.1.3.2 CE\_KEY[n]\_REG

| 偏移地址：0x0004+N*4<br>(N=0,1,2,3,4,5,6,7) |     | 默认值: 0x0000_0000 |                                                              |
|----------------------------------------|-----|------------------|--------------------------------------------------------------|
| 位置                                     | 访问  | 默认值              | 描述                                                           |
| 31:0                                   | R/W | 0                | CE_KEY<br>Key[n] Input Value (n= 0~7)/ PRNG Seed[n] (n= 0~5) |

#### 5.1.3.3 CE\_IV [n]\_REG

| 偏移地址：0x0024+N*4<br>(N=0,1,2,3) |     | 默认值: 0x0000_0000 |                                                                   |
|--------------------------------|-----|------------------|-------------------------------------------------------------------|
| 位置                             | 访问  | 默认值              | 描述                                                                |
| 31:0                           | R/W | 0                | CE_IV_VALUE<br>Initialization Vector (IV[n]) Input Value (n= 0~3) |

#### 5.1.3.4 CE\_CNT[n]\_REG

| 偏移地址：0x0034+N*4<br>(N=0,1,2,3) |     | 默认值: 0x0000_0000 |                                                                   |
|--------------------------------|-----|------------------|-------------------------------------------------------------------|
| 位置                             | 访问  | 默认值              | 描述                                                                |
| 31:0                           | R/W | 0                | CE_IV_VALUE<br>Initialization Vector (IV[n]) Input Value (n= 0~3) |

#### 5.1.3.5 CE\_FCSR\_REG

| 偏移地址：0x0044 |    | 默认值: 0x6000_0F0F |    |
|-------------|----|------------------|----|
| 位置          | 访问 | 默认值              | 描述 |
| 31          | /  | /                | /  |

|       |     |      |                                                                                                                                                                                                         |
|-------|-----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30    | R   | 0x1  | RXFIFO_STATUS<br>RX FIFO Empty<br>0: No room for new word in RX FIFO<br>1: More than one room for new word in RX FIFO (>= 1 word)                                                                       |
| 29:24 | R   | 0x20 | RXFIFO_EMP_CNT<br>RX FIFO Empty Space Word Counter                                                                                                                                                      |
| 23    | /   | /    | /                                                                                                                                                                                                       |
| 22    | R   | 0    | TXFIFO_STATUS<br>TX FIFO Data Available Flag<br>0: No available data in TX FIFO<br>1: More than one data in TX FIFO (>= 1 word)                                                                         |
| 21:16 | R   | 0    | TXFIFO_AVA_CNT<br>TX FIFO Available Word Counter                                                                                                                                                        |
| 15:13 | /   | /    | /                                                                                                                                                                                                       |
| 12:8  | R/W | 0xF  | RXFIFO_INT_TRIG_LEVEL<br>RX FIFO Empty Trigger Level<br>Interrupt and DMA request trigger level for RXFIFO normal condition<br>Trigger Level = RXTL + 1<br>Notes: RX FIFO is used for input the data.   |
| 7:5   | /   | /    | /                                                                                                                                                                                                       |
| 4:0   | R/W | 0xF  | TXFIFO_INT_TRIG_LEVEL<br>TX FIFO Trigger Level<br>Interrupt and DMA request trigger level for TXFIFO normal condition<br>Trigger Level = TXTL + 1<br>Notes: TX FIFO is used for output the result data. |

#### 5.1.3.6 CE\_ICSR\_REG

| 偏移地址 : 0x0048 |     |     | 默认值: 0x6000_0F0F                                                                                                                                                        |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                      |
| 31:12         | /   | /   | /                                                                                                                                                                       |
| 11            | R/W | 0   | HASH_CRC_END_INT_PEND<br>0: No pending<br>1: HASH/CRC end pending                                                                                                       |
| 10            | R/W | 0   | RXFIFO_EMP_PENDING_BIT<br>RX FIFO Empty Pending bit<br>0: No pending<br>1: RX FIFO Empty pending<br>Notes: Write '1' to clear or automatic clear if interrupt condition |

|     |     |   |                                                                                                                                                                                                                                                                       |
|-----|-----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |   | fails.                                                                                                                                                                                                                                                                |
| 9   | /   | / | /                                                                                                                                                                                                                                                                     |
| 8   | R/W | 0 | TXFIFO_AVA_PENDING_BIT<br>TX FIFO Data Available Pending bit<br>0: No TX FIFO pending<br>1: TX FIFO pending<br><br>Notes: Write '1' to clear or automatic clear if interrupt condition fails.                                                                         |
| 7:5 | /   | / | /                                                                                                                                                                                                                                                                     |
| 4   | R/W | 0 | DRQ_ENABLE<br>DRQ Enable<br>0: Disable DRQ (CPU polling mode)<br>1: Enable DRQ (DMA mode)                                                                                                                                                                             |
| 3   | R/W | 0 | HASH_CRC_END_INT_ENABLE<br>0: Disable<br>1: Enable                                                                                                                                                                                                                    |
| 2   | R/W | 0 | RXFIFO_EMP_INT_ENABLE<br>RX FIFO Empty Interrupt Enable<br>0: Disable<br>1: Enable<br><br>Notes: If it is set to '1', when the number of empty room is great or equal ( $\geq$ ) the preset threshold, the interrupt is trigger and the correspond flag is set.       |
| 1   | /   | / | /                                                                                                                                                                                                                                                                     |
| 0   | R/W | 0 | TXFIFO_AVA_INT_ENABLE<br>TX FIFO Data Available Interrupt Enable<br>0: Disable<br>1: Enable<br><br>Notes: If it is set to '1', when available data number is great or equal ( $\geq$ ) the preset threshold, the interrupt is trigger and the correspond flag is set. |

#### 5.1.3.7 CE\_MD [n]\_REG

| 偏移地址 : 0x004C+N*4<br>(N=0,1,2,3,4) |     | 默认值: 0x0000_0000 |                                                                           |
|------------------------------------|-----|------------------|---------------------------------------------------------------------------|
| 位置                                 | 访问  | 默认值              | 描述                                                                        |
| 31:0                               | R/W | 0                | AES-CTS / CRC text length in byte unit<br>The value of '0' means no data. |

### 5.1.3.8 CE\_CTS\_LEN\_REG

| 偏移地址：0x0060 |     |     | 默认值：0x0000_0000                                                           |
|-------------|-----|-----|---------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                        |
| 31:0        | R/W | 0   | AES-CTS / CRC text length in byte unit<br>The value of '0' means no data. |

### 5.1.3.9 CE\_CRC\_POLY\_REG

| 偏移地址：0x0070 |     |     | 默认值：0x0000_0000 |
|-------------|-----|-----|-----------------|
| 位置          | 访问  | 默认值 | 描述              |
| 31:0        | R/W | 0   | CRC_POLY        |

### 5.1.3.10 CE\_RESULT\_REG

| 偏移地址：0x0070 |     |     | 默认值：0x0000_0000        |
|-------------|-----|-----|------------------------|
| 位置          | 访问  | 默认值 | 描述                     |
| 31:0        | R/W | 0   | CRC computation result |

### 5.1.3.11 CE\_MD[n]\_REG

| 偏移地址：0x00A0+(N-5)*4<br>(N=5,6,7) |    |     | 默认值：0x0000_0000                                    |
|----------------------------------|----|-----|----------------------------------------------------|
| 位置                               | 访问 | 默认值 | 描述                                                 |
| 31:0                             | R  | 0   | CE_SHA256_MDn<br>SHA256 Message Digest n (n = 5~7) |

### 5.1.3.12 CE\_RX\_REG

| 偏移地址：0x0200 |    |     | 默认值：0x0000_0000                         |
|-------------|----|-----|-----------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                      |
| 31:0        | W  | 0   | CE_RX_FIFO<br>32-bits RX FIFO for Input |

### 5.1.3.13 CE\_TX\_REG

| 偏移地址：0x0204 |    |     | 默认值：0x0000_0000                          |
|-------------|----|-----|------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                       |
| 31:0        | R  | 0   | CE_TX_FIFO<br>32-bits TX FIFO for Output |

### 5.1.4 CE 模块时钟要求

| 时钟名称    | 描述              | 要求                   |
|---------|-----------------|----------------------|
| ahb_clk | AHB bus clock   | $\geq 24\text{MHz}$  |
| ss_clk  | CE serial clock | $\leq 200\text{MHz}$ |

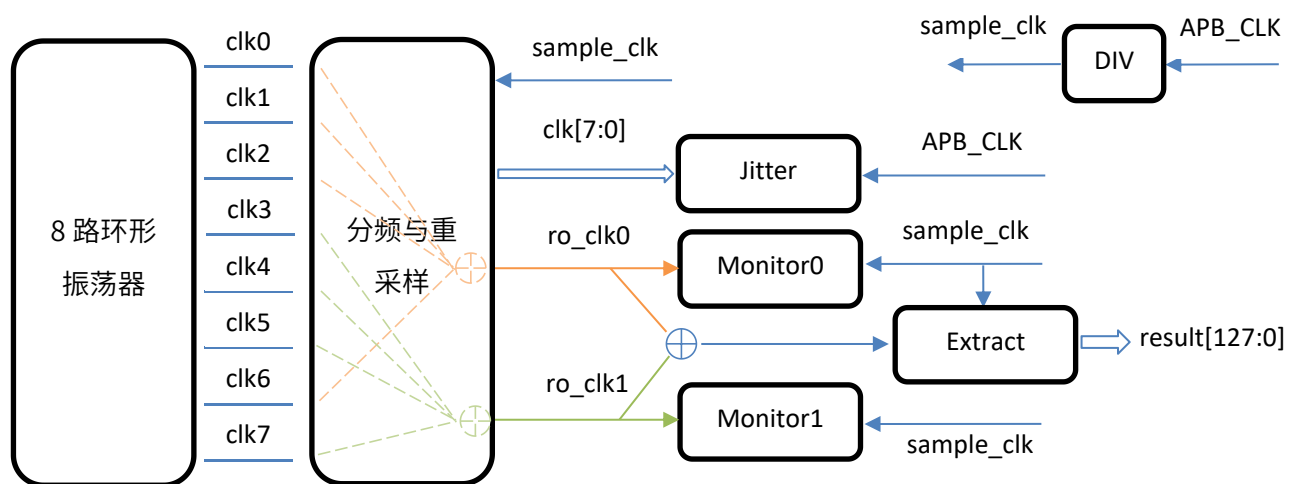
## 5.2 真随机数发生器(TRNG)

### 5.2.1 概述

TRNG(True Random Number Generator)是真随机数发生器，随机源是8路独立的环形振荡器，由模拟器件电源噪声产生频率抖动，用低频时钟重采样，然后进行弹性抽取和熵提取处理，最终输出128bit真随机数。

TRNG模块的数据流框架图如下图所示。

图 5-2 TRNG 模块数据流框架图



- Jitter 模块是一个计数器模块，可以分别统计从环形振荡器送过来的 8 路时钟一段时间内的上升沿个数。其中计数时长 timing (REG 0x08) 的单位时间长度取决于 APB 时钟（通常为 60MHz）。
- Monitor 模块是一个监测模块，功能是自动检测来自模拟的时钟是否符合要求。Monitor0 和 Monitor1 各监测 1 路时钟。特别说明，8 路时钟分成了两组，分别送给两个独立的监测模块，要想通过监测至少要 2 路起振。Monitor 有两个指标，分别是 RTC(游程长度)和 ATP(频数偏移)。
- RTC 是检测连续 0 和连续 1 的长度，超出阈值则报错，门限默认值 1032。
- ATP 是统计设定窗长(默认值 1024)内累计 0 和累计 1 的个数，超过阈值则报错，门限默认值 1009。
- Extract 模块是抽取模块，8 路时钟最终合成 1 路随机的 0/1 序列，在 Extract 模块中进行弹性抽取和熵提取处理，每次抽取输出一个 128bit 真随机数。

### 5.2.2 TRNG 寄存器列表

| 模块名  | 基地址        |
|------|------------|
| TRNG | 0x40044400 |



| 寄存器名                          | 偏移地址   | 描述                                   |
|-------------------------------|--------|--------------------------------------|
| TRNG_CTRL_CFG_REG             | 0x0000 | TRNG Control Register                |
| TRNG_JITTER_CFG_REG           | 0x0004 | TRNG Jitter Control Register         |
| TRNG_JITTER_CNT_TIMING_REG    | 0x0008 | TRNG Jitter Counter Timing Register  |
| TRNG_MONITOR_RCT_REG          | 0x000C | TRNG Monitor RTC Register            |
| TRNG_MONITOR_APT_REG          | 0x0010 | TRNG Monitor APT Register            |
| TRNG_EXTRACT_CFG_REG          | 0x0014 | TRNG Extract Control Register        |
| TRNG_RAND_BIT_URN0_REG        | 0x0018 | TRNG Rand Data0 Register             |
| TRNG_RAND_BIT_URN1_REG        | 0x001C | TRNG Rand Data1 Register             |
| TRNG_RAND_BIT_URN2_REG        | 0x0020 | TRNG Rand Data2 Register             |
| TRNG_RAND_BIT_URN3_REG        | 0x0024 | TRNG Rand Data3 Register             |
| TRNG_JITTER_CNT_RESULT0_REG   | 0x0028 | TRNG Jitter Counter Result0 Register |
| TRNG_JITTER_CNT_RESULT1_REG   | 0x002C | TRNG Jitter Counter Result1 Register |
| TRNG_JITTER_CNT_RESULT2_REG   | 0x0030 | TRNG Jitter Counter Result2 Register |
| TRNG_JITTER_CNT_RESULT3_REG   | 0x0034 | TRNG Jitter Counter Result3 Register |
| TRNG_JITTER_CNT_RESULT4_REG   | 0x0038 | TRNG Jitter Counter Result4 Register |
| TRNG_JITTER_CNT_RESULT5_REG   | 0x003C | TRNG Jitter Counter Result5 Register |
| TRNG_JITTER_CNT_RESULT6_REG   | 0x0040 | TRNG Jitter Counter Result6 Register |
| TRNG_JITTER_CNT_RESULT7_REG   | 0x0044 | TRNG Jitter Counter Result7 Register |
| TRNG_JITTER_COUNTER_READY_REG | 0x0048 | TRNG Jitter Ready Register           |
| TRNG_REG_ACCESS_CTR1          | 0x004C | TRNG Access Counter Low Register     |
| TRNG_REG_ACCESS_CTR2          | 0x0050 | TRNG Access Counter High Register    |

### 5.2.3 TRNG 寄存器描述

#### 5.2.3.1 TRNG\_CTL\_REG

| 偏移地址：0x0000 |    |     | 默认值: 0x0000_OFF0 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:15       | /  | /   | /                |
| 14:12       | R  | 0   | TRNG_READY       |

|      |     |      |                                                                                                        |
|------|-----|------|--------------------------------------------------------------------------------------------------------|
|      |     |      | 指示 TRNG 状态<br>4: 表示两组 RO 全部 Failed<br>3: 第一组 RO Failed<br>2: 第二组 RO Failed<br>1: Busy 指示<br>0: IDLE 指示 |
| 11:4 | R/W | 0xFF | TRNG_RO_CTRL<br>按 Bit 分别对应 8 个环振的开启/关闭状态控制位，高有效                                                        |
| 3    | /   | /    | /                                                                                                      |
| 2:1  | R/W | 0    | TURN_RO_DETUNE<br>微调 RO(环形振荡器)的振荡频率，值越大频率越快                                                            |
| 0    | R/W | 0    | TRNG_EN<br>0: Disable<br>1: Enable                                                                     |

### 5.2.3.2 TRNG\_JITTER\_REG

| 偏移地址 : 0x0004 |     |     | 默认值: 0x0000_0002                                                    |
|---------------|-----|-----|---------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                  |
| 31:3          | /   | /   | /                                                                   |
| 2             | R/W | 0   | TRNG_JITTER_COUNTER_START<br>TRNG Jitter Counter 相关配置完成后，启动执行，上升沿有效 |
| 1             | R/W | 1   | TRNG_COUNTER_2DIV<br>可选 2 分频器使能信号，高有效，默认高                           |
| 0             | R/W | 0   | TRNG_JITTER_WORK_EN<br>使能 Jitter 测量功能，同时控制 8 个分频器电路，高有效             |

### 5.2.3.3 TRNG\_JITTER\_CNT\_TIMING\_REG

| 偏移地址：0x0008 |    |     | 默认值: 0x0000_0000                               |
|-------------|----|-----|------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                             |
| 31:21       | /  | /   | /                                              |
| 20:0        | R  | 0x0 | TRNG_JITTER_COUNTER_TIMING<br><br>定时器，用于计时给定时长 |

### 5.2.3.4 TRNG\_MONITOR\_RCT\_REG

| 偏移地址：0x000C |     |      | 默认值: 0x0000_0408                                                 |
|-------------|-----|------|------------------------------------------------------------------|
| 位置          | 访问  | 默认值  | 描述                                                               |
| 31:12       | /   | /    | /                                                                |
| 11          | R/W | 0    | TRNG_MONITOR_EN<br><br>检测使能信号，高电平有效且默认低                          |
| 10:0        | R/W | 1032 | TRNG_RCT_C<br><br>RCT(游程长度)检测门限，默认值 1032。检测连续 0 或者 1 的长度，超出阈值则报错 |

### 5.2.3.5 TRNG\_MONITOR\_APT\_REG

| 偏移地址：0x0010 |     |      | 默认值: 0x0040_03F1                                                                                |
|-------------|-----|------|-------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值  | 描述                                                                                              |
| 31:23       | /   | /    | /                                                                                               |
| 22:11       | R/W | 1024 | TRNG_APT_W<br><br>APT(频数偏移)检测窗长，默认值 1024。统计连续 trng_APT_W 个 Bit 中 0 和 1 的个数，任一个超过 trng_APT_C 则报错 |
| 10:0        | R/W | 1009 | TRNG_APT_C<br><br>APT(频数偏移)检测门限，默认值 1009                                                        |

### 5.2.3.6 TRNG\_EXTRACT\_CFG\_REG

| 偏移地址：0x0014 |       |     | 默认值: 0x0000_0000                                                                                             |
|-------------|-------|-----|--------------------------------------------------------------------------------------------------------------|
| 位置          | 访问    | 默认值 | 描述                                                                                                           |
| 31:8        | /     | /   | /                                                                                                            |
| 25:12       | R/W   | 0   | TRNG_RESILIENT_RATIO<br>弹性抽取函数的抽取率                                                                           |
| 11:10       | /     | /   | /                                                                                                            |
| 9           | R/W1C | 0   | TRNG_EXTRACT_START<br>写 1 代表启动弹性抽取函数进行随机数生成，硬件自动清零                                                           |
| 8           | R/W   | 0   | TRNG_RESILIENT_TYPE<br>熵提取方法<br>0: CRC32<br>1: 异或抽取                                                          |
| 7:4         | R/W   | 0   | TRNG_RO_SAMPLING_RATIO1<br>获取 trng_ro_sampling_clk 所需要的对主时钟 trng_clk 的再次分频倍数，实际生效分频值=trng_ro_sampling_div1+1 |
| 3:0         | R/W   | 0   | TRNG_RO_SAMPLING_RATIO0<br>获取 trng_ro_sampling_clk 所需要的对主时钟 trng_clk 的首次分频倍数，实际生效分频值=trng_ro_sampling_div0+1 |

### 5.2.3.7 TRNG\_RAND\_BIT\_URN0\_REG

| 偏移地址：0x0018 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:0        | RC | 0   | 随机块内容，只读且读清零     |

### 5.2.3.8 TRNG\_RAND\_BIT\_URN1\_REG

| 偏移地址：0x001C |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:0        | RC | 0   | 随机块内容，只读且读清零     |

### 5.2.3.9 TRNG\_RAND\_BIT\_URN2\_REG

| 偏移地址：0x0020 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:0        | RC | 0   | 随机块内容，只读且读清零     |

### 5.2.3.10 TRNG\_RAND\_BIT\_URN3\_REG

| 偏移地址：0x0024 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:0        | RC | 0   | 随机块内容，只读且读清零     |

### 5.2.3.11 TRNG\_JITTER\_CNT\_RESULT0\_REG

| 偏移地址：0x0028 |    |     | 默认值: 0x0000_0000                                                                        |
|-------------|----|-----|-----------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                      |
| 31:0        | R  | 0   | TRNG_JITTER_COUNTER_RESULT0<br><br>在 TRNG_JITTER_COUNTER_TIMING 时间间隔内,第 0 路分频器输出的上升沿的个数 |

### 5.2.3.12 TRNG\_JITTER\_CNT\_RESULT1\_REG

| 偏移地址：0x002C |    |     | 默认值: 0x0000_0000                                                                        |
|-------------|----|-----|-----------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                      |
| 31:0        | R  | 0   | TRNG_JITTER_counter_RESULT1<br><br>在 TRNG_JITTER_COUNTER_TIMING 时间间隔内,第 1 路分频器输出的上升沿的个数 |

### 5.2.3.13 TRNG\_JITTER\_CNT\_RESULT2\_REG

| 偏移地址：0x0030 |    |     | 默认值: 0x0000_0000                                                                        |
|-------------|----|-----|-----------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                      |
| 31:0        | R  | 0   | TRNG_JITTER_counter_RESULT2<br><br>在 TRNG_JITTER_COUNTER_TIMING 时间间隔内,第 2 路分频器输出的上升沿的个数 |

### 5.2.3.14 TRNG\_JITTER\_CNT\_RESULT3\_REG

| 偏移地址：0x0034 |    |     | 默认值: 0x0000_0000                                                                        |
|-------------|----|-----|-----------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                      |
| 31:0        | R  | 0   | TRNG_JITTER_counter_RESULT3<br><br>在 TRNG_JITTER_COUNTER_TIMING 时间间隔内,第 3 路分频器输出的上升沿的个数 |

### 5.2.3.15 TRNG\_JITTER\_CNT\_RESULT4\_REG

| 偏移地址：0x0038 |    |     | 默认值: 0x0000_0000                                                                        |
|-------------|----|-----|-----------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                      |
| 31:0        | R  | 0   | TRNG_JITTER_counter_RESULT4<br><br>在 TRNG_JITTER_COUNTER_TIMING 时间间隔内,第 4 路分频器输出的上升沿的个数 |

### 5.2.3.16 TRNG\_JITTER\_CNT\_RESULT5\_REG

| 偏移地址：0x003C |    |     | 默认值: 0x0000_0000                                                                        |
|-------------|----|-----|-----------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                      |
| 31:0        | R  | 0   | TRNG_JITTER_counter_RESULT5<br><br>在 TRNG_JITTER_COUNTER_TIMING 时间间隔内,第 5 路分频器输出的上升沿的个数 |

### 5.2.3.17 TRNG\_JITTER\_CNT\_RESULT6\_REG

| 偏移地址：0x0040 |    |     | 默认值: 0x0000_0000                                                                        |
|-------------|----|-----|-----------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                      |
| 31:0        | R  | 0   | TRNG_JITTER_counter_RESULT6<br><br>在 TRNG_JITTER_COUNTER_TIMING 时间间隔内,第 6 路分频器输出的上升沿的个数 |

### 5.2.3.18 TRNG\_JITTER\_CNT\_RESULT7\_REG

| 偏移地址：0x0044 |    |     | 默认值: 0x0000_0000                                                                        |
|-------------|----|-----|-----------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                      |
| 31:0        | R  | 0   | TRNG_JITTER_counter_RESULT7<br><br>在 TRNG_JITTER_COUNTER_TIMING 时间间隔内,第 7 路分频器输出的上升沿的个数 |

### 5.2.3.19 TRNG\_JITTER\_COUNTER\_READY\_REG

| 偏移地址：0x0048 |    |     | 默认值: 0x0000_0FF0                                                                           |
|-------------|----|-----|--------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                         |
| 31:1        | /  | /   | /                                                                                          |
| 0           | R  | 0   | TRNG_JITTER_READY<br><br>指示 TRNG JITTER 状态<br><br>1: READY 指示, 可以取数<br><br>0: Busy/idle 状态 |

### 5.2.3.20 TRNG\_REG\_ACCESS\_CTR1

| 偏移地址：0x004C |    |     | 默认值: 0x0000_0000                                                    |
|-------------|----|-----|---------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                  |
| 31:0        | R  | x   | 统计 trng 寄存器被 APB 访问的次数（低 32 位），访问此寄存器不会增加次数，复位后无确定初值，休眠唤醒后此寄存器值会被保留 |

### 5.2.3.21 TRNG\_REG\_ACCESS\_CTR2

| 偏移地址：0x0050 |    |     | 默认值: 0x0000_0000                                                   |
|-------------|----|-----|--------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                 |
| 31:8        | /  | /   | /                                                                  |
| 7:0         | R  | x   | 统计 trng 寄存器被 APB 访问的次数（高 8 位），访问此寄存器不会增加次数，复位后无确定初值，休眠唤醒后此寄存器值会被保留 |

## 5.3 安全标识(Security ID)

### 5.3.1 概述

XR806包含1024位的eFuse空间，其中128位用于芯片ID，其他位为系统参数预留。Security ID的工作时钟为26MHz，来自APB时钟。

### 5.3.2 SID 寄存器列表

| 模块名               | 基地址             |                                   |
|-------------------|-----------------|-----------------------------------|
| Security ID       | 0x40043C00      |                                   |
| 寄存器名              | 偏移地址            | 描述                                |
| SID_PRCTL_REG     | 0x0040          | SID Program/Read Control Register |
| SID_PKEY_REG      | 0x0050          | SID Program Key Value Register    |
| SID_RKEY_REG      | 0x0060          | SID Read Key Value Register       |
| BURNED_TIMING_REG | 0x0090          | SID Burned Timing Control         |
| SDI_DBG_REG       | 0x0094          | SID Debug Register                |
| SID_VALUE_REG     | 0x0100 – 0x01FF | SID Chip ID Register 0            |

### 5.3.3 SID 寄存器描述

#### 5.3.3.1 SID\_PR\_CTRL\_REG

| 偏移地址：0x0040 |     |     | 默认值: 0x0000_0000                                                                                                                                                     |
|-------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                   |
| 31:29       | /   | /   | /                                                                                                                                                                    |
| 28          | R/W | 0   | EFUSE_CLOCK_GATE_SW<br><br>OSC40M (eFuse_clock) clock gate enable.<br>When read or program the effuse, this bit should be set to 1                                   |
| 27:24       | /   | /   | /                                                                                                                                                                    |
| 23:16       | R/W | 0   | PG_INDEX<br>Program Index<br>The index value of 8-bits electrical fuse hardware macrocell address offset, and the lowest two bits must be zero.                      |
| 15:8        | R/W | 0   | OP_LOCK<br><br>Operation Lock<br>The Read Start Bit (bit1) and Program Start bit (bit0) and software one-short read start (bit4) only can be written when these bits |



|     |     |   |                                                                                                                                                              |
|-----|-----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |   | (bit[15:8]) set to 0xAC                                                                                                                                      |
| 7:5 | /   | / | /                                                                                                                                                            |
| 4   | R/W | 0 | SW_ALL_READ_START<br>Software one-shot whole Efuse Read Start<br><br>Write '1' to start software whole Efuse Read and automatically clear to '0' after read. |
| 3   | /   | / | /                                                                                                                                                            |
| 2   | R   | x | HW_READ_STATUS<br>Hardware read status<br>0: No hardware operation<br>1: Hardware Reading                                                                    |
| 1   | R/W | 0 | EFUSE_READ_START<br><br>Software Read Start<br><br>Write '1' to start software read and automatically clear to '0' after read.                               |
| 0   | R/W | 0 | EFUSE_PRG_START<br><br>Software Program Start<br><br>Write '1' to start software program and automatically clear to '0' after program.                       |

### 5.3.3.2 SID\_PKEY\_REG

| 偏移地址：0x0050 |     | 默认值: 0x0000_0000 |                                          |
|-------------|-----|------------------|------------------------------------------|
| 位置          | 访问  | 默认值              | 描述                                       |
| 31:0        | R/W | 0                | SID_PGKEY_VALUE<br><br>Program key value |

### 5.3.3.3 SID\_RKEY\_REG

| 偏移地址：0x0060 |    | 默认值: 0x0000_0000 |                                          |
|-------------|----|------------------|------------------------------------------|
| 位置          | 访问 | 默认值              | 描述                                       |
| 31:0        | R  | 0                | SID_RDKEY_VALUE<br><br>Program key value |

### 5.3.3.4 SID\_BURN\_TIMING\_REG

| 偏移地址：0x0090 |     |       | 默认值: 0x630B_905C                                                                                       |
|-------------|-----|-------|--------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值   | 描述                                                                                                     |
| 31:28       | R/W | 0x6   | Control Taen of the eFuse Reading cycle<br>value * Tosc<br><br>Taen > Trd + 35ns                       |
| 28:24       | R/W | 0x3   | Control Trd of the eFuse Reading cycle<br>value * Tosc<br><br>Trd > 40ns                               |
| 23:12       | R/W | 0X0B9 | Control Taen of the eFuse program cycle<br>value * Tosc<br><br>Taen > Tpgm + 2300ns                    |
| 11:0        | R/W | 0X05C | Control Tpgm of the eFuse Program cycle<br>value * Tosc<br><br>Tpgm >= 2300ns (2600ns > Tpgm > 2000ns) |

### 5.3.3.5 SID\_DBG\_REG

| 偏移地址：0x0094 |     |     | 默认值: 0x0000_0000                                                                                                                             |
|-------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                           |
| 31:5        | /   | /   | /                                                                                                                                            |
| 4           | R/W | 0   | LDO-EFUSE software switch<br>0: soft switch off<br>1: soft switch on<br>Note: Switch on accumulative total time should less than 1 second!!! |
| 3:0         | /   | /   | /                                                                                                                                            |

### 5.3.3.6 SID\_VALUE\_REG

| 偏移地址：0x0100~0x01FF |    |     | 默认值: 0x0000_0000  |
|--------------------|----|-----|-------------------|
| 位置                 | 访问 | 默认值 | 描述                |
| 31:0               | R  | x   | SID [32n+31: 32n] |

## 6 外设及通用接口模块

### 6.1 外设安全控制(TZASC)

#### 6.1.1 概述

TZASC 可以将总线外设、存储区域的访问权限灵活配置为安全访问或非安全访问。

#### 6.1.2 TZASC 寄存器列表

| 模块名                                  | 基地址               |                                                                                |
|--------------------------------------|-------------------|--------------------------------------------------------------------------------|
| TZASC                                | 0x4008A000        |                                                                                |
| 寄存器名                                 | 偏移地址              | 描述                                                                             |
| AHB_DEV_SEC_CTRL_REG                 | 0x0000            | The device on AHB bus secure control register                                  |
| APB_DEV_SEC_CTRL_REG                 | 0x0004            | The device on APB bus secure control register                                  |
| TZASC_INT_EN_REG                     | 0x0008            | TZASC secure case interrupt enable                                             |
| TZASC_INT_STA_REG                    | 0x000c            | TZASC secure case interrupt pending                                            |
| SEC_SPACE_ADRR_START_REG             | $0x0010+N*0x10$   | The secure space start address<br>(N=0,1,2,3)                                  |
| SEC_SPACE_ADRR_END_REG               | $0x0010+N*0x10+4$ | The secure space end address<br>(N=0,1,2,3)                                    |
| FLASHCTRL_SEC_CTRL_REG               | 0x0070            | Flash controller secure space control register                                 |
| FLASH_PSRAM_SEC_SPACE_ADRR_START_REG | $0x0080+N*0x10$   | The secure space start address only for flash/psram storage address<br>(N=0,1) |
| FLASH_PSRAM_SEC_SPACE_ADRR_END_REG   | $0x0080+N*0x10+4$ | The secure space end address only for flash/psram storage address<br>(N=0,1)   |
| TZASC_CLK_MANAGE_SEC_CASE_REG        | 0x00C0            | The registers for clock manage secure control register                         |

### 6.1.3 TZASC 寄存器描述

#### 6.1.3.1 AHB\_DEV\_SEC\_CTRL\_REG

| 偏移地址：0x0000 |    |     | 默认值：0x0000_0000                                                                                                          |
|-------------|----|-----|--------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                       |
| 31          | RW | 0x0 | JTAG_FORCE_SEC_ENABLE<br>Force JTAG connect enable in CPU secure state. It has high priority.<br>1: enable<br>0: disable |
| 30:29       | /  | /   | /                                                                                                                        |
| 28          | RW | 0x0 | WLAN_AHB_SEC<br>1: the device is secure<br>0: the device is no-secure                                                    |
| 27          | RW | 0x0 | BLE_SEC<br>1: the device is secure<br>0: the device is no-secure                                                         |
| 26:22       | /  | /   | /                                                                                                                        |
| 21          | RW | 0x0 | WLAN_CTRL_SEC<br>1: the device is secure<br>0: the device is no-secure                                                   |
| 20          | RW | 0x0 | SS_SEC<br>1: the device is secure<br>0: the device is no-secure                                                          |
| 19          | RW | 0x0 | SPIO_SEC<br>1: the device is secure<br>0: the device is no-secure                                                        |
| 18          | RW | 0x0 | SDC1_SEC<br>1: the device is secure<br>0: the device is no-secure                                                        |
| 17          | /  | /   | /                                                                                                                        |
| 16          | RW | 0x0 | SYSCTRL_SEC<br>1: the device is secure<br>0: the device is no-secure                                                     |
| 15:14       | /  | /   | /                                                                                                                        |
| 13          | RW | 0x0 | DCHACHE_CTRL_SEC<br>1: the device is secure<br>0: the device is no-secure                                                |

|    |    |     |                                                                         |
|----|----|-----|-------------------------------------------------------------------------|
| 12 | RW | 0x0 | FLASH_CTRL_SEC<br>1: the device is secure<br>0: the device is no-secure |
| 11 | RW | 0x0 | SRAM9_SEC<br>1: the device is secure<br>0: the device is no-secure      |
| 10 | RW | 0x0 | SRAM08_SEC<br>1: the device is secure<br>0: the device is no-secure     |
| 9  | RW | 0x0 | SRAM07_SEC<br>1: the device is secure<br>0: the device is no-secure     |
| 8  | RW | 0x0 | SRAM06_SEC<br>1: the device is secure<br>0: the device is no-secure     |
| 7  | RW | 0x0 | SRAM05_SEC<br>1: the device is secure<br>0: the device is no-secure     |
| 6  | RW | 0x0 | SRAM04_SEC<br>1: the device is secure<br>0: the device is no-secure     |
| 5  | RW | 0x0 | reserved                                                                |
| 4  | RW | 0x0 | SRAM03_SEC<br>1: the device is secure<br>0: the device is no-secure     |
| 3  | RW | 0x0 | SRAM02_SEC<br>1: the device is secure<br>0: the device is no-secure     |
| 2  | RW | 0x0 | SRAM01_SEC<br>1: the device is secure<br>0: the device is no-secure     |
| 1  | RW | 0x0 | SRAM00_SEC<br>1: the device is secure<br>0: the device is no-secure     |
| 0  | RW | 0x0 | ROM_SEC<br>1: the device is secure<br>0: the device is no-secure        |

### 6.1.3.2 APB\_DEV\_SEC\_CTRL\_REG

| 偏移地址：0x0004 |    |     | 默认值：0x0000_0000                                                            |
|-------------|----|-----|----------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                         |
| 31          | /  | /   | /                                                                          |
| 30          | RW | 0x0 | LPUART1_SEC<br>1: the device is secure<br>0: the device is no-secure       |
| 29          |    |     | LPUART0_SEC<br>1: the device is secure<br>0: the device is no-secure       |
| 28:23       | /  | /   | /                                                                          |
| 22          | RW | 0x0 | CAPSEN_SEC<br>1: the device is secure<br>0: the device is no-secure        |
| 21          | RW | 0x0 | FLASH_ENCRYPT_SEC<br>1: the device is secure<br>0: the device is no-secure |
| 20          | RW | 0x0 | ISO7816_SEC<br>1: the device is secure<br>0: the device is no-secure       |
| 19          | RW | 0x0 | KEYSCAN_SEC<br>1: the device is secure<br>0: the device is no-secure       |
| 18          | RW | 0x0 | TRNG_SEC<br>1: the device is secure<br>0: the device is no-secure          |
| 17          | RW | 0x0 | CODEC_SEC<br>1: the device is secure<br>0: the device is no-secure         |
| 16          | RW | 0x0 | UART2_SEC<br>1: the device is secure<br>0: the device is no-secure         |
| 15          | RW | 0x0 | SID_SEC<br>1: the device is secure<br>0: the device is no-secure           |
| 14          | RW | 0x0 | RTC_SEC<br>1: the device is secure                                         |

|    |    |     |                                                                     |
|----|----|-----|---------------------------------------------------------------------|
|    |    |     | 0: the device is no-secure                                          |
| 13 | RW | 0x0 | DMIC_SEC<br>1: the device is secure<br>0: the device is no-secure   |
| 12 | RW | 0x0 | GPIO_SEC<br>1: the device is secure<br>0: the device is no-secure   |
| 11 | RW | 0x0 | IRRX_SEC<br>1: the device is secure<br>0: the device is no-secure   |
| 10 | RW | 0x0 | IRTX_SEC<br>1: the device is secure<br>0: the device is no-secure   |
| 9  | RW | 0x0 | GPADC_SEC<br>1: the device is secure<br>0: the device is no-secure  |
| 8  | RW | 0x0 | DAUDIO_SEC<br>1: the device is secure<br>0: the device is no-secure |
| 7  | RW | 0x0 | PWM_SEC<br>1: the device is secure<br>0: the device is no-secure    |
| 6  | RW | 0x0 | TWI1_SEC<br>1: the device is secure<br>0: the device is no-secure   |
| 5  | RW | 0x0 | TWIO_SEC<br>1: the device is secure<br>0: the device is no-secure   |
| 4  | RW | 0x0 | UART1_SEC<br>1: the device is secure<br>0: the device is no-secure  |
| 3  | RW | 0x0 | UART0_SEC<br>1: the device is secure<br>0: the device is no-secure  |
| 2  | RW | 0x0 | TIMER_SEC<br>1: the device is secure<br>0: the device is no-secure  |
| 1  | RW | 0x0 | CCM_SEC<br>1: the device is secure                                  |

|   |    |     |                                                                    |
|---|----|-----|--------------------------------------------------------------------|
|   |    |     | 0: the device is no-secure                                         |
| 0 | RW | 0x0 | GPRCM_SEC<br>1: the device is secure<br>0: the device is no-secure |

#### 6.1.3.3 TZASC\_INT\_EN\_REG

| 偏移地址 : 0x0008 |    |     | 默认值: 0x0000_0000                                                                       |
|---------------|----|-----|----------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                     |
| 31:1          | /  | /   | /                                                                                      |
| 0             | RW | 0x0 | TZASC_INT_EN<br>TZASC Secure case occurred Interrupt Enable<br>0: Disable<br>1: Enable |

#### 6.1.3.4 TZASC\_INT\_STA\_REG

| 偏移地址 : 0x000C |    |     | 默认值: 0x0000_0000                                                                                                                                                           |
|---------------|----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                         |
| 31:1          | /  | /   | /                                                                                                                                                                          |
| 0             | RW | 0x0 | TZASC_INT_STA<br>This bit is set if TZASC secure case has occurred. Writing 1 to this bit clears it.<br>0: TZASC secure case not occurred<br>1: TZASC secure case occurred |

#### 6.1.3.5 SEC\_SPACE\_ADRR\_START\_REG

| 偏移地址 : 0x10+N*0x10+0x0<br>(N=0,1,2,3) |    |     | 默认值: 0x0000_0000                                                           |
|---------------------------------------|----|-----|----------------------------------------------------------------------------|
| 位置                                    | 访问 | 默认值 | 描述                                                                         |
| 31:2                                  | RW | 0x0 | SEC_SPACE_ADRR_START<br>bit[31:2], The start word address of secure space. |
| 1:0                                   | /  | /   | /                                                                          |

#### 6.1.3.6 SEC\_SPACE\_ADRR\_END\_REG

| 偏移地址 : 0x10+N*0x10+0x4<br>(N=0,1,2,3) |    |     | 默认值: 0x0000_0000   |
|---------------------------------------|----|-----|--------------------|
| 位置                                    | 访问 | 默认值 | 描述                 |
| 31:2                                  | /  | /   | SEC_SPACE_ADRR_END |



|   |    |     |                                                             |
|---|----|-----|-------------------------------------------------------------|
|   |    |     | bit[31:2], The end word address of secure space.            |
| 1 | /  | /   | /                                                           |
| 0 | RW | 0x0 | SEC_SPACE_VLD<br>0: invalid<br>1: the secure space is valid |

#### 6.1.3.7 FLASHCTRL\_SEC\_CTRL\_REG

| 偏移地址 : 0x0070 |    |     | 默认值: 0x0000_0000                                                                                                   |
|---------------|----|-----|--------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                 |
| 31            | RW | 0x0 | FLASHCTRL_SEC_SPACE_ENABLE<br>Enable flash controller secure space valid<br>0: disable<br>1: enable                |
| 30:28         | /  | /   | /                                                                                                                  |
| 27:16         | RW | 0x0 | FLASHCTRL_SEC_SPACE_END_OFFSET_ADDR<br>Bit[11:0], The end offset address of flash controller with secure space     |
| 15:12         | /  | /   | /                                                                                                                  |
| 11:0          | RW | 0x0 | FLASHCTRL_SEC_SPACE_START_OFFSET_ADDR<br>Bit[11:0], The start offset address of flash controller with secure space |

#### 6.1.3.8 FLASH\_PSRAM\_SEC\_SPACE\_ADRR\_START\_REG

| 偏移地址 : 0x80+N*0x10+0x0<br>(N=0,1) |    |     | 默认值: 0x0000_0000                                                                      |
|-----------------------------------|----|-----|---------------------------------------------------------------------------------------|
| 位置                                | 访问 | 默认值 | 描述                                                                                    |
| 31:2                              | RW | 0x0 | FLASH_PSRAMSEC_SPACE_ADRR_START<br>bit[31:2], The start word address of secure space. |
| 1:0                               | /  | /   | /                                                                                     |

#### 6.1.3.9 FLASH\_PSRAM\_SEC\_SPACE\_ADRR\_END\_REG

| 偏移地址 : 0x80+N*0x10+0x4<br>(N=0,1) |    |     | 默认值: 0x0000_0000                                                                   |
|-----------------------------------|----|-----|------------------------------------------------------------------------------------|
| 位置                                | 访问 | 默认值 | 描述                                                                                 |
| 31:2                              | /  | /   | FLASH_PSRAM_SEC_SPACE_ADRR_END<br>bit[31:2], The end word address of secure space. |
| 1                                 | RW | 0x0 | FLASH_PSRAM_SEL<br>0: the secure space for FLASH<br>1: the secure space for PSRAM  |

|   |    |     |                                                                         |
|---|----|-----|-------------------------------------------------------------------------|
| 0 | RW | 0x0 | FLASH_PSRAM_SEC_SPACE_VLD<br>0: invalid<br>1: the secure space is valid |
|---|----|-----|-------------------------------------------------------------------------|

#### 6.1.3.10 TZASC\_CLK\_MANAGE\_SEC\_CASE\_REG

| 偏移地址 : 0x00C0 |    |     | 默认值: 0x0000_0000                                                                                                                              |
|---------------|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                            |
| 31:5          | /  | /   | /                                                                                                                                             |
| 6             | RW | 0x0 | SYS_LFCLK_CTRL_SEC<br>The Register SYS_LFCLK_CTRL (0x40040008) secure case<br>1: the register is secure<br>0: the register is no-secure       |
| 5             | RW | 0x0 | DCXO_CTRL0_SEC<br>The Register DCXO_CTRL0 (0x40040058) secure case<br>1: the register is secure<br>0: the register is no-secure               |
| 4             | RW | 0x0 | PCLK_SPC_CLK_CTRL_SEC<br>The Register PCLK_SPC_CLK_CTRL (0x4004045C) secure case<br>1: the register is secure<br>0: the register is no-secure |
| 3             | RW | 0x0 | CPU_BUS_CLKCFG_SEC<br>The Register CPU_BUS_CLKCFG (0x40040400) secure case<br>1: the register is secure<br>0: the register is no-secure       |
| 2             | RW | 0x0 | DEV_CLK_CTRL_SEC<br>The Register DEV_CLK_CTRL (0x40040034) secure case<br>1: the register is secure<br>0: the register is no-secure           |
| 1             | RW | 0x0 | SYS_CLK1_CTRL_SEC<br>The Register SYS_CLK1_CTRL (0x40040024) secure case<br>1: the register is secure<br>0: the register is no-secure         |
| 0             | RW | 0x0 | SYS_PLL_CTRL_SEC<br>The Register SYS_PLL_CTRL (0x40040020) secure case<br>1: the register is secure<br>0: the register is no-secure           |

## 6.2 UART 接口

### 6.2.1 概述

UART用于与外设、调制解调器(数据载波设备，DCE)或数据集进行串行通信。数据从主设备(CPU)通过APB总线写到UART，它被转换成串行形式并传输到目标设备。串行数据也被UART接收并存储给主处理器(CPU)来读取。

UART包含用于控制字符长度、波特率、奇偶校验产生/检查和中断产生的寄存器。尽管来自UART的中断输出信号只有一个，但是有多个优先级的中断类型可以负责它的断言。每种中断类型都可以通过控制寄存器分别启用/禁用。

UART具有16450和16550操作模式，与一系列标准软件驱动程序兼容。在16550模式下，发送和接收操作都由FIFO缓冲。在16450模式下，这些FIFO被禁用。

UART支持从5到8位的字长度，一个可选的奇偶校验位和1、1½或2停止位，并完全由AMBA APB CPU接口控制。同时，包含一个16位可配置波特率发生器，一个8位临时寄存器，以及单独的发送和接收FIFO。并提供了8条调制解调器控制线和一个诊断环回模式。

中断可以产生自TX缓冲区/FIFO、RX缓冲区/FIFO、调制解调器状态和线路状态条件等。

在需要红外SIR串行数据格式的系统中集成，UART可以配置为具有软件可编程的IrDA SIR模式。不选择该模式时，系统只提供UART (RS232标准)串口数据格式。

UART控制器包含以下特性：

- 兼容行业标准 16550 UARTs
- 64-Bytes 发送和接收数据 FIFO
- 支持 DMA 控制器接口
- 支持软硬件流制
- 支持 IrDA 1.0 SIR
- 支持 RS-485/9-bit 模式

下图描述了UART模块的典型时序。

图 6-1 UART 串行数据时序图

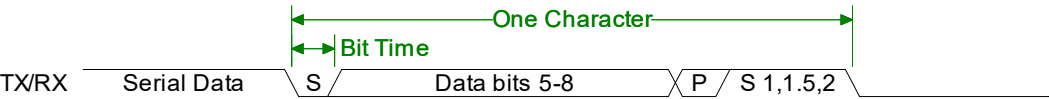
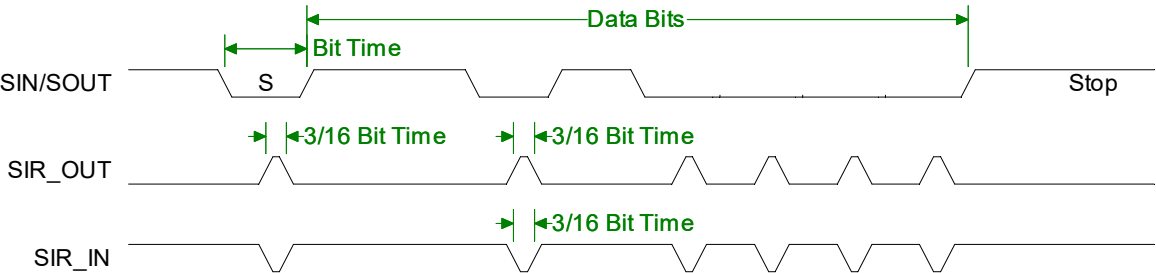


图 6-2 UART IrDA 时序图



## 6.2.2 UART 寄存器列表

| 模块名                  | 基地址        |                                          |
|----------------------|------------|------------------------------------------|
| UART Controller 0    | 0x40040C00 |                                          |
| UART Controller 1    | 0x40041000 |                                          |
| UART Controller 2    | 0x40041400 |                                          |
| 寄存器名                 | 偏移地址       | 描述                                       |
| UART_RBR_REG         | 0x0000     | UART Receive Buffer Register             |
| UART_THR_REG         | 0x0000     | UART Transmit Holding Register           |
| UART_DLL_REG         | 0x0000     | UART Divisor Latch Low Register          |
| UART_DLH_REG         | 0x0004     | UART Divisor Latch High Register         |
| UART_IER_REG         | 0x0004     | UART Interrupt Enable Register           |
| UART_IIR_REG         | 0x0008     | UART Interrupt Identity Register         |
| UART_FCR_REG         | 0x0008     | UART FIFO Control Register               |
| UART_LCR_REG         | 0x000C     | UART Line Control Register               |
| UART_MCR_REG         | 0x0010     | UART Modem Control Register              |
| UART_LSR_REG         | 0x0014     | UART Line Status Register                |
| UART_MSR_REG         | 0x0018     | UART Modem Status Register               |
| UART_SCH_REG         | 0x001C     | UART Scratch Register                    |
| UART_USR_REG         | 0x007C     | UART Status Register                     |
| UART_TFL_REG         | 0x0080     | UART Transmit FIFO Level                 |
| UART_RFL_REG         | 0x0084     | UART_RFL                                 |
| UART_HALT_REG        | 0x00A4     | UART Halt TX Register                    |
| UART_DBG_DLL_REG     | 0x00B0     | UART Debug DLL Register                  |
| UART_DBG_DLH_REG     | 0x00B4     | UART Debug DLH Register                  |
| UART_485_CTL_REG     | 0x00C0     | UART RS485 Control and Status Register   |
| RS485_ADDR_MATCH_REG | 0x00C4     | UART RS485 Address Match Register        |
| BUS_IDLE_CHK_REG     | 0x00C8     | UART RS485 Bus Idle Check Register       |
| TX_DLY_REG           | 0x00CC     | UART TX Delay Register                   |
| UART_BDCR_REG        | 0x00D4     | UART Baudrate Detection Control Register |

|                |        |                                               |
|----------------|--------|-----------------------------------------------|
| UART_BDCLR_REG | 0x00D8 | UART Baudrate Detection Counter Low Register  |
| UART_BDCHR_REG | 0x00DC | UART Baudrate Detection Counter High Register |

## 6.2.3 UART 寄存器描述

### 6.2.3.1 UART\_RBR\_REG

| 偏移地址 : 0x0000 |    | 默认值: 0x0000_0000 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------|----|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值              | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 31:8          | /  | /                | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7:0           | R  | 0                | <p>RBR</p> <p>Receiver Buffer Register</p> <p>Data byte received on the serial input port (sin) in UART mode, or the serial infrared input (sir_in) in infrared mode. The data in this register is valid only if the Data Ready (DR) bit in the Line Status Register (LCR) is set.</p> <p>If in FIFO mode and FIFOs are enabled (FCR [0] set to one), this register accesses the head of the receive FIFO. If the receive FIFO is full and this register is not read before the next data character arrives, then the data already in the FIFO is preserved, but any incoming data are lost and an overrun error occurs.</p> |

### 6.2.3.2 UART\_THR\_REG

| 偏移地址 : 0x0000 |    | 默认值: 0x0000_0000 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|----|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值              | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 31:8          | /  | /                | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7:0           | W  | 0                | <p>THR</p> <p>Transmit Holding Register</p> <p>Data to be transmitted on the serial output port (sout) in UART mode or the serial infrared output (sir_out_n) in infrared mode. Data should only be written to the THR when the THR Empty (THRE) bit (LSR[5]) is set.</p> <p>If in FIFO mode and FIFOs are enabled (FCR[0] = 1) and THRE is set, 16 number of characters of data may be written to the THR before the FIFO is full. Any attempt to write data when the FIFO is full results in the write data being lost.</p> |

### 6.2.3.3 UART\_DLL\_REG

| 偏移地址 : 0x0000 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 31:8          | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7:0           | R/W | 0   | <p>DLL</p> <p>Divisor Latch Low</p> <p>Lower 8 bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART. This register may only be accessed when the DLAB bit (LCR[7]) is set and the UART is not busy (USR[0] is zero).</p> <p>The output baud rate is equal to the serial clock (sclk) frequency divided by sixteen times the value of the baud rate divisor, as follows: baud rate = (serial clock freq) / (16 * divisor).</p> <p>Note that with the Divisor Latch Registers (DLL and DLH) set to zero, the baud clock is disabled and no serial communications occur. Also, once the DLL is set, at least 8 clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.</p> |

### 6.2.3.4 UART\_DLH\_REG

| 偏移地址 : 0x0004 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 31:8          | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7:0           | R/W | 0   | <p>DLH</p> <p>Divisor Latch High</p> <p>Upper 8 bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART. This register may only be accessed when the DLAB bit (LCR[7]) is set and the UART is not busy (USR[0] is zero).</p> <p>The output baud rate is equal to the serial clock (sclk) frequency divided by sixteen times the value of the baud rate divisor, as follows: baud rate = (serial clock freq) / (16 * divisor).</p> <p>Note that with the Divisor Latch Registers (DLL and DLH) set to zero, the baud clock is disabled and no serial communications occur. Also, once the DLH is set, at least 8 clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.</p> |

### 6.2.3.5 UART\_IER\_REG

| 偏移地址 : 0x0004 |     |     | 默认值: 0x0000_0000 |
|---------------|-----|-----|------------------|
| 位置            | 访问  | 默认值 | 描述               |
| 31:8          | /   | /   | /                |
| 7             | R/W | 0   | PTIME            |

|     |     |   |                                                                                                                                                                                                                                                                                                                      |
|-----|-----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |   | <p>Programmable THRE Interrupt Mode Enable</p> <p>This is used to enable/disable the generation of THRE Interrupt.</p> <p>0: Disable</p> <p>1: Enable</p>                                                                                                                                                            |
| 6:5 | /   | / | /                                                                                                                                                                                                                                                                                                                    |
| 4   | R/W | 0 | <p>RS485_INT_EN</p> <p>RS485 Interrupt Enable</p> <p>0:Disable</p> <p>1:Enable</p>                                                                                                                                                                                                                                   |
| 3   | R/W | 0 | <p>EDSSI</p> <p>Enable Modem Status Interrupt</p> <p>This is used to enable/disable the generation of Modem Status Interrupt. This is the fourth highest priority interrupt.</p> <p>0: Disable</p> <p>1: Enable</p>                                                                                                  |
| 2   | R/W | 0 | <p>ELSI</p> <p>Enable Receiver Line Status Interrupt</p> <p>This is used to enable/disable the generation of Receiver Line Status Interrupt. This is the highest priority interrupt.</p> <p>0: Disable</p> <p>1: Enable</p>                                                                                          |
| 1   | R/W | 0 | <p>ETBEI</p> <p>Enable Transmit Holding Register Empty Interrupt</p> <p>This is used to enable/disable the generation of Transmitter Holding Register Empty Interrupt. This is the third highest priority interrupt.</p> <p>0: Disable</p> <p>1: Enable</p>                                                          |
| 0   | R/W | 0 | <p>ERBFI</p> <p>Enable Received Data Available Interrupt</p> <p>This is used to enable/disable the generation of Received Data Available Interrupt and the Character Timeout Interrupt (if in FIFO mode and FIFOs enabled). These are the second highest priority interrupts.</p> <p>0: Disable</p> <p>1: Enable</p> |

### 6.2.3.6 UART\_IIR\_REG

| 偏移地址 : 0x0008 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 31:8          | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7:6           | R  | 0   | FEFLAG<br><br>FIFOs Enable Flag<br>This is used to indicate whether the FIFOs are enabled or disabled.<br>00: Disable<br>11: Enable                                                                                                                                                                                                                                                                                                                                                     |
| 5:4           | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3:0           | R  | 0x1 | IID<br><br>Interrupt ID<br>This indicates the highest priority pending interrupt which can be one of the following types:<br>0000: modem status<br>0001: no interrupt pending<br>0010: THR empty<br>0011: RS485 Interrupt<br>0100: received data available<br>0110: receiver line status<br>0111: busy detect<br>1100: character timeout<br><br>Bit 3 indicates an interrupt can only occur when the FIFOs are enabled and used to distinguish a Character Timeout condition interrupt. |

表 6-1 UART 中断 ID 优先级

| 中断 ID | 优先级     | 中断类型                    | 中断源                                                                                                                         | 中断复位                                                                                                                                           |
|-------|---------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 0001  | -       | None                    | None                                                                                                                        | -                                                                                                                                              |
| 0110  | Highest | Receiver line status    | Overrun/parity/framing errors or break interrupt.                                                                           | Reading the line status register                                                                                                               |
| 0011  | Second  | RS485 Interrupt         | In RS485 mode, receives address data and match setting address.                                                             | Writes 1 to addr flag to reset                                                                                                                 |
| 0100  | Third   | Received data available | Receiver data available (non-FIFO mode or FIFOs disabled) or RCVR FIFO trigger level reached (FIFO mode and FIFOs enabled). | Reading the receiver buffer register (non-FIFO mode or FIFOs disabled) or the FIFO drops below the trigger level (FIFO mode and FIFOs enabled) |
| 1100  | Fourth  | Character timeout       | No characters in or out of the RCVR FIFO during the last 4                                                                  | Reading the receiver buffer                                                                                                                    |



|      |         |                                 |                                                                                                                                                                                            |                                                                                                                                                                                                |
|------|---------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |         | indication                      | character times and there is at least 1character in it during this time.                                                                                                                   | register                                                                                                                                                                                       |
| 0010 | Fifth   | Transmit holding register empty | Transmitter holding register empty (Program THRE Mode disabled) or XMIT FIFO at or below threshold (Program THRE Mode enabled)                                                             | Reading the IIR register (if source of interrupt); or, writing into THR (FIFOs or THRE Mode not selected or disabled) or XMIT FIFO above threshold (FIFOs and THRE Mode selected and enabled). |
| 0000 | Sixth   | Modem status                    | Clear to send or data set ready or ring indicator or data carrier detect. Note that if auto flow control mode is enabled, a change in CTS (that is, DCTS set) does not cause an interrupt. | Reading the Modem status Register                                                                                                                                                              |
| 0111 | Seventh | Busy detect indication          | UART_16550_COMPATIBLE = NO and master has tried to write to the Line Control Register while the UART is busy (USR[0] is set to one).                                                       | Reading the UART status register                                                                                                                                                               |

### 6.2.3.7 UART\_FCR\_REG

| 偏移地址 : 0x0008 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 31:8          | /  | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7:6           | W  | 0   | <p>RT</p> <p>RCVR Trigger</p> <p>This is used to select the trigger level in the receiver FIFO at which the Received Data Available Interrupt is generated. In auto flow control mode it is used to determine when the rts_n signal is de-asserted. It also determines when the dma_rx_req_n signal is asserted in certain modes of operation.</p> <p>00: 1 character in the FIFO</p> <p>01: FIFO 1/4 full</p> <p>10: FIFO 1/2 full</p> <p>11: FIFO-2 less than full</p> |
| 5:4           | W  | 0   | <p>TFT</p> <p>TX Empty Trigger</p> <p>Writes have no effect when THRE_MODE_USER = Disabled. This is used to select the empty threshold level at which the THRE Interrupts are generated when the mode is active. It also determines when the dma_tx_req_n signal is asserted when in certain modes of operation.</p> <p>00: FIFO empty</p> <p>01: 2 characters in the FIFO</p> <p>10: FIFO 1/4 full</p>                                                                  |

|   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |   |   | 11: FIFO ½ full                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3 | W | 0 | <p>DMAM</p> <p>DMA Mode</p> <p>0: Mode 0</p> <p>In this mode, if PTE is high and TX FIFO is enable, the TX DMA request will send when TFL is less than or equals to FIFO Trigger Level. If PTE is high and TX FIFO is disabled, the TX DMA request will send when THRE is empty. If PTE is low, the TX DMA request will send when the TX FIFO is empty.</p> <p>If dma_pte_rx is high and RX FIFO is enabled, the RX DRQ will send when RFL is equals to or more than FIFO Trigger Level.</p> <p>1: Mode 1</p> <p>In this mode, if TX FIFO is enabled and the PTE is high, the TX DMA request will send when TFL is less than or equals to FIFO Trigger Level. If PTE is low, the TX DMA request will send when TX FIFO is empty and the request stops only when TX FIFO is empty.</p> <p>If RFL is equals to or more than FIFO Trigger Level, the RX DRQ will be set 1, in otherwise, it will be set 0.</p> |
| 2 | W | 0 | <p>XFIFOR</p> <p>XMIT FIFO Reset</p> <p>This resets the control portion of the transmit FIFO and treats the FIFO as empty. This also de-asserts the DMA TX request.</p> <p>It is 'self-clearing'. It is not necessary to clear this bit.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1 | W | 0 | <p>RFIFOR</p> <p>RCVR FIFO Reset</p> <p>This resets the control portion of the receive FIFO and treats the FIFO as empty. This also de-asserts the DMA RX request.</p> <p>It is 'self-clearing'. It is not necessary to clear this bit.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0 | W | 0 | <p>FIFOE</p> <p>Enable FIFOs</p> <p>This enables/disables the transmit (XMIT) and receive (RCVR) FIFOs. Whenever the value of this bit is changed both the XMIT and RCVR controller portion of FIFOs is reset.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

#### 6.2.3.8 UART\_LCR\_REG

| 偏移地址 : 0x000C |     |     | 默认值: 0x0000_0000                                                                                                                                                                                  |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                |
| 31:8          | /   | /   | /                                                                                                                                                                                                 |
| 7             | R/W | 0   | <p>DLAB</p> <p>Divisor Latch Access Bit</p> <p>It is writeable only when UART is not busy (USR[0] is zero) and always readable. This bit is used to enable reading and writing of the Divisor</p> |

|     |     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|-----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |   | <p>Latch register (DLL and DLH) to set the baud rate of the UART. This bit must be cleared after initial baud rate setup in order to access other registers.</p> <p>0: Select RX Buffer Register (RBR) / TX Holding Register(THR) and Interrupt Enable Register (IER)</p> <p>1: Select Divisor Latch LS Register (DLL) and Divisor Latch MS Register (DLM)</p>                                                                                                                                                                                                                                           |
| 6   | R/W | 0 | <p>BC</p> <p>Break Control Bit</p> <p>This is used to cause a break condition to be transmitted to the receiving device. If set to one the serial output is forced to the spacing (logic 0) state. When not in Loopback Mode, as determined by MCR[4], the sout line is forced low until the Break bit is cleared. If SIR_MODE = Enabled and active (MCR[6] set to one) the sir_out_n line is continuously pulsed. When in Loopback Mode, the break condition is internally looped back to the receiver and the sir_out_n line is forced low.</p>                                                        |
| 5:4 | R/W | 0 | <p>EPS</p> <p>Even Parity Select</p> <p>It is writeable only when UART is not busy (USR[0] is zero) and always writable readable. This is used to select between even and odd parity, when parity is enabled (PEN set to one). Setting the LCR[5] is used to reverse the LCR[4].</p> <p>00: Odd Parity</p> <p>01: Even Parity</p> <p>1X: Reverse LCR[4]</p> <p>In RS485 mode, it is the 9th bit--address bit.</p> <p>11:9th bit = 1, indicates that this is a address byte</p> <p>10:9th bit = 0, indicates that this is a data byte</p> <p>Note: When use this function, PEN(LCR[3]) must set to 1.</p> |
| 3   | R/W | 0 | <p>PEN</p> <p>Parity Enable</p> <p>It is writeable only when UART is not busy (USR[0] is zero) and always readable. This bit is used to enable and disable parity generation and detection in transmitted and received serial character respectively.</p> <p>0: parity disabled</p> <p>1: parity enabled</p>                                                                                                                                                                                                                                                                                             |
| 2   | R/W | 0 | <p>STOP</p> <p>Number of stop bits</p> <p>It is writeable only when UART is not busy (USR[0] is zero) and always readable. This is used to select the number of stop bits per character that the peripheral transmits and receives. If set to zero, one stop bit is transmitted in the serial data. If set to one and the data bits are set to 5 (LCR[1:0] set to zero) one and a half stop bits is transmitted. Other/Wise, two stop bits are transmitted. Note that regardless of the number of stop bits selected the receiver checks only the first stop bit.</p>                                    |

|     |     |   |                                                                                                                                                                                                                                                                                                                                                  |
|-----|-----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |   | 0: 1 stop bit<br>1: 1.5 stop bits when DLS (LCR[1:0]) is zero, else 2 stop bit                                                                                                                                                                                                                                                                   |
| 1:0 | R/W | 0 | DLS<br><br>Data Length Select<br><br>It is writeable only when UART is not busy (USR[0] is zero) and always readable. This is used to select the number of data bits per character that the peripheral transmits and receives. The number of bit that may be selected areas follows:<br><br>00: 5 bits<br>01: 6 bits<br>10: 7 bits<br>11: 8 bits |

### 6.2.3.9 UART\_MCR\_REG

| 偏移地址 : 0x0010 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 31:8          | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7:6           | R/W | 0   | UART_FUNCTION<br><br>Select IrDA or RS485<br>0:Uart Mode<br>1:IrDA SIR Mode<br>2:RS485 Mode<br>3:Reverse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5             | R/W | 0   | AFCE<br><br>Auto Flow Control Enable<br><br>When FIFOs are enabled and the Auto Flow Control Enable (AFCE) bit is set, Auto Flow Control features are enabled.<br>0: Auto Flow Control Mode disabled<br>1: Auto Flow Control Mode enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4             | R/W | 0   | LOOP<br><br>Loop Back Mode<br>0: Normal Mode<br>1: Loop Back Mode<br><br>This is used to put the UART into a diagnostic mode for test purposes. If operating in UART mode (SIR_MODE != Enabled or not active, MCR[6] set to zero), data on the sout line is held high, while serial data output is looped back to the sin line, internally. In this mode all the interrupts are fully functional. Also, in loopback mode, the modem control inputs (dsr_n, cts_n, ri_n, dcd_n) are disconnected and the modem control outputs (dtr_n, rts_n, out1_n, out2_n) are looped back to the inputs, internally. If operating in infrared mode (SIR_MODE == Enabled AND active, MCR[6] set to one), data on the |

|     |     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|-----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |   | sir_out_n line is held low, while serial data output is inverted and looped back to the sir_in line.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3:2 | /   | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1   | R/W | 0 | <p>RTS</p> <p>Request to Send</p> <p>This is used to directly control the Request to Send (rts_n) output. The Request To Send (rts_n) output is used to inform the modem or data set that the UART is ready to exchange data. When Auto RTS Flow Control is not enabled (MCR[5] set to zero), the rts_n signal is set low by programming MCR[1] (RTS) to a high. In Auto Flow Control, AFCE_MODE == Enabled and active (MCR[5] set to one) and FIFOs enable (FCR[0] set to one), the rts_n output is controlled in the same way, but is also gated with the receiver FIFO threshold trigger (rts_n is inactive high when above the threshold). The rts_n signal is de-asserted when MCR[1] is set low.</p> <p>0: rts_n de-asserted (logic 1)</p> <p>1: rts_n asserted (logic 0)</p> <p>Note that in Loopback mode (MCR[4] set to one), the rts_n output is held inactive high while the value of this location is internally looped back to an input.</p> |
| 0   | R/W | 0 | <p>DTR</p> <p>Data Terminal Ready</p> <p>This is used to directly control the Data Terminal Ready (dtr_n) output. The value written to this location is inverted and driven out on dtr_n.</p> <p>0: dtr_n de-asserted (logic 1)</p> <p>1: dtr_n asserted (logic 0)</p> <p>The Data Terminal Ready output is used to inform the modem or data set that the UART is ready to establish communications.</p> <p>Note that in Loopback mode (MCR[4] set to one), the dtr_n output is held inactive high while the value of this location is internally looped back to an input.</p>                                                                                                                                                                                                                                                                                                                                                                            |

### 6.2.3.10 UART\_LSR\_REG

| 偏移地址 : 0x0014 |    |     | 默认值: 0x0000_0060                                                                                                                                                                                                                                                                                           |
|---------------|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                         |
| 31:8          | /  | /   | /                                                                                                                                                                                                                                                                                                          |
| 7             | R  | 0   | <p>FIFOERR</p> <p>RX Data Error in FIFO</p> <p>When FIFOs are disabled, this bit is always 0. When FIFOs are enabled, this bit is set to 1 when there is at least one PE, FE, or BI in the RX FIFO. It is cleared by a read from the LSR register provided there are no subsequent errors in the FIFO.</p> |

|   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|---|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | R | 1 | <p>TEMT</p> <p>Transmitter Empty</p> <p>If the FIFOs are disabled, this bit is set to "1" whenever the TX Holding Register and the TX Shift Register are empty. If the FIFOs are enabled, this bit is set whenever the TX FIFO and the TX Shift Register are empty. In both cases, this bit is cleared when a byte is written to the TX data channel.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5 | R | 1 | <p>THRE</p> <p>TX Holding Register Empty</p> <p>If the FIFOs are disabled, this bit is set to "1" whenever the TX Holding Register is empty and ready to accept new data and it is cleared when the CPU writes to the TX Holding Register.</p> <p>If the FIFOs are enabled, this bit is set to "1" whenever the TX FIFO is empty and it is cleared when at least one byte is written to the TX FIFO.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4 | R | 0 | <p>BI</p> <p>Break Interrupt</p> <p>This is used to indicate the detection of a break sequence on the serial input data.</p> <p>If in UART mode (SIR_MODE == Disabled), it is set whenever the serial input, sin, is held in a logic '0' state for longer than the sum of start time + data bits + parity + stop bits.</p> <p>If in infrared mode (SIR_MODE == Enabled), it is set whenever the serial input, sir_in, is continuously pulsed to logic '0' for longer than the sum of start time + data bits + parity + stop bits. A break condition on serial input causes one and only one character, consisting of all zeros, to be received by the UART.</p> <p>In the FIFO mode, the character associated with the break condition is carried through the FIFO and is revealed when the character is at the top of the FIFO. Reading the LSR clears the BI bit. In the non-FIFO mode, the BI indication occurs immediately and persists until the LSR is read.</p> |
| 3 | R | 0 | <p>FE</p> <p>Framing Error</p> <p>This is used to indicate the occurrence of a framing error in the receiver. A framing error occurs when the receiver does not detect a valid STOP bit in the received data.</p> <p>In the FIFO mode, since the framing error is associated with a character received, it is revealed when the character with the framing error is at the top of the FIFO. When a framing error occurs, the UART tries to resynchronize. It does this by assuming that the error was due to the start bit of the next character and then continues receiving the other bit i.e. data, and/or parity and stop. It should be noted that the Framing Error (FE) bit (LSR[3]) is set if a break interrupt has occurred, as indicated by Break Interrupt (BI) bit (LSR[4]).</p> <p>0: no framing error<br/>1: framing error</p>                                                                                                                            |

|   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |   |   | Reading the LSR clears the FE bit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2 | R | 0 | <p>PE</p> <p>Parity Error</p> <p>This is used to indicate the occurrence of a parity error in the receiver if the Parity Enable (PEN) bit (LCR[3]) is set. In the FIFO mode, since the parity error is associated with a character received, it is revealed when the character with the parity error arrives at the top of the FIFO. It should be noted that the Parity Error (PE) bit (LSR[2]) is set if a break interrupt has occurred, as indicated by Break Interrupt (BI) bit (LSR[4]).</p> <p>0: no parity error<br/>1: parity error</p> <p>Reading the LSR clears the PE bit.</p>                                 |
| 1 | R | 0 | <p>OE</p> <p>Overrun Error</p> <p>This occurs if a new data character was received before the previous data was read. In the non-FIFO mode, the OE bit is set when a new character arrives in the receiver before the previous character was read from the RBR. When this happens, the data in the RBR is overR/Written. In the FIFO mode, an overrun error occurs when the FIFO is full and a new character arrives at the receiver. The data in the FIFO is retained and the data in the receive shift register is lost.</p> <p>0: no overrun error<br/>1: overrun error</p> <p>Reading the LSR clears the OE bit.</p> |
| 0 | R | 0 | <p>DR</p> <p>Data Ready</p> <p>This is used to indicate that the receiver contains at least one character in the RBR or the receiver FIFO.</p> <p>0: no data ready<br/>1: data ready</p> <p>This bit is cleared when the RBR is read in non-FIFO mode, or when the receiver FIFO is empty, in FIFO mode.</p>                                                                                                                                                                                                                                                                                                             |

#### 6.2.3.11 UART\_MSR\_REG

| 偏移地址 : 0x0018 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                            |
|---------------|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                          |
| 31:8          | /  | /   | /                                                                                                                                                                                                                                                                           |
| 7             | R  | 0   | <p>DCD</p> <p>Line State of Data Carrier Detect</p> <p>This is used to indicate the current state of the modem control line dcd_n. This bit is the complement of dcd_n. When the Data Carrier Detect input (dcd_n) is asserted it is an indication that the carrier has</p> |

|   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |   |   | <p>been detected by the modem or data set.</p> <p>0: dcd_n input is de-asserted (logic 1)</p> <p>1: dcd_n input is asserted (logic 0)</p>                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6 | R | 0 | <p>RI</p> <p>Line State of Ring Indicator</p> <p>This is used to indicate the current state of the modem control line ri_n. This bit is the complement of ri_n. When the Ring Indicator input (ri_n) is asserted it is an indication that a telephone ringing signal has been received by the modem or data set.</p> <p>0: ri_n input is de-asserted (logic 1)</p> <p>1: ri_n input is asserted (logic 0)</p>                                                                                                                             |
| 5 | R | 0 | <p>DSR</p> <p>Line State of Data Set Ready</p> <p>This is used to indicate the current state of the modem control line dsr_n. This bit is the complement of dsr_n. When the Data Set Ready input (dsr_n) is asserted it is an indication that the modem or data set is ready to establish communications with UART.</p> <p>0: dsr_n input is de-asserted (logic 1)</p> <p>1: dsr_n input is asserted (logic 0)</p> <p>In Loopback Mode (MCR[4] set to one), DSR is the same as MCR[0] (DTR).</p>                                          |
| 4 | R | 0 | <p>CTS</p> <p>Line State of Clear To Send</p> <p>This is used to indicate the current state of the modem control line cts_n. This bit is the complement of cts_n. When the Clear to Send input (cts_n) is asserted it is an indication that the modem or data set is ready to exchange data with UART.</p> <p>0: cts_n input is de-asserted (logic 1)</p> <p>1: cts_n input is asserted (logic 0)</p> <p>In Loopback Mode (MCR[4] = 1), CTS is the same as MCR[1] (RTS).</p>                                                              |
| 3 | R | 0 | <p>DDCD</p> <p>Delta Data Carrier Detect</p> <p>This is used to indicate that the modem control line dcd_n has changed since the last time the MSR was read.</p> <p>0: no change on dcd_n since last read of MSR</p> <p>1: change on dcd_n since last read of MSR</p> <p>Reading the MSR clears the DD CD bit.</p> <p>Note: If the DD CD bit is not set and the dcd_n signal is asserted (low) and a reset occurs (software or other/Wise), then the DD CD bit is set when the reset is removed if the dcd_n signal remains asserted.</p> |
| 2 | R | 0 | <p>TERI</p> <p>Trailing Edge Ring Indicator</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



|   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|---|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |   |   | <p>This is used to indicate that a change on the input ri_n (from an active-low to an inactive-high state) has occurred since the last time the MSR was read.</p> <p>0: no change on ri_n since last read of MSR<br/>1: change on ri_n since last read of MSR</p> <p>Reading the MSR clears the TERI bit.</p>                                                                                                                                                                                                                                                                                             |
| 1 | R | 0 | <p>DDSR</p> <p>Delta Data Set Ready</p> <p>This is used to indicate that the modem control line dsr_n has changed since the last time the MSR was read.</p> <p>0: no change on dsr_n since last read of MSR<br/>1: change on dsr_n since last read of MSR</p> <p>Reading the MSR clears the DDSR bit. In Loopback Mode (MCR[4] = 1), DDSR reflects changes on MCR[0] (DTR).</p> <p>Note: If the DDSR bit is not set and the dsr_n signal is asserted (low) and a reset occurs (software or other/Wise), then the DDSR bit is set when the reset is removed if the dsr_n signal remains asserted.</p>      |
| 0 | R | 0 | <p>DCTS</p> <p>Delta Clear to Send</p> <p>This is used to indicate that the modem control line cts_n has changed since the last time the MSR was read.</p> <p>0: no change on ctsdsr_n since last read of MSR<br/>1: change on ctsdsr_n since last read of MSR</p> <p>Reading the MSR clears the DCTS bit. In Loopback Mode (MCR[4] = 1), DCTS reflects changes on MCR[1] (RTS).</p> <p>Note: If the DCTS bit is not set and the cts_n signal is asserted (low) and a reset occurs (software or other/Wise), then the DCTS bit is set when the reset is removed if the cts_n signal remains asserted.</p> |

#### 6.2.3.12 UART\_SCH\_REG

| 偏移地址 : 0x001C |     | 默认值: 0x0000_0000 |                                                                                                                                                                |
|---------------|-----|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值              | 描述                                                                                                                                                             |
| 31:8          | /   | /                | /                                                                                                                                                              |
| 7:0           | R/W | 0                | <p>SCRATCH_REG</p> <p>Scratch Register</p> <p>This register is for programmers to use as a temporary storage space. It has no defined purpose in the UART.</p> |

### 6.2.3.13 UART\_USR\_REG

| 偏移地址 : 0x007C |    |     | 默认值: 0x0000_0006                                                                                                                                                                                                                                        |
|---------------|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                      |
| 31:5          | /  | /   | /                                                                                                                                                                                                                                                       |
| 4             | R  | 0   | <p>RFF</p> <p>Receive FIFO Full</p> <p>This is used to indicate that the receive FIFO is completely full.</p> <p>0: Receive FIFO not full</p> <p>1: Receive FIFO Full</p> <p>This bit is cleared when the RX FIFO is no longer full.</p>                |
| 3             | R  | 0   | <p>RFNE</p> <p>Receive FIFO Not Empty</p> <p>This is used to indicate that the receive FIFO contains one or more entries.</p> <p>0: Receive FIFO is empty</p> <p>1: Receive FIFO is not empty</p> <p>This bit is cleared when the RX FIFO is empty.</p> |
| 2             | R  | 1   | <p>TFE</p> <p>Transmit FIFO Empty</p> <p>This is used to indicate that the transmit FIFO is completely empty.</p> <p>0: Transmit FIFO is not empty</p> <p>1: Transmit FIFO is empty</p> <p>This bit is cleared when the TX FIFO is no longer empty.</p> |
| 1             | R  | 1   | <p>TFNF</p> <p>Transmit FIFO Not Full</p> <p>This is used to indicate that the transmit FIFO is not full.</p> <p>0: Transmit FIFO is full</p> <p>1: Transmit FIFO is not full</p> <p>This bit is cleared when the TX FIFO is full.</p>                  |
| 0             | R  | 0   | <p>BUSY</p> <p>UART Busy Bit</p> <p>0: Idle or inactive</p> <p>1: Busy</p>                                                                                                                                                                              |

### 6.2.3.14 UART\_TFL\_REG

| 偏移地址：0x0080 |    |     | 默认值：0x0000_0000                                                                                   |
|-------------|----|-----|---------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                |
| 31:7        | /  | /   | /                                                                                                 |
| 6:0         | R  | 0   | TFL<br><br>Transmit FIFO Level<br>This indicates the number of data entries in the transmit FIFO. |

### 6.2.3.15 UART\_RFL\_REG

| 偏移地址：0x0084 |    |     | 默认值：0x0000_0000                                                                                 |
|-------------|----|-----|-------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                              |
| 31:7        | /  | /   | /                                                                                               |
| 6:0         | R  | 0   | RFL<br><br>Receive FIFO Level<br>This indicates the number of data entries in the receive FIFO. |

### 6.2.3.16 UART\_RBR\_REG

| 偏移地址：0x00A4 |     |     | 默认值：0x0000_0000                                                                                                                                                                      |
|-------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                   |
| 31:8        | /   | /   |                                                                                                                                                                                      |
| 7           | R/W | 0   | DMA_PTE_RX<br><br>RX_DRQ 的发送。<br>对于 DMA1 模式,当 RFL 大于等于 TRIG 或者接收超时发送 DRQ, 否则 DRQ 为低。<br>对于 DMA0 模式,当 DMA_PTE_RX = 1 且 FIFO 开时, 当 RFL 大于等于 TRIG 时发送 REQ, 否则不发送。否则只要存在 RX 有效数据就发送 DRQ。 |
| 6           | R/W | 0   | PTE<br><br>TX_REQ 的发送。<br>对于 DMA1 模式 0(FIFO 开, FCR[3]=1),如果 PTE 信号为高, 当 TFL 小于等于 TRIG, 发送 DMA 请求。当 PTE 信号为低, 当 FIFO 为空时发送 DMA 请求。DMA 请求只会在 FIFO 满的时候掉下来。                             |

|   |       |   |                                                                                                                                                                                                                                                                                                                                                          |
|---|-------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |       |   | 对于 DMA0 模式(FIFO 关或 FCR[3]=0),如果 PTE 信号为高且 FIFO 开,当 TFL 小于等于 TRIG, 发送 DMA 请求,如果条件不满足则 DRQ 掉下来。如果 PTE 信号为高且 FIFO 关, THRE 空时, 发送 DMA 请求,如果条件不满足则 DRQ 掉下来。当 PTE 信号为低,当 FIFO 为空时发送 DMA 请求。                                                                                                                                                                    |
| 5 | R/W   | 0 | <p>SIR_RX_INVERT</p> <p>SIR Receiver Pulse Polarity Invert</p> <p>0: Not invert receiver signal</p> <p>1: Invert receiver signal</p>                                                                                                                                                                                                                     |
| 4 | R/W   | 0 | <p>SIR_TX_INVERT</p> <p>SIR Transmit Pulse Polarity Invert</p> <p>0: Not invert transmit pulse</p> <p>1: Invert transmit pulse</p>                                                                                                                                                                                                                       |
| 3 | /     | / | /                                                                                                                                                                                                                                                                                                                                                        |
| 2 | R/WAC | 0 | <p>CHANGE_UPDATE</p> <p>After the user using HALT[1] to change the baud rate or LCR configuration, write 1 to update the configuration and waiting this bit self-clear to 0 to finish update process. Write 0 to this bit has no effect.</p> <p>1: Update trigger, Self clear to 0 when finish update.</p>                                               |
| 1 | R/W   | 0 | <p>CHCFG_AT_BUSY</p> <p>This is an enable bit for the user to change LCR register configuration (except for the DLAB bit) and baud rate register (DLH and DLL) when the UART is busy (USB[0] is 1).</p> <p>1: Enable change when busy</p>                                                                                                                |
| 0 | R/W   | 0 | <p>HALT_TX</p> <p>Halt TX</p> <p>This register is use to halt transmissions for testing, so that the transmit FIFO can be filled by the master when FIFOs are implemented and enabled.</p> <p>0 : Halt TX disabled</p> <p>1 : Halt TX enabled</p> <p>Note: If FIFOs are not enabled, the setting of the halt TX register has no effect on operation.</p> |

### 6.2.3.17 UART\_TXDLY\_REG

| 偏移地址 : 0x00CC |     |     | 默认值: 0x0000_0000                                                                                                                                      |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                    |
| 31:8          | /   | /   | /                                                                                                                                                     |
| 7:0           | R/W | 0   | DLY<br><br>The delay time between the last stop bit and the next start bit. The unit is Tclk. It is use to control the space between two bytes in TX. |

### 6.2.3.18 UART\_BDC\_REG

| 偏移地址 : 0x00D4 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 31:2          | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1             | R/W | 0   | Auto Baudrate Detect Enable<br><br>1: enable<br>0: disable<br><br>Note: Set this bit to 1 to start the UART Baudrate Detection. The detection circuit will detect the start bit and count the cycles of PCLK in one bit duration. The number of the cycles will be written in the PCLK_NUM field. Set this bit to 0 to stop the UART Baudrate Detection.                                                                                       |
| 0             | R/W | 0   | Mode<br><br>1: Normal mode<br>0: Compare mode<br><br>Note:<br>In normal mode, when the detection circuit is enabled, the number of the PCLK cycles will be latched in each one bit duration and be written into the PCLK_NUM field.<br><br>In compare mode, the number of the PCLK cycles will be latched in each one bit duration, but only the value which equals to the value latched in last time will be written into the PCLK_NUM field. |

### 6.2.3.19 UART\_BDCL\_REG

| 偏移地址 : 0x00D8 |     |     | 默认值: 0x0000_0000 |
|---------------|-----|-----|------------------|
| 位置            | 访问  | 默认值 | 描述               |
| 31:8          | /   | /   | /                |
| 7:0           | R/W | 0   | PCLK_NUM_LOW     |

|  |  |  |                                                                         |
|--|--|--|-------------------------------------------------------------------------|
|  |  |  | The low 8 bits of the number of the cycles counted in one bit duration. |
|--|--|--|-------------------------------------------------------------------------|

#### 6.2.3.20 UART\_ABCR\_REG

| 偏移地址 : 0x00DC |     |     | 默认值: 0x0000_0000                                                                              |
|---------------|-----|-----|-----------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                            |
| 31:8          | /   | /   | /                                                                                             |
| 7:0           | R/W | 0   | PCLK_NUM_HIGH<br><br>The high 8 bits of the number of the cycles counted in one bit duration. |

## 6.3 LPUART 接口

### 6.3.1 概述

LPUART主要作用是用作系统休眠时，发特定字符或字符串组合唤醒系统，由于休眠时只能使用32K时钟源，最高波特率只支持到9600。

### 6.3.2 LPUART 寄存器列表

| 模块名                    | 基地址        |                                  |
|------------------------|------------|----------------------------------|
| LPUART0                | 0x40045800 |                                  |
| LPUART1                | 0x40045C00 |                                  |
| 寄存器名                   | 偏移地址       | 描述                               |
| LPUART_GP_SR_CON_REG   | 0x0000     | LPUART Control Register          |
| LPUART_BAUD_CONFIG_REG | 0x0004     | LPUART Baudrate Config Register  |
| LPUART_RX_CONFIG_REG   | 0x0010     | LPUART RX Config Register        |
| LPUART_RX_DATA_REG     | 0x0014     | LPUART RX Data Register          |
| LPUART_INTR_EN_REG     | 0x0020     | LPUART Interrupt Enable Register |
| LPUART_INTR_STATUS_REG | 0x0024     | LPUART Interrupt Status Register |
| LPUART_INTR_CLR_REG    | 0x0028     | LPUART Interrupt Clear Register  |
| LPUART_RX_CMP_REG1     | 0x002C     | LPUART RX Compare Register1      |
| LPUART_RX_CMP_REG2     | 0x0030     | LPUART RX Compare Register2      |

### 6.3.3 LPUART 寄存器描述

#### 6.3.3.1 LUART\_GP\_SR\_CON\_REG

| 偏移地址：0x0000 |    |      | 默认值: 0x0000_0000                       |
|-------------|----|------|----------------------------------------|
| 位置          | 访问 | 默认值  | 描述                                     |
| 31:9        | /  | /    | /                                      |
| 8           | W  | 1'b0 | cfg_soft_rst_p<br>软复位。<br>软件置 1，硬件自清零。 |
| 7:0         | /  | /    | /                                      |

#### 6.3.3.2 LUART\_BAUD\_CONFIG\_REG

| 偏移地址：0x0004 |     |       | 默认值: 0x0302_0006                                                                               |
|-------------|-----|-------|------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值   | 描述                                                                                             |
| 31:24       | R/W | 8'h03 | cfg_denominator[7:0]<br>配置除数。由于采用双沿设计，如果时钟为 32K，目标波特率为 9600，则有 $64K/9.6K=20/3=6$ 余 2，则该寄存器配 3。 |
| 23:16       | R/W | 8'h02 | cfg_remainder[7:0]<br>配置余数。由于采用双沿设计，如果时钟为 32K，目标波特率为 9600，则有 $64K/9.6K=20/3=6$ 余 2，则该寄存器配 2。   |
| 15:0        | R/W | 8'h06 | cfg_quotient[15:0]<br>配置商值。由于采用双沿设计，如果时钟为 32K，目标波特率为 9600，则有 $64K/9.6K=20/3=6$ 余 2，则该寄存器配 6。   |

#### 6.3.3.3 LUART\_RX\_CONFIG\_REG

| 偏移地址：0x0010 |     |      | 默认值: 0x0000_0000                                         |
|-------------|-----|------|----------------------------------------------------------|
| 位置          | 访问  | 默认值  | 描述                                                       |
| 31:13       | /   | /    | /                                                        |
| 12          | R/W | 1'b0 | cfg_rx_msb_first<br>1'b0: 先收取的作为 LSB<br>1'b1: 先收取的作为 MSB |
| 11          | /   | /    | /                                                        |
| 10:8        | R/W | 3'h0 | cfg_rx_data_width[2:0]<br>选择 RX 数据位的长度。                  |

|     |     |      |                                                                                                                                                                                                                                                              |
|-----|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |      | 3'b000: 4bit data<br>3'b001: 5bit data<br>3'b010: 6bit data<br>3'b011: 7bit data<br>3'b100: 8bit data<br>3'b101: 9bit data<br>other: 8bit data<br>值得注意的是，RXFIFO 的位宽是 9bit，如果 rx data 少于 9bit，高位将被补零。                                                         |
| 7   | /   | /    | /                                                                                                                                                                                                                                                            |
| 6:4 | R/W | 3'h0 | cfg_rx_parity_mode[2:0]<br>选择 RX 校验位类型。<br>3'b000: 没有校验位。<br>3'b001: even。每个字节传送整个过程中 bit 为 1 的个数是偶数个。通过校验位来调整个数。<br>3'b010: odd。每个字节传送整个过程中 bit 为 1 的个数是奇数。通过校验位来调整个数。<br>3'b011: space。即校验位总为 0。<br>3'b100: mark。即校验位总为 1。<br>3'b101/3'b110/3'b111: 没有校验位。 |
| 3   | /   | /    | /                                                                                                                                                                                                                                                            |
| 2   | R   | 1'b0 | rx_partiy_err_rpt<br>报告最近一次 RX 校验结果是否正确。<br>1'b0: 校验结果正确。<br>1'b1: 校验结果错误。                                                                                                                                                                                   |
| 1   | /   | /    | /                                                                                                                                                                                                                                                            |
| 0   | R/W | 1'b0 | cfg_rx_en<br>用于使能 RX。硬件在实现时，该信号被用作 RX 模块的时钟使能。<br>1'b1: enable RX 模块<br>1'b0: disable RX 模块                                                                                                                                                                  |

#### 6.3.3.4 LUART\_RX\_DATA\_REG

| 偏移地址：0x0014 |    | 默认值: 0x0000_0000 |    |
|-------------|----|------------------|----|
| 位置          | 访问 | 默认值              | 描述 |
| 31:9        | /  | /                | /  |



|     |   |      |                                           |
|-----|---|------|-------------------------------------------|
| 8:0 | R | 9'h0 | rx_data[8:0]<br>读取该寄存器将触发从 RXFIFO 读取新的数据。 |
|-----|---|------|-------------------------------------------|

### 6.3.3.5 LUART\_INTR\_EN\_REG

| 偏移地址 : 0x0020 |     |      | 默认值: 0x0000_0000                                     |
|---------------|-----|------|------------------------------------------------------|
| 位置            | 访问  | 默认值  | 描述                                                   |
| 31:10         | /   | /    | /                                                    |
| 9             | R/W | 1'b0 | cfg_rx_data_int_en<br>在非 RX FIFO 模式下, RX 接收到数据中断使能位。 |
| 8             | /   | /    | /                                                    |
| 7             | R/W | 1'b0 | cfg_rx_data_cmp_int_en<br>RX DATA Compare 中断使能位。     |
| 6:0           | /   | /    | /                                                    |

### 6.3.3.6 LUART\_INTR\_STATUS\_REG

| 偏移地址 : 0x0024 |    |      | 默认值: 0x0000_0000                                   |
|---------------|----|------|----------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                 |
| 31:10         | /  | /    | /                                                  |
| 9             | R  | 1'b0 | rx_data_int_flag<br>在非 RX FIFO 模式下, RX 接收到数据中断状态位。 |
| 8             | /  | /    | /                                                  |
| 7             | R  | 1'b0 | rx_data_cmp_int_flag<br>RX DATA Compare 中断状态位。     |
| 6:0           | /  | /    | /                                                  |

### 6.3.3.7 LUART\_INTR\_CLR\_REG

| 偏移地址 : 0x0028 |    |      | 默认值: 0x0000_0000                                                 |
|---------------|----|------|------------------------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                               |
| 31:10         | /  | /    | /                                                                |
| 9             | W  | 1'b0 | cfg_rx_data_int_clr<br>在非 RX FIFO 模式下, RX 接收到数据中断清零位。软件置位, 硬件清零。 |
| 8             | /  | /    | /                                                                |

|     |   |      |                                                             |
|-----|---|------|-------------------------------------------------------------|
| 7   | W | 1'b0 | cfg_rx_data_cmp_int_clr<br>RX DATA Compare 中断清零位。软件置位，硬件清零。 |
| 6:0 | / | /    | /                                                           |

### 6.3.3.8 LUART\_RX\_CMP\_REG1

| 偏移地址：0x002C |     |      | 默认值: 0x0000_0000                                                                                                                          |
|-------------|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值  | 描述                                                                                                                                        |
| 31:30       | /   | /    | /                                                                                                                                         |
| 29:21       | R/W | 9'b0 | cfg_rx_cmp2[8:0]<br>与 RX Data 进行第 3 个比较的寄存器。                                                                                              |
| 20:12       | R/W | 9'b0 | cfg_rx_cmp1[8:0]<br>与 RX Data 进行第 2 个比较的寄存器。                                                                                              |
| 11:3        | R/W | 9'b0 | cfg_rx_cmp0[8:0]<br>与 RX Data 进行第 1 个比较的寄存器。                                                                                              |
| 2:0         | R/W | 3'b0 | cfg_rx_cmp_num[2:0]<br>配置用于比较的个数。<br>0：比较 1 个数<br>1：比较 1 个数<br>2：比较 2 个数<br>3：比较 3 个数<br>4：比较 4 个数<br>5：比较 5 个数<br>6：比较 1 个数<br>7：比较 1 个数 |

### 6.3.3.9 LUART\_RX\_CMP\_REG2

| 偏移地址：0x0030 |     |      | 默认值: 0x0000_0000                             |
|-------------|-----|------|----------------------------------------------|
| 位置          | 访问  | 默认值  | 描述                                           |
| 31:21       | /   | /    | /                                            |
| 20:12       | R/W | 9'b0 | cfg_rx_cmp4[8:0]<br>与 RX Data 进行第 5 个比较的寄存器。 |
| 11:3        | R/W | 9'b0 | cfg_rx_cmp3[8:0]<br>与 RX Data 进行第 4 个比较的寄存器。 |

|     |   |   |   |
|-----|---|---|---|
| 2:0 | / | / | / |
|-----|---|---|---|

6.4 SPI 接口

6.4.1 概述

SPI支持4种不同的数据传输格式。软件可以通过设置SPI传输控制寄存器的bit1(极性)和bit0(相位)来选择SPI工作的四种模式之一。SPI控制器主程序使用SPI\_SCLK信号来传输数据进出移位寄存器。使用四种可编程时钟相位和极性的任一组合，数据都可以被时钟采样。

在Phase 0, polar 0和Phase 1, polar 1操作期间，输出数据在时钟下降沿上变化，输入数据在时钟上升沿上移位。

在Phase 1, polar 0和Phase 0, polar 1操作期间，输出数据在时钟上升沿上变化，输入数据在时钟下降沿上移位。

POL定义了SPI\_SCLK处于空闲状态时的信号极性。当POL为'1'时SPI\_SCLK为高电平，当POL为'0'时SPI\_SCLK为低电平。PHA决定SPI\_SCLK的前沿是用于数据建立还是采集。当PHA为'1'时，时钟沿用于数据建立，当PHA为'0'时，时钟沿用于数据采集。

下表给出了SPI四种模式的说明。

图 6- 3 SPI Modes with Clock Polarity and Phase

| SPI Mode | POL | PHA | Leading Edge    | Trailing Edge   |
|----------|-----|-----|-----------------|-----------------|
| 0        | 0   | 0   | Rising, Sample  | Falling, Setup  |
| 1        | 0   | 1   | Rising, Setup   | Falling, Sample |
| 2        | 1   | 0   | Falling, Sample | Rising, Setup   |
| 3        | 1   | 1   | Falling, Setup  | Rising, Sample  |

图 6- 4 SPI Phase 0 Transfer Format

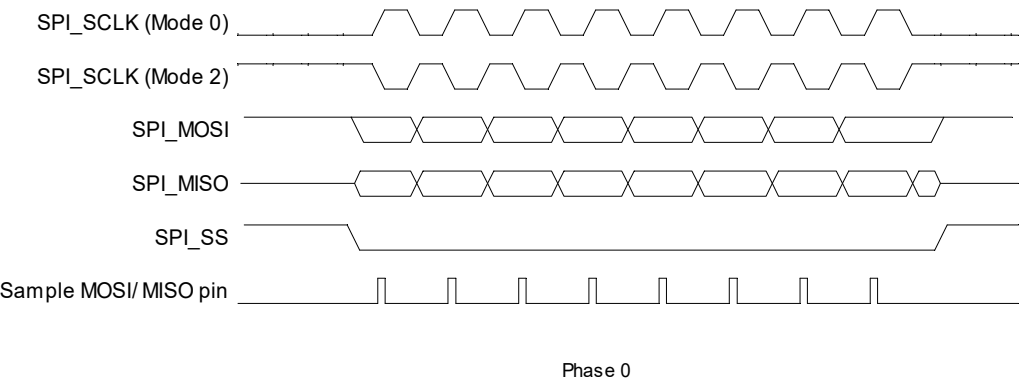
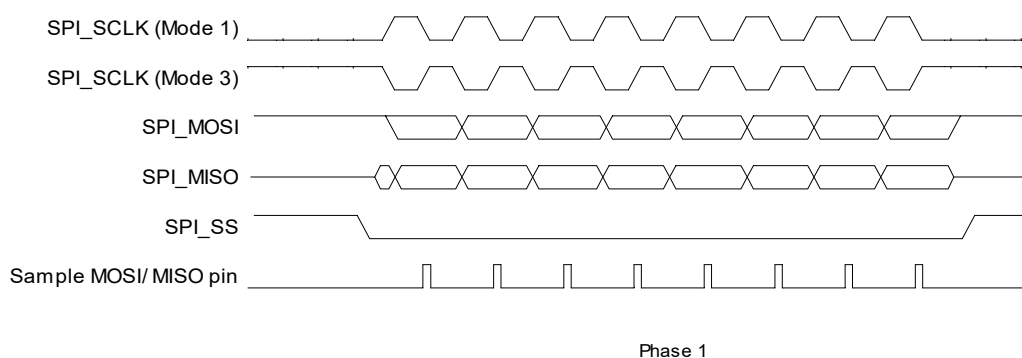


图 6- 5 SPI Phase 1 Transfer Format



## 6.4.2 SPI 寄存器列表

| 模块名                    | 基地址        |                                      |
|------------------------|------------|--------------------------------------|
| SPI Controller 0       | 0x40005000 |                                      |
| 寄存器名                   | 偏移地址       | 描述                                   |
| SPI_VER_REG            | 0x0000     | SPI Version Number Register          |
| SPI_CTRL_REG           | 0x0004     | SPI Global Control Register          |
| SPI_TCTRL_REG          | 0x0008     | SPI Transfer Control register        |
| SPI_IER_REG            | 0x0010     | SPI Interrupt Control register       |
| SPI_STA_REG            | 0x0014     | SPI Interrupt Status register        |
| SPI_FCTL_REG           | 0x0018     | SPI FIFO Control register            |
| SPI_FST_REG            | 0x001C     | SPI FIFO Status register             |
| SPI_WAIT_REG           | 0x0020     | SPI Wait Clock Counter register      |
| SPI_CCTR_REG           | 0x0024     | SPI Clock Rate Control register      |
| SPI_BC_REG             | 0x0030     | SPI Burst Counter register           |
| SPI_TC_REG             | 0x0034     | SPI Transmit Counter Register        |
| SPI_BCC_REG            | 0x0038     | SPI Burst Control register           |
| SPI_NDMA_MODE_CTRL_REG | 0x0088     | SPI Normal DMA Mode Control Register |
| SPI_TXD_REG            | 0x0200     | SPI TX Data register                 |
| SPI_RXD_REG            | 0x0300     | SPI RX Data register                 |

### 6.4.3 SPI 寄存器描述

#### 6.4.3.1 SPI\_VER\_REG

| 偏移地址 : 0x0000 |    |         | 默认值: 0x0009_0000 |
|---------------|----|---------|------------------|
| 位置            | 访问 | 默认值     | 描述               |
| 31:0          | R  | 0x90000 | SPI_VER          |

#### 6.4.3.2 SPI\_CTRL\_REG

| 偏移地址 : 0x0004 |       |     | 默认值: 0x0000_0080                                                                                                                                                                                                                                            |
|---------------|-------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问    | 默认值 | 描述                                                                                                                                                                                                                                                          |
| 31            | R/WAC | 0   | SPI_RST<br><br>SPI Software Reset<br>Write 1 to this bit will clear the SPI controller, and auto clear to 0 when reset operation completes<br>Write 0 has no effect.                                                                                        |
| 30:8          | /     | /   | /                                                                                                                                                                                                                                                           |
| 7             | R/W   | 1   | TP_EN<br><br>Transmit Pause Enable<br>In master mode, it is used to control transmit state machine to stop smart burst sending when RX FIFO is full.<br>1: stop transmit data when RXFIFO full<br>0: normal operation<br>Note: cannot be written when XCH=1 |
| 6:2           | /     | /   | /                                                                                                                                                                                                                                                           |
| 1             | R/W   | 0   | SPI_MODE<br><br>Master Slave selection<br>0: slave mode<br>1: master mode<br>Note: cannot be written when XCH=1                                                                                                                                             |
| 0             | R/W   | 0   | SPI_EN<br><br>SPI Module Enable Control<br>0: SPI Controller Disable<br>1: SPI Controller Enable                                                                                                                                                            |

### 6.4.3.3 SPI\_TCTRL\_REG

| 偏移地址 : 0x0008 |     |     | 默认值: 0x0000_0087                                                                                                                                                                                                                                                                                                                                                              |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                            |
| 31            | R/W | 0   | <p>XCH</p> <p>Exchange Burst</p> <p>In master mode it is used to start SPI burst.</p> <p>0: Idle</p> <p>1: Initiates exchange</p> <p>Write 1 to this bit will start the SPI burst and this bit will auto clear after finishing the bursts transfer specified by BC(Burst Counter). Write 1 to SPI_RST will also clear this bit.</p> <p>Write 0 to this bit has no effect.</p> |
| 30:15         | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                             |
| 14            | R/W | 0   | <p>SDDM</p> <p>Sending Data Delay Mode</p> <p>0: normal sending</p> <p>1: delay sending</p> <p>Set this bit to 1 to make the data that should be send with a delay of half cycle of SPI_CLK in dual io mode for SPI mode 0</p>                                                                                                                                                |
| 13            | R/W | 0   | <p>SDM</p> <p>Master Sample Data Mode</p> <p>1: normal sample mode</p> <p>0: delay sample mode</p> <p>In normal sample mode, SPI master samples the data at the correct edge for each SPI mode;</p> <p>In delay sample mode, SPI master samples data at the edge that is half cycle delayed by the correct edge defined in respective SPI mode.</p>                           |
| 12            | R/W | 0   | <p>FBS</p> <p>First Transmit Bit Select</p> <p>0: MSB first</p> <p>1: LSB first</p>                                                                                                                                                                                                                                                                                           |
| 11            | R/W | 0   | <p>SDC</p> <p>Master Sample Data Control</p> <p>Set this bit to 1 to make the internal read sample point with a delay of half cycle of SPI_CLK. It is used in high speed(&gt;48MHz) read operation to reduce the error caused by the time delay of SPI_CLK propagating between master and slave.</p>                                                                          |

|     |     |     |                                                                                                                                                                                                                                                                                                                                                           |
|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |     | 0: normal operation, do not delay internal read sample point<br>1: delay internal read sample point                                                                                                                                                                                                                                                       |
| 10  | R/W | 0x0 | RPSM<br><br>Rapids mode select<br>Select Rapids mode for high speed write.<br>0: normal write mode<br>1: rapids write mode<br>Note: Can't be written when XCH=1.                                                                                                                                                                                          |
| 9   | R/W | 0x0 | DDB<br><br>Dummy Burst Type<br>0: The bit value of dummy SPI burst is zero<br>1: The bit value of dummy SPI burst is one<br>Note: Can't be written when XCH=1.                                                                                                                                                                                            |
| 8   | R/W | 0x0 | DHB<br><br>Discard Hash Burst<br>In master mode it controls whether discarding unused SPI bursts<br>0: Receiving all SPI bursts in BC period(Burst Counter period)<br>1: Discard unused SPI bursts, only fetching the SPI bursts during dummy burst period. The bursts number is specified by TC(Transmit Counter).<br>Note: Can't be written when XCH=1. |
| 7   | R/W | 0x1 | SS_LEVEL<br><br>When control SS signal manually (SPI_CTRL_REG.SS_OWNER==1), set this bit to '1' or '0' to control the level of SS signal.<br>0: set SS to low<br>1: set SS to high<br>Note: Can't be written when XCH=1.                                                                                                                                  |
| 6   | R/W | 0x0 | SS_OWNER<br><br>SS Output Owner Select<br>Usually, controller sends SS signal automatically with data together. When this bit is set to 1, software must manually write SPI_CTL_REG.SS_LEVEL to 1 or 0 to control the level of SS signal.<br>0: SPI controller<br>1: Software<br>Note: Can't be written when XCH=1.                                       |
| 5:4 | R/W | 0x0 | SS_SEL                                                                                                                                                                                                                                                                                                                                                    |

|   |     |     |                                                                                                                                                                                                                                                                              |
|---|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |     | <p>SPI Chip Select</p> <p>Select one of four external SPI Master/Slave Devices</p> <p>00: SPI_SS0 will be asserted</p> <p>01: SPI_SS1 will be asserted</p> <p>10: SPI_SS2 will be asserted</p> <p>11: SPI_SS3 will be asserted</p> <p>Note: Can't be written when XCH=1.</p> |
| 3 | R/W | 0x0 | <p>SSCTL</p> <p>In master mode, this bit selects the output wave form for the SPI_SSx signal. Only valid when SS_OWNER = 0.</p> <p>0: SPI_SSx remains asserted between SPI bursts</p> <p>1: Negate SPI_SSx between SPI bursts</p> <p>Note: Can't be written when XCH=1.</p>  |
| 2 | R/W | 0x1 | <p>SPOL</p> <p>SPI Chip Select Signal Polarity Control</p> <p>0: Active high polarity (0 = Idle)</p> <p>1: Active low polarity (1 = Idle)</p> <p>Note: Can't be written when XCH=1.</p>                                                                                      |
| 1 | R/W | 0x1 | <p>CPOL</p> <p>SPI Clock Polarity Control</p> <p>0: Active high polarity (0 = Idle)</p> <p>1: Active low polarity (1 = Idle)</p> <p>Note: Can't be written when XCH=1.</p>                                                                                                   |
| 0 | R/W | 0x1 | <p>CPHA</p> <p>SPI Clock/Data Phase Control</p> <p>0: Phase 0 (Leading edge for sample data)</p> <p>1: Phase 1 (Leading edge for setup data)</p> <p>Note: Can't be written when XCH=1.</p>                                                                                   |

#### 6.4.3.4 SPI\_IER\_REG

| 偏移地址 : 0x0010 |     |     | 默认值: 0x0000_0000 |
|---------------|-----|-----|------------------|
| 位置            | 访问  | 默认值 | 描述               |
| 31:14         | R   | 0x0 | Reserved.        |
| 13            | R/W | 0x0 | SS_INT_EN        |



|    |     |     |                                                                                                               |
|----|-----|-----|---------------------------------------------------------------------------------------------------------------|
|    |     |     | SSI Interrupt Enable<br>Chip Select Signal (SSx) from valid state to invalid state<br>0: Disable<br>1: Enable |
| 12 | R/W | 0x0 | TC_INT_EN<br><br>Transfer Completed Interrupt Enable<br>0: Disable<br>1: Enable                               |
| 11 | R/W | 0x0 | TF_UDR_INT_EN<br><br>TXFIFO under run Interrupt Enable<br>0: Disable<br>1: Enable                             |
| 10 | R/W | 0x0 | TF_OVF_INT_EN<br><br>TX FIFO Overflow Interrupt Enable<br>0: Disable<br>1: Enable                             |
| 9  | R/W | 0x0 | RF_UDR_INT_EN<br><br>RXFIFO under run Interrupt Enable<br>0: Disable<br>1: Enable                             |
| 8  | R/W | 0x0 | RF_OVF_INT_EN<br><br>RX FIFO Overflow Interrupt Enable<br>0: Disable<br>1: Enable                             |
| 7  | R   | 0x0 | Reserved.                                                                                                     |
| 6  | R/W | 0x0 | TF_FUL_INT_EN<br><br>TX FIFO Full Interrupt Enable<br>0: Disable<br>1: Enable                                 |
| 5  | R/W | 0x0 | TX_EMP_INT_EN<br><br>TX FIFO Empty Interrupt Enable<br>0: Disable<br>1: Enable                                |

|   |     |     |                                                                                        |
|---|-----|-----|----------------------------------------------------------------------------------------|
| 4 | R/W | 0x0 | TX_ERQ_INT_EN<br><br>TX FIFO Empty Request Interrupt Enable<br>0: Disable<br>1: Enable |
| 3 | R   | 0x0 | Reserved                                                                               |
| 2 | R/W | 0x0 | RF_FUL_INT_EN<br><br>RX FIFO Full Interrupt Enable<br>0: Disable<br>1: Enable          |
| 1 | R/W | 0x0 | RX_EMP_INT_EN<br><br>RX FIFO Empty Interrupt Enable<br>0: Disable<br>1: Enable         |
| 0 | R/W | 0x0 | RF_RDY_INT_EN<br><br>RX FIFO Ready Request Interrupt Enable<br>0: Disable<br>1: Enable |

#### 6.4.3.5 SPI\_STA\_REG

| 偏移地址 : 0x0014 |       |     | 默认值: 0x0000_0032                                                                                                                                                                                                                                                                                                                                               |
|---------------|-------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问    | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                             |
| 31:14         | /     | 0x0 | /                                                                                                                                                                                                                                                                                                                                                              |
| 13            | R/W1C | 0x0 | SSI<br><br>SS Invalid Interrupt<br>When SSI is 1, it indicates that SS has changed from valid state to invalid state. Writing 1 to this bit clears it.                                                                                                                                                                                                         |
| 12            | R/W1C | 0x0 | TC<br><br>Transfer Completed<br>In master mode, it indicates that all bursts specified by BC has been exchanged. In other condition, When set, this bit indicates that all the data in TXFIFO has been loaded in the Shift register, and the Shift register has shifted out all the bits. Writing 1 to this bit clears it.<br>0: Busy<br>1: Transfer Completed |
| 11            | R/W1C | 0x0 | TF_UDF                                                                                                                                                                                                                                                                                                                                                         |

|    |       |     |                                                                                                                                                                                                       |
|----|-------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |       |     | <p>TXFIFO under run</p> <p>This bit is set when if the TXFIFO is under run. Writing 1 to this bit clears it.</p> <p>0: TXFIFO is not under run</p> <p>1: TXFIFO is under run</p>                      |
| 10 | R/W1C | 0x0 | <p>TF_OVF</p> <p>TXFIFO Overflow</p> <p>This bit is set when if the TXFIFO is overflow. Writing 1 to this bit clears it.</p> <p>0: TXFIFO is not overflow</p> <p>1: TXFIFO is overflowed</p>          |
| 9  | R/W1C | 0x0 | <p>RX_UDF</p> <p>RXFIFO Under run</p> <p>When set, this bit indicates that RXFIFO has under run. Writing 1 to this bit clears it.</p>                                                                 |
| 8  | R/W1C | 0x0 | <p>RX_OVF</p> <p>RXFIFO Overflow</p> <p>When set, this bit indicates that RXFIFO has overflowed. Writing 1 to this bit clears it.</p> <p>0: RXFIFO is available.</p> <p>1: RXFIFO has overflowed.</p> |
| 7  | /     | /   | /                                                                                                                                                                                                     |
| 6  | R/W1C | 0x0 | <p>TX_FULL</p> <p>TXFIFO Full</p> <p>This bit is set when if the TXFIFO is full. Writing 1 to this bit clears it.</p> <p>0: TXFIFO is not Full</p> <p>1: TXFIFO is Full</p>                           |
| 5  | R/W1C | 0x1 | <p>TX_EMP</p> <p>TXFIFO Empty</p> <p>This bit is set if the TXFIFO is empty. Writing 1 to this bit clears it.</p> <p>0: TXFIFO contains one or more words.</p> <p>1: TXFIFO is empty</p>              |
| 4  | R/W1C | 0x1 | <p>TX_READY</p> <p>TXFIFO Ready</p> <p>0: TX_WL &gt; TX_TRIG_LEVEL</p>                                                                                                                                |

|   |       |     |                                                                                                                                                                                                                                                     |
|---|-------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |       |     | <p>1: TX_WL &lt;= TX_TRIG_LEVEL</p> <p>This bit is set any time if TX_WL &lt;= TX_TRIG_LEVEL. Writing “1” to this bit clears it. Where TX_WL is the water level of RXFIFO</p>                                                                       |
| 3 | /     | /   | reserved                                                                                                                                                                                                                                            |
| 2 | R/W1C | 0x0 | <p>RX_FULL</p> <p>RXFIFO Full</p> <p>This bit is set when the RXFIFO is full. Writing 1 to this bit clears it.</p> <p>0: Not Full</p> <p>1: Full</p>                                                                                                |
| 1 | R/W1C | 0x1 | <p>RX_EMP</p> <p>RXFIFO Empty</p> <p>This bit is set when the RXFIFO is empty. Writing 1 to this bit clears it.</p> <p>0: Not empty</p> <p>1: empty</p>                                                                                             |
| 0 | R/W1C | 0x0 | <p>RX_RDY</p> <p>RXFIFO Ready</p> <p>0: RX_WL &lt; RX_TRIG_LEVEL</p> <p>1: RX_WL &gt;= RX_TRIG_LEVEL</p> <p>This bit is set any time if RX_WL &gt;= RX_TRIG_LEVEL. Writing “1” to this bit clears it. Where RX_WL is the water level of RXFIFO.</p> |

#### 6.4.3.6 SPI\_FCTL\_REG

| 偏移地址 : 0x0018 |       |     | 默认值: 0x0040_0001                                                                                                                                                                                                                                                                                  |
|---------------|-------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问    | 默认值 | 描述                                                                                                                                                                                                                                                                                                |
| 31            | R/WAC | 0x0 | <p>TX_FIFO_RST</p> <p>TX FIFO Reset</p> <p>Write ‘1’ to this bit will reset the control portion of the TX FIFO and auto clear to ‘0’ when completing reset operation, write to ‘0’ has no effect.</p>                                                                                             |
| 30            | R/W   | 0x0 | <p>TF_TEST_ENB</p> <p>TX Test Mode Enable</p> <p>0: disable</p> <p>1: enable</p> <p>Note: In normal mode, TX FIFO can only be read by SPI controller, write ‘1’ to this bit will switch TX FIFO read and write function to AHB bus. This bit is used to test the TX FIFO, don’t set in normal</p> |

|       |       |      |                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |       |      | operation and don't set RF_TEST and TF_TEST at the same time.                                                                                                                                                                                                                                                                                   |
| 29:25 | /     | /    | /                                                                                                                                                                                                                                                                                                                                               |
| 24    | R/W   | 0x0  | TF_DRQ_EN<br><br>TX FIFO DMA Request Enable<br>0: Disable<br>1: Enable                                                                                                                                                                                                                                                                          |
| 23:16 | R/W   | 0x40 | TX_TRIG_LEVEL<br><br>TX FIFO Empty Request Trigger Level                                                                                                                                                                                                                                                                                        |
| 15    | R/WAC | 0x0  | RF_RST<br><br>RXFIFO Reset<br><br>Write '1' to this bit will reset the control portion of the receiver FIFO, and auto clear to '0' when completing reset operation, write '0' to this bit has no effect.                                                                                                                                        |
| 14    | R/W   | 0x0  | RF_TEST<br><br>RX Test Mode Enable<br>0: Disable<br>1: Enable<br><br>Note: In normal mode, RX FIFO can only be written by SPI controller, write '1' to this bit will switch RX FIFO read and write function to AHB bus. This bit is used to test the RX FIFO, don't set in normal operation and don't set RF_TEST and TF_TEST at the same time. |
| 13:9  | /     | /    | /                                                                                                                                                                                                                                                                                                                                               |
| 8     | R/W   | 0x0  | RF_DRQ_EN<br><br>RX FIFO DMA Request Enable<br>0: Disable<br>1: Enable                                                                                                                                                                                                                                                                          |
| 7:0   | R/W   | 0x1  | RX_TRIG_LEVEL<br><br>RX FIFO Ready Request Trigger Level                                                                                                                                                                                                                                                                                        |

#### 6.4.3.7 SPI\_FST\_REG

| 偏移地址：0x001C |    | 默认值: 0x0000_0000 |       |
|-------------|----|------------------|-------|
| 位置          | 访问 | 默认值              | 描述    |
| 31          | R  | 0x0              | TB_WR |

|       |   |     |                                                                                                                                                                                             |
|-------|---|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |   |     | TX FIFO Write Buffer Write Enable                                                                                                                                                           |
| 30:28 | R | 0x0 | TB_CNT<br><br>TX FIFO Write Buffer Counter<br>These bits indicate the number of words in TX FIFO Write Buffer                                                                               |
| 27:24 | R | 0x0 | Reserved                                                                                                                                                                                    |
| 23:16 | R | 0x0 | TF_CNT<br><br>TX FIFO Counter<br><br>These bits indicate the number of words in TX FIFO<br>0: 0 byte in TX FIFO<br>1: 1 byte in TX FIFO<br>...<br>64 64 bytes in TX FIFO<br>other: Reserved |
| 15    | R | 0x0 | RB_WR<br><br>RX FIFO Read Buffer Write Enable                                                                                                                                               |
| 14:12 | R | 0x0 | RB_CNT<br><br>RX FIFO Read Buffer Counter<br>These bits indicate the number of words in RX FIFO Read Buffer                                                                                 |
| 11:8  | R | 0x0 | Reserved                                                                                                                                                                                    |
| 7:0   | R | 0x0 | RF_CNT<br><br>RX FIFO Counter<br>These bits indicate the number of words in RX FIFO<br>0: 0 byte in RX FIFO<br>1: 1 byte in RX FIFO<br>...<br>64: 64 bytes in RX FIFO<br>other: Reserved    |

#### 6.4.3.8 SPI\_WAIT\_REG

| 偏移地址：0x0020 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:20       | /  | /   | /                |

|       |     |     |                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19:16 | R/W | 0x0 | <p>SWC</p> <p>Dual mode direction switch wait clock counter (for master mode only).</p> <p>0: No wait states inserted</p> <p>n: n SPI_SCLK wait states inserted</p> <p>Note: These bits control the number of wait states to be inserted before start dual data transfer in dual SPI mode. The SPI module counts SPI_SCLK by SWC for delaying next word data transfer.</p> <p>Note: Can't be written when XCH=1.</p> |
| 15:0  | R/W | 0x0 | <p>WCC</p> <p>Wait Clock Counter (In Master mode)</p> <p>These bits control the number of wait states to be inserted in data transfers. The SPI module counts SPI_SCLK by WCC for delaying next word data transfer.</p> <p>0: No wait states inserted</p> <p>N: N SPI_SCLK wait states inserted</p>                                                                                                                  |

#### 6.4.3.9 SPI\_CCTR\_REG

| 偏移地址 : 0x0024 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                      |
|---------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                    |
| 31:13         | /   | /   | /                                                                                                                                                                                                                     |
| 12            | R/W | 0x0 | <p>DRS</p> <p>Divide Rate Select (Master Mode Only)</p> <p>0: Select Clock Divide Rate 1</p> <p>1: Select Clock Divide Rate 2</p>                                                                                     |
| 11:8          | R/W | 0x0 | <p>CDR1</p> <p>Clock Divide Rate 1 (Master Mode Only)</p> <p>The SPI_SCLK is determined according to the following equation:<br/> <math display="block">\text{SPI\_CLK} = \text{SRC\_CLK} / 2^n</math> </p>           |
| 7:0           | R/W | 0x2 | <p>CDR2</p> <p>Clock Divide Rate 2 (Master Mode Only)</p> <p>The SPI_SCLK is determined according to the following equation:<br/> <math display="block">\text{SPI\_CLK} = \text{SRC\_CLK} / (2 * (n + 1))</math> </p> |

#### 6.4.3.10 SPI\_BC\_REG

| 偏移地址：0x0030 |     |     | 默认值: 0x0000_0000                                                                                                                                                              |
|-------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                            |
| 31:24       | /   | /   | /                                                                                                                                                                             |
| 23:0        | R/W | 0x0 | <p>MBC</p> <p>Master Burst Counter</p> <p>In master mode, this field specifies the total burst number .</p> <p>0: 0 burst</p> <p>1: 1 burst</p> <p>...</p> <p>N: N bursts</p> |

#### 6.4.3.11 SPI\_TC\_REG

| 偏移地址：0x0034 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                            |
|-------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                          |
| 31:24       | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                           |
| 23:0        | R/W | 0x0 | <p>MWTC</p> <p>Master Write Transmit Counter</p> <p>In master mode, this field specifies the burst number that should be sent to TXFIFO before automatically sending dummy burst . For saving bus bandwidth, the dummy burst (all zero bits or all one bits) is sent by SPI Controller automatically.</p> <p>0: 0 burst</p> <p>1: 1 burst</p> <p>...</p> <p>N: N bursts</p> |

#### 6.4.3.12 SPI\_BCC\_REG

| 偏移地址：0x0038 |     |     | 默认值: 0x0000_0000                                                                                         |
|-------------|-----|-----|----------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                       |
| 31:29       | R   | 0x0 | Reserved                                                                                                 |
| 28          | R/W | 0x0 | <p>DRM</p> <p>Master Dual Mode RX Enable</p> <p>0: RX use single-bit mode</p> <p>1: RX use dual mode</p> |



|       |     |     |                                                                                                                                                                                                                                                                                                                                                               |
|-------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |     |     | Note: Can't be written when XCH=1.                                                                                                                                                                                                                                                                                                                            |
| 27:24 | R/W | 0x0 | <p>DBC</p> <p>Master Dummy Burst Counter</p> <p>In master mode, this field specifies the burst number that should be sent before receive in dual SPI mode. The data is don't care by the device.</p> <p>0: 0 burst</p> <p>1: 1 burst</p> <p>...</p> <p>N: N bursts</p> <p>Note: Can't be written when XCH=1.</p>                                              |
| 23:0  | R/W | 0x0 | <p>STC</p> <p>Master Single Mode Transmit Counter</p> <p>In master mode, this field specifies the burst number that should be sent in single mode before automatically sending dummy burst. This is the first transmit counter in all bursts.</p> <p>0: 0 burst</p> <p>1: 1 burst</p> <p>...</p> <p>N: N bursts</p> <p>Note: Can't be written when XCH=1.</p> |

#### 6.4.3.13 SPI\_NDMA\_MODE\_CTRL\_REG

| 偏移地址 : 0x0088 |     |      | 默认值: 0x0000_00A5                                                                                                                                                   |
|---------------|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值  | 描述                                                                                                                                                                 |
| 31:8          | /   | /    | /                                                                                                                                                                  |
| 7:0           | R/W | 0xA5 | <p>NDMA_MODE_CTL</p> <p>0xEA: NDMA handshake mode</p> <p>Note: NDMA wait mode doesn't care this value. 0xA5 can be used in handshake mode, but 0xEA is better.</p> |

#### 6.4.3.14 SPI\_TXD\_REG

| 偏移地址 : 0x0200 |     |     | 默认值: 0x0000_0000 |
|---------------|-----|-----|------------------|
| 位置            | 访问  | 默认值 | 描述               |
| 31:0          | R/W | 0x0 | TDATA            |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <p>Transmit Data</p> <p>This register can be accessed in byte, half-word or word unit by AHB. In byte accessing method, if there are rooms in TXFIFO, one burst data is written to TXFIFO and the depth is increased by 1. In half-word accessing method, two SPI burst data are written and the TXFIFO depth is increase by 2. In word accessing method, four SPI burst data are written and the TXFIFO depth is increased by 4.</p> <p>Note: This address is writing-only if TF_TEST is '0' , and if TF_TEST is set to '1' , this address is readable and writable to test the TX FIFO through the AHB bus.</p> |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 6.4.3.15 SPI\_RXD\_REG

| 偏移地址 : 0x0300 |    |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 31:0          | R  | 0x0 | <p>RDATA</p> <p>Receive Data</p> <p>This register can be accessed in byte, half-word or word unit by AHB. In byte accessing method, if there are data in RXFIFO, the top word is returned and the RXFIFO depth is decreased by 1. In half-word accessing method, two SPI bursts are returned and the RXFIFO depth is decrease by 2. In word accessing method, the four SPI bursts are returned and the RXFIFO depth is decreased by 4.</p> <p>Note: This address is read-only if RF_TEST is '0' , and if RF_TEST is set to '1' , this address is readable and writable to test the RX FIFO through the AHB bus.</p> |

## 6.5 TWI 接口

### 6.5.1 概述

TWI控制器作为CPU主机与串行TWI总线之间的接口而设计。它可以支持所有标准的TWI传输，包括从设备和主设备。与TWI总线的通信是使用中断或轮询握手按字节进行的。TWI控制器可在标准模式(100K bps)或快速模式下工作，支持高达400K bps的数据速率。特定应用下支持多主和10位寻址模式。从模式下也支持通用呼叫寻址。

TWI控制器具有以下特性：

- 支持软件配置为从设备或主设备
- 支持重复 START 信号
- 支持多主系统
- 允许使用 TWI 总线进行 10 位寻址
- 执行仲裁和时钟同步
- 低地址和普通呼叫地址检测
- 在地址检测上中断
- 支持速度高达 400K bps（高速模式）

- 允许在宽范围的时钟输入频率内进行操作

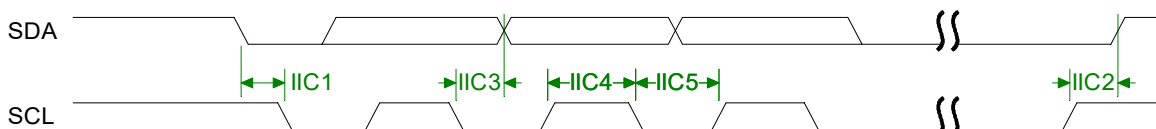
传输的数据总是以8位(字节)为单位，后面跟着一个确认位。每次传输可以传输的字节数是不受限制的。数据首先通过MSB串行传输。在数据传输的每个字节之间，接收设备将保持时钟线SCL低，迫使发送器进入等待状态，同时等待微处理器的响应。

数据传输必须有应答。除了每个字节之间的SCL保持之外，包括与应答相关的时钟周期在内，时钟线一直由主设备驱动。在每个字节发送后，发送器释放SDA线，以允许接收端拉低SDA线并发送一个应答信号(或将其保持高电平以发送一个“无应答”)给发送器。

当一个从端接收器不应答从端地址(因为没有资源可用而无法接收)时，数据线必须由从端保持高电平，这样主端就可以生成一个STOP条件来终止传输。从接收器也可以通过保持应答信号为高电平，以表示不期望在传输期间发送更多的数据。此时，主设备应该产生停止条件以中止转移。

下图说明了在TWI串行总线上SDA信号线和SCL信号线的关系。

图 6-6 TWI 时序图



## 6.5.2 TWI 寄存器列表

| 模块名              | 基地址        |                              |
|------------------|------------|------------------------------|
| TWI Controller 0 | 0x40041C00 |                              |
| TWI Controller 1 | 0X40042000 |                              |
| 寄存器名             | 偏移地址       | 描述                           |
| TWI_ADDR_REG     | 0x0000     | TWI Slave address            |
| TWI_XADDR_REG    | 0x0004     | TWI Extended slave address   |
| TWI_DATA_REG     | 0x0008     | TWI Data byte                |
| TWI_CNTR_REG     | 0x000C     | TWI Control register         |
| TWI_STAT_REG     | 0x0010     | TWI Status register          |
| TWI_CCR_REG      | 0x0014     | TWI Clock control register   |
| TWI_SRST_REG     | 0x0018     | TWI Software reset           |
| TWI_EFR_REG      | 0x001C     | TWI Enhance Feature register |
| TWI_LCR_REG      | 0x0020     | TWI Line Control register    |

### 6.5.3 TWI 寄存器描述

#### 6.5.3.1 TWI\_ADDR\_REG

| 偏移地址：0x0000 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 31:8        | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7:1         | R/W | 0   | <p>SLA</p> <p>Slave address</p> <p>- 7-bit addressing</p> <p>SLA6, SLA5, SLA4, SLA3, SLA2, SLA1, SLA0</p> <p>- 10-bit addressing</p> <p>1, 1, 1, 1, 0, SLAX[9:8]</p> <p>For 7-bit addressing:</p> <p>SLA6 – SLA0 is the 7-bit address of the TWI when in slave mode. When the TWI receives this address after a START condition, it will generate an interrupt and enter slave mode. (SLA6 corresponds to the first bit received from the TWI bus.) If GCE is set to ‘1’, the TWI will also recognize the general call address (00h).</p> <p>For 10-bit addressing:</p> <p>When the address received starts with 11110b, the TWI recognizes this as the first part of a 10-bit address and if the next two bits match ADDR[2:1] (i.e. SLAX9 and SLAX8 of the device’s extended address), it sends an ACK. (The device does not generate an interrupt at this point.) If the next byte of the address matches the XADDR register (SLAX7 – SLAX0), the TWI generates an interrupt and goes into slave mode.</p> |
| 0           | R/W | 0   | <p>GCE</p> <p>General call address enable</p> <p>0: Disable</p> <p>1: Enable</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

#### 6.5.3.2 TWI\_XADDR\_REG

| 偏移地址：0x0004 |     |     | 默认值: 0x0000_0000                                         |
|-------------|-----|-----|----------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                       |
| 31:8        | /   | /   | /                                                        |
| 7:0         | R/W | 0   | <p>SLAX</p> <p>Extend Slave Address</p> <p>SLAX[7:0]</p> |

### 6.5.3.3 TWI\_DATA\_REG

| 偏移地址：0x0008 |     |     | 默认值: 0x0000_0000                                       |
|-------------|-----|-----|--------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                     |
| 31:8        | /   | /   | /                                                      |
| 7:0         | R/W | 0   | TWI_DATA<br><br>Data byte for transmitting or received |

### 6.5.3.4 TWI\_ADDR\_REG

| 偏移地址：0x000C |       |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|-------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问    | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 31:8        | /     | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 7           | R/W   | 0   | INT_EN<br><br>Interrupt Enable<br>1' b0: The interrupt line always low<br>1' b1: The interrupt line will go high when INT_FLAG is set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6           | R/W   | 0   | BUS_EN<br><br>TWI Bus Enable<br>1' b0: The TWI bus inputs ISDA/ISCL are ignored and the TWI Controller will not respond to any address on the bus<br>1' b1: The TWI will respond to calls to its slave address – and to the general call address if the GCE bit in the ADDR register is set.<br>Notes: In master operation mode, this bit should be set to '1'                                                                                                                                                                                                                                                                                                             |
| 5           | R/WAC | 0   | M_STA<br><br>Master Mode Start<br><br>When M_STA is set to '1', TWI Controller enters master mode and will transmit a START condition on the bus when the bus is free. If the M_STA bit is set to '1' when the TWI Controller is already in master mode and one or more bytes have been transmitted, then a repeated START condition will be sent. If the M_STA bit is set to '1' when the TWI is being accessed in slave mode, the TWI will complete the data transfer in slave mode then enter master mode when the bus has been released.<br><br>The M_STA bit is cleared automatically after a START condition has been sent: writing a '0' to this bit has no effect. |

|     |       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|-------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | R/W1C | 0 | <p>M_STP</p> <p>Master Mode Stop</p> <p>If M_STP is set to ‘1’ in master mode, a STOP condition is transmitted on the TWI bus. If the M_STP bit is set to ‘1’ in slave mode, the TWI will behave as if a STOP condition has been received, but no STOP condition will be transmitted on the TWI bus. If both M_STA and M_STP bits are set, the TWI will first transmit the STOP condition (if in master mode) then transmit the START condition.</p> <p>The M_STP bit is cleared automatically: writing a ‘0’ to this bit has no effect.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3   | R/W1C | 0 | <p>INT_FLAG</p> <p>Interrupt Flag</p> <p>INT_FLAG is automatically set to ‘1’ when any of 28 (out of the possible 29) states is entered (see ‘STAT Register’ below). The only state that does not set INT_FLAG is state F8h. If the INT_EN bit is set, the interrupt line goes high when IFLG is set to ‘1’. If the TWI is operating in slave mode, data transfer is suspended when INT_FLAG is set and the low period of the TWI bus clock line (SCL) is stretched until ‘1’ is written to INT_FLAG. The TWI clock line is then released and the interrupt line goes low.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2   | R/W   | 0 | <p>A_ACK</p> <p>Assert Acknowledge</p> <p>When A_ACK is set to ‘1’, an Acknowledge (low level on SDA) will be sent during the acknowledge clock pulse on the TWI bus if:</p> <ol style="list-style-type: none"> <li>1. Either the whole of a matching 7-bit slave address or the first or the second byte of a matching 10-bit slave address has been received.</li> <li>2. The general call address has been received and the GCE bit in the ADDR register is set to ‘1’.</li> <li>3. A data byte has been received in master or slave mode.</li> </ol> <p>When A_ACK is ‘0’, a Not Acknowledge (high level on SDA) will be sent when a data byte is received in master or slave mode.</p> <p>If A_ACK is cleared to ‘0’ in slave transmitter mode, the byte in the DATA register is assumed to be the ‘last byte’. After this byte has been transmitted, the TWI will enter state C8h then return to the idle state (status code F8h) when INT_FLAG is cleared.</p> <p>The TWI will not respond as a slave unless A_ACK is set.</p> |
| 1:0 | /     | / | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

### 6.5.3.5 TWI\_STAT\_REG

| 偏移地址 : 0x0010 |    |      | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 31:8          | /  | /    | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7:0           | R  | 0xF8 | <p>STA</p> <p>Status Information Byte</p> <p>Code Status</p> <p>0x00: Bus error</p> <p>0x08: START condition transmitted</p> <p>0x10: Repeated START condition transmitted</p> <p>0x18: Address + Write bit transmitted, ACK received</p> <p>0x20: Address + Write bit transmitted, ACK not received</p> <p>0x28: Data byte transmitted in master mode, ACK received</p> <p>0x30: Data byte transmitted in master mode, ACK not received</p> <p>0x38: Arbitration lost in address or data byte</p> <p>0x40: Address + Read bit transmitted, ACK received</p> <p>0x48: Address + Read bit transmitted, ACK not received</p> <p>0x50: Data byte received in master mode, ACK transmitted</p> <p>0x58: Data byte received in master mode, not ACK transmitted</p> <p>0x60: Slave address + Write bit received, ACK transmitted</p> <p>0x68: Arbitration lost in address as master, slave address + Write bit received, ACK transmitted</p> <p>0x70: General Call address received, ACK transmitted</p> <p>0x78: Arbitration lost in address as master, General Call address received, ACK transmitted</p> <p>0x80: Data byte received after slave address received, ACK transmitted</p> <p>0x88: Data byte received after slave address received, not ACK transmitted</p> <p>0x90: Data byte received after General Call received, ACK transmitted</p> <p>0x98: Data byte received after General Call received, not ACK transmitted</p> <p>0xA0: STOP or repeated START condition received in slave mode</p> <p>0xA8: Slave address + Read bit received, ACK transmitted</p> <p>0xB0: Arbitration lost in address as master, slave address + Read bit received, ACK transmitted</p> <p>0xB8: Data byte transmitted in slave mode, ACK received</p> <p>0xC0: Data byte transmitted in slave mode, ACK not received</p> <p>0xC8: Last byte transmitted in slave mode, ACK received</p> <p>0xD0: Second Address byte + Write bit transmitted, ACK received</p> <p>0xD8: Second Address byte + Write bit transmitted, ACK not received</p> <p>0xF8: No relevant status information, INT_FLAG=0</p> <p>Others: Reserved</p> |

### 6.5.3.6 TWI\_CCR\_REG

| 偏移地址 : 0x0014 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 31:7          | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6:3           | R/W | 0   | CLK_M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2:0           | R/W | 0   | CLK_N<br><br>The TWI bus is sampled by the TWI at the frequency defined by F0:<br>$F_{\text{samp}} = F_0 = F_{\text{in}} / 2^{\text{CLK\_N}}$<br><br>The TWI OSCL output frequency, in master mode, is F1 / 10:<br>$F_1 = F_0 / (\text{CLK\_M} + 1)$<br>$F_{\text{oscl}} = F_1 / 10 = F_{\text{in}} / (2^{\text{CLK\_N}} * (\text{CLK\_M} + 1) * 10)$<br><br>For Example:<br>Fin = 48Mhz (APB clock input)<br>For 400kHz full speed 2Wire, CLK_N = 2, CLK_M=2<br>$F_0 = 48\text{M} / 2^2 = 12\text{Mhz}$ , $F_1 = F_0 / (10 * (2 + 1)) = 0.4\text{Mhz}$<br>For 100Khz standard speed 2Wire, CLK_N=2, CLK_M=11<br>$F_0 = 48\text{M} / 2^2 = 12\text{Mhz}$ , $F_1 = F_0 / (10 * (11 + 1)) = 0.1\text{Mhz}$ |

### 6.5.3.7 TWI\_SRST\_REG

| 偏移地址 : 0x0018 |       |     | 默认值: 0x0000_0000                                                                                                                |
|---------------|-------|-----|---------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问    | 默认值 | 描述                                                                                                                              |
| 31:1          | /     | /   | /                                                                                                                               |
| 0             | R/W1C | 0   | SOFT_RST<br><br>Soft Reset<br><br>Write '1' to this bit to reset the TWI and clear to '0' when completing Soft Reset operation. |

### 6.5.3.8 TWI\_EFR\_REG

| 偏移地址 : 0x001C |     |     | 默认值: 0x0000_0000                                        |
|---------------|-----|-----|---------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                      |
| 31:2          | /   | /   | /                                                       |
| 0:1           | R/W | 0   | DBN<br><br>Data Byte number follow Read Command Control |



|  |  |  |                                                                                                                                                                                                |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | No Data Byte to be wrote after read command<br>Only 1 byte data to be wrote after read command<br>2 bytes data can be wrote after read command<br>3 bytes data can be wrote after read command |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 6.5.3.9 TWI\_LCR\_REG

| 偏移地址 : 0x0020 |     |     | 默认值: 0x0000_003A                                                                                                                                                                                                  |
|---------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                |
| 31:6          | /   | /   | /                                                                                                                                                                                                                 |
| 5             | R   | 1   | SCL_STATE<br><br>Current state of TWI_SCL<br><br>0 – low<br>1 - high                                                                                                                                              |
| 4             | R   | 1   | SDA_STATE<br><br>Current state of TWI_SDA<br><br>0 – low<br>1 - high                                                                                                                                              |
| 3             | R/W | 1   | SCL_CTL<br><br>TWI_SCL line state control bit<br>When line control mode is enabled (bit[2] set), value of this bit decide the output level of TWI_SCL<br><br>0 – output low level<br>1 – output high level        |
| 2             | R/W | 0   | SCL_CTL_EN<br><br>TWI_SCL line state control enable<br>When this bit is set, the state of TWI_SCL is control by the value of bit[3].<br>0-disable TWI_SCL line control mode<br>1-enable TWI_SCL line control mode |
| 1             | R/W | 1   | SDA_CTL<br><br>TWI_SDA line state control bit<br>When line control mode is enabled (bit[0] set), value of this bit decide the output level of TWI_SDA<br><br>0 – output low level                                 |

|   |     |   |                                                                                                                                                                                                                                      |
|---|-----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |   | 1 – output high level                                                                                                                                                                                                                |
| 0 | R/W | 0 | <p>SDA_CTL_EN</p> <p>TWI_SDA line state control enable</p> <p>When this bit is set, the state of TWI_SDA is control by the value of bit[1].</p> <p>0-disable TWI_SDA line control mode</p> <p>1-enable TWI_SDA line control mode</p> |

## 6.6 CIR 接口

### 6.6.1 概述

#### 6.6.1.1 CIR 接收控制器 (IRRX)

IRRX 包括以下特点:

- 全物理层实现
- 支持 CIR 远程控制
- 64x8 位 FIFO 数据缓冲区
- 可配置 FIFO 阈值

当在空中发现 CIR 信号时, CIR 接收机在可配置的频率下采样输入信号, 并记录这些采样数据到 RX FIFO。CIR 接收机采用游程码(RLC)对脉冲宽度进行编码。编码数据在一个 64 级 8 位宽的 RX FIFO 缓冲, MSB 位用于记录接收 CIR 信号的极性。高电平表示为逻辑“1”, 低电平表示为逻辑“0”。其余 7 位用于表示 RLC 的长度, 最大长度为 128。如果一个电平(高电平或低电平)的持续周期超过 128, 则再使用另一个字节。

在空气中, 总是有一些噪音。可以通过设置一个阈值来滤除噪声, 降低系统负荷, 提高系统稳定性。

#### 6.6.1.2 CIR 发射控制器 (IRTX)

IRTX 包括以下特点:

- 支持工业标准 AMBA 外围总线(APB), 全兼容 AMBA 规范 2.0 修订版
- 全物理层实现
- 128 字节的 FIFO 数据缓冲区
- 可配置的载波频率
- 支持中断和 DMA 支持
- 支持 DMA 握手和等待模式

CIR 模式使用可变脉宽调制技术以实现远程控制应用下各种格式的红外编码。

下图为 NEC 协议 IRTX 调制技术示例: 逻辑“1”传输时间为 2.25ms, 逻辑“0”传输时间为 1.12ms。



|                |        |                                               |
|----------------|--------|-----------------------------------------------|
| CIR_IDC_H_REG  | 0x000C | CIR Transmit Idle Duration threshold Register |
| CIR_IDC_L_REG  | 0x0010 | CIR Transmit Idle Duration threshold Register |
| CIR_TICR_H_REG | 0x0014 | CIR Transmit Idle Counter Register            |
| CIR_TICR_L_REG | 0x0018 | CIR Transmit Idle Counter Register            |
| CIR_TEL_REG    | 0x0020 | CIR TX FIFO empty Level Register              |
| CIR_TXINT_REG  | 0x0024 | CIR Transmit Interrupt Control Register       |
| CIR_TAC_REG    | 0x0028 | CIR Transmit FIFO Available Counter Register  |
| CIR_TXSTA_REG  | 0x002C | CIR Transmit Status Register                  |
| CIR_TXT_REG    | 0x0030 | CIR Transmit Threshold Register               |
| CIR_DMA_REG    | 0x0034 | CIR DMA Control Register                      |
| CIR_TXFIFO_REG | 0x0080 | CIR Transmit FIFO Data Register               |

### 6.6.3 CIR Receiver 寄存器描述

#### 6.6.3.1 CIR\_CTL\_REG

| 偏移地址：0x0000 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                |
|-------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                              |
| 31:9        | /   | /   | /                                                                                                                                                                                                                                                                                               |
| 8           | R/W | 0x0 | CGPO<br>General Program Output (GPO) Control in CIR mode for TX Pin<br>0: Low level<br>1: High level                                                                                                                                                                                            |
| 7:6         | R/W | 0x0 | Active Pulse Accept Mode<br>00: Start to accept data after detect de-bounce pulse edge<br>01: Start to accept data after detect de-bounce pulse edge<br>10: Start to accept data after detect low voltage de-bounce pulse<br>11: Start to accept data after detect high voltage de-bounce pulse |
| 5:4         | R/W | 0x0 | CIR ENABLE<br>00~10: Reserved<br>11: CIR mode enable                                                                                                                                                                                                                                            |
| 3:2         | /   | /   | /                                                                                                                                                                                                                                                                                               |
| 1           | R/W | 0x0 | RXEN<br>Receiver Block Enable<br>0: Disable                                                                                                                                                                                                                                                     |

|   |     |     |                                                                                                                                              |
|---|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |     | 1: Enable                                                                                                                                    |
| 0 | R/W | 0x0 | GEN<br>Global Enable<br>A disable on this bit overrides any other block or channel enables and flushes all FIFOs.<br>0: Disable<br>1: Enable |

#### 6.6.3.2 CIR\_RXCTL\_REG

| 偏移地址 : 0x0010 |     |     | 默认值: 0x0000_0004                                                                                     |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                   |
| 31:3          | /   | /   | /                                                                                                    |
| 2             | R/W | 0x1 | RPPI<br>Receiver Pulse Polarity Invert<br>0: Not invert receiver signal<br>1: Invert receiver signal |
| 1:0           | /   | /   | /                                                                                                    |

#### 6.6.3.3 CIR\_RXFIFO\_REG

| 偏移地址 : 0x0020 |    |     | 默认值: 0x0000_0000   |
|---------------|----|-----|--------------------|
| 位置            | 访问 | 默认值 | 描述                 |
| 31:8          | /  | /   | /                  |
| 7:0           | R  | 0x0 | Receiver Byte FIFO |

#### 6.6.3.4 CIR\_RXINT\_REG

| 偏移地址 : 0x002C |     |     | 默认值: 0x0000_0000                                                                                                                                                                |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                              |
| 31:16         | /   | /   | /                                                                                                                                                                               |
| 15:8          | R/W | 0x0 | RAL<br>RX FIFO Available Received Byte Level for interrupt and DMA request<br>TRIGGER_LEVEL = RAL + 1                                                                           |
| 5             | R/W | 0x0 | DRQ_EN<br>RX FIFO DMA Enable<br>0: Disable<br>1: Enable<br><br>When set to '1', the Receiver FIFO DRQ is asserted if reaching RAL. The DRQ is de-asserted when condition fails. |

|     |     |     |                                                                                                                                                                                                    |
|-----|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | R/W | 0x0 | RAI_EN<br>RX FIFO Available Interrupt Enable<br>0: Disable<br>1: Enable<br><br>When set to '1', the Receiver FIFO IRQ is asserted if reaching RAL.<br>The IRQ is de-asserted when condition fails. |
| 3:2 | /   | /   | /                                                                                                                                                                                                  |
| 1   | R/W | 0x0 | RPEI_EN<br>Receiver Packet End Interrupt Enable<br>0: Disable<br>1: Enable                                                                                                                         |
| 0   | R/W | 0x0 | ROI_EN<br>Receiver FIFO Overrun Interrupt Enable<br>0: Disable<br>1: Enable                                                                                                                        |

#### 6.6.3.5 CIR\_RXSTA\_REG

| 偏移地址 : 0x0030 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                  |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                |
| 31:15         | /   | /   | /                                                                                                                                                                                                 |
| 14:8          | R   | 0x0 | RAC<br>RX FIFO Available Counter<br>0: No available data in RX FIFO<br>1: 1 byte available data in RX FIFO<br>2: 2 byte available data in RX FIFO<br>...<br>64: 64 byte available data in RX FIFO |
| 7             | R   | 0x0 | STAT<br>Status of CIR<br><br>0x0 – Idle<br><br>0x1 – busy                                                                                                                                         |
| 6:5           | /   | /   | /                                                                                                                                                                                                 |
| 4             | R/W | 0x0 | RA<br>RX FIFO Available<br>0: RX FIFO not available according its level<br>1: RX FIFO available according its level<br><br>This bit is cleared by writing a '1' .                                 |
| 3:2           | /   | /   | /                                                                                                                                                                                                 |

|   |     |     |                                                                                                                                                                                                                                                                                                                                      |
|---|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | R/W | 0x0 | <p>RPE</p> <p>Receiver Packet End Flag</p> <p>0: STO was not detected. In CIR mode, one CIR symbol is receiving or not detected.</p> <p>1: STO field or packet abort symbol (7' b0000,000 and 8' b0000,0000 for MIR and FIR) is detected. In CIR mode, one CIR symbol is received.</p> <p>This bit is cleared by writing a '1' .</p> |
| 0 | R/W | 0x0 | <p>ROI</p> <p>Receiver FIFO Overrun</p> <p>0: Receiver FIFO not overrun</p> <p>1: Receiver FIFO overrun</p> <p>This bit is cleared by writing a '1' .</p>                                                                                                                                                                            |

#### 6.6.3.6 CIR\_CONFIG\_REG

| 偏移地址 : 0x0034 |     |      | 默认值: 0x0000_1828                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值  | 描述                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 31:25         | /   | /    | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 24            | R/W | 0x0  | <p>SCS2</p> <p>Bit2 of Sample Clock Select for CIR</p> <p>This bit is defined by SCS bits below.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 23            | R/W | 0x0  | <p>ATHC</p> <p>Active Threshold Control for CIR</p> <p>0x0 –ATHR in Unit of (Sample Clock)</p> <p>0x1 –ATHR in Unit of (128*Sample Clocks)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 22:16         | R/W | 0x0  | <p>ATHR</p> <p>Active Threshold for CIR</p> <p>These bits control the duration of CIR from Idle to Active State. The duration can be calculated by ((ATHR + 1)*(ATHC? Sample Clock: 128*Sample Clock)).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 15:8          | R/W | 0x18 | <p>ITHR</p> <p>Idle Threshold for CIR</p> <p>The Receiver uses it to decide whether the CIR command has been received. If there is no CIR signal on the air, the receiver is staying in IDLE status. One active pulse will bring the receiver from IDLE status to Receiving status. After the CIR is end, the inputting signal will keep the specified level (high or low level) for a long time. The receiver can use this idle signal duration to decide that it has received the CIR command. The corresponding flag is asserted. If the corresponding interrupt is enable, the interrupt line is asserted to CPU.</p> <p>When the duration of signal keeps one status (high or low level) for the specified duration ( (ITHR + 1)*128 sample_clk ), this means that the previous CIR command has been finished.</p> |
| 7:2           | R/W | 0xa  | <p>NTHR</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |        |        | <p>Noise Threshold for CIR</p> <p>When the duration of signal pulse (high or low level) is less than NTHR, the pulse is taken as noise and should be discarded by hardware.</p> <p>0: all samples are recorded into RX FIFO</p> <p>1: If the signal is only one sample duration, it is taken as noise and discarded.</p> <p>2: If the signal is less than (<math>\leq</math>) two sample duration, it is taken as noise and discarded.</p> <p>...</p> <p>61: if the signal is less than (<math>\leq</math>) sixty-one sample duration, it is taken as noise and discarded.</p>                                                                                              |      |        |        |              |   |   |   |           |   |   |   |            |   |   |   |            |   |   |   |            |   |   |   |        |   |   |   |          |   |   |   |          |   |   |   |          |
|------|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|--------|--------------|---|---|---|-----------|---|---|---|------------|---|---|---|------------|---|---|---|------------|---|---|---|--------|---|---|---|----------|---|---|---|----------|---|---|---|----------|
| 1:0  | R/W    | 0x0    | <p>SCS</p> <p>Sample Clock Select for CIR</p> <table border="1"> <thead> <tr> <th>SCS2</th><th>SCS[1]</th><th>SCS[0]</th><th>Sample Clock</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>0</td><td>ir_clk/64</td></tr> <tr> <td>0</td><td>0</td><td>1</td><td>ir_clk/128</td></tr> <tr> <td>0</td><td>1</td><td>0</td><td>ir_clk/256</td></tr> <tr> <td>0</td><td>1</td><td>1</td><td>ir_clk/512</td></tr> <tr> <td>1</td><td>0</td><td>0</td><td>ir_clk</td></tr> <tr> <td>1</td><td>0</td><td>1</td><td>Reserved</td></tr> <tr> <td>1</td><td>1</td><td>0</td><td>Reserved</td></tr> <tr> <td>1</td><td>1</td><td>1</td><td>Reserved</td></tr> </tbody> </table> | SCS2 | SCS[1] | SCS[0] | Sample Clock | 0 | 0 | 0 | ir_clk/64 | 0 | 0 | 1 | ir_clk/128 | 0 | 1 | 0 | ir_clk/256 | 0 | 1 | 1 | ir_clk/512 | 1 | 0 | 0 | ir_clk | 1 | 0 | 1 | Reserved | 1 | 1 | 0 | Reserved | 1 | 1 | 1 | Reserved |
| SCS2 | SCS[1] | SCS[0] | Sample Clock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |        |        |              |   |   |   |           |   |   |   |            |   |   |   |            |   |   |   |            |   |   |   |        |   |   |   |          |   |   |   |          |   |   |   |          |
| 0    | 0      | 0      | ir_clk/64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |        |        |              |   |   |   |           |   |   |   |            |   |   |   |            |   |   |   |            |   |   |   |        |   |   |   |          |   |   |   |          |   |   |   |          |
| 0    | 0      | 1      | ir_clk/128                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |        |        |              |   |   |   |           |   |   |   |            |   |   |   |            |   |   |   |            |   |   |   |        |   |   |   |          |   |   |   |          |   |   |   |          |
| 0    | 1      | 0      | ir_clk/256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |        |        |              |   |   |   |           |   |   |   |            |   |   |   |            |   |   |   |            |   |   |   |        |   |   |   |          |   |   |   |          |   |   |   |          |
| 0    | 1      | 1      | ir_clk/512                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |        |        |              |   |   |   |           |   |   |   |            |   |   |   |            |   |   |   |            |   |   |   |        |   |   |   |          |   |   |   |          |   |   |   |          |
| 1    | 0      | 0      | ir_clk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |        |        |              |   |   |   |           |   |   |   |            |   |   |   |            |   |   |   |            |   |   |   |        |   |   |   |          |   |   |   |          |   |   |   |          |
| 1    | 0      | 1      | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |        |        |              |   |   |   |           |   |   |   |            |   |   |   |            |   |   |   |            |   |   |   |        |   |   |   |          |   |   |   |          |   |   |   |          |
| 1    | 1      | 0      | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |        |        |              |   |   |   |           |   |   |   |            |   |   |   |            |   |   |   |            |   |   |   |        |   |   |   |          |   |   |   |          |   |   |   |          |
| 1    | 1      | 1      | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |        |        |              |   |   |   |           |   |   |   |            |   |   |   |            |   |   |   |            |   |   |   |        |   |   |   |          |   |   |   |          |   |   |   |          |

#### 6.6.4 CIR Transmit 寄存器描述

##### 6.6.4.1 CIR\_TGLR\_REG

| 偏移地址 : 0x0000 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                          |
|---------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                        |
| 31:8          | /   | /   | /                                                                                                                                                                                                                                                         |
| 7             | R/W | 0   | <p>IMS</p> <p>Internal Modulation Select</p> <p>0: the transmitting signal is not modulated</p> <p>1: the transmitting signal is modulated internally</p>                                                                                                 |
| 6:5           | R/W | 0   | <p>DRMC</p> <p>duty ratio of modulated carrier is high level /low level.</p> <p>0: low level is the one time of high level</p> <p>1: low level is the two times of high level</p> <p>2: low level is the three times of high level</p> <p>3: reserved</p> |



|     |     |   |                                                                                                                                                                                                                                                                                      |
|-----|-----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4:3 | /   | / | /                                                                                                                                                                                                                                                                                    |
| 2   | R/W | 0 | TPPI<br>Transmit Pulse Polarity Invert<br>0: Not invert transmit pulse<br>1: Invert transmit pulse                                                                                                                                                                                   |
| 1   | R/W | 0 | TR<br>Transmit Reset<br>When this bit is set, the transmitting is reset. The FIFO will be flush, the TIC field and CSS field will be clean during Transmit Reset. This field will automatically clean when the Transmit Reset is finished, and the CIR transmitter will state Idle . |
| 0   | R/W | 0 | TXEN<br>Transmit Block Enable<br>0: Disable the CIR Transmitter<br>1: Enable the CIR Transmitter                                                                                                                                                                                     |

#### 6.6.4.2 CIR\_X\_REG

| 偏移地址 : 0x0004 |     |      | 默认值: 0x0000_009E                                                                                                                                                                                                                                                                                                                 |
|---------------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值  | 描述                                                                                                                                                                                                                                                                                                                               |
| 31:8          | /   | /    | /                                                                                                                                                                                                                                                                                                                                |
| 7:0           | R/W | 0x9E | RFMC<br>Reference Frequency of modulated carrier.<br>Reference Frequency of modulated carrier based on a division of a fixed functional clock(FCLK).The range of the modulated carrier is usually 30KHz to 60KHz.The most consumer electronics is 38KHz. The default modulated carrier is 38KHz when FCLK is 6MHz. RFMC= FCLK/N. |

#### 6.6.4.3 CIR\_TCR\_REG

| 偏移地址 : 0x0008 |     |     | 默认值: 0x0000_0002                                                                                                                                                                      |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                    |
| 31:8          | /   | /   | /                                                                                                                                                                                     |
| 7             | R/W | 0   | CSS<br>Cyclical Pulse Start/Stop Control<br>Start to transmit when set to '1',<br>0: Stop when cleared to '0'. From start to stop, all data in FIFO must be transmitted.<br>1: Start. |
| 6:4           | /   | /   | /                                                                                                                                                                                     |
| 3:1           | R/W | 1   | RCS<br>Reference Clock Select for CIR Transmit(>=1)                                                                                                                                   |

|   |     |   |                                                                                                                                                                                                                                                                                                                                                            |
|---|-----|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |   | 001: CIR Transmit reference clock is ir_clk/2<br>010: CIR Transmit reference clock is ir_clk/4<br>011: CIR Transmit reference clock is ir_clk/8<br>100: CIR Transmit reference clock is ir_clk/64<br>101: CIR Transmit reference clock is ir_clk/128<br>110: CIR Transmit reference clock is ir_clk/256<br>111: CIR Transmit reference clock is ir_clk/512 |
| 0 | R/W | 0 | TTS<br>Type of the transmission signal<br>0: The transmitting wave is single non-cyclical pulse.<br>1: The transmitting wave is cyclical short-pulse.                                                                                                                                                                                                      |

#### 6.6.4.4 CIR\_IDC\_H\_REG

| 偏移地址：0x000C |     |     | 默认值: 0x0000_0000                                                                                  |
|-------------|-----|-----|---------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                |
| 31:8        | /   | /   | /                                                                                                 |
| 3:0         | R/W | 0   | IDC_H<br>Idle Duration Counter threshold(High 4 bit)<br>Idle Duration = 128*IDC*Ts (IDC = 0~4095) |

#### 6.6.4.5 CIR\_IDC\_L\_REG

| 偏移地址：0x0010 |     |     | 默认值: 0x0000_0000                                                                                 |
|-------------|-----|-----|--------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                               |
| 31:8        | /   | /   | /                                                                                                |
| 7:0         | R/W | 0   | IDC_L<br>Idle Duration Counter threshold(Low 8 bit)<br>Idle Duration = 128*IDC*Ts (IDC = 0~4095) |

#### 6.6.4.6 CIR\_TICR\_H\_REG

| 偏移地址：0x0014 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                              |
|-------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                            |
| 31:8        | /   | /   | /                                                                                                                                                                                                                                                                                             |
| 7:0         | R/W | 0   | TIC_H<br>Transmit Idle Counter_H(High 8 bit)<br>Count in 128*Ts (Sample Duration, 1/Fs) when transmitter is idle, and it should be reset when the transmitter active.<br>When this counter reaches the maximum value (0xFFFF), it will stop automatically, and should not be cleared to zero. |

#### 6.6.4.7 CIR\_TICR\_L\_REG

| 偏移地址：0x0018 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                              |
|-------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                            |
| 31:8        | /   | /   | /                                                                                                                                                                                                                                                                                                             |
| 7:0         | R/W | 0   | <p>TIC_L</p> <p>Transmit Idle Counter_L(Low 8 bit)</p> <p>Count in 128*Ts (Sample Duration, 1/Fs) when transmitter is idle, and it should be reset when the transmitter active.</p> <p>When this counter reach the maximum value (0xFFFF), it will stop automatically, and should not be cleared to zero.</p> |

#### 6.6.4.8 CIR\_TEL\_REG

| 偏移地址：0x0020 |     |     | 默认值: 0x0000_0000                                                                      |
|-------------|-----|-----|---------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                    |
| 31:8        | /   | /   | /                                                                                     |
| 7:0         | R/W | 0   | <p>TEL</p> <p>TX FIFO empty Level for DRQ and IRQ.</p> <p>TRIGGER_LEVEL = TEL + 1</p> |

#### 6.6.4.9 CIR\_TXINT\_REG

| 偏移地址：0x0024 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                     |
|-------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                   |
| 31:3        | /   | /   | /                                                                                                                                                                                                                                                    |
| 2           | R/W | 0   | <p>DRQ_EN</p> <p>TX FIFO DMA Enable</p> <p>0: Disable</p> <p>1: Enable</p> <p>When set to '1', the Tx FIFO DRQ is asserted if the number of the transmitting data in the FIFO is less then the RAL. The DRQ is de-asserted when condition fails.</p> |
| 1           | R/W | 0   | <p>TAI_EN</p> <p>TX FIFO Available Interrupt Enable</p> <p>0:Disable</p> <p>1:Enable</p>                                                                                                                                                             |
| 0           | R/W | 0   | <p>TPEI_EN</p> <p>Transmit Packet End Interrupt Enable for Cyclical Pulse</p> <p>0:Disable</p> <p>1:Enable</p>                                                                                                                                       |

|  |  |  |                                                                                                         |
|--|--|--|---------------------------------------------------------------------------------------------------------|
|  |  |  | TUI_EN<br>Transmitter FIFO under run Interrupt Enable for Non-cyclical Pulse<br>0: Disable<br>1: Enable |
|--|--|--|---------------------------------------------------------------------------------------------------------|

#### 6.6.4.10 CIR\_TAC\_REG

| 偏移地址 : 0x0028 |    |     | 默认值: 0x0000_0128                                                                                                                                                                                              |
|---------------|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                                            |
| 31:6          | /  | /   | /                                                                                                                                                                                                             |
| 7:0           | R  | 128 | TAC<br>TX FIFO Available Space Counter<br>0: No available space in TX FIFO<br>1: 1 byte available space in TX FIFO<br>2: 2 byte available space in TX FIFO<br>...<br>128: 128 byte available space in TX FIFO |

#### 6.6.4.11 CIR\_TXSTA\_REG

| 偏移地址 : 0x002C |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                 |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                               |
| 31:4          | /   | /   | /                                                                                                                                                                                                                                                                                                                |
| 3             | R   | 0   | STCT<br>Status of CIR Transmitter<br>0x0:Idle<br>0x1:Active<br>This bit will automatically set when the controller begins transmit the data in the FIFO. The “1” will last when the data in the FIFO. It will automatically be cleaned to “0” when all data in the FIFO is transmitted.<br>The bit is for debug. |
| 2             | R   | 0   | DRQ<br>DMA Request Flag<br>When set to ‘1’, the Tx FIFO DRQ is asserted if the number of the transmitting data in the FIFO is less then the RAL. The DRQ is de-asserted when condition fails.<br>This bit is for debug.                                                                                          |
| 1             | R/W | 0   | TAI<br>TX FIFO Available Interrupt Flag                                                                                                                                                                                                                                                                          |

|   |     |   |                                                                                                                                                                                                                                                                                                                                                                                       |
|---|-----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |   | 0: TX FIFO not available by its level<br>1: TX FIFO available by its level<br>This bit is cleared by writing a '1' .                                                                                                                                                                                                                                                                  |
| 0 | R/W | 0 | TPE<br>Transmitter Packet End Flag for Cyclical Pulse<br>0: Transmissions of address, control and data fields not completed<br>1: Transmissions of address, control and data fields completed<br><br>TUR<br>Transmitter FIFO Under Run Flag for Non-cyclical Pulse<br>0: No transmitter FIFO under run<br>1: Transmitter FIFO under run<br><br>This bit is cleared by writing a '1' . |

#### 6.6.4.12 CIR\_TXT\_REG

| 偏移地址 : 0x0030 |     |     | 默认值: 0x0000_0000                                                                                                                                                                             |
|---------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                           |
| 31:8          | /   | /   | /                                                                                                                                                                                            |
| 7:0           | R/W | 0   | NCTT<br>Non-cyclical Pulse Transmit Threshold<br>The controller will trigger transmitting the data in the FIFO when the data byte number has reach the Transmit Threshold set in this field. |

#### 6.6.4.13 CIR\_DMA\_CTL\_REG

| 偏移地址 : 0x0034 |     |     | 默认值: 0x0000_00A5                                                                                  |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                |
| 31:8          | /   | /   | /                                                                                                 |
| 7:6           | R/W | 2   | 00:dma_active is low<br>01:dma_active is high<br>10:dma_req<br>11:active is control by controller |
| 5             | R/W | 1   | 0: active fall do not care ack<br>1: active fall must after detect ack is high                    |
| 4:0           | R/W | 5   | dma_active high after last signal is high                                                         |

#### 6.6.4.14 CIR\_TXFIFO\_REG

| 偏移地址：0x0080 |    |     | 默认值: 0x0000_0000                                                                                                                                      |
|-------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问 | 默认值 | 描述                                                                                                                                                    |
| 31:8        | /  | /   | /                                                                                                                                                     |
| 7:0         | W  | 0   | Transmit Byte FIFO<br>When the transmitting is trigger, the data in the FIFO will be transmitted until the data number has been transmitted finished. |

## 6.7 PWM 接口

### 6.7.1 概述

PWM模块带8个PWM通道，分4个PWM对：PWM01对、PWM23对、PWM45对、PWM67对，PWM01对由PWM0和PWM1通道构成，PWM23对由PWM2和PWM3通道构成，PWM45对由PWM4和PWM5通道构成，PWM67对由PWM6和PWM7通道构成。PWM对主要由3部分组成：1个时钟控制器、2个定时器逻辑模块、1个可编程死区发生器。

每个PWM通道支持PWM输出和捕捉输入两大功能。PWM通道的时钟源有24M\_PLL和APB1（100M）可选，经过时钟控制器处理后送至PWM通道的定时器逻辑模块。PWM通道配置为PWM输出功能，其定时器逻辑模块可输出单脉冲波形或周期波形，频率范围在0Hz~24/100MHz之间。

PWM通道的定时器逻辑模块也可复用于捕捉输入功能，PWM通道捕捉到外部上升沿时锁存16-bit加法计数器当前值至上升沿锁存寄存器CRLR，PWM通道捕捉到外部下降沿时锁存16-bit加法计数器当前值至下降沿锁存寄存器CFLR，根据2个寄存器CRLR和CFLR的值可精确计算出外部时钟的频率。

PWM对可配置执行互补输出或插入死区时间的PWM波形对输出。当PWM对里的2个通道的prescale相同、period寄存器值相同且active state相反时，PWM对输出互补波形；当PWM对的可编程死区发生器被使能，PWM对输出插入死区时间的波形对，波形对的死区插入时间可控。

PWM通道配置为PWM输出功能或捕捉输入功能，PWM通道都可配置产生中断。配置为PWM输出功能时，每当16-bit加法计数器达到Entire cycle值，PWM通道可被使能产生中断；配置为捕捉输入功能时，每当PWM通道捕捉到外部上升沿，PWM通道可被使能产生1个中断，当PWM通道捕捉到下降沿，PWM通道可被使能产生1个中断，捕捉到上升沿或下降沿都可触发中断产生。

PWM控制器包含如下特性：

- 8 个 PWM 通道，分 4 个 PWM 对
- 支持脉冲、周期和互补对输出
- 支持捕捉输入
- 带可编程死区发生器，死区时间可控
- 0-24/100M 输出频率范围，0%-100%占空比可调，最小分辨率 1/65536
- 支持 PWM 输出和捕捉输入中断

## 6.7.2 PWM 寄存器列表

| 模块名             | 基地址                                       |                                    |
|-----------------|-------------------------------------------|------------------------------------|
| PWM             | 0x40042800                                |                                    |
| 寄存器名            | 偏移地址                                      | 描述                                 |
| PWM_PIER_REG    | 0x0000                                    | PWM IRQ Enable Register            |
| PWM_PISR_REG    | 0x0004                                    | PWM IRQ Status Register            |
| PWM_CIER_REG    | 0x0010                                    | Capture IRQ Enable Register        |
| PWM_CISR_REG    | 0x0014                                    | Capture IRQ Status Register        |
| PWM_PCCR01_REG  | 0x0020                                    | PWM01 Clock Configuration Register |
| PWM_PCCR23_REG  | 0x0024                                    | PWM23 Clock Configuration Register |
| PWM_PCCR45_REG  | 0x0028                                    | PWM45 Clock Configuration Register |
| PWM_PCCR67_REG  | 0x002C                                    | PWM67 Clock Configuration Register |
| PWM_PDZCR01_REG | 0x0030                                    | PWM01 Dead Zone Control Register   |
| PWM_PDZCR23_REG | 0x0034                                    | PWM23 Dead Zone Control Register   |
| PWM_PDZCR45_REG | 0x0038                                    | PWM45 Dead Zone Control Register   |
| PWM_PDZCR67_REG | 0x003C                                    | PWM67 Dead Zone Control Register   |
| PWM_PER_REG     | 0x0040                                    | PWM Enable Register                |
| PWM_CER_REG     | 0x0044                                    | Capture Enable Register            |
| PWM_PCR_REG     | $0x0060 + 0x0 + N * 0x20 (N = 0 \sim 7)$  | PWM Control Register               |
| PWM_PPR_REG     | $0x0060 + 0x4 + N * 0x20 (N = 0 \sim 7)$  | PWM Period Register                |
| PWM_PCNTR_REG   | $0x0060 + 0x8 + N * 0x20 (N = 0 \sim 7)$  | PWM Count Register                 |
| PWM_CCR_REG     | $0x0060 + 0xC + N * 0x20 (N = 0 \sim 7)$  | Capture Control Register           |
| PWM_CRLR_REG    | $0x0060 + 0x10 + N * 0x20 (N = 0 \sim 7)$ | Capture Rise Lock Register         |
| PWM_CFLR_REG    | $0x0060 + 0x14 + N * 0x20 (N = 0 \sim 7)$ | Capture Fall Lock Register         |

### 6.7.3 PWM 寄存器描述

#### 6.7.3.1 PWM\_PIER\_REG

| 偏移地址：0x0000 |     |     | 默认值: 0x0000_0000                                                                                                       |
|-------------|-----|-----|------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                     |
| 31:8        | /   | /   | /                                                                                                                      |
| 7           | R/W | 0x0 | PCIE7.<br>PWM channel 7 Interrupt Enable.<br>0: PWM channel 7 Interrupt Disable;<br>1: PWM channel 7 Interrupt Enable. |
| 6           | R/W | 0x0 | PCIE6.<br>PWM channel 6 Interrupt Enable.<br>0: PWM channel 6 Interrupt Disable;<br>1: PWM channel 6 Interrupt Enable. |
| 5           | R/W | 0x0 | PCIE5.<br>PWM channel 5 Interrupt Enable.<br>0: PWM channel 5 Interrupt Disable;<br>1: PWM channel 5 Interrupt Enable. |
| 4           | R/W | 0x0 | PCIE4.<br>PWM channel 4 Interrupt Enable.<br>0: PWM channel 4 Interrupt Disable;<br>1: PWM channel 4 Interrupt Enable. |
| 3           | R/W | 0x0 | PCIE3.<br>PWM channel 3 Interrupt Enable.<br>0: PWM channel 3 Interrupt Disable;<br>1: PWM channel 3 Interrupt Enable. |
| 2           | R/W | 0x0 | PCIE2.<br>PWM channel 2 Interrupt Enable.<br>0: PWM channel 2 Interrupt Disable;<br>1: PWM channel 2 Interrupt Enable. |
| 1           | R/W | 0x0 | PCIE1.<br>PWM channel 1 Interrupt Enable.<br>0: PWM channel 1 Interrupt Disable;<br>1: PWM channel 1 Interrupt Enable. |
| 0           | R/W | 0x0 | PCIE0.<br>PWM channel 0 Interrupt Enable.<br>0: PWM channel 0 Interrupt Disable;<br>1: PWM channel 0 Interrupt Enable. |



### 6.7.3.2 PWM\_PISR\_REG

| 偏移地址：0x0004 |       |     | 默认值：0x0000_0000                                                                                                                                                                                                                                                                                                                                  |
|-------------|-------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问    | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                               |
| 31:8        | /     | /   | /                                                                                                                                                                                                                                                                                                                                                |
| 7           | R/W1C | 0x0 | <p>PIS7.</p> <p>PWM channel 7 Interrupt Status. When PWM channel 7 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit.</p> <p>Reads 0: PWM channel 7 interrupt is not pending.<br/>1: PWM channel 7 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear PWM channel 7 interrupt status.</p> |
| 6           | R/W1C | 0x0 | <p>PIS6.</p> <p>PWM channel 6 Interrupt Status. When PWM channel 6 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit.</p> <p>Reads 0: PWM channel 6 interrupt is not pending.<br/>1: PWM channel 6 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear PWM channel 6 interrupt status.</p> |
| 5           | R/W1C | 0x0 | <p>PIS5.</p> <p>PWM channel 5 Interrupt Status. When PWM channel 5 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit.</p> <p>Reads 0: PWM channel 5 interrupt is not pending.<br/>1: PWM channel 5 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear PWM channel 5 interrupt status.</p> |
| 4           | R/W1C | 0x0 | <p>PIS4.</p> <p>PWM channel 4 Interrupt Status. When PWM channel 4 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit.</p> <p>Reads 0: PWM channel 4 interrupt is not pending.<br/>1: PWM channel 4 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear PWM channel 4 interrupt status.</p> |
| 3           | R/W1C | 0x0 | <p>PIS3.</p> <p>PWM channel 3 Interrupt Status. When PWM channel 3 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit.</p> <p>Reads 0: PWM channel 3 interrupt is not pending.<br/>1: PWM channel 3 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear PWM channel 3 interrupt status.</p> |

|   |       |     |                                                                                                                                                                                                                                                                                                                                                  |
|---|-------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | R/W1C | 0x0 | <p>PIS2.</p> <p>PWM channel 2 Interrupt Status. When PWM channel 2 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit.</p> <p>Reads 0: PWM channel 2 interrupt is not pending.<br/>1: PWM channel 2 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear PWM channel 2 interrupt status.</p> |
| 1 | R/W1C | 0x0 | <p>PIS1.</p> <p>PWM channel 1 Interrupt Status. When PWM channel 1 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit.</p> <p>Reads 0: PWM channel 1 interrupt is not pending.<br/>1: PWM channel 1 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear PWM channel 1 interrupt status.</p> |
| 0 | R/W1C | 0x0 | <p>PIS0.</p> <p>PWM channel 0 Interrupt Status. When PWM channel 0 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit.</p> <p>Reads 0: PWM channel 0 interrupt is not pending.<br/>1: PWM channel 0 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear PWM channel 0 interrupt status.</p> |

### 6.7.3.3 PWM\_CIER\_REG

| 偏移地址 : 0x0010 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                        |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                      |
| 31:16         | /   | /   | /                                                                                                                                                                                                                                                       |
| 15            | R/W | 0x0 | <p>CFIE7.</p> <p>If this enable bit is set 1, when capture channel 7 captures falling edge, it generates a capture channel 7 pending.</p> <p>0: Capture channel 7 fall lock Interrupt disable;<br/>1: Capture channel 7 fall lock Interrupt enable.</p> |
| 14            | R/W | 0x0 | <p>CRIE7.</p> <p>If this enable bit is set 1, when capture channel 7 captures rising edge, it generates a capture channel 7 pending.</p> <p>0: Capture channel 7 rise lock Interrupt disable;<br/>1: Capture channel 7 rise lock Interrupt enable.</p>  |
| 13            | R/W | 0x0 | <p>CFIE6.</p> <p>If this enable bit is set 1, when capture channel 6 captures falling edge, it generates a capture channel 6 pending.</p> <p>0: Capture channel 6 fall lock Interrupt disable;</p>                                                      |

|    |     |     |                                                                                                                                                                                                                                                            |
|----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |     |     | 1: Capture channel 6 fall lock Interrupt enable.                                                                                                                                                                                                           |
| 12 | R/W | 0x0 | <p>CRIE6.</p> <p>If this enable bit is set 1, when capture channel 6 captures rising edge, it generates a capture channel 6 pending.</p> <p>0: Capture channel 6 rise lock Interrupt disable;</p> <p>1: Capture channel 6 rise lock Interrupt enable.</p>  |
| 11 | R/W | 0x0 | <p>CFIE5.</p> <p>If this enable bit is set 1, when capture channel 5 captures falling edge, it generates a capture channel 5 pending.</p> <p>0: Capture channel 5 fall lock Interrupt disable;</p> <p>1: Capture channel 5 fall lock Interrupt enable.</p> |
| 10 | R/W | 0x0 | <p>CRIE5.</p> <p>If this enable bit is set 1, when capture channel 5 captures rising edge, it generates a capture channel 5 pending.</p> <p>0: Capture channel 5 rise lock Interrupt disable;</p> <p>1: Capture channel 5 rise lock Interrupt enable.</p>  |
| 9  | R/W | 0x0 | <p>CFIE4.</p> <p>If this enable bit is set 1, when capture channel 4 captures falling edge, it generates a capture channel 4 pending.</p> <p>0: Capture channel 4 fall lock Interrupt disable;</p> <p>1: Capture channel 4 fall lock Interrupt enable.</p> |
| 8  | R/W | 0x0 | <p>CRIE4.</p> <p>If this enable bit is set 1, when capture channel 4 captures rising edge, it generates a capture channel 4 pending.</p> <p>0: Capture channel 4 rise lock Interrupt disable;</p> <p>1: Capture channel 4 rise lock Interrupt enable.</p>  |
| 7  | R/W | 0x0 | <p>CFIE3.</p> <p>If this enable bit is set 1, when capture channel 3 captures falling edge, it generates a capture channel 3 pending.</p> <p>0: Capture channel 3 fall lock Interrupt disable;</p> <p>1: Capture channel 3 fall lock Interrupt enable.</p> |
| 6  | R/W | 0x0 | <p>CRIE3.</p> <p>If this enable bit is set 1, when capture channel 3 captures rising edge, it generates a capture channel 3 pending.</p> <p>0: Capture channel 3 rise lock Interrupt disable;</p> <p>1: Capture channel 3 rise lock Interrupt enable.</p>  |
| 5  | R/W | 0x0 | <p>CFIE2.</p> <p>If this enable bit is set 1, when capture channel 2 captures falling edge, it generates a capture channel 2 pending.</p> <p>0: Capture channel 2 fall lock Interrupt disable;</p> <p>1: Capture channel 2 fall lock Interrupt enable.</p> |
| 4  | R/W | 0x0 | <p>CRIE2.</p> <p>If this enable bit is set 1, when capture channel 2 captures rising edge,</p>                                                                                                                                                             |

|   |     |     |                                                                                                                                                                                                                                         |
|---|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |     | it generates a capture channel 2 pending.<br>0: Capture channel 2 rise lock Interrupt disable;<br>1: Capture channel 2 rise lock Interrupt enable.                                                                                      |
| 3 | R/W | 0x0 | CFIE1.<br>If this enable bit is set 1, when capture channel 1 captures falling edge, it generates a capture channel 1 pending.<br>0: Capture channel 1 fall lock Interrupt disable;<br>1: Capture channel 1 fall lock Interrupt enable. |
| 2 | R/W | 0x0 | CRIE1.<br>If this enable bit is set 1, when capture channel 1 captures rising edge, it generates a capture channel 1 pending.<br>0: Capture channel 1 rise lock Interrupt disable;<br>1: Capture channel 1 rise lock Interrupt enable.  |
| 1 | R/W | 0x0 | CFIE0.<br>If this enable bit is set 1, when capture channel 0 captures falling edge, it generates a capture channel 0 pending.<br>0: Capture channel 0 fall lock Interrupt disable;<br>1: Capture channel 0 fall lock Interrupt enable. |
| 0 | R/W | 0x0 | CRIE0.<br>If this enable bit is set 1, when capture channel 0 captures rising edge, it generates a capture channel 0 pending.<br>0: Capture channel 0 rise lock Interrupt disable;<br>1: Capture channel 0 rise lock Interrupt enable.  |

#### 6.7.3.4 PWM\_CISR\_REG

| 偏移地址 : 0x0014 |       |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|-------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问    | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 31:8          | /     | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 15            | R/W1C | 0x0 | CFIS7.<br>Capture channel 7 falling lock interrupt status.<br>When capture channel 7 captures falling edge, if capture channel 7 fall lock interrupt (CFIE7) is enabled, this bit is set 1 by hardware.<br>Write 1 to clear this bit.<br>Reads 0: Capture channel 7 interrupt is not pending.<br>1: Capture channel 7 interrupt is pending.<br>Writes 0: no effect.<br>1: Clear capture channel 7 interrupt status. |
| 14            | R/W1C | 0x0 | CRIS7.<br>Capture channel 7 rising lock interrupt status.<br>When capture channel 7 captures rising edge, if capture channel 7 rise lock interrupt (CRIE7) is enabled, this bit is set 1 by hardware.<br>Write 1 to clear this bit.                                                                                                                                                                                 |

|    |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----|-------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |       |     | <p>Reads 0: Capture channel 7 interrupt is not pending.<br/>1: Capture channel 7 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear capture channel 7 interrupt status.</p>                                                                                                                                                                                                                                                          |
| 13 | R/W1C | 0x0 | <p>CFIS6.</p> <p>Capture channel 6 falling lock interrupt status.</p> <p>When capture channel 6 captures falling edge, if capture channel 6 fall lock interrupt (CFIE6) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 6 interrupt is not pending.<br/>1: Capture channel 6 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear capture channel 6 interrupt status.</p> |
| 12 | R/W1C | 0x0 | <p>CRIS6.</p> <p>Capture channel 6 rising lock interrupt status.</p> <p>When capture channel 6 captures rising edge, if capture channel 6 rise lock interrupt (CRIE6) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 6 interrupt is not pending.<br/>1: Capture channel 6 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear capture channel 6 interrupt status.</p>   |
| 11 | R/W1C | 0x0 | <p>CFIS5.</p> <p>Capture channel 5 falling lock interrupt status.</p> <p>When capture channel 5 captures falling edge, if capture channel 5 fall lock interrupt (CFIE5) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 5 interrupt is not pending.<br/>1: Capture channel 5 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear capture channel 5 interrupt status.</p> |
| 10 | R/W1C | 0x0 | <p>CRIS5.</p> <p>Capture channel 5 rising lock interrupt status.</p> <p>When capture channel 5 captures rising edge, if capture channel 5 rise lock interrupt (CRIE5) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 5 interrupt is not pending.<br/>1: Capture channel 5 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear capture channel 5 interrupt status.</p>   |
| 9  | R/W1C | 0x0 | <p>CFIS4.</p> <p>Capture channel 4 falling lock interrupt status.</p> <p>When capture channel 4 captures falling edge, if capture channel 4 fall lock interrupt (CFIE4) is enabled, this bit is set 1 by hardware.</p>                                                                                                                                                                                                                           |

|   |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---|-------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |       |     | <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 4 interrupt is not pending.<br/>1: Capture channel 4 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear capture channel 4 interrupt status.</p>                                                                                                                                                                                                                        |
| 8 | R/W1C | 0x0 | <p>CRIS4.</p> <p>Capture channel 4 rising lock interrupt status.</p> <p>When capture channel 4 captures rising edge, if capture channel 4 rise lock interrupt (CRIE4) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 4 interrupt is not pending.<br/>1: Capture channel 4 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear capture channel 4 interrupt status.</p>   |
| 7 | R/W1C | 0x0 | <p>CFIS3.</p> <p>Capture channel 3 falling lock interrupt status.</p> <p>When capture channel 3 captures falling edge, if capture channel 3 fall lock interrupt (CFIE3) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 3 interrupt is not pending.<br/>1: Capture channel 3 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear capture channel 3 interrupt status.</p> |
| 6 | R/W1C | 0x0 | <p>CRIS3.</p> <p>Capture channel 3 rising lock interrupt status.</p> <p>When capture channel 3 captures rising edge, if capture channel 3 rise lock interrupt (CRIE3) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 3 interrupt is not pending.<br/>1: Capture channel 3 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear capture channel 3 interrupt status.</p>   |
| 5 | R/W1C | 0x0 | <p>CFIS2.</p> <p>Capture channel 2 falling lock interrupt status.</p> <p>When capture channel 2 captures falling edge, if capture channel 2 fall lock interrupt (CFIE2) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 2 interrupt is not pending.<br/>1: Capture channel 2 interrupt is pending.</p> <p>Writes 0: no effect.<br/>1: Clear capture channel 2 interrupt status.</p> |
| 4 | R/W1C | 0x0 | <p>CRIS2.</p> <p>Capture channel 2 rising lock interrupt status.</p> <p>When capture channel 2 captures rising edge, if capture channel 2</p>                                                                                                                                                                                                                                                                                                    |

|   |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|-------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |       |     | <p>rise lock interrupt (CRIE2) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 2 interrupt is not pending.</p> <p>1: Capture channel 2 interrupt is pending.</p> <p>Writes 0: no effect.</p> <p>1: Clear capture channel 2 interrupt status.</p>                                                                                                                                          |
| 3 | R/W1C | 0x0 | <p>CFIS1.</p> <p>Capture channel 1 falling lock interrupt status.</p> <p>When capture channel 1 captures falling edge, if capture channel 1 fall lock interrupt (CFIE1) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 1 interrupt is not pending.</p> <p>1: Capture channel 1 interrupt is pending.</p> <p>Writes 0: no effect.</p> <p>1: Clear capture channel 1 interrupt status.</p> |
| 2 | R/W1C | 0x0 | <p>CRIS1.</p> <p>Capture channel 1 rising lock interrupt status.</p> <p>When capture channel 1 captures rising edge, if capture channel 1 rise lock interrupt (CRIE1) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 1 interrupt is not pending.</p> <p>1: Capture channel 1 interrupt is pending.</p> <p>Writes 0: no effect.</p> <p>1: Clear capture channel 1 interrupt status.</p>   |
| 1 | R/W1C | 0x0 | <p>CFIS0.</p> <p>Capture channel 0 falling lock interrupt status.</p> <p>When capture channel 0 captures falling edge, if capture channel 0 fall lock interrupt (CFIE0) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 0 interrupt is not pending.</p> <p>1: Capture channel 0 interrupt is pending.</p> <p>Writes 0: no effect.</p> <p>1: Clear capture channel 0 interrupt status.</p> |
| 0 | R/W1C | 0x0 | <p>CRIS0.</p> <p>Capture channel 0 rising lock interrupt status.</p> <p>When capture channel 0 captures rising edge, if capture channel 0 rise lock interrupt (CRIE0) is enabled, this bit is set 1 by hardware.</p> <p>Write 1 to clear this bit.</p> <p>Reads 0: Capture channel 0 interrupt is not pending.</p> <p>1: Capture channel 0 interrupt is pending.</p> <p>Writes 0: no effect.</p> <p>1: Clear capture channel 0 interrupt status.</p>   |

### 6.7.3.5 PWM\_PCCR01\_REG

| 偏移地址 : 0x0020 |     |     | 默认值: 0x0000_0000                                                                                                                                                         |
|---------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                       |
| 31:9          | /   | /   | /                                                                                                                                                                        |
| 8:7           | R/W | 0x0 | PWM01_CLK_SRC.<br>Select PWM01 clock source.<br>00: 24M_PLL<br>01: APB1<br>Others: /                                                                                     |
| 6             | R/W | 0x0 | PWM01_CLK_SRC_BYPASS_TO_PWM1.<br>Bypass PWM01 clock source to PWM1 output.<br>0: not bypass<br>1: bypass                                                                 |
| 5             | R/W | 0x0 | PWM01_CLK_SRC_BYPASS_TO_PWM0.<br>Bypass PWM01 clock source to PWM0 output.<br>0: not bypass<br>1: bypass                                                                 |
| 4             | R/W | 0x0 | PWM01_CLK_GATING.<br>Gating clock for PWM01.<br>0: Mask<br>1: Pass                                                                                                       |
| 0:3           | R/W | 0x0 | PWM01_CLK_DIV_M.<br>PWM01 clock divide M<br>0000: /1<br>0001: /2<br>0010: /4<br>0011: /8<br>0100: /16<br>0101: /32<br>0110: /64<br>0111: /128<br>1000: /256<br>others: / |

### 6.7.3.6 PWM\_PCCR23\_REG

| 偏移地址 : 0x0024 |    |     | 默认值: 0x0000_0000 |
|---------------|----|-----|------------------|
| 位置            | 访问 | 默认值 | 描述               |
| 31:9          | /  | /   | /                |



|     |     |     |                                                                                                                                                                          |
|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8:7 | R/W | 0x0 | PWM23_CLK_SRC_SEL.<br>Select PWM23 clock source.<br>00: 24M_PLL<br>01: APB1<br>Others: /                                                                                 |
| 6   | R/W | 0x0 | PWM23_CLK_SRC_BYPASS_TO_PWM3.<br>Bypass PWM23 clock source to PWM3 output.<br>0: not bypass<br>1: bypass                                                                 |
| 5   | R/W | 0x0 | PWM23_CLK_SRC_BYPASS_TO_PWM2.<br>Bypass PWM23 clock source to PWM2 output.<br>0: not bypass<br>1: bypass                                                                 |
| 4   | R/W | 0x0 | PWM23_CLK_GATING<br>Gating clock for PWM23.<br>0: Mask<br>1: Pass                                                                                                        |
| 0:3 | R/W | 0x0 | PWM23_CLK_DIV_M.<br>PWM23 clock divide M<br>0000: /1<br>0001: /2<br>0010: /4<br>0011: /8<br>0100: /16<br>0101: /32<br>0110: /64<br>0111: /128<br>1000: /256<br>others: / |

### 6.7.3.7 PWM\_PCCR45\_REG

| 偏移地址 : 0x0028 |     |     | 默认值: 0x0000_0000                                                                         |
|---------------|-----|-----|------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                       |
| 31:9          | /   | /   | /                                                                                        |
| 8:7           | R/W | 0x0 | PWM45_CLK_SRC_SEL.<br>Select PWM45 clock source.<br>00: 24M_PLL<br>01: APB1<br>Others: / |
| 6             | R/W | 0x0 | PWM45_CLK_SRC_BYPASS_TO_PWM5.                                                            |

|     |     |     |                                                                                                                                                                          |
|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |     | Bypass PWM45 clock source to PWM5 output.<br>0: not bypass<br>1: bypass                                                                                                  |
| 5   | R/W | 0x0 | PWM45_CLK_SRC_BYPASS_TO_PWM4.<br>Bypass PWM45 clock source to PWM4 output.<br>0: not bypass<br>1: bypass                                                                 |
| 4   | R/W | 0x0 | PWM45_CLK_GATING.<br>Gating clock for PWM45.<br>0: Mask<br>1: Pass                                                                                                       |
| 0:3 | R/W | 0x0 | PWM45_CLK_DIV_M.<br>PWM45 clock divide M<br>0000: /1<br>0001: /2<br>0010: /4<br>0011: /8<br>0100: /16<br>0101: /32<br>0110: /64<br>0111: /128<br>1000: /256<br>others: / |

#### 6.7.3.8 PWM\_PCCR67\_REG

| 偏移地址 : 0x002C |     |     | 默认值: 0x0000_0000                                                                                         |
|---------------|-----|-----|----------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                       |
| 31:9          | /   | /   | /                                                                                                        |
| 8:7           | R/W | 0x0 | PWM67_CLK_SRC_SEL.<br>Select PWM67 clock source.<br>00: 24M_PLL<br>01: APB1<br>Others: /                 |
| 6             | R/W | 0x0 | PWM67_CLK_SRC_BYPASS_TO_PWM7.<br>Bypass PWM67 clock source to PWM7 output.<br>0: not bypass<br>1: bypass |
| 5             | R/W | 0x0 | PWM67_CLK_SRC_BYPASS_TO_PWM6.<br>Bypass PWM67 clock source to PWM6 output.<br>0: not bypass              |

|     |     |     |                                                                                                                                                                          |
|-----|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |     | 1: bypass                                                                                                                                                                |
| 4   | R/W | 0x0 | PWM67_CLK_GATING.<br>Gating clock for PWM67.<br>0: Mask<br>1: Pass                                                                                                       |
| 0:3 | R/W | 0x0 | PWM67_CLK_DIV_M.<br>PWM67 clock divide M<br>0000: /1<br>0001: /2<br>0010: /4<br>0011: /8<br>0100: /16<br>0101: /32<br>0110: /64<br>0111: /128<br>1000: /256<br>others: / |

#### 6.7.3.9 PWM\_PDZCR01\_REG

| 偏移地址 : 0x0030 |     |     | 默认值: 0x0000_0000                                                                        |
|---------------|-----|-----|-----------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                      |
| 15:8          | R/W | 0x0 | PDZINTV01.<br>PWM01 Dead Zone interval value.                                           |
| 7:1           | /   | /   | /                                                                                       |
| 0             | R/W | 0x0 | PWM01_DZ_EN.<br>PWM01 Dead Zone enable.<br>0: Dead Zone disable<br>1: Dead Zone enable. |

#### 6.7.3.10 PWM\_PDZCR23\_REG

| 偏移地址 : 0x0034 |     |     | 默认值: 0x0000_0000                                                                        |
|---------------|-----|-----|-----------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                      |
| 15:8          | R/W | 0x0 | PWM23_DZ_INTV.<br>PWM23 Dead Zone interval value.                                       |
| 7:1           | /   | /   | /                                                                                       |
| 0             | R/W | 0x0 | PWM23_DZ_EN.<br>PWM23 Dead Zone enable.<br>0: Dead Zone disable<br>1: Dead Zone enable. |

### 6.7.3.11 PWM\_PDZCR45\_REG

| 偏移地址：0x0038 |     |     | 默认值: 0x0000_0000                                                                        |
|-------------|-----|-----|-----------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                      |
| 15:8        | R/W | 0x0 | PWM45_DZ_INTV.<br>PWM45 Dead Zone interval value.                                       |
| 7:1         | /   | /   | /                                                                                       |
| 0           | R/W | 0x0 | PWM45_DZ_EN.<br>PWM45 Dead Zone enable.<br>0: Dead Zone disable<br>1: Dead Zone enable. |

### 6.7.3.12 PWM\_PDZCR67\_REG

| 偏移地址：0x003C |     |     | 默认值: 0x0000_0000                                                                        |
|-------------|-----|-----|-----------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                      |
| 15:8        | R/W | 0x0 | PWM67_DZ_INTV.<br>PWM67 Dead Zone interval value.                                       |
| 7:1         | /   | /   | /                                                                                       |
| 0           | R/W | 0x0 | PWM67_DZ_EN.<br>PWM67 Dead Zone enable.<br>0: Dead Zone disable<br>1: Dead Zone enable. |

### 6.7.3.13 PWM\_PER\_REG

| 偏移地址：0x0040 |     |     | 默认值: 0x0000_0000                                                                                                                                            |
|-------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                                          |
| 31:1        | /   | /   | /                                                                                                                                                           |
| 7           | R/W | 0x0 | PWM7_EN.<br>When enable PWM, the 16-bit up-counter starts working and PWM channel7 is permitted to output PWM waveform.<br>0: PWM disable<br>1: PWM enable. |
| 6           | R/W | 0x0 | PWM6_EN.<br>When enable PWM, the 16-bit up-counter starts working and PWM channel6 is permitted to output PWM waveform.<br>0: PWM disable<br>1: PWM enable. |
| 5           | R/W | 0x0 | PWM5_EN.<br>When enable PWM, the 16-bit up-counter starts working and PWM                                                                                   |

|   |     |     |                                                                                                                                                             |
|---|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |     | channel5 is permitted to output PWM waveform.<br>0: PWM disable<br>1: PWM enable.                                                                           |
| 4 | R/W | 0x0 | PWM4_EN.<br>When enable PWM, the 16-bit up-counter starts working and PWM channel4 is permitted to output PWM waveform.<br>0: PWM disable<br>1: PWM enable. |
| 3 | R/W | 0x0 | PWM3_EN.<br>When enable PWM, the 16-bit up-counter starts working and PWM channel3 is permitted to output PWM waveform.<br>0: PWM disable<br>1: PWM enable. |
| 2 | R/W | 0x0 | PWM2_EN.<br>When enable PWM, the 16-bit up-counter starts working and PWM channel2 is permitted to output PWM waveform.<br>0: PWM disable<br>1: PWM enable. |
| 1 | R/W | 0x0 | PWM1_EN.<br>When enable PWM, the 16-bit up-counter starts working and PWM channel1 is permitted to output PWM waveform.<br>0: PWM disable<br>1: PWM enable. |
| 0 | R/W | 0x0 | PWM0_EN.<br>When enable PWM, the 16-bit up-counter starts working and PWM channel0 is permitted to output PWM waveform.<br>0: PWM disable<br>1: PWM enable. |

#### 6.7.3.14 PWM\_CER\_REG

| 偏移地址 : 0x0044 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                              |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                            |
| 31:8          | /   | /   | /                                                                                                                                                                                                             |
| 7             | R/W | 0x0 | CAP7_EN.<br>When enable capture function, the 16-bit up-counter starts working and capture channel7 is permitted to capture external falling edge or rising edge.<br>0: Capture disable<br>1: Capture enable. |
| 6             | R/W | 0x0 | CAP6_EN.<br>When enable capture function, the 16-bit up-counter starts working and capture channel6 is permitted to capture external falling edge or rising edge.                                             |

|   |     |     |                                                                                                                                                                                                               |
|---|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |     |     | 0: Capture disable<br>1: Capture enable.                                                                                                                                                                      |
| 5 | R/W | 0x0 | CAP5_EN.<br>When enable capture function, the 16-bit up-counter starts working and capture channel5 is permitted to capture external falling edge or rising edge.<br>0: Capture disable<br>1: Capture enable. |
| 4 | R/W | 0x0 | CAP4_EN.<br>When enable capture function, the 16-bit up-counter starts working and capture channel4 is permitted to capture external falling edge or rising edge.<br>0: Capture disable<br>1: Capture enable. |
| 3 | R/W | 0x0 | CAP3_EN.<br>When enable capture function, the 16-bit up-counter starts working and capture channel3 is permitted to capture external falling edge or rising edge.<br>0: Capture disable<br>1: Capture enable. |
| 2 | R/W | 0x0 | CAP2_EN.<br>When enable capture function, the 16-bit up-counter starts working and capture channel2 is permitted to capture external falling edge or rising edge.<br>0: Capture disable<br>1: Capture enable. |
| 1 | R/W | 0x0 | CAP1_EN.<br>When enable capture function, the 16-bit up-counter starts working and capture channel1 is permitted to capture external falling edge or rising edge.<br>0: Capture disable<br>1: Capture enable. |
| 0 | R/W | 0x0 | CAP0_EN.<br>When enable capture function, the 16-bit up-counter starts working and capture channel is permitted to capture external falling edge or rising edge.<br>0: Capture disable<br>1: Capture enable.  |

#### 6.7.3.15 PWM\_PCR\_REG

| 偏移地址 : $0x60+0x0+N*0x20$<br>(N= 0~7) |    | 默认值: 0x0000_0000 |    |
|--------------------------------------|----|------------------|----|
| 位置                                   | 访问 | 默认值              | 描述 |
| 31:13                                | /  | /                | /  |

|     |       |      |                                                                                                                                                                                                                                                                 |
|-----|-------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11  | R     | 0x0  | PWM_PERIOD_RDY.<br>PWM period register ready.<br>0: PWM period register is ready to write<br>1: PWM period register is busy.                                                                                                                                    |
| 10  | R/W1S | 0x0  | PWM_PUL_START.<br>PWM pulse output start.<br>0: no effect<br>1: output 1 pulse.<br>After finishing configuring for outputting pulse, set this bit once and then PWM would output one pulse. After the pulse is finished, the bit will be cleared automatically. |
| 9   | R/W   | 0x0  | PWM_MODE.<br>PWM output mode select.<br>0: cycle mode<br>1: pulse mode.                                                                                                                                                                                         |
| 8   | R/W   | 0x0  | PWM_ACT_STA.<br>PWM active state.<br>0: Low Level<br>1: High Level.                                                                                                                                                                                             |
| 7:0 | R/W   | 0x00 | PWM_PRESCAL_K.<br>PWM pre-scale K, actual pre-scale is (K+1).<br>K = 0, actual pre-scale: 1<br>K = 1, actual pre-scale: 2<br>K = 2, actual pre-scale: 3<br>K = 3, actual pre-scale: 4<br>.....<br>K = 255, actual pre-scale: 256.                               |

### 6.7.3.16 PWM\_PPR\_REG

| 偏移地址 :0x60+0x4+N*0x20<br>(N= 0~7) |     |        | 默认值: 0x0000_0000                                                                                                                                                                                                              |
|-----------------------------------|-----|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                | 访问  | 默认值    | 描述                                                                                                                                                                                                                            |
| 31:16                             | R/W | 0x0000 | PWM_ENTIRE_CYCLE.<br>Number of the entire cycles in the PWM clock.<br>0 = 1 cycle<br>1 = 2 cycles<br>.....<br>N = N+1 cycles<br>If the register need to be modified dynamically, the PCLK should be faster than the PWM CLK . |
| 15:0                              | R/W | 0x0000 | PWM_ACT_CYCLE.                                                                                                                                                                                                                |

|  |  |  |                                                                                                       |
|--|--|--|-------------------------------------------------------------------------------------------------------|
|  |  |  | Number of the active cycles in the PWM clock.<br>0 = 0 cycle<br>1 = 1 cycles<br>.....<br>N = N cycles |
|--|--|--|-------------------------------------------------------------------------------------------------------|

#### 6.7.3.17 PWM\_PCNTR\_REG

| 偏移地址 : 0x60+0x8+N*0x20<br>(N= 0~7) |    |        | 默认值: 0x0000_0000                                                                                               |
|------------------------------------|----|--------|----------------------------------------------------------------------------------------------------------------|
| 位置                                 | 访问 | 默认值    | 描述                                                                                                             |
| 31:16                              | /  | /      | /                                                                                                              |
| 15:0                               | R  | 0x0000 | On PWM output or capture input, reading this register could get the current value of the PWM 16bit up-counter. |

#### 6.7.3.18 PWM\_CCR\_REG

| 偏移地址 : 0x60+0xC+N*0x20<br>(N= 0~7) |     |     | 默认值: 0x0000_0000                                                                                                                                                                 |
|------------------------------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                 | 访问  | 默认值 | 描述                                                                                                                                                                               |
| 31:3                               | /   | /   | /                                                                                                                                                                                |
| 2                                  | R/W | 0x0 | CRLF.<br>When capture channel captures rising edge, the 16-bit up-counter's current value is latched to CRLR and then this bit is set 1 by hardware. Write 1 to clear this bit.  |
| 1                                  | R/W | 0x0 | CFLF.<br>When capture channel captures falling edge, the 16-bit up-counter's current value is latched to CFLR and then this bit is set 1 by hardware. Write 1 to clear this bit. |
| 0                                  | R/W | 0x0 | CAPINV.<br>Inversing the signal inputted form capture channel before capture channel's 16bit counter.<br>0: not inverse<br>1: inverse                                            |

#### 6.7.3.19 PWM\_CRLR\_REG

| 偏移地址 : 0x60+0x10+N*0x20<br>(N= 0~7) |    |     | 默认值: 0x0000_0000 |
|-------------------------------------|----|-----|------------------|
| 位置                                  | 访问 | 默认值 | 描述               |
| 31:16                               | /  | /   | /                |



|      |   |        |                                                                                                               |
|------|---|--------|---------------------------------------------------------------------------------------------------------------|
| 15:0 | R | 0x0000 | When capture channel captures rising edge, the 16-bit up-counter's current value is latched to this register. |
|------|---|--------|---------------------------------------------------------------------------------------------------------------|

### 6.7.3.20 PWM\_CFLR\_REG

| 偏移地址 : 0x60+0x14+N*0x20<br>(N= 0~7) |    |        | 默认值: 0x0000_0000                                                                                               |
|-------------------------------------|----|--------|----------------------------------------------------------------------------------------------------------------|
| 位置                                  | 访问 | 默认值    | 描述                                                                                                             |
| 31:16                               | /  | /      | /                                                                                                              |
| 15:0                                | R  | 0x0000 | When capture channel captures falling edge, the 16-bit up-counter's current value is latched to this register. |

## 6.8 智能卡接口(ISO7816)

### 6.8.1 概述

智能卡读卡器(SCR)是在系统和智能卡之间传输数据的通信控制器。控制器可以执行完整的智能卡会话，包括卡激活、卡去激活、冷/热复位、应答复位(ATR)响应接收、数据传输等。

主要特性如下：

- 支持 ISO/IEC 7816-3:1997(E)和 EMV2000(4.0)规范
- 执行完整智能卡会话所需的功能，包括：
  - 卡片激活和去激活
  - 冷/热复位
  - 应答复位(ATR)响应接收
  - 数据传输
- 支持可调时钟速率和比特率
- 可配置的自动字节重复
- 支持常用的通信协议
  - T=0，异步半双工字符传输
  - T=1，异步半双工块传输
- 128 位 FIFO 用于数据发送和接收。
- 支持 FIFO 带阈值的接收和传输缓冲区(高达 128 位)
- 支持可配置的定时功能
  - 智能卡启动时间
  - 智能卡复位时间
  - 保护时间
  - 超时计时器

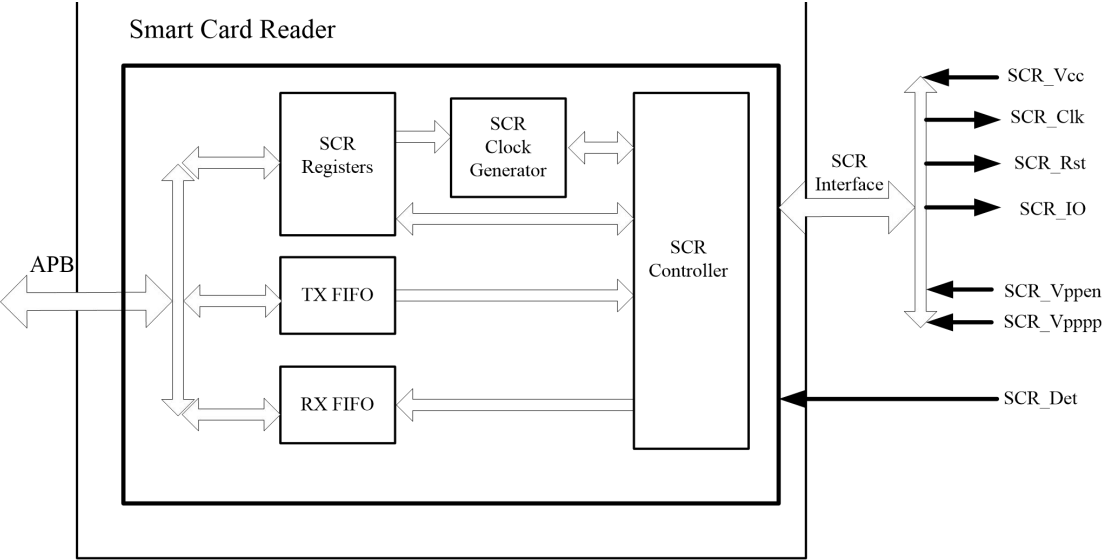
- 支持同步和任何其他非 ISO 7816 和非 EMV 卡

6.8.2 功能描述

6.8.2.1 模块框图

SCR逻辑框图如下图所示。

图 6-9 SCR 逻辑框图



6.8.2.2 接口信号

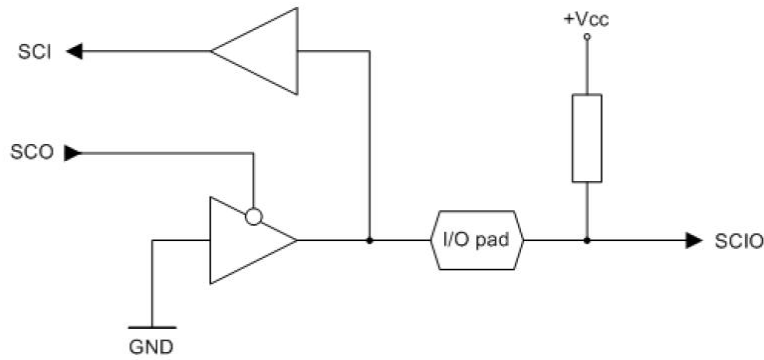
SCR 的接口信号如下表所示。

表 6-2 SCR 接口信号

| 信号名称      | 描述                              | 类型  |
|-----------|---------------------------------|-----|
| SCR_CLK   | Clock of SCR                    | O   |
| SCR_RST   | Reset signal                    | O   |
| SCR_IO    | Data signal                     | I/O |
| SCR_DET   | Card detect signal              | I   |
| SCR_PWREN | Smart card power enable         | O   |
| SCR_VPPEN | Smart card program power enable | O   |
| SCR_VPPPP | Smart Card Vpp Pause and Prog   | O   |

6.8.2.3 管脚应用电路

图 6-10 SCR IO Pad 框图



**注意**  
SCIO 管脚需要接上拉电阻。

#### 6.8.2.4 电气特性

#### 6.8.2.5 SCR 冷复位

图 6-11 SCR 冷复位时序框图

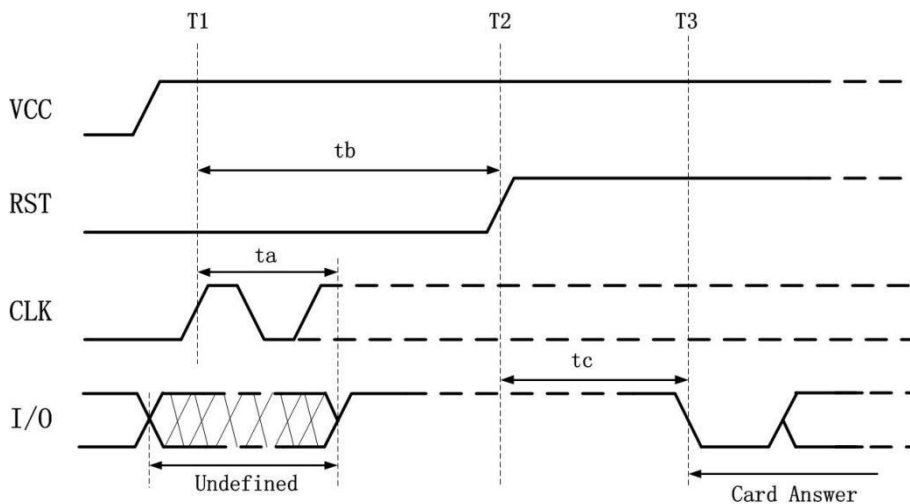


表 6-3 SCR 冷复位时序要求

| 参数              | 名称 | 最小值 | 典型值 | 最大值   | 单位   |
|-----------------|----|-----|-----|-------|------|
| Reset Hold Time | tb | 400 | -   | -     | Tclk |
| /               | ta | -   | -   | 200   | Tclk |
| /               | tc | 400 | -   | 40000 | Tclk |

激活结束时(RST处于低电平状态, VCC通电, 接口设备中I/O处于接收模式, CLK提供一个合适和稳定的时钟信号), 卡准备好了冷复位。卡在冷复位前内部处于不定态。

如图6-11所示, 时钟信号在T1时间作用于CLK。在CLK (T1 + ta)后200个时钟周期(时延ta), 卡应该将I/O设置为高电平时钟信号。冷复位需要在CLK信号保持 (T1 + tb) 时间之后, 将RST保持在低电平状态至少400个时钟周期 (延迟tb) 。当RST状态为低电平时, 接口设备忽略I/O上的状态。

在时间 $T_b$ 时，RST被置于高电平状态。I/O上的响应应该在RST上( $T_2 + t_c$ )信号上升沿之后的400到40000时钟周期(延迟)之间开始。如果应答没有在40000钟内开始，接口设备应执行去激活操作。

### 6.8.2.6 SCR 热复位

图 6-12 SCR 热复位时序框图

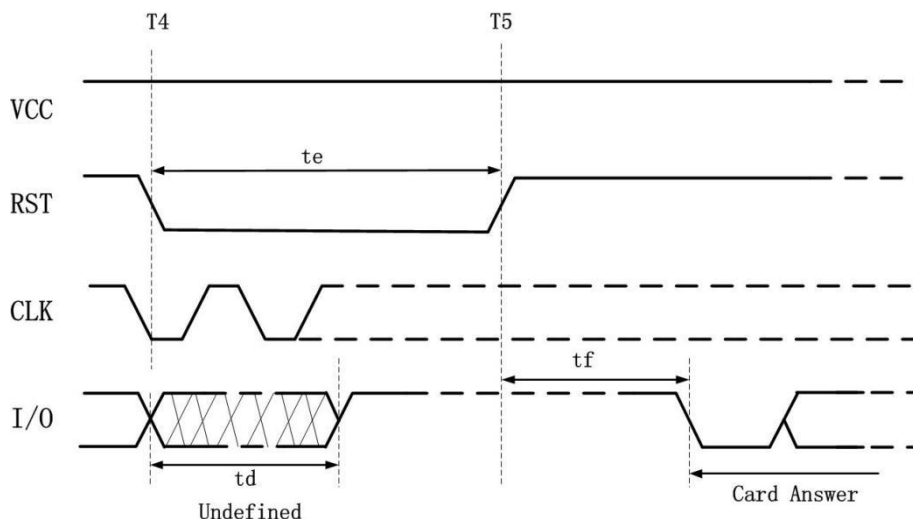


表 6-4 SCR 热复位时序要求

| 参数 | 名称 | 最小值 | 典型值 | 最大值   | 单位   |
|----|----|-----|-----|-------|------|
| /  | td | -   | -   | 200   | Tclk |
| /  | te | 400 | -   | -     | Tclk |
| /  | tf | 400 | -   | 40000 | Tclk |

热复位与冷复位不同，接口设备可能会在任意时刻复位卡，即使在回答复位时，也不能在接收前强制复位字符TS和T0。小于4 464 ( $= 12 \times 372$ ) 个时钟周期时，不应启动热复位字符T0的前缘之后。

如图6-12所示，当VCC及CLK信号稳定后，接口设备在 $T_4$ 时刻通过将RST设置为低电平（延迟 $t_e$ 时间）来发起热复位。将RST保持低电平状态后的200个时钟周期内，卡将I/O设置为高电平状态（延迟 $t_d$ 时间）。当RST在低电平状态时，接口设备将忽略I/O上的状态。

在 $T_5$ 时刻，RST被置于高电平状态。I/O上的应答将在RST上升沿保持（ $T_5 + t_f$ ）时间后的400到40000时钟周期（延迟）之间开始。如果答案没有在40000钟内开始，接口设备应执行去激活操作。

### 6.8.3 ISO7816 寄存器列表

| 模块名           | 基地址        |                                               |
|---------------|------------|-----------------------------------------------|
| SCR           | 0x40044C00 |                                               |
| 寄存器名          | 偏移地址       | 描述                                            |
| SCR_CSR_REG   | 0x0000     | Smart Card Reader Control and Status Register |
| SCR_INTEN_REG | 0x0004     | Smart Card Reader Interrupt Enable Register 1 |

|               |        |                                                    |
|---------------|--------|----------------------------------------------------|
| SCR_INTST_REG | 0x0008 | Smart Card Reader Interrupt Status Register 1      |
| SCR_FCSR_REG  | 0x000C | Smart Card Reader FIFO Control and Status Register |
| SCR_FCNT_REG  | 0x0010 | Smart Card Reader RX and TX FIFO Counter Register  |
| SCR_RPT_REG   | 0x0014 | Smart Card Reader RX and TX Repeat Register        |
| SCR_DIV_REG   | 0x0018 | Smart Card Reader Clock and Baud Divisor Register  |
| SCR_LTIM_REG  | 0x001C | Smart Card Reader Line Time Register               |
| SCR_CTIM_REG  | 0x0020 | Smart Card Reader Character Time Register          |
| SCR_LCTLR_REG | 0x0030 | Smart Card Reader Line Control Register            |
| SCR_FSM_REG   | 0x003C | Smart Card Reader FSM Register                     |
| SCR_DT_REG    | 0x0040 | Smart Card Reader Debounce Time Register           |
| SCR_FIFO_REG  | 0x0100 | Smart Card Reader RX and TX FIFO Access Point      |
| SCR_VER_REG   | 0x03FC | Smart Card Reader Version Register                 |

## 6.8.4 ISO7816 寄存器描述

### 6.8.4.1 SCR\_CSR\_REG

| 偏移地址：0x0000 |     |     | 默认值: 0x0000_0000                                                                                                                                |
|-------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                                              |
| 31          | R   | 0x0 | SCDET<br>Smart Card Detected<br><br>This bit is set to '1' when the scdetect input is active at least for a debounce time.                      |
| 30:25       | /   | /   | /                                                                                                                                               |
| 24          | R/W | 0x0 | SCDETPOL<br>Smart Card Detect Polarity<br><br>This bit set polarity of scdetect signal.<br>0: Low Active<br>1: High Active                      |
| 23:22       | R/W | 0x0 | Protocol Selection (PTLSEL)<br><br>0x0 – T=0.<br><br>0x1 – T=1, no character repeating and no guard time is used when T=1 protocol is selected. |

|       |     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |     |     | 0x2 – Reserved<br>0x3 – Reserved                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 21    | R/W | 0x0 | ATRSTFLUSH<br>ATR Start Flush FIFO<br>When enabled, both FIFOs are flushed before the ATR is started.                                                                                                                                                                                                                                                                                                                                             |
| 20    | R/W | 0x0 | TSRXE<br>TS Receive Enable<br>When set to ‘1’, the TS character (the first ATR character) will be store in RXFIFO during card session.                                                                                                                                                                                                                                                                                                            |
| 19    | R/W | 0x0 | CLKSTPPOL<br>Clock Stop Polarity<br>The value of the scclk output during the clock stop state.                                                                                                                                                                                                                                                                                                                                                    |
| 18    | R/W | 0x0 | PECRXE<br>Parity Error Character Receive Enable<br>Enables storage of the characters received with wrong parity in RX FIFO.                                                                                                                                                                                                                                                                                                                       |
| 17    | R/W | 0x0 | MSBF<br>MSB First<br>When high, inverse bit ordering convention (msb to lsb) is used.                                                                                                                                                                                                                                                                                                                                                             |
| 16    | R/W | 0x0 | DATAPOL<br>Data Polarity<br>When high, inverse level convention is used (A=’ 1’ , Z=’ 0’ ).                                                                                                                                                                                                                                                                                                                                                       |
| 15:12 | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 11    | R/W | 0x0 | DEACT<br>Setting of this bit initializes the deactivation sequence. When the deactivation is finished, the DEACT bit is automatically cleared.                                                                                                                                                                                                                                                                                                    |
| 10    | R/W | 0x0 | ACT<br>Activation. Setting of this bit initializes the activation sequence. When the activation is finished, the ACT bit is automatically cleared.                                                                                                                                                                                                                                                                                                |
| 9     | R/W | 0x0 | WARMRST<br>Warm Reset Command. Writing ‘1’ to this bit initializes Warm Reset of the Smart Card. This bit is always read as ‘0’ .                                                                                                                                                                                                                                                                                                                 |
| 8     | R/W | 0x0 | CLKSTOP<br>Clock Stop. When this bit is asserted and the smart card I/O line is in ‘Z’ state, the SCR core stops driving of the smart card clock signal after the CLKSTOPDELAY time expires. The smart card clock is restarted immediately after the CLKSTOP signal is deasserted. New character transmission can be started after CLKSTARTDELAY time. The expiration of both times is signaled by the CLKSTOPRUN bit in the interrupt registers. |
| 7:3   | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|   |     |     |                                                                                                                                                                                                    |
|---|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | R/W | 0x0 | GINTEN<br>Global Interrupt Enable. When high, IRQ output assertion is enabled.                                                                                                                     |
| 1 | R/W | 0x0 | RXEN<br>Receiving Enable. When enabled the characters sent by the Smart Card are received by the UART and stored in RX FIFO. Receiving is internally disabled while a transmission is in progress. |
| 0 | R/W | 0x0 | TXEN<br>Transmission Enable. When enabled the characters are read from TX FIFO and transmitted through UART to the Smart Card.                                                                     |

#### 6.8.4.2 SCR\_INTEN\_REG

| 偏移地址：0x0004 |     |     | 默认值: 0x0000_0000                                              |
|-------------|-----|-----|---------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                            |
| 31:24       | /   | /   | /                                                             |
| 23          | R/W | 0x0 | SCDEA<br>Smart Card Deactivation Interrupt Enable.            |
| 22          | R/W | 0x0 | SCACT<br>Smart Card Activation Interrupt Enable.              |
| 21          | R/W | 0x0 | SCINS<br>Smart Card Inserted Interrupt Enable.                |
| 20          | R/W | 0x0 | SCREM<br>Smart Card Removed Interrupt Enable.                 |
| 19          | R/W | 0x0 | ATRDONE<br>ATR Done Interrupt Enable.                         |
| 18          | R/W | 0x0 | ATRFail<br>ATR Fail Interrupt Enable.                         |
| 17          | R/W | 0x0 | C2CFULL<br>Two Consecutive Characters Limit Interrupt Enable. |
| 16          | R/W | 0x0 | CLKSTOPRUN<br>Smart Card Clock Stop/Run Interrupt Enable.     |
| 15:13       | /   | /   | /                                                             |
| 12          | R/W | 0x0 | RXPERR<br>RX Parity Error Interrupt Enable.                   |
| 11          | R/W | 0x0 | RXDONE<br>RX Done Interrupt Enable.                           |
| 10          | R/W | 0x0 | RXFIFOTH<br>RX FIFO Threshold Interrupt Enable.               |

|     |     |     |                                                  |
|-----|-----|-----|--------------------------------------------------|
| 9   | R/W | 0x0 | RXFIFOFULL<br>RX FIFO Full Interrupt Enable.     |
| 8:5 | /   | /   | /                                                |
| 4   | R/W | 0x0 | TXPERR<br>TX Parity Error Interrupt Enable.      |
| 3   | R/W | 0x0 | TXDONE<br>TX Done Interrupt Enable.              |
| 2   | R/W | 0x0 | TXFIFOTHD<br>TX FIFO Threshold Interrupt Enable. |
| 1   | R/W | 0x0 | TXFIFOEMPTY<br>TX FIFO Empty Interrupt Enable.   |
| 0   | R/W | 0x0 | TXFIFODONE<br>TX FIFO Done Interrupt Enable.     |

#### 6.8.4.3 SCR\_INTST\_REG

| 偏移地址：0x0008 |       |     | 默认值: 0x0000_0000                                                                                                                                                    |
|-------------|-------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问    | 默认值 | 描述                                                                                                                                                                  |
| 31:24       | /     | /   | /                                                                                                                                                                   |
| 23          | R/W1C | 0x0 | SCDEA<br>Smart Card Deactivation Interrupt. When enabled, this interrupt is asserted after the Smart Card deactivation sequence is complete.<br>Write '1' to clear. |
| 22          | R/W1C | 0x0 | SCACT<br>Smart Card Activation Interrupt. When enabled, this interrupt is asserted after the Smart Card activation sequence is complete.<br>Write '1' to clear.     |
| 21          | R/W1C | 0x0 | SCINS<br>Smart Card Inserted Interrupt. When enabled, this interrupt is asserted after the smart card insertion.<br>Write '1' to clear.                             |
| 20          | R/W1C | 0x0 | SCREM<br>Smart Card Removed Interrupt. When enabled, this interrupt is asserted after the smart card removal.<br>Write '1' to clear.                                |
| 19          | R/W1C | 0x0 | ATRDONE<br>ATR Done Interrupt. When enabled, this interrupt is asserted after the ATR sequence is successfully completed.<br>Write '1' to clear.                    |
| 18          | R/W1C | 0x0 | ATRFail                                                                                                                                                             |



|       |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|-------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |       |     | <p>ATR Fail Interrupt. When enabled, this interrupt is asserted if the ATR sequence fails.</p> <p>Write '1' to clear.</p>                                                                                                                                                                                                                                                                                                                                                                                        |
| 17    | R/W1C | 0x0 | <p>C2CFULL</p> <p>Two Consecutive Characters Limit Interrupt. When enabled, this interrupt is asserted if the time between two consecutive characters, transmitted between the Smart Card and the Reader in both directions, is equal the Two Characters Delay Limit described below. The C2CFULL interrupt is internally enabled from the ATR start to the deactivation or ATR restart initialization. It is recommended to use this counter to detect unresponsive Smart Cards.</p> <p>Write '1' to clear.</p> |
| 16    | R/W1C | 0x0 | <p>CLKSTOPRUN</p> <p>Smart Card Clock Stop/Run Interrupt. When enabled, this interrupt is asserted in two cases:</p> <p>When the smart card clock is stopped.</p> <p>When the new character can be started after the clock restart.</p> <p>To distinguish between the two interrupt cases, we recommend reading the CLKSTOP bit in SCR_CTRL1 register.</p> <p>Write '1' to clear.</p>                                                                                                                            |
| 15:13 | /     | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 12    | R/W1C | 0x0 | <p>RXPERR</p> <p>RX Parity Error Interrupt. When enabled, this interrupt is asserted after the character with wrong parity was received when the number of repeated receptions exceeds RXREPEAT value or T=1 protocol is used.</p> <p>Write '1' to clear.</p>                                                                                                                                                                                                                                                    |
| 11    | R/W1C | 0x0 | <p>RXDONE</p> <p>RX Done Interrupt. When enabled, this interrupt is asserted after a character was received from the Smart Card.</p> <p>Write '1' to clear.</p>                                                                                                                                                                                                                                                                                                                                                  |
| 10    | R/W1C | 0x0 | <p>RXFIFOTH</p> <p>RX FIFO Threshold Interrupt. When enabled, this interrupt is asserted if the number of bytes in RX FIFO is equal or exceeds the RX FIFO threshold.</p> <p>Write '1' to clear.</p>                                                                                                                                                                                                                                                                                                             |
| 9     | R/W1C | 0x0 | <p>RXFIFOFULL</p> <p>RX FIFO Full Interrupt. When enabled, this interrupt is asserted if the RX FIFO is filled up.</p> <p>Write '1' to clear.</p>                                                                                                                                                                                                                                                                                                                                                                |
| 8:5   | /     | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4     | R/W1C | 0x0 | <p>TXPERR</p> <p>TX Parity Error Interrupt. When enabled, this interrupt is asserted if the Smart Card signals wrong character parity during the guard time after the character transmission was repeated TXREPEAT times or T=1 protocol is used.</p> <p>Write '1' to clear.</p>                                                                                                                                                                                                                                 |

|   |       |     |                                                                                                                                                                                         |
|---|-------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | R/W1C | 0x0 | TXDONE<br>TX Done Interrupt. When enabled, this interrupt is asserted after one character was transmitted to the smart card.<br>Write '1' to clear.                                     |
| 2 | R/W1C | 0x0 | TXFIFOTH<br>TX FIFO Threshold Interrupt. When enabled, this interrupt is asserted if the number of bytes in TX FIFO is equal or less than the TX FIFO threshold.<br>Write '1' to clear. |
| 1 | R/W1C | 0x0 | TXFIFOEMPTY<br>TX FIFO Empty Interrupt. When enabled, this interrupt is asserted if the TX FIFO is emptied out.<br>Write '1' to clear.                                                  |
| 0 | R/W1C | 0x0 | TXFIFODONE<br>TX FIFO Done Interrupt. When enabled, this interrupt is asserted after all bytes from TX FIFO were transferred to the Smart Card.<br>Write '1' to clear.                  |

#### 6.8.4.4 SCR\_FCSR\_REG

| 偏移地址 : 0x000C |     |     | 默认值: 0x0000_0100                                                                   |
|---------------|-----|-----|------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                 |
| 31:11         | /   | /   | /                                                                                  |
| 10            | R/W | 0x0 | RXFIFOFLUSH<br>Flush RX FIFO. RX FIFO is flushed, when '1' is written to this bit. |
| 9             | R   | 0x0 | RXFIFOFULL<br>RX FIFO Full.                                                        |
| 8             | R   | 0x1 | RXFIFOEMPTY<br>RX FIFO Empty.                                                      |
| 7:3           | /   | /   | /                                                                                  |
| 2             | R/W | 0x0 | TXFIFOFLUSH<br>Flush TX FIFO. TX FIFO is flushed, when '1' is written to this bit. |
| 1             | R   | 0x0 | TXFIFOFULL<br>TX FIFO Full.                                                        |
| 0             | R   | 0x1 | TXFIFOEMPTY<br>TX FIFO Empty.                                                      |

#### 6.8.4.5 SCR\_FIFOCNT\_REG

| 偏移地址 : 0x0010 |     |     | 默认值: 0x0000_0000                                                                                                                                                                       |
|---------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                     |
| 31:24         | R/W | 0x0 | RXFTH<br>RX FIFO Threshold<br>These bits set the interrupt threshold of RX FIFO. The interrupt is asserted when the number of bytes it receives is equal to, or exceeds the threshold. |
| 23:16         | R/W | 0x0 | TXFTH<br>TX FIFO Threshold<br>These bits set the interrupt threshold of TX FIFO. The interrupt is asserted when the number of bytes in TX FIFO is equal to or less than the threshold. |
| 15:8          | R   | 0x0 | RXFCNT<br>RX FIFO Counter<br>These bits provide the number of bytes stored in the RXFIFO.                                                                                              |
| 7:0           | R   | 0x0 | TXFCNT<br>TX FIFO Counter<br>These bits provide the number of bytes stored in the TXFIFO.                                                                                              |

#### 6.8.4.6 SCR\_REPEAT\_REG

| 偏移地址 : 0x0014 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                           |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                         |
| 15:8          | /   | /   | /                                                                                                                                                                                                                                                          |
| 7:4           | R/W | 0x0 | RXRPT<br>RX Repeat<br>This is a 4-bit register that specifies the number of attempts to request character re-transmission after wrong parity was detected. The re-transmission of the character is requested using the error signal during the guard time. |
| 3:0           | R/W | 0x0 | TXRPT<br>TX Repeat<br>This is a 4-bit register that specifies the number of attempts to re-transmit the character after the Smart Card signals the wrong parity during the guard time.                                                                     |

#### 6.8.4.7 SCR\_CLKDIV\_REG

| 偏移地址 : 0x0018 |     |     | 默认值: 0x0000_0000                                                                   |
|---------------|-----|-----|------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                 |
| 31:16         | R/W | 0x0 | BAUDDIV<br>Baud Clock Divisor. This 16-bit register defines the divisor value used |

|      |     |     |                                                                                                                                                                                                                                                                                                               |
|------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |     |     | to generate the Baud Clock impulses from the system clock.<br>$BAUD = \frac{f_{sysclk}}{2 * (BAUDDIV + 1)}$                                                                                                                                                                                                   |
| 15:0 | R/W | 0x0 | SCCDIV<br>Smart Card Clock Divisor. This 16-bit register defines the divisor value used to generate the Smart Card Clock from the system clock.<br>$f_{sclk} = \frac{f_{sysclk}}{2 * (SCCDIV + 1)}$<br>$f_{sclk}$ is the frequency of Smart Card Clock Signal.<br>$f_{sysclk}$ is the frequency of APB Clock. |

#### 6.8.4.8 SCR\_LTIM\_REG

| 偏移地址 : 0x001C |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                   |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                 |
| 31:24         | /   | /   | /                                                                                                                                                                                                                                                                                  |
| 23:16         | R/W | 0x0 | ATR<br>ATR Start Limit. This 16-bit register defines the maximum time between the rising edge of the scrstn signal and the start of ATR response.<br>$\text{ATR Start Limit} = 128 * \text{ATR} * T_{sclk}$                                                                        |
| 15:8          | R/W | 0x0 | RST<br>Reset Duration. This 16-bit register sets the duration of the Smart Card reset sequence. This value is same for the cold and warm reset.<br>$\text{Cold/Warm Reset Duration} = 128 * \text{RST} * T_{sclk}$                                                                 |
| 7:0           | R/W | 0x0 | ACT<br>Activation/Deactivation Time. This 16-bit register sets the duration of each part of the activation and deactivation sequence.<br>$\text{Activation/Deactivation Duration} = 128 * \text{ACT} * T_{sclk}$<br>$T_{sclk} = \frac{1}{f_{sclk}}$ is the Smart Card Clock Cycle. |

#### 6.8.4.9 SCR\_CTIM\_REG

| 偏移地址 : 0x0020 |     |     | 默认值: 0x0000_0000                                                                                                                                     |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                   |
| 31:16         | R/W | 0x0 | CHARLIMIT<br>Character Limit. This 16-bit register sets the maximum time between the leading edges of two consecutive characters. The value is ETUs. |
| 15:8          | /   | /   | /                                                                                                                                                    |

|     |     |     |                                                                                                                                                                                                                                                     |
|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0 | R/W | 0x0 | <b>GUARDTIME</b><br>Character Guard time. This 8-bit register sets a delay at the end of each character transmitted from the Smart Card Reader to the Smart Card. The value is in ETUs. The parity error is besides signaled during the guard time. |
|-----|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 6.8.4.10 SCR\_LCTLR\_REG

| 偏移地址 : 0x0030 |     |     | 默认值: 0x0000_0000                                                                                                                                                                         |
|---------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                       |
| 31:8          | /   | /   | /                                                                                                                                                                                        |
| 7             | R/W | 0x0 | <b>DSCVPPPP</b><br>Direct Smart Card Vpp Pause/Prog. It provides direct access to SCVPPPP output.                                                                                        |
| 6             | R/W | 0x0 | <b>DSCVPPEN</b><br>Direct Smart Card Vpp Enable. It provides direct access to SCVPPEN output.                                                                                            |
| 5             | R/W | 0x0 | <b>AUTOADEAVPP</b><br>Automatic Vpp Handling. When high, it enables automatic handling of DSVPPEN and DSVPPPP signals during activation and deactivation sequence.                       |
| 4             | R/W | 0x0 | <b>DSCVCC</b><br>Direct Smart Card VCC. When DIRACCPADS='1', the DSCVCC bit provides direct access to SCVCC pad.                                                                         |
| 3             | R/W | 0x0 | <b>DSCRST</b><br>Direct Smart Card Clock. When DIRACCPADS='1', the DSCRST bit provides direct access to SCRST pad.                                                                       |
| 2             | R/W | 0x0 | <b>DSCCLK</b><br>Direct Smart Card Clock. When DIRACCPADS='1', the DSCCLK bit provides direct access to SCCLK pad.                                                                       |
| 1             | R/W | 0x0 | <b>DSCIO</b><br>Direct Smart Card Input/Output. When DIRACCPADS='1', the DSCIO bit provides direct access to SCIO pad.                                                                   |
| 0             | R/W | 0x0 | <b>DIRACCPADS</b><br>Direct Access to Smart Card Pads. When high, it disables a serial interface functionality and enables direct control of the smart card pads using following 4 bits. |

#### 6.8.4.11 SCR\_FSM\_REG

| 偏移地址 : 0x003C |    |     | 默认值: 0x0000_0000  |
|---------------|----|-----|-------------------|
| 位置            | 访问 | 默认值 | 描述                |
| 31:24         | R  | 0x0 | ATR_Structure_FSM |

|       |   |     |         |
|-------|---|-----|---------|
| 23:16 | R | 0x0 | ATR_FSM |
| 15:8  | R | 0x0 | ACT_FSM |
| 7:0   | R | 0x0 | SCR_FSM |

#### 6.8.4.12 SCR\_DT\_REG

| 偏移地址：0x0040 |     | 默认值: 0x0000_03FF |                                                                                       |
|-------------|-----|------------------|---------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值              | 描述                                                                                    |
| 31:0        | R/W | 0x3ff            | SCR_Debounce_Time<br>Set the debounce time value.The time uint is the cycle of SCCLK. |

#### 6.8.4.13 SCR\_FIFO\_REG

| 偏移地址：0x0100 |     | 默认值: 0x0000_0000 |                                                                                                                                                                                              |
|-------------|-----|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值              | 描述                                                                                                                                                                                           |
| 31:8        | /   | /                | /                                                                                                                                                                                            |
| 7:0         | R/W | 0x0              | FIFO_DATA<br>This 8-bit register provides access to the RX and TX FIFO buffers. The TX FIFO is accessed during the APB write transfer. The RX FIFO is accessed during the APB read transfer. |

#### 6.8.4.14 SCR\_VER\_REG

| 偏移地址：0x03FC |    | 默认值: 0x0001_0000 |                                |
|-------------|----|------------------|--------------------------------|
| 位置          | 访问 | 默认值              | 描述                             |
| 31:24       | /  | /                | /                              |
| 24:16       | R  | 0x1              | GEN_VER<br>Generation version. |
| 15:8        | R  | 0x0              | SUB_VER<br>Sub version.        |
| 7:0         | R  | 0x0              | PRJ_VER<br>Project version.    |

## 6.9 通用模数转换器(GPADC)

### 6.9.1 概述

GPADC是一个带有8通道多路复用器的12位采样模拟数字转换器。该ADC是一种逐次逼近寄存器（SAR）转换器。

主要特性如下：

- 12 位分辨率和 10 位有效位的 SAR 型模数转换器

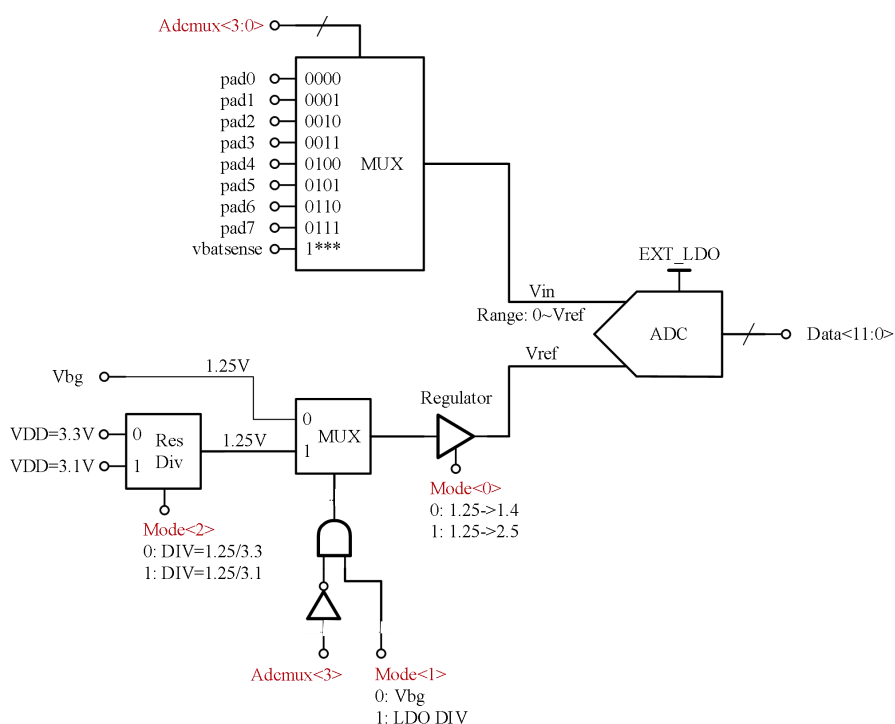
- 8 通道多路复用器，第 9 通道专门用于 VBAT 电压检测
- FIFO 数据寄存器深度为 64
- 电源电压：VDD\_EXT （1.8~3.3V），模拟输入范围：0 ~ 2.5V
- 最大采样频率:1MHz
- 支持自校准
- 支持数据比较和中断
- 支持四种操作模式
  - 单次转换模式
  - 单次循环转换模式
  - 连续转换模式
  - 突发转换模式

## 6.9.2 功能描述

### 6.9.2.1 模块框图

GPADC逻辑框图如下图所示。

图 6-13 GPADC 逻辑框图



GPADC的其他特性描述如下。

- 测量 0~7 通道时，Vref 的产生受控于 Mode<2:0>，如下表所示。

| Mode<2:0> | Description for Vref Generation      |
|-----------|--------------------------------------|
| 000       | Vref=1.4V, regulated from Vbg(1.25V) |

|     |                                      |
|-----|--------------------------------------|
| 001 | Vref=2.5V, regulated from Vbg(1.25V) |
| 010 | Vref=1.4V, generated from 3.3V VDD   |
| 011 | Vref=2.5V, generated from 3.3V VDD   |
| 100 | Same to case 000                     |
| 101 | Same to case 001                     |
| 110 | Vref=1.4V, generated from 3.1V VDD   |
| 111 | Vref=2.5V, generated from 3.1V VDD   |

- 测量 8 通道 VBAT 电压时，Vref 的产生仅受控于 Mode<0>，如下表所示。

| Mode<0> | Description for Vref Generation      |
|---------|--------------------------------------|
| 0       | Vref=1.4V, regulated from Vbg(1.25V) |
| 1       | Vref=2.5V, regulated from Vbg(1.25V) |

- Worst case 情况下，Vref 源的切换需要 10us 进行建立。
- VBAT sense = 1/3 \* VBAT。
- 增加低功耗模式。

### 6.9.2.2 接口信号

表 6-5 GPADC 接口信号

| 信号名称   | 描述           | 管脚说明     | 类型       |
|--------|--------------|----------|----------|
| GPADC0 | Analog Input | IO       | Analog   |
| GPADC1 | Analog Input | IO       | Analog   |
| GPADC2 | Analog Input | IO       | Analog   |
| GPADC3 | Analog Input | IO       | Analog   |
| GPADC4 | Analog Input | IO       | Analog   |
| GPADC5 | Analog Input | IO       | Analog   |
| GPADC6 | Analog Input | IO       | Analog   |
| GPADC7 | Reserved     | Reserved | Reserved |
| GPADC8 | VBAT Input   | VBAT     | Analog   |

### 6.9.2.3 工作模式

- 单次转换模式

GPADC 在指定的通道完成一次转换，转换数据更新在对应通道的数据寄存器中。

- 单次循环转换模式

GPADC 在所有指定的通道完成一个周期转换，转换数据更新在对应通道的数据寄存器中。



- 连续转换模式

GPADC 在所有指定的通道连续转换直到软件停止，转换数据更新在对应通道的数据寄存器中。

- 突发转换模式

GPADC 采样和转换在指定通道进行，并将结果顺序地存入 FIFO, FIFO 深度为 32 并支持中断控制。

### 6.9.3 GPADC 寄存器列表

| 模块名                 | 基地址              |                                              |
|---------------------|------------------|----------------------------------------------|
| GPADC               | 0x40043000       |                                              |
| 寄存器名                | 偏移地址             | 描述                                           |
| GP_SR_CON_REG       | 0x0000           | GPADC Sample Rate Configure Register         |
| GP_CTRL_REG         | 0x0004           | GPADC Control Register                       |
| GP_CS_EN_REG        | 0x0008           | GPADC Compare and Select Enable Register     |
| GP_FIFO_INTC_REG    | 0x000C           | GPADC FIFO Interrupt Control Register        |
| GP_FIFO_INTS_REG    | 0x0010           | GPADC FIFO Interrupt Status Register         |
| GP_FIFO_DATA_REG    | 0x0014           | GPADC FIFO Data Register                     |
| GP_CDATA_REG        | 0x0018           | GPADC Calibration Data Register              |
| GP_DATAH_INTC_REG   | 0x0020           | GPADC Data Low Interrupt Configure Register  |
| GP_DATAH_INTC_REG   | 0x0024           | GPADC Data High Interrupt Configure Register |
| GP_DATA_INTC_REG    | 0x0028           | GPADC Data Interrupt Configure Register      |
| GP_DATAH_INTC_REG   | 0x0030           | GPADC Data Low Interrupt Status Register     |
| GP_DATAH_INTC_REG   | 0x0034           | GPADC Data High Interrupt Status Register    |
| GP_DATA_INTC_REG    | 0x0038           | GPADC Data Interrupt Status Register         |
| GP_CHn_CMP_DATA_REG | $0x0040 + 4 * n$ | GPADC CH n Compare Data Register             |
| GP_CHn_DATA_REG     | $0x0080 + 4 * n$ | GPADC CH n Data Register                     |

### 6.9.4 GPADC 寄存器描述

#### 6.9.4.1 GP\_SR\_CON\_REG

| 偏移地址 : 0x0000 |     |            | 默认值: 0x0207_0033 |
|---------------|-----|------------|------------------|
| 位置            | 访问  | 默认值        | 描述               |
| 31:16         | R/W | 0x207(50K) | FS_DIV.          |

|      |     |           |                                                                                                                                                                                                                                                            |
|------|-----|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |     |           | ADC Sample Frequency Divider<br>采样周期=采样时间+量化时间+空闲<br>采用时间由 TACQ 配置，量化时间固定为 17 个 HOSC 周期<br>当 FS_DIV>TACQ+17 时，实际采样率为 HOSC/(FS_DIV+1)<br>当 FS_DIV≤TACQ+17 时，实际采样率为 HOSC/(TACQ+18)<br>CLK_IN/(n+1)<br>Note: HOSC = 26MHz, n = 0x207; HOSC = 40MHz, n = 0x31F |
| 15:0 | R/W | 0x33(2uS) | TACQ.<br>ADC acquire time<br>CLK_IN/(n+1)<br>Note: HOSC = 26MHz, n = 0x33; HOSC = 40MHz, n = 0x4F                                                                                                                                                          |

#### 6.9.4.2 GP\_CTRL\_REG

| 偏移地址 : 0x0004 |      |     | 默认值: 0x0080_0000                                                                                                                                     |
|---------------|------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问   | 默认值 | 描述                                                                                                                                                   |
| 31:24         | R/ W | 0x0 | ADC_FIRST_DLY.<br>ADC First Convert Delay setting, ADC conversion of each channel is delayed by N samples                                            |
| 23:22         | /    | /   | /                                                                                                                                                    |
| 21:20         | R/W  | 0x0 | ADC_OP_BIAS.(Adjust the bandwidth of the ADC amplifier)<br>ADC OP Bias                                                                               |
| 19:18         | R/W  | 0x0 | GPADC Work Mode<br>00: Single conversion mode<br>01: Single-cycle conversion mode<br>10: Continuous conversion mode<br>11: Burst conversion mode     |
| 17            | R/W  | 0x0 | ADC_CALI_EN.<br>ADC Calibration<br>1: start Calibration, it is clear to 0 after calibration                                                          |
| 16            | R/W  | 0x0 | ADC_EN.<br>ADC Function Enable<br>0: Disable<br>1: Enable<br>Note: the work mode and the channel number must be set before the ADC_EN bit being set. |
| 15:6          | /    | /   | /                                                                                                                                                    |
| 5             | R/W  | 0x0 | VBAT_DET_EN<br>0: disable<br>1: enable                                                                                                               |

|     |     |     |                                                                                                                                                                                                                                                                                           |
|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |     | Note: the 8th channel is designed for VBAT Voltage Detection and this bit should be set to 1 before enable the CH8                                                                                                                                                                        |
| 4   | R/W | 0x0 | LP_EN<br>0: disable<br>1: enable                                                                                                                                                                                                                                                          |
| 3:1 | R/W | 0x0 | VREF_MODE_SEL<br>000: Vref is 1.4V from BG, choose when vdd is 1.6V~3V<br>001: Vref is 2.5V from BG, choose when vdd is 3V~3.6V<br>010: Vref is 2.5V from divider of EXT_LDO, choose when vdd=3.3V<br>110: Vref is 2.5V from divider of EXT_LDO, choose when vdd=3.1V<br>others: reserved |
| 0   | R/W | 0x0 | ADC_LDO_EN<br>0: disable<br>1: enable<br><br>Note: before enabling the ADC function, you must enable the LDO-ADC firstly.                                                                                                                                                                 |

#### 6.9.4.3 GP\_CS\_EN\_REG

| 偏移地址 : 0x0008       |     |     | 默认值: 0x0000_0000                                                            |
|---------------------|-----|-----|-----------------------------------------------------------------------------|
| 位置                  | 访问  | 默认值 | 描述                                                                          |
| 31:24               | /   | /   | /                                                                           |
| [n+16]<br>(n = 0-8) | R/W | 0x0 | ADC_CHn_CMP_EN<br>Channel n Compare Enable<br>0: Disable<br>1: Enable       |
| 15:8                | /   | /   | /                                                                           |
| [n]<br>(n = 0-8)    | R/W | 0x0 | ADC_CHn_SELECT.<br>Analog input channel n Select<br>0: Disable<br>1: Enable |

#### 6.9.4.4 GP\_FIFO\_INTC\_REG

| 偏移地址 : 0x000C |     |     | 默认值: 0x0000_1F00                                                      |
|---------------|-----|-----|-----------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                    |
| 31:19         | /   | /   | /                                                                     |
| 18            | R/W | 0x0 | FIFO_DATA_DRQ_EN<br>ADC FIFO Data DRQ Enable<br>0:Disable<br>1:Enable |

|       |       |      |                                                                                                 |
|-------|-------|------|-------------------------------------------------------------------------------------------------|
| 17    | R/W   | 0x0  | FIFO_OVERRUN_IRQ_EN.<br>ADC FIFO Over Run IRQ Enable<br>0: Disable<br>1: Enable                 |
| 16    | R/W   | 0x0  | FIFO_DATA_IRQ_EN.<br>ADC FIFO Data Available IRQ Enable<br>0: Disable<br>1: Enable              |
| 15:14 | /     | /    | /                                                                                               |
| 13:8  | R/W   | 0x1F | FIFO_TRIG_LEVEL.<br>Interrupt and DMA request trigger level for ADC<br>Trigger Level = TXTL + 1 |
| 7:5   | /     | /    | /                                                                                               |
| 4     | R/W1C | 0x0  | FIFO_FLUSH.<br>ADC FIFO Flush<br>Write '1' to flush TX FIFO, self clear to '0'                  |
| 3:0   | /     | /    | /                                                                                               |

#### 6.9.4.5 GP\_FIFO\_INTS\_REG

| 偏移地址：0x0010 |       | 默认值: 0x0000_0000 |                                                                                                                                                                                                      |
|-------------|-------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问    | 默认值              | 描述                                                                                                                                                                                                   |
| 31:18       | /     | /                | /                                                                                                                                                                                                    |
| 17          | R/W1C | 0x0              | FIFO_OVERRUN_PENDING.<br>ADC FIFO Over Run IRQ pending<br>0: No Pending IRQ<br>1: FIFO Overrun Pending IRQ<br>Write '1' to clear this interrupt or automatic clear if interrupt condition fails      |
| 16          | R/W1C | 0x0              | FIFO_DATA_PENDING.<br>ADC FIFO Data Available pending Bit<br>0: NO Pending IRQ<br>1: FIFO Available Pending IRQ<br>Write '1' to clear this interrupt or automatic clear if interrupt condition fails |
| 15:14       | /     | /                | /                                                                                                                                                                                                    |
| 13:8        | R     | 0x0              | RXA_CNT.<br>ADC FIFO available Sample Word Counter                                                                                                                                                   |

|     |   |   |   |
|-----|---|---|---|
| 7:0 | / | / | / |
|-----|---|---|---|

#### 6.9.4.6 GP\_FIFO\_DATA\_REG

| 偏移地址：0x0014 |     |     | 默认值: 0x0000_0000                   |
|-------------|-----|-----|------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                 |
| 31:12       | /   | /   | /                                  |
| 11:0        | R/W | 0xX | GP_FIFO_DATA<br>GPADC Data in FIFO |

#### 6.9.4.7 GP\_CDATA\_REG

| 偏移地址：0x0018 |     |       | 默认值: 0x0000_0000                   |
|-------------|-----|-------|------------------------------------|
| 位置          | 访问  | 默认值   | 描述                                 |
| 11:0        | R/W | 0x000 | GP_CDATA<br>GPADC Calibration Data |

#### 6.9.4.8 GP\_DATA1\_INTC\_REG

| 偏移地址：0x0020      |    |     | 默认值: 0x0000_0000                          |
|------------------|----|-----|-------------------------------------------|
| 位置               | 访问 | 默认值 | 描述                                        |
| 31:8             | /  | /   | /                                         |
| [n]<br>(n = 0-8) | RW | 0   | CHn_LOW_IRQ_EN<br>0: Disable<br>1: Enable |

#### 6.9.4.9 GP\_DATAH\_INTC\_REG

| 偏移地址：0x0024      |    |     | 默认值: 0x0000_0000                           |
|------------------|----|-----|--------------------------------------------|
| 位置               | 访问 | 默认值 | 描述                                         |
| 31:8             | /  | /   | /                                          |
| [n]<br>(n = 0-8) | RW | 0   | CHn_HIGH_IRQ_EN<br>0: Disable<br>1: Enable |

#### 6.9.4.10 GP\_DATA\_INTC\_REG

| 偏移地址：0x0028 |    |     | 默认值: 0x0000_0000 |
|-------------|----|-----|------------------|
| 位置          | 访问 | 默认值 | 描述               |
| 31:8        | /  | /   | /                |

|                  |    |   |                                            |
|------------------|----|---|--------------------------------------------|
| [n]<br>(n = 0-8) | RW | 0 | CHn_DATA_IRQ_EN<br>0: Disable<br>1: Enable |
|------------------|----|---|--------------------------------------------|

#### 6.9.4.11 GP\_DATAH\_INTS\_REG

| 偏移地址 : 0x0030    |    |     | 默认值: 0x0000_0000                                                                                                                                            |
|------------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置               | 访问 | 默认值 | 描述                                                                                                                                                          |
| 31:8             | /  | /   | /                                                                                                                                                           |
| [n]<br>(n = 0-8) | RW | 0   | CHn_LOW_PENGDING<br>1: Channel n Voltage Low Available Pending IRQ<br><br>Write '1' to clear this interrupt or automatic clear if interrupt condition fails |

#### 6.9.4.12 GP\_DATAH\_INTS\_REG

| 偏移地址 : 0x0034    |    |     | 默认值: 0x0000_0000                                                                                                                                              |
|------------------|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置               | 访问 | 默认值 | 描述                                                                                                                                                            |
| 31:8             | /  | /   | /                                                                                                                                                             |
| [n]<br>(n = 0-8) | RW | 0   | CHn_HIGH_PENGDING<br>1: Channel n Voltage High Available Pending IRQ<br><br>Write '1' to clear this interrupt or automatic clear if interrupt condition fails |

#### 6.9.4.13 GP\_DATA\_INTS\_REG

| 偏移地址 : 0x0038    |    |     | 默认值: 0x0000_0000                                                                                                                                                           |
|------------------|----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置               | 访问 | 默认值 | 描述                                                                                                                                                                         |
| 31:8             | /  | /   | /                                                                                                                                                                          |
| [n]<br>(n = 0-8) | RW | 0   | CHn_DATA_PENGDING<br>0: NO Pending IRQ<br>1: Channel n Data Available Pending IRQ<br><br>Write '1' to clear this interrupt or automatic clear if interrupt condition fails |

#### 6.9.4.14 GP\_CHn\_CMP\_DATA\_REG

| 偏移地址 : 0x0040 + 4 * n<br>(n = 0-8) |    |     | 默认值: 0x0BFF_0400 |
|------------------------------------|----|-----|------------------|
| 位置                                 | 访问 | 默认值 | 描述               |
| 31:28                              | /  | /   | /                |

|       |     |       |                                                  |
|-------|-----|-------|--------------------------------------------------|
| 27:16 | R/W | 0xBFF | CHn_CMP_HIG_DATA<br>Channel n Voltage High Value |
| 15:12 | /   | /     | /                                                |
| 11:0  | R/W | 0x400 | CHn_CMP_LOW_DATA<br>Channel n Voltage Low Value  |

#### 6.9.4.15 GP\_CHn\_DATA\_REG

| 偏移地址 : $0x0080 + 4 * n$<br>( $n = 0-8$ ) |    |       | 默认值: 0x0000_0000              |
|------------------------------------------|----|-------|-------------------------------|
| 位置                                       | 访问 | 默认值   | 描述                            |
| 31:12                                    | /  | /     | /                             |
| 11:0                                     | R  | 0x000 | GP_CHn_DATA<br>Channel n Data |

## 6.10 通用 I/O(GPIO)

### 6.10.1 概述

XR806 提供了多达 26 个 GPIO(通用 IO)引脚，所有端口都有备用功能，可以通过配置选择实现在 I/O 引脚上多路复用 GPIO 功能。

XR806 中有两种类型的 GPIO 设计：GPIO 和 AGPIO。每个 GPIO 可以配置以下选项：

- 输入/ 输出/ 浮空(Hi-Z)模式
- 输入模式：上拉或下拉
- 输出模式：主动驱动
- 上拉/下拉控制：上拉和下拉内阻  $90K\Omega$ （在 PVT 条件下 $\pm 30\%$ 的变化）
- 具有 5 种触发模式的外部中断 IO：高电平、低电平、上升沿、下降沿、双边沿
- 14 个唤醒 IO 可以设置在 hibernate 状态（只用 RTC 域工作）下通过外部中断唤醒系统
- 所有的 IO 可以设置在 STANDBY 模式（RTC 和 AO 域工作）下通过外部中断唤醒系统

数字 IO AGPIO 功能除上述功能外，还可以作为专用的内控信号用于选择数字和模拟功能，这些 IO 即多路复用的 8 通道 ADC（通道 8 是内部连接测量 VBAT 电压）。



**注意**

PA23 在芯片上电启动过程中必须为高电平，才能保证整个芯片可以正常上电并运行系统。

## 6.10.2 GPIO 功能描述

表 6- 6 XR806Bxxx GPIO 复用关系表

| GPIO               | FUNC2        | FUNC3       | FUNC4     | FUNC5          | FUNC8   | FUNC9      |
|--------------------|--------------|-------------|-----------|----------------|---------|------------|
| PA00               | FEM_CTRL1    | AUDIO_PWMP  | TWI1_SCL  | IR_RX          | KEY_Y0  | PWM5/ECT5  |
| PA01               | FEM_CTRL2    | AUDIO_PWMN  | TWI1_SDA  | FLASH_CS1      | KEY_Y1  | PWM6/ECT6  |
| PA10/WUPIO0        | ADC_CH0      | SPI0_MOSI   | -         | UART1_RX       | KEY_Y2  | IR_TX      |
| PA11/WUPIO1        | ADC_CH1      | SPI0_MISO   | I2S_MCLK  | UART1_TX       | KEY_Y3  | IR_RX      |
| PA12/WUPIO2        | ADC_CH2      | PWM4/ECT4   | I2S_BCLK  | IR_TX          | KEY_Y4  | TWI0_SCL   |
| PA13/WUPIO3        | ADC_CH3      | PWM5/ECT5   | I2S_DI    | UART2_TX       | KEY_Y5  | TWI0_SDA   |
| PA14/WUPIO4        | ADC_CH4      | PWM6/ECT6   | I2S_DO    | UART2_RX       | KEY_Y6  | -          |
| PA15/WUPIO5        | ADC_CH5      | SPI0_CS0    | I2S_LRCLK | UART2_CTS      | KEY_Y7  | TWI1_SCL   |
| PA16/WUPIO6        | ADC_CH6      | SPI0_CLK    | -         | UART2_RTS      | KEY_X0  | TWI1_SDA   |
| PA17/WUPIO7        | TWI0_SCL     | AUDIO_PWMP  | 32KOSCO   | IR_TX          | KEY_X1  | -          |
| PA18/WUPIO8        | TWI0_SDA     | AUDIO_PWMN  | FEM_CTRL2 | FLASH_CS1      | KEY_X2  | -          |
| PA19/WUPIO9        | UART2_RTS    | CARD_DATA   | PWM0/ECT0 | SPI0_MOSI      | KEY_X3  | AUDIO_PWMP |
| PA20/WUPIO10       | UART2_CTS    | CARD_CLK    | PWM1/ECT1 | SPI0_MISO      | KEY_X4  | AUDIO_PWMN |
| PA21/WUPIO11       | UART2_RX     | CARD_RST    | PWM2/ECT2 | SPI0_CS0       | KEY_X5  | I2S_DO     |
| PA22/WUPIO12       | UART2_TX     | CARD_DETECT | PWM3/ECT3 | SPI0_CLK       | KEY_X6  | I2S_LRCLK  |
| PA23/WUPIO13 /TEST | DCXO_PUP_OUT | IR_RX       | FEM_CTRL1 | FEM_CTRL2      | KEY_X7  | I2S_MCLK   |
| PB00 <sup>1</sup>  | UART0_TX     | JTAG_TMS    | PWM4/ECT4 | SWD_TMS        | KEY_Y8  | -          |
| PB01 <sup>1</sup>  | UART0_RX     | JTAG_TCK    | PWM5/ECT5 | SWD_TCK        | KEY_Y9  | -          |
| PB02               | UART0_CTS    | JTAG_TDO    | PWM6/ECT6 | FLASH_WP/IO2   | KEY_Y10 | SWD_TMS    |
| PB03               | UART0_RTS    | JTAG_TDI    | PWM7/ECT7 | FLASH_HOLD/IO3 | KEY_Y11 | SWD_TCK    |
| PB04               | SPI0_MOSI    | PWM0/ECT0   | UART1_RTS | FLASH_MOSI/IO0 | KEY_Y12 | I2S_BCLK   |
| PB05               | SPI0_MISO    | PWM1/ECT1   | UART1_CTS | FLASH_MISO/IO1 | KEY_Y13 | I2S_DI     |
| PB06               | SPI0_CS0     | PWM2/ECT2   | UART1_RX  | FLASH_CS0      | KEY_Y14 | I2S_DO     |
| PB07               | SPI0_CLK     | PWM3/ECT3   | UART1_TX  | FLASH_CLK      | KEY_Y15 | I2S_LRCLK  |
| PB14 <sup>1</sup>  | UART1_TX     | UART2_TX    | TWI1_SCL  | UART0_CTS      | KEY_Y0  | PWM5/ECT5  |
| PB15 <sup>1</sup>  | UART1_RX     | UART2_RX    | TWI1_SDA  | UART0_RTS      | KEY_Y1  | PWM6/ECT6  |



| GPIO                  | FUNC2        | FUNC3       | FUNC4     | FUNC5          | FUNC8   | FUNC9      |
|-----------------------|--------------|-------------|-----------|----------------|---------|------------|
| PA00                  | FEM_CTRL1    | AUDIO_PWMP  | TWI1_SCL  | IR_RX          | KEY_Y0  | PWM5/ECT5  |
| PA01                  | FEM_CTRL2    | AUDIO_PWMN  | TWI1_SDA  | FLASH_CS1      | KEY_Y1  | PWM6/ECT6  |
| PA10/WUPIO0           | ADC_CH0      | SPI0_MOSI   | -         | UART1_RX       | KEY_Y2  | IR_TX      |
| PA11/WUPIO1           | ADC_CH1      | SPI0_MISO   | I2S_MCLK  | UART1_TX       | KEY_Y3  | IR_RX      |
| PA12/WUPIO2           | ADC_CH2      | PWM4/ECT4   | I2S_BCLK  | IR_TX          | KEY_Y4  | TWI0_SCL   |
| PA13/WUPIO3           | ADC_CH3      | PWM5/ECT5   | I2S_DI    | UART2_TX       | KEY_Y5  | TWI0_SDA   |
| PA14/WUPIO4           | ADC_CH4      | PWM6/ECT6   | I2S_DO    | UART2_RX       | KEY_Y6  | -          |
| PA15/WUPIO5           | ADC_CH5      | SPI0_CS0    | I2S_LRCLK | UART2_CTS      | KEY_Y7  | TWI1_SCL   |
| PA16/WUPIO6           | ADC_CH6      | SPI0_CLK    | -         | UART2_RTS      | KEY_X0  | TWI1_SDA   |
| PA17/WUPIO7           | TWI0_SCL     | AUDIO_PWMP  | 32KOSCO   | IR_TX          | KEY_X1  | -          |
| PA18/WUPIO8           | TWI0_SDA     | AUDIO_PWMN  | FEM_CTRL2 | FLASH_CS1      | KEY_X2  | -          |
| PA19/WUPIO9           | UART2_RTS    | CARD_DATA   | PWM0/ECT0 | SPI0_MOSI      | KEY_X3  | AUDIO_PWMP |
| PA20/WUPIO10          | UART2_CTS    | CARD_CLK    | PWM1/ECT1 | SPI0_MISO      | KEY_X4  | AUDIO_PWMN |
| PA21/WUPIO11          | UART2_RX     | CARD_RST    | PWM2/ECT2 | SPI0_CS0       | KEY_X5  | I2S_DO     |
| PA22/WUPIO12          | UART2_TX     | CARD_DETECT | PWM3/ECT3 | SPI0_CLK       | KEY_X6  | I2S_LRCLK  |
| PA23/WUPIO13<br>/TEST | DCXO_PUP_OUT | IR_RX       | FEM_CTRL1 | FEM_CTRL2      | KEY_X7  | I2S_MCLK   |
| PB00 <sup>1</sup>     | UART0_TX     | JTAG_TMS    | PWM4/ECT4 | SWD_TMS        | KEY_Y8  | -          |
| PB01 <sup>1</sup>     | UART0_RX     | JTAG_TCK    | PWM5/ECT5 | SWD_TCK        | KEY_Y9  | -          |
| PB02                  | UART0_CTS    | JTAG_TDO    | PWM6/ECT6 | FLASH_WP/IO2   | KEY_Y10 | SWD_TMS    |
| PB03                  | UART0_RTS    | JTAG_TDI    | PWM7/ECT7 | FLASH_HOLD/IO3 | KEY_Y11 | SWD_TCK    |
| PB04                  | SPI0_MOSI    | PWM0/ECT0   | UART1_RTS | FLASH_MOSI/IO0 | KEY_Y12 | I2S_BCLK   |
| PB05                  | SPI0_MISO    | PWM1/ECT1   | UART1_CTS | FLASH_MISO/IO1 | KEY_Y13 | I2S_DI     |
| PB06                  | SPI0_CS0     | PWM2/ECT2   | UART1_RX  | FLASH_CS0      | KEY_Y14 | I2S_DO     |
| PB07                  | SPI0_CLK     | PWM3/ECT3   | UART1_TX  | FLASH_CLK      | KEY_Y15 | I2S_LRCLK  |
| PB14 <sup>1</sup>     | UART1_TX     | UART2_TX    | TWI1_SCL  | UART0_CTS      | KEY_Y0  | PWM5/ECT5  |
| PB15 <sup>1</sup>     | UART1_RX     | UART2_RX    | TWI1_SDA  | UART0_RTS      | KEY_Y1  | PWM6/ECT6  |



#### NOTE

1. 支持 1.8V/3.3V/5V IO 电平标准;其他 IO 只支持 1.8V/3.3V 电平标准;

| GPI0               | FUNC2        | FUNC3       | FUNC4     | FUNC5          | FUNC8   | FUNC9      |
|--------------------|--------------|-------------|-----------|----------------|---------|------------|
| PA00               | FEM_CTRL1    | AUDIO_PWMP  | TWI1_SCL  | IR_RX          | KEY_Y0  | PWM5/ECT5  |
| PA01               | FEM_CTRL2    | AUDIO_PWMN  | TWI1_SDA  | FLASH_CS1      | KEY_Y1  | PWM6/ECT6  |
| PA10/WUPIO0        | ADC_CH0      | SPI0_MOSI   | -         | UART1_RX       | KEY_Y2  | IR_TX      |
| PA11/WUPIO1        | ADC_CH1      | SPI0_MISO   | I2S_MCLK  | UART1_TX       | KEY_Y3  | IR_RX      |
| PA12/WUPIO2        | ADC_CH2      | PWM4/ECT4   | I2S_BCLK  | IR_TX          | KEY_Y4  | TWI0_SCL   |
| PA13/WUPIO3        | ADC_CH3      | PWM5/ECT5   | I2S_DI    | UART2_TX       | KEY_Y5  | TWI0_SDA   |
| PA14/WUPIO4        | ADC_CH4      | PWM6/ECT6   | I2S_DO    | UART2_RX       | KEY_Y6  | -          |
| PA15/WUPIO5        | ADC_CH5      | SPI0_CS0    | I2S_LRCLK | UART2_CTS      | KEY_Y7  | TWI1_SCL   |
| PA16/WUPIO6        | ADC_CH6      | SPI0_CLK    | -         | UART2_RTS      | KEY_X0  | TWI1_SDA   |
| PA17/WUPIO7        | TWI0_SCL     | AUDIO_PWMP  | 32KOSCO   | IR_TX          | KEY_X1  | -          |
| PA18/WUPIO8        | TWI0_SDA     | AUDIO_PWMN  | FEM_CTRL2 | FLASH_CS1      | KEY_X2  | -          |
| PA19/WUPIO9        | UART2_RTS    | CARD_DATA   | PWM0/ECT0 | SPI0_MOSI      | KEY_X3  | AUDIO_PWMP |
| PA20/WUPIO10       | UART2_CTS    | CARD_CLK    | PWM1/ECT1 | SPI0_MISO      | KEY_X4  | AUDIO_PWMN |
| PA21/WUPIO11       | UART2_RX     | CARD_RST    | PWM2/ECT2 | SPI0_CS0       | KEY_X5  | I2S_DO     |
| PA22/WUPIO12       | UART2_TX     | CARD_DETECT | PWM3/ECT3 | SPI0_CLK       | KEY_X6  | I2S_LRCLK  |
| PA23/WUPIO13 /TEST | DCXO_PUP_OUT | IR_RX       | FEM_CTRL1 | FEM_CTRL2      | KEY_X7  | I2S_MCLK   |
| PB00 <sup>1</sup>  | UART0_TX     | JTAG_TMS    | PWM4/ECT4 | SWD_TMS        | KEY_Y8  | -          |
| PB01 <sup>1</sup>  | UART0_RX     | JTAG_TCK    | PWM5/ECT5 | SWD_TCK        | KEY_Y9  | -          |
| PB02               | UART0_CTS    | JTAG_TDO    | PWM6/ECT6 | FLASH_WP/IO2   | KEY_Y10 | SWD_TMS    |
| PB03               | UART0_RTS    | JTAG_TDI    | PWM7/ECT7 | FLASH_HOLD/IO3 | KEY_Y11 | SWD_TCK    |
| PB04               | SPI0_MOSI    | PWM0/ECT0   | UART1_RTS | FLASH_MOSI/IO0 | KEY_Y12 | I2S_BCLK   |
| PB05               | SPI0_MISO    | PWM1/ECT1   | UART1_CTS | FLASH_MISO/IO1 | KEY_Y13 | I2S_DI     |
| PB06               | SPI0_CS0     | PWM2/ECT2   | UART1_RX  | FLASH_CS0      | KEY_Y14 | I2S_DO     |
| PB07               | SPI0_CLK     | PWM3/ECT3   | UART1_TX  | FLASH_CLK      | KEY_Y15 | I2S_LRCLK  |
| PB14 <sup>1</sup>  | UART1_TX     | UART2_TX    | TWI1_SCL  | UART0_CTS      | KEY_Y0  | PWM5/ECT5  |
| PB15 <sup>1</sup>  | UART1_RX     | UART2_RX    | TWI1_SDA  | UART0_RTS      | KEY_Y1  | PWM6/ECT6  |

2. 其他功能描述可参考《XR806 PIN MUX.pdf》；

表 6- 7 XR806Axxx GPIO 复用关系表

| GPIO                  | FUNC2        | FUNC3       | FUNC4     | FUNC5          | FUNC6   |
|-----------------------|--------------|-------------|-----------|----------------|---------|
| PA11/WUPIO1           | ADC_CH1      | SPI0_MISO   | I2S_MCLK  | UART1_TX       | EINTA11 |
| PA12/WUPIO2           | ADC_CH2      | PWM4/ECT4   | I2S_BCLK  | IR_TX          | EINTA12 |
| PA13/WUPIO3           | ADC_CH3      | PWM5/ECT5   | I2S_DI    | UART2_TX       | EINTA13 |
| PA19/WUPIO9           | UART2_RTS    | CARD_DATA   | PWM0/ECT0 | SPI0_MOSI      | EINTA19 |
| PA20/WUPIO10          | UART2_CTS    | CARD_CLK    | PWM1/ECT1 | SPI0_MISO      | EINTA20 |
| PA21/WUPIO11          | UART2_RX     | CARD_RST    | PWM2/ECT2 | SPI0_CS0       | EINTA21 |
| PA22/WUPIO12          | UART2_TX     | CARD_DETECT | PWM3/ECT3 | SPI0_CLK       | EINTA22 |
| PA23/WUPIO13/<br>TEST | DCXO_PUP_OUT | IR_RX       | FEM_CTRL1 | FEM_CTRL2      | EINTA23 |
| PB00 <sup>1</sup>     | UART0_TX     | JTAG_TMS    | PWM4/ECT4 | SWD_TMS        | EINTB0  |
| PB01 <sup>1</sup>     | UART0_RX     | JTAG_TCK    | PWM5/ECT5 | SWD_TCK        | EINTB1  |
| PB02                  | UART0_CTS    | JTAG_TDO    | PWM6/ECT6 | FLASH_WP/IO2   | EINTB2  |
| PB03                  | UART0_RTS    | JTAG_TDI    | PWM7/ECT7 | FLASH_HOLD/IO3 | EINTB3  |
| PB04                  | SPI0_MOSI    | PWM0/ECT0   | UART1_RTS | FLASH_MOSI/IO0 | EINTB4  |
| PB05                  | SPI0_MISO    | PWM1/ECT1   | UART1_CTS | FLASH_MISO/IO1 | EINTB5  |
| PB06                  | SPI0_CS0     | PWM2/ECT2   | UART1_RX  | FLASH_CS0      | EINTB6  |
| PB07                  | SPI0_CLK     | PWM3/ECT3   | UART1_TX  | FLASH_CLK      | EINTB7  |
| PB14 <sup>1</sup>     | UART1_TX     | UART2_TX    | TWI1_SCL  | UART0_CTS      | EINTB14 |
| PB15 <sup>1</sup>     | UART1_RX     | UART2_RX    | TWI1_SDA  | UART0_RTS      | EINTB15 |



**NOTE**

1. 支持 1.8V/3.3V/5V IO 电平标准;其他 IO 只支持 1.8V/3.3V 电平标准;
2. 其他功能描述可参考《XR806 PIN MUX.pdf》;

### 6.10.3 GPIO 寄存器列表

| 模块名             | 基地址             |                                       |
|-----------------|-----------------|---------------------------------------|
| GPIO Controller | 0x40050000      |                                       |
| 寄存器名            | 偏移地址            | 描述                                    |
| Pn_CFG0_REG     | n*0x0024 + 0x00 | Port n Configure Register 0 (n = 0,1) |

|                  |                            |                                             |
|------------------|----------------------------|---------------------------------------------|
| Pn_CFG1_REG      | $n * 0x0024 + 0x04$        | Port n Configure Register 1 (n = 0,1)       |
| Pn_CFG2_REG      | $n * 0x0024 + 0x08$        | Port n Configure Register 2 (n = 0,1)       |
| Pn_CFG3_REG      | $n * 0x0024 + 0x0C$        | Port n Configure Register 3 (n = 0,1)       |
| Pn_DATA_REG      | $n * 0x0024 + 0x10$        | Port n Data Register (n = 0,1)              |
| Pn_DRV0_REG      | $n * 0x0024 + 0x14$        | Port n Multi-Driving Register 0 (n = 0,1)   |
| Pn_DRV1_REG      | $n * 0x0024 + 0x18$        | Port n Multi-Driving Register 1 (n = 0,1)   |
| Pn_PULL0_REG     | $n * 0x0024 + 0x1C$        | Port n Pull Register 0 (n = 0,1)            |
| Pn_PULL1_REG     | $n * 0x0024 + 0x20$        | Port n Pull Register 1 (n = 0,1)            |
| Pn_EINT_CFG0_REG | $0x0200 + n * 0x20 + 0x00$ | PIO External Interrupt Configure Register 0 |
| Pn_EINT_CFG1_REG | $0x0200 + n * 0x20 + 0x04$ | PIO External Interrupt Configure Register 1 |
| Pn_EINT_CFG2_REG | $0x0200 + n * 0x20 + 0x08$ | PIO External Interrupt Configure Register 2 |
| Pn_EINT_CFG3_REG | $0x0200 + n * 0x20 + 0x0C$ | PIO External Interrupt Configure Register 3 |
| Pn_EINT_CTL_REG  | $0x0200 + n * 0x20 + 0x10$ | PIO External Interrupt Control Register     |
| Pn_EINT_STA_REG  | $0x0200 + n * 0x20 + 0x14$ | PIO External Interrupt Status Register      |
| Pn_EINT_DBC_REG  | $0x0200 + n * 0x20 + 0x18$ | PIO External Interrupt Debounce Register    |

## 6.10.4 GPIO 寄存器描述

### 6.10.4.1 Pn\_CFG0\_REG

| 偏移地址 : $n * 0x24 + 0x00$<br>n=0,1  |     |     | 默认值: 0x0000_0007         |
|------------------------------------|-----|-----|--------------------------|
| 位置                                 | 访问  | 默认值 | 描述                       |
| $[4 * m + 3 : 4 * m]$<br>(m = 0~7) | R/W | 0x7 | GPIO Group n IO m Config |

### 6.10.4.2 Pn\_CFG1\_REG

| 偏移地址 : $n * 0x24 + 0x04$<br>n=0,1           |     |     | 默认值: 0x0000_0007                                                                                                                                  |
|---------------------------------------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                          | 访问  | 默认值 | 描述                                                                                                                                                |
| $[4 * (m-8) + 3 : 4 * (m-8)]$<br>(m = 8~15) | R/W | 0x7 | GPIO Group n IO m Config<br><br>Note: The function numbers please refer to the Pin Multiplexing. If this IO is not exist, this field is reserved. |

#### 6.10.4.3 Pn\_CFG2\_REG

| 偏移地址 : $n \times 0x24 + 0x08$<br>$n=0,1$                        |     |     | 默认值: 0x0000_0007                                                                                                                                  |
|-----------------------------------------------------------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                                              | 访问  | 默认值 | 描述                                                                                                                                                |
| $[4 \times (m-16) + 3:4 \times (m-16)]$<br>( $m = 16 \sim 23$ ) | R/W | 0x7 | GPIO Group n IO m Config<br><br>Note: The function numbers please refer to the Pin Multiplexing. If this IO is not exist, this field is reserved. |

#### 6.10.4.4 Pn\_CFG3\_REG

| 偏移地址 : $n \times 0x24 + 0x0C$<br>$n=0,1$                        |     |     | 默认值: 0x0000_0007                                                                                                                                  |
|-----------------------------------------------------------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                                              | 访问  | 默认值 | 描述                                                                                                                                                |
| $[4 \times (m-24) + 3:4 \times (m-24)]$<br>( $m = 24 \sim 31$ ) | R/W | 0x7 | GPIO Group n IO m Config<br><br>Note: The function numbers please refer to the Pin Multiplexing. If this IO is not exist, this field is reserved. |

#### 6.10.4.5 Pn\_DATA\_REG

| 偏移地址 : $n \times 0x24 + 0x10$<br>$n=0,1$ |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                   |
|------------------------------------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                       | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                 |
| 31:0                                     | R/W | 0   | DATA<br><br>If the port is configured as input, the corresponding bit is the pin state. If the port is configured as output, the pin state is the same as the corresponding bit. The read bit value is the value setup by software. If the port is configured as functional pin, the undefined value will be read. |

#### 6.10.4.6 Pn\_DRV0\_REG

| 偏移地址 : $n \times 0x24 + 0x14$<br>$n=0,1$             |     |     | 默认值: 0x0000_0001                                                                                                                                                                 |
|------------------------------------------------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                                   | 访问  | 默认值 | 描述                                                                                                                                                                               |
| $[2 \times m + 1:2 \times m]$<br>( $m = 15 \sim 0$ ) | R/W | 0x1 | DRV_STRENGTH<br>00: level 0                      01: level 1<br>10: level 2                      11: level 3<br><br>Note: This field is only valid when the related IO is exist. |

#### 6.10.4.7 Pn\_DRV1\_REG

| 偏移地址 :n*0x24 + 0x18<br>n=0,1         |     |     | 默认值: 0x0000_0001                                                                                                                                                                 |
|--------------------------------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                   | 访问  | 默认值 | 描述                                                                                                                                                                               |
| [2*(m-16)+1:2*(m-16)]<br>(m = 31~16) | R/W | 0x1 | DRV_STRENGTH<br>00: level 0                      01: level 1<br>10: level 2                      11: level 3<br><br>Note: This field is only valid when the related IO is exist. |

#### 6.10.4.8 Pn\_PULL0\_REG

| 偏移地址 :n*0x24 + 0x1C<br>n=0,1 |     |     | 默认值: 0x0000_0000                                                                                                                                                     |
|------------------------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                           | 访问  | 默认值 | 描述                                                                                                                                                                   |
| [2*m+1:2*m]<br>(m = 15~0)    | R/W | 0x0 | PULL<br>00: NO PULL                      01: Pull-Up<br>10: Pull-Down                      11: /<br><br>Note: This field is only valid when the related IO is exist. |

#### 6.10.4.9 Pn\_PULL1\_REG

| 偏移地址 :n*0x24 + 0x20<br>n=0,1         |     |     | 默认值: 0x0000_0000                                                                                                                                                     |
|--------------------------------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                   | 访问  | 默认值 | 描述                                                                                                                                                                   |
| [2*(m-16)+1:2*(m-16)]<br>(m = 31~16) | R/W | 0x0 | PULL<br>00: NO PULL                      01: Pull-Up<br>10: Pull-Down                      11: /<br><br>Note: This field is only valid when the related IO is exist. |

#### 6.10.4.10 Pn\_EINT\_CFG0\_REG

| 偏移地址 :0x200 + 0x20*n + 0x00 |     |     | 默认值: 0x0000_0007                                                       |
|-----------------------------|-----|-----|------------------------------------------------------------------------|
| 位置                          | 访问  | 默认值 | 描述                                                                     |
| [4*m+3:4*m]<br>(m = 0~7)    | R/W | 0x0 | EINT_MODE<br><br>0: Positive Edge<br>1: Negative Edge<br>2: High Level |

|  |  |  |                                                                                                                 |
|--|--|--|-----------------------------------------------------------------------------------------------------------------|
|  |  |  | 3: Low Level<br>4: Double Edge<br>others: /<br><br>Note: This field is only valid when the related IO is exist. |
|--|--|--|-----------------------------------------------------------------------------------------------------------------|

#### 6.10.4.11 Pn\_EINT\_CFG1\_REG

| 偏移地址 :0x200 + 0x20*n + 0x04         |     |     | 默认值: 0x0000_0000                                                                                                                                                                          |
|-------------------------------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                  | 访问  | 默认值 | 描述                                                                                                                                                                                        |
| $[4*(m-8)+3:4*(m-8)]$<br>(m = 8~15) | R/W | 0x0 | EINT_MODE<br><br>0: Positive Edge<br>1: Negative Edge<br>2: High Level<br>3: Low Level<br>4: Double Edge<br>others: /<br><br>Note: This field is only valid when the related IO is exist. |

#### 6.10.4.12 Pn\_EINT\_CFG2\_REG

| 偏移地址 :0x200 + 0x20*n + 0x08            |     |     | 默认值: 0x0000_0000                                                                                                                                                                          |
|----------------------------------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                     | 访问  | 默认值 | 描述                                                                                                                                                                                        |
| $[4*(m-16)+3:4*(m-16)]$<br>(m = 16~23) | R/W | 0x0 | EINT_MODE<br><br>0: Positive Edge<br>1: Negative Edge<br>2: High Level<br>3: Low Level<br>4: Double Edge<br>others: /<br><br>Note: This field is only valid when the related IO is exist. |

#### 6.10.4.13 Pn\_EINT\_CFG3\_REG

| 偏移地址 :0x200 + 0x20*n + 0x0C            |     |     | 默认值: 0x0000_0000                  |
|----------------------------------------|-----|-----|-----------------------------------|
| 位置                                     | 访问  | 默认值 | 描述                                |
| $[4*(m-24)+3:4*(m-24)]$<br>(m = 24~31) | R/W | 0x0 | EINT_MODE<br><br>0: Positive Edge |

|  |  |  |                                                                                                                                                      |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | 1: Negative Edge<br>2: High Level<br>3: Low Level<br>4: Double Edge<br>others: /<br><br>Note: This field is only valid when the related IO is exist. |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 6.10.4.14 Pn\_EINT\_CTRL\_REG

| 偏移地址 :0x200 + 0x20*n + 0x0C |     |     | 默认值: 0x0000_0000                                                                                      |
|-----------------------------|-----|-----|-------------------------------------------------------------------------------------------------------|
| 位置                          | 访问  | 默认值 | 描述                                                                                                    |
| m<br>(m = 31~0)             | R/W | 0x0 | EINT_IRQEN<br>0: disable<br>1: enable<br>Note: This field is only valid when the related IO is exist. |

#### 6.10.4.15 Pn\_EINT\_STA\_REG

| 偏移地址 :0x200 + 0x20*n + 0x14 |     |     | 默认值: 0x0000_0000                                                                                                                   |
|-----------------------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------|
| 位置                          | 访问  | 默认值 | 描述                                                                                                                                 |
| [m]<br>(m = 31~0)           | R/W | 0x0 | EINT_IRQST<br>0: nothing<br>1: irq pending<br>write 1 to clear<br><br>Note: This field is only valid when the related IO is exist. |

#### 6.10.4.16 Pn\_EINT\_DBC\_REG

| 偏移地址 :0x200 + 0x20*n + 0x18 |     |     | 默认值: 0x0000_0000                                                     |
|-----------------------------|-----|-----|----------------------------------------------------------------------|
| 位置                          | 访问  | 默认值 | 描述                                                                   |
| 31:7                        | /   | /   | /                                                                    |
| 6:4                         | R/W | 0   | Debounce Clock Pre-Scale n<br>The selected clock is prescaled by 2^n |
| 3:1                         | /   | /   | /                                                                    |
| 0                           | R/W | 0x0 | PIO_INT_CLK_SEL<br><br>0: LFCLK<br>1: HFCLK                          |



## 7 音频接口(Audio)

### 7.1 数字音频功放驱动器(Audio-PWM)

#### 7.1.1 概述

XR806具有一个数字差分PWM（DPWM）线路输出驱动器，用于驱动外部D类音频放大器。DPWM线路输出驱动器通过上采样和调制PCM输入来差分驱动Audio\_PWMP和Audio\_PWMN引脚。DPWM线性输出驱动器的输出幅度受控于VDDIO电压。

主要特性如下：

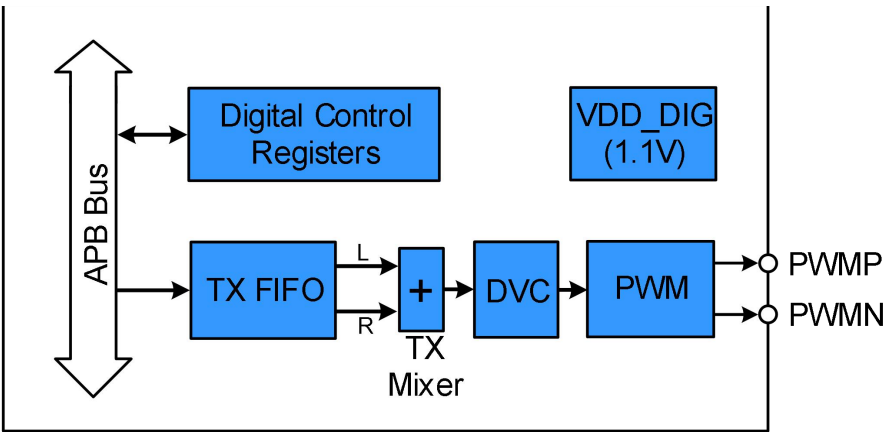
- 单声道数字音频 PWM 线路输出驱动器(CLD)
  - SNR >= 95dB (A-Weighting)
  - THD+N <= -80dB (A-Weighting)
  - 支持8KHz到48KHz的CLD采样率
- 一个 64x18 位 FIFO 用于 CLD 数据传输
- 可编辑的 FIFO 阈值
- 支持 DMA 和中断

#### 7.1.2 功能描述

##### 7.1.2.1 模块框图

Audio-PWM Class-D驱动器逻辑框图如下图所示。

图 7-1 Audio-PWM Class-D 驱动器逻辑框图



##### 7.1.2.2 接口信号

表 7-1 Audio-PWM 接口信号

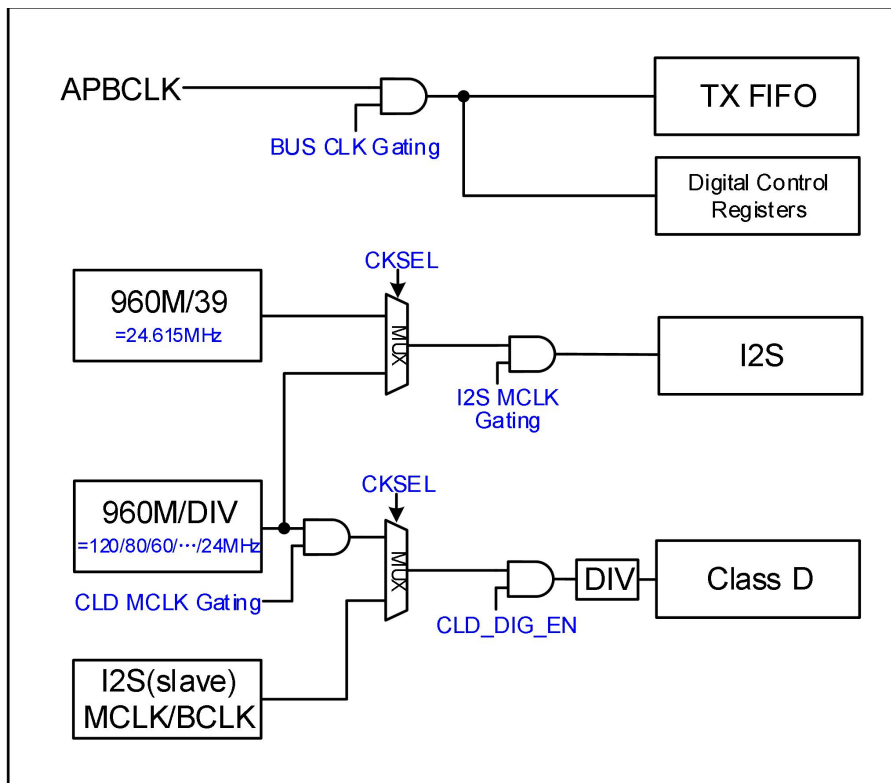
| 信号名称 | 类型 | 描述                                                   |
|------|----|------------------------------------------------------|
| PWMP | O  | Differential positive output to Digital PWM Class-D. |
| PWMN | O  | Differential negative output to Digital PWM Class-D. |

### 7.1.2.3 时钟复位及中断说明

#### ● 时钟

下图描述了数字音频PWM Class-D驱动器的时钟源。用户可以在时钟控制器单元(CCM)获取时钟设置、配置和门控信息。

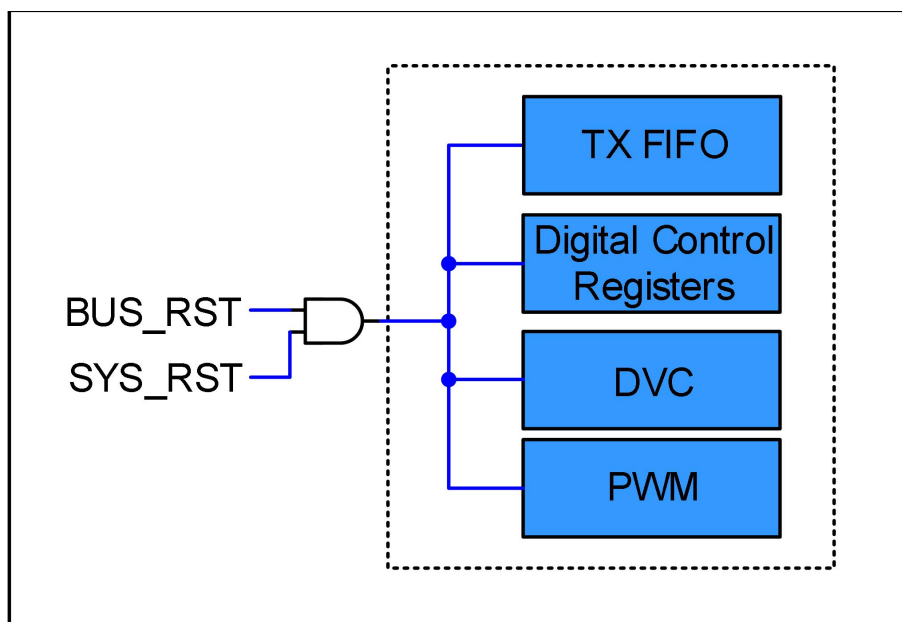
图 7- 2 Audio-PWM Class-D 驱动器时钟系统



#### ● 复位

下图描述了数字音频PWM Class-D驱动器的复位源。SYS\_RST和BUS\_RST可以复位编解码的数字部分。SYS\_RST由系统复位电路提供，BUS\_RST产生于CCM模块对应的寄存器。

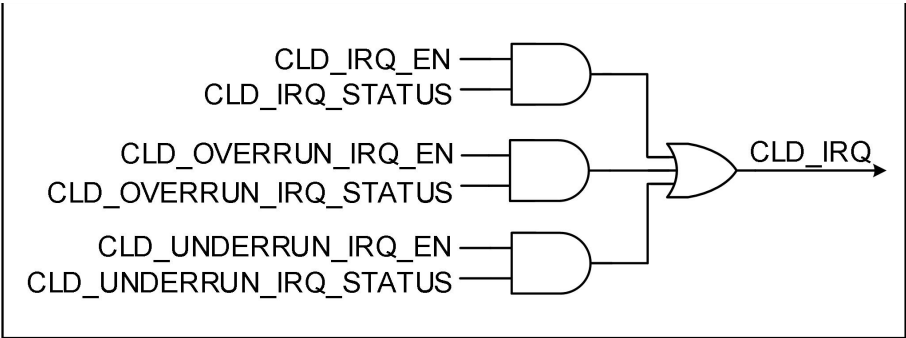
图 7- 3 Audio-PWM Class-D 驱动器复位系统



● 中断

数字音频PWM Class-D驱动器的中断系统如下图所示。

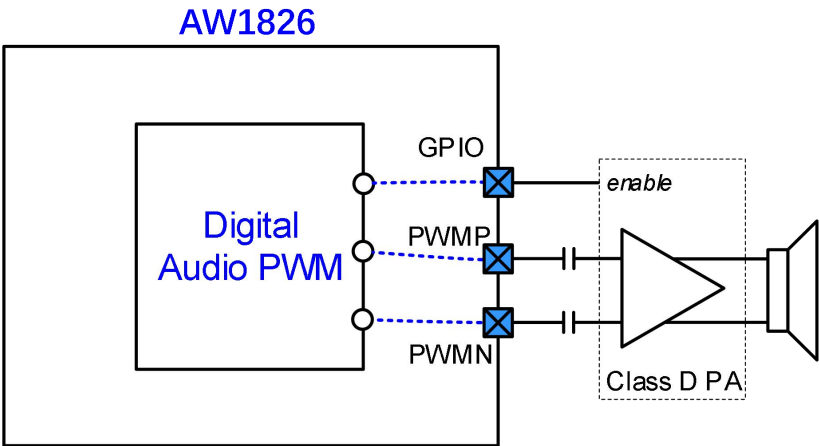
图 7- 4 Audio-PWM Class-D 驱动器中断系统



7.1.2.4 典型应用图示

下图给出了数字音频PWM Class-D驱动器的典型应用框图供参考。

图 7- 5 Audio-PWM Class-D 驱动器应用框图



7.1.3 Audio-PWM Class-D Driver 寄存器列表

| 模块名                      | 基地址        |                                  |
|--------------------------|------------|----------------------------------|
| Audio PWM Class-D Driver | 0x40044000 |                                  |
| 寄存器名                     | 偏移地址       | 描述                               |
| CLD_DIG_CTRL_REG         | 0x00       | Class-D Digital Control Register |
| CLD_FIFO_CTRL_REG        | 0x04       | Class-D FIFO Control Register    |
| CLD_FIFO_STA_REG         | 0x08       | Class-D FIFO Status Register     |
| CLD_TXDATA_REG           | 0x0C       | Class-D TX DATA Register         |
| CLD_TXCNT_REG            | 0x10       | Class-D TX FIFO Counter Register |

## 7.1.4 Audio-PWM Class-D Driver 寄存器描述

### 7.1.4.1 CLD\_DIG\_CTRL\_REG

| 偏移地址：0x0000 |     |     | 默认值: 0x0096_CC03                                                                                                     |
|-------------|-----|-----|----------------------------------------------------------------------------------------------------------------------|
| 位置          | 访问  | 默认值 | 描述                                                                                                                   |
| 31          | R/W | 0x0 | Class-D clock enable<br>0: Disable<br>1: Enable                                                                      |
| 30          | R/W | 0x0 | PWM module reset control<br>0: reset asserted<br>1: reset de-asserted                                                |
| 29          | R/W | 0x0 | Reserved                                                                                                             |
| 28          | R/W | 0x0 | PWM_DIG_EN.<br>PWM Digital Enable<br>0: Disable<br>1: Enable                                                         |
| 27:26       | R/W | 0x0 | Reserved                                                                                                             |
| 25:24       | R/W | 0x0 | PWM MODULATION MODE<br>00:156x (48k series)<br>01:136x (44.1k series)<br>10:128x (Slave Mode)<br>11:64x (Slave Mode) |
| 23          | R/W | 0x1 | Class-D dither enable<br>0:disable<br>1:enable                                                                       |
| 22:21       | R/W | 0x0 | Class-D dither level selection<br>00:1<br>01:3<br>10:7<br>11:15                                                      |
| 20          | R/W | 0x1 | Class-D mute detection enable<br>0:disable<br>1:enable                                                               |
| 19:17       | R/W | 0x3 | Class-D mute detection in time: @1/fs<br>000:16<br>001:32<br>010:64<br>011:128<br>100:256                            |

|       |     |      |                                                                                                                                                                                                                                        |
|-------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |     |      | 101:512<br>110:1024<br>111:2048                                                                                                                                                                                                        |
| 16:14 | R/W | 0x3  | clsd_dsm_tun<br>class-d modulating gain control<br>000:2/8<br>001:3/8<br>010:4/8<br>011:5/8<br>100:6/8<br>101:7/8<br>110: 8/8<br>111: 9/9                                                                                              |
| 13:12 | R/W | 0x0  | CLD_PTN_SEL<br>CLD Pattern Select<br>00: Normal(Audio sample from CLD mixer)<br>01: -6 dB sin wave<br>10: -60 dB sin wave<br>11: zero data                                                                                             |
| 11    | R/W | 0x1  | CLD_DVC_ZCD_EN<br>Class-D DVC Zero Cross Detect Enable<br>0:disable<br>1:enable                                                                                                                                                        |
| 10:4  | R/W | 0x40 | CLD_DVC_VOL (CLD DVC)<br>Class-D DVC volume<br>(-47.25dB To 47.25dB, 0.75dB/Step)<br>0x00: Mute<br>0x01: -47.25dB<br>.....<br>0x3E = -1.5dB<br>0x3F = -0.75dB<br>0x40 = 0dB<br>0x41= 0.75dB<br>0x42 = 1.5dB<br>.....<br>0x7F = 47.25dB |
| 3:2   | R/W | 0x0  | Reserved                                                                                                                                                                                                                               |
| 1:0   | R/W | 0x3  | TX_MIX_CTRL<br>00: TXL<br>01: TXR<br>10: TXL+TXR                                                                                                                                                                                       |

|  |  |  |                 |
|--|--|--|-----------------|
|  |  |  | 11: (TXL+TXR)/2 |
|--|--|--|-----------------|

#### 7.1.4.2 CLD\_FIFO\_CTRL\_REG

| 偏移地址 : 0x0004 |     |      | 默认值: 0x0000_1000                                                                                                                                                                                                                                                                             |
|---------------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值  | 描述                                                                                                                                                                                                                                                                                           |
| 31:27         | /   | /    | /                                                                                                                                                                                                                                                                                            |
| 26            | R/W | 0x0  | SEND_LASAT.<br>Audio sample select when TX FIFO under run<br>0: Sending zero<br>1: Sending last audio sample                                                                                                                                                                                 |
| 25            | R/W | 0x0  | reserved                                                                                                                                                                                                                                                                                     |
| 24            | R/W | 0x0  | FIFO_MODE.<br>For 18-bits transmitted audio sample:<br>0: FIFO_I[17:0] = {TXDATA[31:14]}<br>1: Reserved<br>For 16-bits transmitted audio sample:<br>0: FIFO_I[17:0] = {TXDATA[15:0], 2'b0}<br>1: FIFO_I[17:0] = {TXDATA[31:16], 2'b0}                                                        |
| 23            | /   | /    | /                                                                                                                                                                                                                                                                                            |
| 22:21         | R/W | 0x0  | CLD_DRQ_CLR_CNT.<br>When TX FIFO available room less than or equal N, DRQ Request will be de-asserted. N is defined here:<br>00: DRQ De-asserted when WLEVEL > TXTL<br>01: 4<br>10: 8<br>11: 16                                                                                              |
| 20:13         | /   | /    | /                                                                                                                                                                                                                                                                                            |
| 12:7          | R/W | 0x20 | TX_TRIG_LEVEL.<br>TX FIFO Empty Trigger Level (TXTL[5:0])<br>Interrupt and DMA request trigger level for TX FIFO normal condition.<br>IRQ/DRQ Generated when WLEVEL ≤ TXTL<br>Notes:<br>WLEVEL represents the number of valid samples in the TX FIFO<br>Only TXTL[5:0] valid when TXMODE = 0 |
| 6             | R/W | 0x0  | CLD_MONO_EN.<br>Class-D Mono Enable<br>0: Stereo, 32 levels FIFO<br>1: mono, 64 levels FIFO<br>When enabled, L & R channel send same data                                                                                                                                                    |

|   |      |     |                                                                                       |
|---|------|-----|---------------------------------------------------------------------------------------|
| 5 | R/W  | 0x0 | TX_SAMPLE_BITS.<br>Transmitting Audio Sample Resolution<br>0: 16 bits<br>1: 18 bits   |
| 4 | R/W  | 0x0 | CLD_DRQ_EN.<br>Class-D FIFO Empty DRQ Enable<br>0: Disable<br>1: Enable               |
| 3 | R/W  | 0x0 | CLD_IRQ_EN.<br>Class-D FIFO Empty IRQ Enable<br>0: Disable<br>1: Enable               |
| 2 | R/W  | 0x0 | FIFO_UNDERRUN_IRQ_EN.<br>Class-D FIFO Under Run IRQ Enable<br>0: Disable<br>1: Enable |
| 1 | R/W  | 0x0 | FIFO_OVERRUN_IRQ_EN.<br>Class-D FIFO Over Run IRQ Enable<br>0: Disable<br>1: Enable   |
| 0 | R/WC | 0x0 | FIFO_FLUSH.<br>Class-D FIFO Flush<br>Write '1' to flush TX FIFO, self clear to '0'    |

#### 7.1.4.3 CLD\_FIFO\_STA\_REG

| 偏移地址 : 0x0008 |       |      | 默认值: 0x0080_4008                                                                                                           |
|---------------|-------|------|----------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问    | 默认值  | 描述                                                                                                                         |
| 31:24         | /     | /    | /                                                                                                                          |
| 23            | R     | 0x1  | TX_EMPTY.<br>TX FIFO Empty<br>0: More than one sample point in TX FIFO (>= 1 word)<br>1: No anyone sample point in TX FIFO |
| 22:15         | /     | /    | /                                                                                                                          |
| 14:8          | R     | 0x40 | TXE_CNT.<br>TX FIFO Empty Space Word Counter                                                                               |
| 7:4           | /     | /    | /                                                                                                                          |
| 3             | R/W1C | 0x1  | TXE_INT.<br>TX FIFO Empty Pending Interrupt                                                                                |

|   |       |     |                                                                                                                                                        |
|---|-------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |       |     | 0: No Pending IRQ<br>1: FIFO Empty Pending Interrupt<br>Write '1' to clear this interrupt or automatic clear if interrupt condition fails.             |
| 2 | R/W1C | 0x0 | TXU_INT.<br>TX FIFO Under run Pending Interrupt<br>0: No Pending Interrupt<br>1: FIFO Under run Pending Interrupt<br>Write '1' to clear this interrupt |
| 1 | R/W1C | 0x0 | TXO_INT.<br>TX FIFO Overrun Pending Interrupt<br>0: No Pending Interrupt<br>1: FIFO Overrun Pending Interrupt<br>Write '1' to clear this interrupt     |
| 0 | /     | /   | /                                                                                                                                                      |

#### 7.1.4.4 AC\_CLD\_TXDATA\_REG

| 偏移地址 : 0x000C |    |     | 默认值: 0x0000_0000                                                                                                                                                              |
|---------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                            |
| 31:0          | W  | 0x0 | TX_DATA.<br>Transmitting left, right channel sample data should be written this register one by one. The left channel sample data is first and then the right channel sample. |

#### 7.1.4.5 CLD\_TXCNT\_REG

| 偏移地址 : 0x0010 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 31:0          | R/W | 0x0 | TX_CNT.<br>TX Sample Counter<br>The audio sample number of sending into TXFIFO. When one sample is put into TXFIFO by DMA or by host IO, the TX sample counter register increases by one. The TX sample counter register can be set to any initial value at any time. After been updated by the initial value, the counter register should count on base of this initial value.<br>Notes: It is used for Audio Synchronization |

## 7.2 数字音频接口控制器(DAUDIO)

### 7.2.1 概述

数字音频接口控制器主要用于在系统存储器和编解码芯片之间传输音频数据流。控制器支持标准I2S模式、左对齐模式、右对齐模式、PCM模式和TDM模式。

DAudio的主要特性如下：



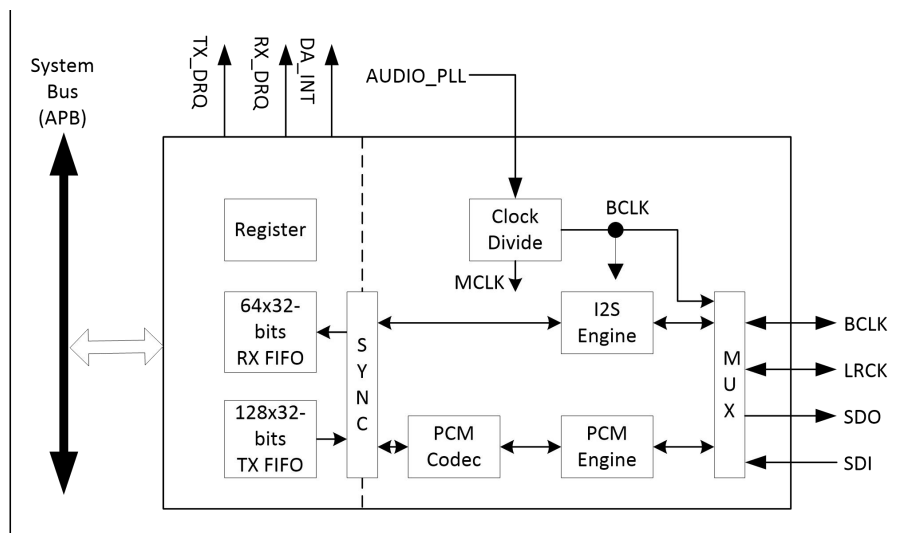
- 符合标准 Philips Inter-IC sound (I2S)总线规范
- 兼容左对齐、右对齐、PCM 模式和时分多路复用(TDM)格式
- 当同时使用 LRCK 和 LRCKR 时，每个接口支持不同的采样周期宽度
- 支持全双工同步工作模式
- 支持主/从模式
- 支持可调接口电压
- 支持 8 位到 32 位的可调音频采样分辨率
- 最多支持 8 个槽位以及 8 位到 32 位的可调宽度
- 支持采样率从 8KHz 到 192KHz
- 支持 1 个数据输出引脚
- 支持 8 位 u-law 和 8 位 A-law 非线性编码
- 一个 32 位宽深度 128 的数据发送 FIFO，一个 32 位宽深度 64 的数据接收 FIFO
- 支持可编程 PCM 帧宽：1 BCLK 宽度(短帧)和 2 BCLKs 宽度(长帧)

## 7.2.2 功能描述

### 7.2.2.1 模块框图

DAudio的内部逻辑框图如下图所示。

图 7-6 DAudio 内部逻辑框图



### 7.2.2.2 接口信号

下表描述了 DAI (DAudio Interface) 的外部信号。BCLK 和 LRCK 为双向 I/O。当 DAI 配置为主设备时，BCLK 和 LRCK 为输出引脚；当 DAI 配置为从设备时，BCLK 和 LRCK 为输入引脚。MCLK 是用于外部设备的输出引脚。SDO 是串行数据输出脚，SDI 是串行数据输入脚。关于通用 I/O 端口的信息，请参见端口控制器。

表 7-2 DAudio 接口信号

| 信号名称 | 类型  | 描述                                             |
|------|-----|------------------------------------------------|
| MCLK | O   | Digital Audio Interface MCLK Output            |
| LRCK | I/O | Digital Audio Interface Serial Clock           |
| BCLK | I/O | Digital Audio Interface Sample Rate Clock/Sync |
| SDO  | O   | Digital Audio Interface Serial Data Output     |
| SDI  | I   | Digital Audio Interface Serial Data input      |

7.2.2.3 时钟源

每个DAI控制器有三个不同的时钟。用户可以选择其中一个作为DAI时钟源。下表描述了DAI的时钟源。用户可以看到时钟控制器单元的时钟设置，配置和门控信息。

表 7-3 DAI 时钟源

| 时钟名称      | 描述                                                                                           |
|-----------|----------------------------------------------------------------------------------------------|
| Audio_PLL | 24.576Mhz or 22.5792Mhz generated by AUDIO-PLL to produce 48KHz or 44.1KHz serial frequency. |

7.2.2.4 时序说明

DAI支持标准I2S模式、左对齐I2S模式、右对齐I2S模式、PCM模式和TDM模式。软件可以通过设置DAI控制寄存器来选择DAI工作的其中一个。

图 7-7 标准 I2S/TDM-I2S 模式时序

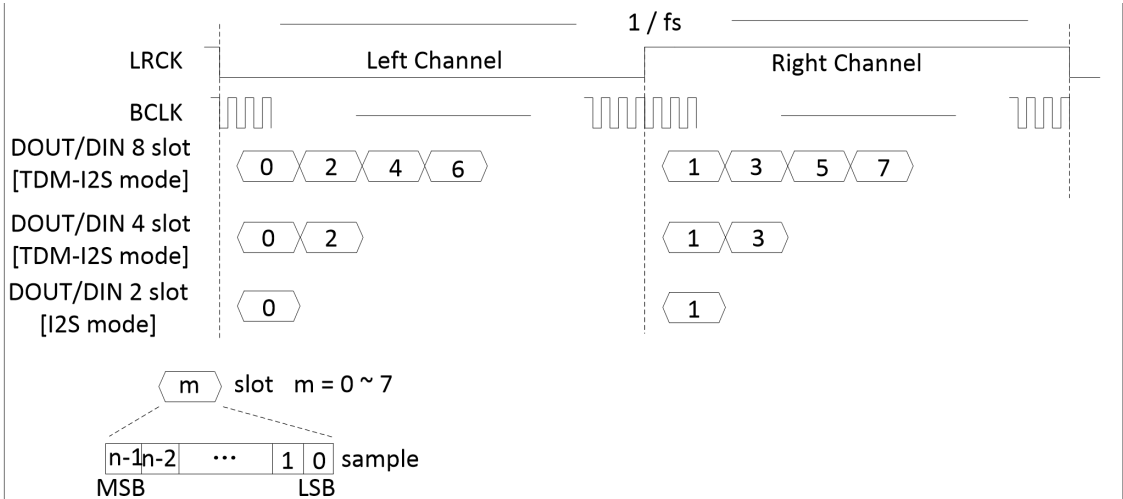


图 7-8 左对齐/TDM-Left 模式时序

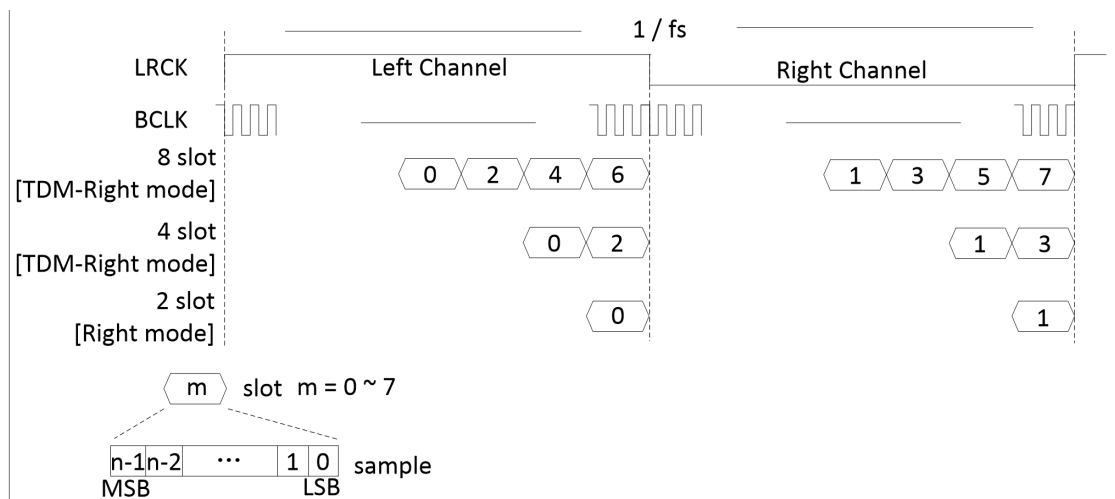
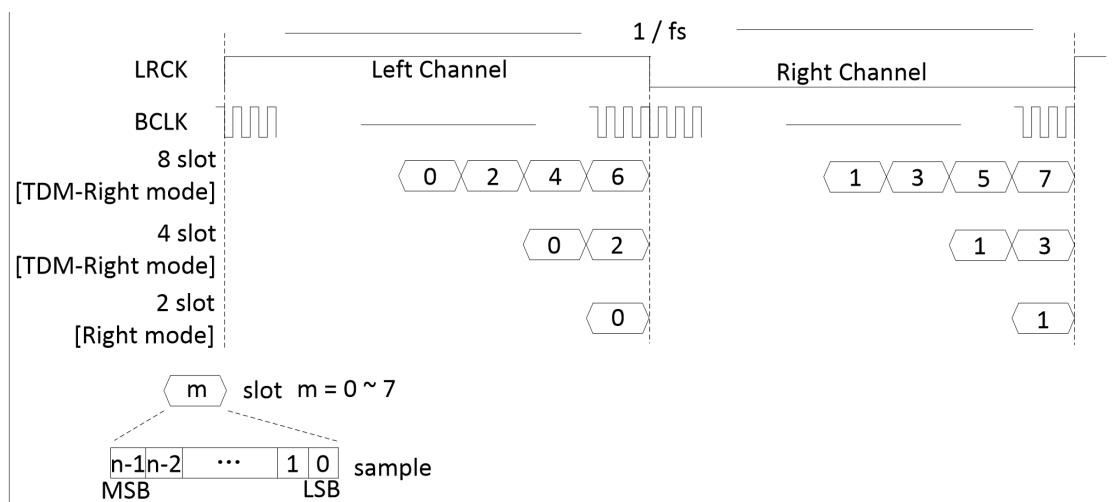


图 7-9 右对齐/TDM-Right 模式时序



### 7.2.2.5 软件操作

DAI的软件操作分为五个步骤：系统设置、DAI初始化、通道设置、DMA设置和使能/关闭模块。下面几节将详细描述这五种设置。

#### ● 系统设置

系统设置的第一步是正确编程GPIO。因为DAI端口是多路复用引脚，用户可以在引脚复用规范中找到这一功能。DAI的时钟源配置需要遵守以下流程。

首先，必须通过CCMU中的PLL\_AUDIO\_CTRL\_REG的PLL\_ENABLE位重置音频锁相环。

第二步，必须在PLL\_AUDIO\_CTRL\_REG中设置音频锁的频率，音频锁相环的配置可查阅时钟源说明。

第三步，当检测到PLL\_AUDIO\_CTRL\_REG的LOCK位变为1时，必须通过DAI0\_CLK\_REG/DAI1\_CLK\_REG打开DAI门控。

最后，必须重置DAI的BUS\_RST\_REG3的位[13:12]，并通过BUS\_GATING\_REG2的位[13:12]打开DAI总线开关。

#### ● DAI 初始化

系统设置完成后，就可以设置DAI的寄存器了。

首先，必须通过将对应寄存器位写0来关闭全局启用位(DAI\_CTL[0])、TX启用位(DAI\_CTL[2])和RX启用位(DAI\_CTL[1])。

之后，必须通过将DAI\_FCTL[25:24]寄存器位写0来清除TX/RX FIFO。

最后，通过将DAI\_TXCNT/DAI\_RXCNT寄存器位写0来清除TX/RX FIFO计数器。

以上便完成了DAI的初始化过程。

#### ● 通道与 DMA 设置

在使用和控制DAI之前，必须先配置TWI。通过TWI配置DAI主模式和从模式，相关配置可以参考DAI的协议以及TWI章节寄存器说明。然后，可以设置平移模式、采样精度、槽宽、帧模式和触发电平，相关寄存器可以在规范中找到。

DAI支持三种传输数据方法。最常见的方式是DMA，在这个应用中只需启用DAI的DRQ功能即可。

#### ● 使能/关闭模块

首先，通过配置寄存器位DAI\_CTL[2:1]来使能TX/RX。然后，通过将DAI\_CTL中的Globe enable位写为1来启用DAI，将Globe Enable写为0来关闭DAI。

### 7.2.2.6 电气特性

以下对 DAI 主模式、从模式的时序及要求做简要说明。

#### ● DAI 主模式时序要求

图 7-10 DAI 主模式时序图

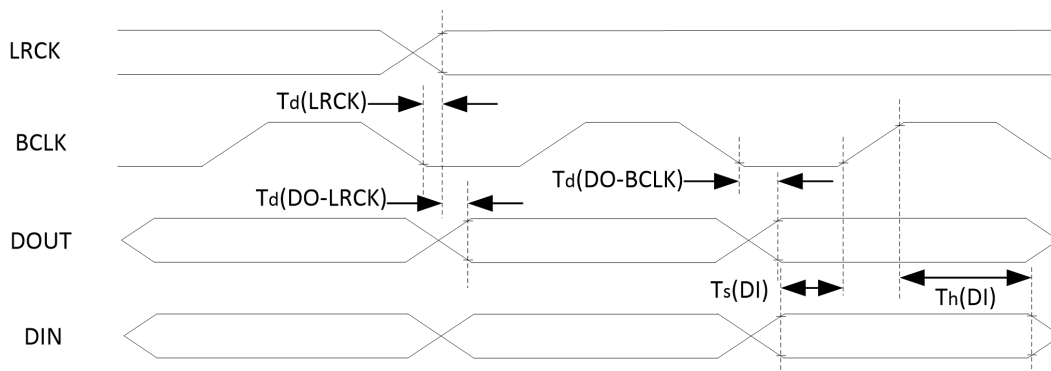


表 7-4 DAI 主模式时序要求

| 参数             | 描述                          | 最小值 | 最大值 | 单位 |
|----------------|-----------------------------|-----|-----|----|
| $T_d(LRCK)$    | LRCK delay                  | -   | 10  | ns |
| $T_d(DO-LRCK)$ | LRCK to DOUT delay(For LJF) | -   | 10  | ns |
| $T_d(DO-BCLK)$ | BCLK to DOUT delay          | -   | 10  | ns |
| $T_s(DI)$      | DIN setup                   | 4   | -   | ns |
| $T_h(DI)$      | DIN hold                    | 4   | -   | ns |
| $T_r$          | BCLK Rise time              | -   | 8*  | ns |
| $T_f$          | BCLK Fall time              | -   | 8   | ns |

● DAI 从模式时序要求

图 7-11 DAI 从模式时序图

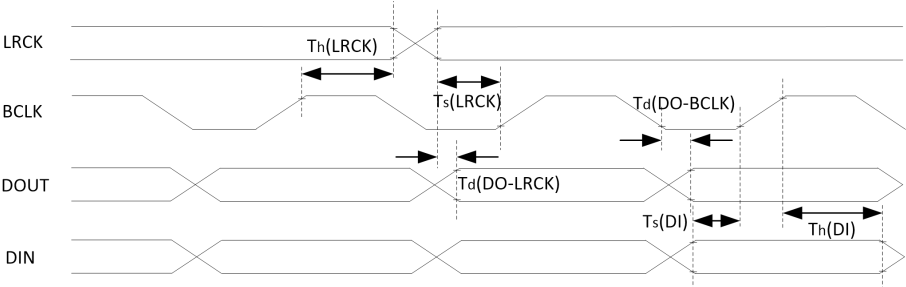


表 7-5 DAI 主模式时序要求

| 参数             | 描述                          | 最小值 | 最大值 | 单位 |
|----------------|-----------------------------|-----|-----|----|
| $T_s(LRCK)$    | LRCK setup                  | 4   | -   | ns |
| $T_h(LRCK)$    | LRCK hold                   | 4   | -   | ns |
| $T_d(DO-LRCK)$ | LRCK to DOUT delay(For LJF) | -   | 10  | ns |
| $T_d(DO-BCLK)$ | BCLK to DOUT delay          | -   | 10  | ns |
| $T_s(DI)$      | DIN setup                   | 4   | -   | ns |
| $T_h(DI)$      | DIN hold                    | 4   | -   | ns |
| $T_r$          | BCLK Rise time              | -   | 4   | ns |

7.2.3 DAUDIO 寄存器列表

| 模块名           | 基地址        |                                                |
|---------------|------------|------------------------------------------------|
| Digital Audio | 0x40042C00 |                                                |
| 寄存器名          | 偏移地址       | 描述                                             |
| DA_CTL        | 0x0000     | Digital Audio Control Register                 |
| DA_FMT0       | 0x0004     | Digital Audio Format Register 0                |
| DA_FMT1       | 0x0008     | Digital Audio Format Register 1                |
| DA_ISTA       | 0x000C     | Digital Audio Interrupt Status Register        |
| DA_RXFIFO     | 0x0010     | Digital Audio RX FIFO Register                 |
| DA_FCTL       | 0x0014     | Digital Audio FIFO Control Register            |
| DA_FSTA       | 0x0018     | Digital Audio FIFO Status Register             |
| DA_INT        | 0x001C     | Digital Audio DMA & Interrupt Control Register |

|             |        |                                                  |
|-------------|--------|--------------------------------------------------|
| DA_TXFIFO   | 0x0020 | Digital Audio TX FIFO Register                   |
| DA_CLKD     | 0x0024 | Digital Audio Clock Divide Register              |
| DA_TXCNT    | 0x0028 | Digital Audio TX Sample Counter Register         |
| DA_RXCNT    | 0x002C | Digital Audio RX Sample Counter Register         |
| DA_CHCFG    | 0x0030 | Digital Audio Channel Configuration register     |
| DA_TX0CHCFG | 0x0034 | Digital Audio TX0 Channel Configuration register |
| DA_TX1CHSEL | 0x0038 | Digital Audio TX1 Channel Select Register        |
| DA_TX2CHSEL | 0x003C | Digital Audio TX2 Channel Select Register        |
| DA_TX3CHSEL | 0x0040 | Digital Audio TX3 Channel Select Register        |
| DA_TX0CHMAP | 0x0044 | Digital Audio TX0 Channel Mapping Register       |
| DA_TX1CHMAP | 0x0048 | Digital Audio TX1 Channel Mapping Register       |
| DA_TX2CHMAP | 0x004C | Digital Audio TX2 Channel Mapping Register       |
| DA_TX3CHMAP | 0x0050 | Digital Audio TX3 Channel Mapping Register       |
| DA_RXCHSEL  | 0x0054 | Digital Audio RX Channel Select register         |
| DA_RXCHMAP  | 0x0058 | Digital Audio RX Channel Mapping Register        |

## 7.2.4 DAUDIO 寄存器描述

### 7.2.4.1 DA\_CTRL\_REG

| 偏移地址：0x0000 |     |     | 默认值: 0x0006_0000                  |
|-------------|-----|-----|-----------------------------------|
| 位置          | 访问  | 默认值 | 描述                                |
| 31:19       | /   | /   | /                                 |
| 18          | R/W | 1   | BCLK_OUT<br>0: input<br>1: output |
| 17          | R/W | 1   | LRCK_OUT<br>0: input<br>1: output |
| 16:9        | /   | /   | /                                 |
| 8           | R/W | 0   | SDO0_EN                           |

|     |     |   |                                                                                                                                                                                |
|-----|-----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |   | 0: Disable, Hi-Z state<br>1: Enable                                                                                                                                            |
| 7   | /   | / | /                                                                                                                                                                              |
| 6   | R/W | 0 | OUT Mute<br>0: normal transfer<br>1: force DOUT to output 0                                                                                                                    |
| 5:4 | R/W | 0 | MODE_SEL<br>Mode Selection<br>0: PCM mode (offset 0: DSP_B; offset 1: DSP_A)<br>1: Left mode (offset 0: LJ mode; offset 1: I2S mode)<br>2: Right-Justified mode<br>3: Reserved |
| 3   | R/W | 0 | LOOP<br>Loop back test<br>0: Normal mode<br>1: Loop back test<br>When set '1', connecting the SDO0 with the SDI                                                                |
| 2   | R/W | 0 | TXEN<br>Transmitter Block Enable<br>0: Disable<br>1: Enable                                                                                                                    |
| 1   | R/W | 0 | RXEN<br>Receiver Block Enable<br>0: Disable<br>1: Enable                                                                                                                       |
| 0   | R/W | 0 | GEN<br>Globe Enable<br>A disable on this bit overrides any other block or channel enables.<br>0: Disable<br>1: Enable                                                          |

#### 7.2.4.2 DA\_FMT0\_REG

| 偏移地址 : 0x0004 |     |     | 默认值: 0x0000_0033                                                                                                                  |
|---------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                |
| 31            | /   | /   | /                                                                                                                                 |
| 30            | R/W | 0   | LRCK_WIDTH<br>(only apply in PCM mode ) LRCK width<br>0: LRCK = 1 BCLK width (short frame)<br>1: LRCK = 2 BCLK width (long frame) |

|       |     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29:20 | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 19    | R/W | 0   | <p>LRCK_POLARITY/LRCKR_POLARITY</p> <p>When apply in I2S / Left-Justified / Right-Justified mode:</p> <p>0: Left channel when LRCK is low</p> <p>1: Left channel when LRCK is high</p> <p>When apply in PCM mode:</p> <p>0: PCM LRCK/LRCKR asserted at the negative edge</p> <p>1: PCM LRCK/LRCKR asserted at the positive edge</p>                                                                                                        |
| 18    | /   | /   | /                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 17:8  | R/W | 0   | <p>LRCK_PERIOD</p> <p>It is used to program the number of BCLKs per channel of sample frame. This value is interpreted as follow:</p> <p>PCM mode: Number of BCLKs within (Left + Right) channel width</p> <p>I2S / Left-Justified / Right-Justified mode: Number of BCLKs within each individual channel width (Left or Right)</p> <p>N+1</p> <p>For example:</p> <p>n = 7: 8 BCLK width</p> <p>...</p> <p>n = 1023: 1024 BCLKs width</p> |
| 7     | R/W | 0   | <p>BCLK_POLARITY</p> <p>0: normal mode, negative edge drive and positive edge sample</p> <p>1: invert mode, positive edge drive and negative edge sample</p>                                                                                                                                                                                                                                                                               |
| 6:4   | R/W | 3   | <p>SR</p> <p>Sample Resolution</p> <p>0: Reserved</p> <p>1: 8-bit</p> <p>2: 12-bit</p> <p>3: 16-bit</p> <p>4: 20-bit</p> <p>5: 24-bit</p> <p>6: 28-bit</p> <p>7: 32-bit</p>                                                                                                                                                                                                                                                                |
| 3     | R/W | 0   | <p>EDGE_TRANSFER</p> <p>0: SDO drive data and SDI sample data at the different BCLK edge</p> <p>1: SDO drive data and SDI sample data at the same BCLK edge</p> <p>BCLK_POLARITY = 0, use negative edge</p> <p>BCLK_POLARITY = 1, use positive edge</p>                                                                                                                                                                                    |
| 2:0   | R/W | 0x3 | <p>SW</p> <p>Slot Width Select</p> <p>0: Reserved</p>                                                                                                                                                                                                                                                                                                                                                                                      |



|  |  |  |                                                                                        |
|--|--|--|----------------------------------------------------------------------------------------|
|  |  |  | 1: 8-bit<br>2: 12-bit<br>3: 16-bit<br>4: 20-bit<br>5: 24-bit<br>6: 28-bit<br>7: 32-bit |
|--|--|--|----------------------------------------------------------------------------------------|

#### 7.2.4.3 DA\_CTRL\_REG

| 偏移地址 : 0x0008 |     |     | 默认值: 0x0000_0030                                                                                                                                                                                                   |
|---------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                 |
| 31:8          | /   | /   |                                                                                                                                                                                                                    |
| 7             | R/W | 0   | RX MLS<br>MSB / LSB First Select<br>0: MSB First<br>1: LSB First                                                                                                                                                   |
| 6             | R/W | 0   | TX MLS<br>MSB / LSB First Select<br>0: MSB First<br>1: LSB First                                                                                                                                                   |
| 5:4           | R/W | 3   | SEXT<br>Sign Extend in slot [sample resolution < slot width]<br>0: Zeros or audio gain padding at LSB position<br>1: Sign extension at MSB position<br>2: Reserved<br>3: Transfer 0 after each sample in each slot |
| 3:2           | R/W | 0   | RX_PDM<br>PCM Data Mode<br>0: Linear PCM<br>1: reserved<br>2: 8-bits u-law<br>3: 8-bits A-law                                                                                                                      |
| 1:0           | R/W | 0   | TX_PDM<br>PCM Data Mode<br>0: Linear PCM<br>1: reserved<br>2: 8-bits u-law<br>3: 8-bits A-law                                                                                                                      |

#### 7.2.4.4 DA\_ISTA\_REG

| 偏移地址 : 0x000C |     |     | 默认值: 0x0000_0010                                                                                                                                                                                                                                    |
|---------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                  |
| 31:7          | /   | /   | /                                                                                                                                                                                                                                                   |
| 6             | R/W | 0   | TXU_INT<br>TX FIFO Under run Pending Interrupt<br>0: No Pending Interrupt<br>1: FIFO Under run Pending Interrupt<br>Write 1 to clear this interrupt                                                                                                 |
| 5             | R/W | 0   | TXO_INT<br>TX FIFO Overrun Pending Interrupt<br>0: No Pending Interrupt<br>1: FIFO Overrun Pending Interrupt<br>Write '1' to clear this interrupt                                                                                                   |
| 4             | R/W | 1   | TXE_INT<br>TX FIFO Empty Pending Interrupt<br>0: No Pending IRQ<br>1: FIFO Empty Pending Interrupt when data in TX FIFO are less than TX trigger level<br>Write '1' to clear this interrupt or automatic clear if interrupt condition fails.        |
| 3             | /   | /   | /                                                                                                                                                                                                                                                   |
| 2             | R/W | 0   | RXU_INT<br>RX FIFO Under run Pending Interrupt<br>0: No Pending Interrupt<br>1:FIFO Under run Pending Interrupt<br>Write 1 to clear this interrupt                                                                                                  |
| 1             | R/W | 0   | RXO_INT<br>RX FIFO Overrun Pending Interrupt<br>0: No Pending IRQ<br>1: FIFO Overrun Pending IRQ<br>Write '1' to clear this interrupt                                                                                                               |
| 0             | R/W | 0   | RXA_INT<br>RX FIFO Data Available Pending Interrupt<br>0: No Pending IRQ<br>1: Data Available Pending IRQ when data in RX FIFO are more than RX trigger level<br>Write '1' to clear this interrupt or automatic clear if interrupt condition fails. |

#### 7.2.4.5 DA\_RXFIFO\_REG

| 偏移地址 : 0x0010 |    |     | 默认值: 0x0000_0000                                                                                                                                   |
|---------------|----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                 |
| 31:0          | R  | 0   | RX_DATA<br>RX Sample<br>Host can get one sample by reading this register. The left channel sample data is first and then the right channel sample. |

#### 7.2.4.6 DA\_FCTL\_REG

| 偏移地址 : 0x0014 |     |      | 默认值: 0x0004_00F0                                                                                                                                                                                                                          |
|---------------|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值  | 描述                                                                                                                                                                                                                                        |
| 31            | R/W | 0    | HUB_EN<br>Audio Hub Enable<br>0 : Disable<br>1 : Enable                                                                                                                                                                                   |
| 30:26         | /   | /    | /                                                                                                                                                                                                                                         |
| 25            | R/W | 0    | FTX<br>Write '1' to flush TX FIFO, self clear to '0'.                                                                                                                                                                                     |
| 24            | R/W | 0    | FRX<br>Write '1' to flush RX FIFO, self clear to '0'.                                                                                                                                                                                     |
| 23:19         | /   | /    | /                                                                                                                                                                                                                                         |
| 18:12         | R/W | 0x40 | TXTL<br>TX FIFO Empty Trigger Level<br>Interrupt and DMA request trigger level for TXFIFO normal condition<br>Trigger Level = TXTL                                                                                                        |
| 11:10         | /   | /    | /                                                                                                                                                                                                                                         |
| 9:4           | R/W | 0xF  | RXTL<br>RX FIFO Trigger Level<br>Interrupt and DMA request trigger level for RXFIFO normal condition<br>Trigger Level = RXTL + 1                                                                                                          |
| 3             | /   | /    | /                                                                                                                                                                                                                                         |
| 2             | R/W | 0    | TXIM<br>TX FIFO Input Mode (Mode 0, 1)<br>0: Valid data at the MSB of TXFIFO register<br>1: Valid data at the LSB of TXFIFO register<br>Example for 20-bits transmitted audio sample:<br>Mode 0: FIFO_I[31:0] = {APB_WDATA[31:12], 12'h0} |

|     |     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|-----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |     |   | Mode 1: FIFO_I[31:0] = {APB_WDATA[19:0], 12'h0}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 1:0 | R/W | 0 | <p>RXOM</p> <p>RX FIFO Output Mode (Mode 0, 1, 2, 3)</p> <p>00: Expanding '0' at LSB of DA_RXFIFO register.</p> <p>01: Expanding received sample sign bit at MSB of DA_RXFIFO register.</p> <p>10: Truncating received samples at high half-word of DA_RXFIFO register and low half-word of DA_RXFIFO register is filled by '0'.</p> <p>11: Truncating received samples at low half-word of DA_RXFIFO register and high half-word of DA_RXFIFO register is expanded by its sign bit.</p> <p>Example for 20-bits received audio sample:</p> <p>Mode 0: APB_RDATA[31:0] = {FIFO_O[31:12], 12'h0}</p> <p>Mode 1: APB_RDATA [31:0] = {12{FIFO_O[31]}, FIFO_O[31:12]}</p> <p>Mode 2: APB_RDATA [31:0] = {FIFO_O[31:16], 16'h0}</p> <p>Mode 3: APB_RDATA [31:0] = {16{FIFO_O[31]}, FIFO_O[31:16]}</p> |

#### 7.2.4.7 DA\_FSTA\_REG

| 偏移地址 : 0x0018 |    |      | 默认值: 0x1080_0000                                                                                                                                  |
|---------------|----|------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值  | 描述                                                                                                                                                |
| 31:29         | /  | /    | /                                                                                                                                                 |
| 28            | R  | 1    | <p>TXE</p> <p>TX FIFO Empty</p> <p>0: No room for new sample in TX FIFO</p> <p>1: More than one room for new sample in TX FIFO (&gt;= 1 word)</p> |
| 27:24         | /  | /    | /                                                                                                                                                 |
| 23:16         | R  | 0x80 | <p>TXE_CNT</p> <p>TX FIFO Empty Space Word Counter</p>                                                                                            |
| 15:9          | /  | /    | /                                                                                                                                                 |
| 8             | R  | 0    | <p>RXA</p> <p>RX FIFO Available</p> <p>0: No available data in RX FIFO</p> <p>1: More than one sample in RX FIFO (&gt;= 1 word)</p>               |
| 7             | /  | /    | /                                                                                                                                                 |
| 6:0           | R  | 0    | <p>RXA_CNT</p> <p>RX FIFO Available Sample Word Counter</p>                                                                                       |

### 7.2.4.8 DA\_INT\_REG

| 偏移地址 : 0x001C |     |     | 默认值: 0x0000_0000                                                                                                                                                    |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                  |
| 31:8          | /   | /   | /                                                                                                                                                                   |
| 7             | R/W | 0   | TX_DRQ<br>TX FIFO Empty DRQ Enable<br>0: Disable<br>1: Enable                                                                                                       |
| 6             | R/W | 0   | TXUI_EN<br>TX FIFO Under run Interrupt Enable<br>0: Disable<br>1: Enable                                                                                            |
| 5             | R/W | 0   | TXOI_EN<br>TX FIFO Overrun Interrupt Enable<br>0: Disable<br>1: Enable<br><br>When set to '1', an interrupt happens when writing new audio data if TX FIFO is full. |
| 4             | R/W | 0   | TXEI_EN<br>TX FIFO Empty Interrupt Enable<br>0: Disable<br>1: Enable                                                                                                |
| 3             | R/W | 0   | RX_DRQ<br>RX FIFO Data Available DRQ Enable<br>0: Disable<br>1: Enable<br><br>When set to '1', RXFIFO DMA Request line is asserted if Data is available in RX FIFO. |
| 2             | R/W | 0   | RXUI_EN<br>RX FIFO Under run Interrupt Enable<br>0: Disable<br>1: Enable                                                                                            |
| 1             | R/W | 0   | RXOI_EN<br>RX FIFO Overrun Interrupt Enable<br>0: Disable<br>1: Enable                                                                                              |
| 0             | R/W | 0   | RXAI_EN<br>RX FIFO Data Available Interrupt Enable<br>0: Disable                                                                                                    |

|  |  |  |           |
|--|--|--|-----------|
|  |  |  | 1: Enable |
|--|--|--|-----------|

#### 7.2.4.9 DA\_TXFIFO\_REG

| 偏移地址 : 0x0020 |    |     | 默认值: 0x0000_0000                                                                                                                                                                          |
|---------------|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问 | 默认值 | 描述                                                                                                                                                                                        |
| 31:0          | W  | 0   | TX_DATA<br>TX Sample<br>Transmitting left, right channel sample data should be written this register one by one. The left channel sample data is first and then the right channel sample. |

#### 7.2.4.10 DA\_CLKD\_REG

| 偏移地址 : 0x0024 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                     |
| 31:9          | /   | /   | /                                                                                                                                                                                                                                                                                                                                                      |
| 8             | R/W | 0   | MCLKO_EN<br>0: Disable MCLK Output<br>1: Enable MCLK Output<br>Notes: Whether in Slave or Master mode, when this bit is set to 1, MCLK should be output.                                                                                                                                                                                               |
| 7:4           | R/W | 0   | BCLKDIV<br>BCLK Divide Ratio from PLL2<br>0: reserved<br>1: Divide by 1<br>2: Divide by 2<br>3: Divide by 4<br>4: Divide by 6<br>5: Divide by 8<br>6: Divide by 12<br>7: Divide by 16<br>8: Divide by 24<br>9: Divide by 32<br>10: Divide by 48<br>11: Divide by 64<br>12: Divide by 96<br>13: Divide by 128<br>14: Divide by 176<br>15: Divide by 192 |
| 3:0           | R/W | 0   | MCLKDIV<br>MCLK Divide Ratio from PLL2 Output<br>0: reserved                                                                                                                                                                                                                                                                                           |

|  |  |  |                                                                                                                                                                                                                                                                                               |
|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | 1: Divide by 1<br>2: Divide by 2<br>3: Divide by 4<br>4: Divide by 6<br>5: Divide by 8<br>6: Divide by 12<br>7: Divide by 16<br>8: Divide by 24<br>9: Divide by 32<br>10: Divide by 48<br>11: Divide by 64<br>12: Divide by 96<br>13: Divide by 128<br>14: Divide by 176<br>15: Divide by 192 |
|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 7.2.4.11 DA\_TXCNT\_REG

| 偏移地址 : 0x0028 |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                               |
|---------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                             |
| 31:0          | R/W | 0   | TX_CNT<br>TX Sample Counter<br>The audio sample number of sending into TXFIFO. When one sample is put into TXFIFO by DMA or by host IO, the TX sample counter register increases by one. The TX sample counter register can be set to any initial valve at any time. After been updated by the initial value, the counter register should count on base of this initial value. |

#### 7.2.4.12 DA\_RXCNT\_REG

| 偏移地址 : 0x002C |     |     | 默认值: 0x0000_0000                                                                                                                                                                                                                                                                                                                                                          |
|---------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                                                                                                                                                                                                                                                                                        |
| 31:0          | R/W | 0   | RX_CNT<br>RX Sample Counter<br>The audio sample number of writing into RXFIFO. When one sample is written by Digital Audio Engine, the RX sample counter register increases by one. The RX sample counter register can be set to any initial valve at any time. After been updated by the initial value, the counter register should count on base of this initial value. |

#### 7.2.4.13 DA\_CHCFG\_REG

| 偏移地址 : 0x0030 |    |     | 默认值: 0x0000_0000 |
|---------------|----|-----|------------------|
| 位置            | 访问 | 默认值 | 描述               |
| 31:10         | /  | /   | /                |

|     |     |   |                                                                                                                                                |
|-----|-----|---|------------------------------------------------------------------------------------------------------------------------------------------------|
| 9   | R/W | 0 | TX_SLOT_HIZ<br>0: normal mode for the last half cycle of BCLK in the slot<br>1: turn to hi-z state for the last half cycle of BCLK in the slot |
| 8   | R/W | 0 | TXn_STATE<br>0: transfer level 0 when not transferring slot<br>1: turn to hi-z state (TDM) when not transferring slot                          |
| 7   | /   | / | /                                                                                                                                              |
| 6:4 | R/W | 0 | RX_SLOT_NUM<br>RX Channel/Slot Number which between CPU/DMA and FIFO<br>0: 1 channel or slot<br>...<br>7: 8 channels or slots                  |
| 3   | /   | / | /                                                                                                                                              |
| 2:0 | R/W | 0 | TX_SLOT_NUM<br>TX Channel/Slot Number which between CPU/DMA and FIFO<br>0: 1 channel or slot<br>...<br>7: 8 channels or slots                  |

#### 7.2.4.14 DA\_TXnCHSEL\_REG

| 偏移地址 : $0x34 + N \times 4$<br>(N = 0, 1, 2, 3) |     |     | 默认值: 0x0000_0000                                                                                                                                                                              |
|------------------------------------------------|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 位置                                             | 访问  | 默认值 | 描述                                                                                                                                                                                            |
| 31:13                                          | /   | /   | /                                                                                                                                                                                             |
| 12                                             | R/W | 0   | TXn_OFFSET<br>TXn offset tune, TXn data offset to LRCK<br>0: no offset<br>1: data is offset by 1 BCLKs to LRCK                                                                                |
| 11:4                                           | R/W | 0   | TXn_CHEN<br>TXn Channel (slot) enable, bit[11:4] refer to slot [7:0]. When one or more slot(s) is(are) disabled, the affected slot(s) is(are) set to disable state<br>0: disable<br>1: enable |
| 3                                              | /   | /   | /                                                                                                                                                                                             |
| 2:0                                            | R/W | 0   | TXn_CHSEL<br>TXn Channel (slot) number Select for each output<br>0: 1 channel / slot<br>...                                                                                                   |



|  |  |  |                       |
|--|--|--|-----------------------|
|  |  |  | 7: 8 channels / slots |
|--|--|--|-----------------------|

#### 7.2.4.15 DA\_TXnCHMAP\_REG

| 偏移地址： 0x44 + n*4<br>(n = 0, 1, 2, 3) |     |     | 默认值: 0x0000_0000                                                             |
|--------------------------------------|-----|-----|------------------------------------------------------------------------------|
| 位置                                   | 访问  | 默认值 | 描述                                                                           |
| 31                                   | /   | /   | /                                                                            |
| 30:28                                | R/W | 0   | TXn_CH7_MAP<br>TXn Channel7 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 27                                   | /   | /   | /                                                                            |
| 26:24                                | R/W | 0   | TXn_CH6_MAP<br>TXn Channel6 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 23                                   | /   | /   | /                                                                            |
| 22:20                                | R/W | 0   | TXn_CH5_MAP<br>TXn Channel5 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 19                                   | /   | /   | /                                                                            |
| 18:16                                | R/W | 0   | TXn_CH4_MAP<br>TXn Channel4 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 15                                   | /   | /   | /                                                                            |
| 14:12                                | R/W | 0   | TXn_CH3_MAP<br>TXn Channel3 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 11                                   | /   | /   | /                                                                            |

|      |     |   |                                                                              |
|------|-----|---|------------------------------------------------------------------------------|
| 10:8 | R/W | 0 | TXn_CH2_MAP<br>TXn Channel2 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 7    | /   | / | /                                                                            |
| 6:4  | R/W | 0 | TXn_CH1_MAP<br>TXn Channel1 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 3    | /   | / | /                                                                            |
| 2:0  | R/W | 0 | TXn_CH0_MAP<br>TXn Channel0 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |

#### 7.2.4.16 DA\_RXCHSEL\_REG

| 偏移地址 : 0x0054 |     |     | 默认值: 0x0000_0000                                                                                             |
|---------------|-----|-----|--------------------------------------------------------------------------------------------------------------|
| 位置            | 访问  | 默认值 | 描述                                                                                                           |
| 31:13         | /   | /   | /                                                                                                            |
| 12            | R/W | 0   | RX_OFFSET<br>RX offset tune, RX data offset to LRCK<br>0: no offset<br>1: data is offset by 1 BCLKs to LRCK  |
| 11:3          | /   | /   |                                                                                                              |
| 2:0           | R/W | 0   | RX_CHSEL<br>RX Channel (slot) number Select for input<br>0: 1 channel / slot<br>...<br>7: 8 channels / slots |

#### 7.2.4.17 DA\_RXCHMAP\_REG

| 偏移地址 : 0x0058 |    |     | 默认值: 0x0000_0000 |
|---------------|----|-----|------------------|
| 位置            | 访问 | 默认值 | 描述               |
| 31            | /  | /   | /                |

|       |     |   |                                                                            |
|-------|-----|---|----------------------------------------------------------------------------|
| 30:28 | R/W | 0 | RX_CH7_MAP<br>RX Channel7 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 27    | /   | / | /                                                                          |
| 26:24 | R/W | 0 | RX_CH6_MAP<br>RX Channel6 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 23    | /   | / | /                                                                          |
| 22:20 | R/W | 0 | RX_CH5_MAP<br>RX Channel5 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 19    | /   | / | /                                                                          |
| 18:16 | R/W | 0 | RX_CH4_MAP<br>RX Channel4 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 15    | /   | / | /                                                                          |
| 14:12 | R/W | 0 | RX_CH3_MAP<br>RX Channel3 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 11    | /   | / | /                                                                          |
| 10:8  | R/W | 0 | RX_CH2_MAP<br>RX Channel2 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |
| 7     | /   | / | /                                                                          |
| 6:4   | R/W | 0 | RX_CH1_MAP                                                                 |

|     |     |   |                                                                            |
|-----|-----|---|----------------------------------------------------------------------------|
|     |     |   | TX Channel1 Mapping<br>0: 1st sample<br>...<br>7: 8th sample               |
| 3   | /   | / | /                                                                          |
| 2:0 | R/W | 0 | RX_CH0_MAP<br>RX Channel0 Mapping<br>0: 1st sample<br>...<br>7: 8th sample |

## 8 常见问题说明

无。

# 附录 A： 术语表

表 A-1 术语表

|          |                                       |            |
|----------|---------------------------------------|------------|
| <b>A</b> |                                       |            |
| AES      | Advanced Encryption Standard          | 高级加密标注     |
| AGC      | Automatic Gain Control                | 自动增益控制     |
| AHB      | AMBA High-speed Bus                   | 先进高性能总线    |
| APB      | Advanced Peripheral Bus               | 高级外设总线     |
| <b>C</b> |                                       |            |
| CIR      | Consumer IR                           | 用户红外辐射接口   |
| CRC      | Cyclic Redundancy Check               | 循环冗余校验     |
| CSI      | CMOS Sensor Interface                 | CMOS 传感器接口 |
| <b>D</b> |                                       |            |
| DES      | Data Encryption Standard              | 数据加密标准     |
| DLL      | Delay-Locked Loop                     | 延迟锁相环      |
| DVFS     | Dynamic Voltage and Frequency Scaling | 动态调频调压     |
| <b>E</b> |                                       |            |
| eMMC     | Embedded Multi-Media Card             | 嵌入式多媒体卡    |
| <b>I</b> |                                       |            |
| I2S      | IIS                                   | 集成电路内置音频总线 |
| <b>L</b> |                                       |            |
| LSB      | Least Significant Bit                 | 最低有效位      |
| <b>M</b> |                                       |            |
| MAC      | Media Access Control                  | 介质访问控制层    |
| MSB      | Most Significant Bit                  | 最高有效位      |
| <b>P</b> |                                       |            |
| PCM      | Pulse Code Modulation                 | 脉冲编码调制     |
| <b>Q</b> |                                       |            |
| SPI      | Synchronous Peripheral Interface      | 串行外设接口     |
| <b>W</b> |                                       |            |
| WLAN     | Wireless Local Area Network           | 无线局域网      |

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